

NP Transition Matrix

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Chapter 1

Folder instructions

This directory collects the source code of the original programs and the development folders.

1.1 Instructions

This folder is intended to store the sources of the original code and the C++ implementation of the project. Distributing and changing the source is possible under the terms of the GNU GPLv3 license (see *License* below). Use of this code and of any derived implementation should credit the original authors. Scientific publications should do so by citing the following references:

- Saija et al. 2001, ApJ, 559, 993, DOI:10.1086/322350
- Borghese, Denti, Saija 2007, Scattering from Model Nonspherical Particles (ISBN 978-3-540-37413-8), DOI:10.1007/978-3-540-37414-5

Instructions on how to set up and use the code are instead given in the project's `build` folder.

1.2 License

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Chapter 2

Folder instructions

This directory contains scripts and tools to evaluate the code functionality.

2.1 Instructions

The code migration stage can be considered successfully fulfilled with the solution of the single sphere and of the cluster of spheres cases in *C++*. To test the reliability of the code, the *C++* version needs to produce consistent output with respect to the original *FORTRAN* code. Since the output files are generally too large for manual inspection and they can be affected by different approximations or numeric noise, a series of scripts designed to compare the two versions and to pin-point possible differences is provided in this folder. The scripts are written in *python*, therefore they need an available *python* environment to work.

2.2 Create models from YAML input

The execution of `NP_TMcode` applications requires machine readable input files to describe the particle model and the radiation field properties. These files are ASCII files that can be edited with common text editors, but they obey a strict format which complies with the requirements of the original code implementation. In order to make the model creation easier, `NP_TMcode` provides a model editor script named `model_maker.py`. This script reads a model description from a `YAML` file and creates the necessary input configuration for the `NP_TMcode` applications. To run the script, just issue:

```
$PATH_TO_SCRIPT/model_maker.py YAML_CONFIGURATION_FILE
```

where `YAML_CONFIGURATION_FILE` must be a valid `YAML` model description (examples are given in the `test_data` folder). More details are given issuing:

```
$PATH_TO_SCRIPT/model_maker.py -help
```

2.3 Compare the results of **FORTRAN** and **C++** codes

1. Follow the instructions to build and run the *FORTRAN* and *C++* versions of the code.
2. Run the `pycompare.py` script providing the following minimal arguments:

```
$PATH_TO_SCRIPT/pycompare.py -ffile FORTRAN_OUTPUT -cfile C++_OUTPUT
```

(The above assumes that `PATH_TO_SCRIPT` is a variable that expands to the path where the script is located). The required output files are called by default `OSPH` and `OCLU` by *FORTRAN* and `c_OSPH` and `c_OCLU` by *C++*, depending on whether the sphere or the cluster case was executed.

1. Check the output of the script to verify that it detects 0 errors and finishes in a `SUCCESS` state.
2. In case of necessity, add the `--html` argument to produce an *HTML* log showing the possible differences and a classification of their severity.
3. Issuing:

```
$PATH_TO_SCRIPT/pycompare.py -help
```

or just:

```
$PATH_TO_SCRIPT/pycompare.py
```

will print a help screen, giving a brief explanation of all the possible options.

2.4 Estimate the recommended orders for running a model

This script applies the Wiscombe criterion and its modifications to estimate the recommended internal and external orders to run a model. To use it, issue:

```
$PATH_TO_SCRIPT/pywiscombe.py -li|le -wave=WAVELENGTH -rad=RADIUS
```

Wavelength and radius must be expressed in meters, with the radius being equal to the minimum radius encircling the particle. More details are given with:

```
$PATH_TO_SCRIPT/pywiscombe.py -help
```


2.5 Estimate the execution time from a terminal log

Performance estimates can be obtained from the code logging system, assuming the user chose to save the terminal output to some log file. To obtain the time spent by the code in performing a specific operation, the syntax is:

```
$PATH_TO_SCRIPT/pytiming.py -logname=LOG_FILE [-filter=FILTER -threads=NUM_THREADS]
```

where `LOG_FILE` must be the name of a file containing the output that would normally go to terminal, `FILTER` must be the starting character sequence of the log line identifying the operation that should be taken into account, and `NUM_THREADS` is the number of processes that were used to perform the whole calculation loop. In case no filter is given, the script provides an estimate of the total amount of time spent in doing the calculation. This estimate, however, is known not to be reliable, because it ignores thread concurrency effects. A more accurate estimate of the total time spent in executing the code is always saved in a file named `c_timing.log`.

2.6 Extract parameters from the code output

The legacy and HDF5 outputs of `NP_TMcode` applications contain all the values computed by the code. However, in many cases the user is interested in extracting plots of cross-sections, forces and torques as functions of wavelength. The `parse_output.py` script performs this task (currently only on the legacy output). To use the script after running a model, issue:

```
$PATH_TO_SCRIPT/parse_output.py -in CODE_RESULT_FILE -out OUTPUT_BASE_NAME -  
app=[SPH|CLU|INCLU]
```

This will parse the output of `np_sphere` (`app=SPH`), `np_cluster` (`app=CLU`) or `np_inclusion` (`app=INCLU`) and put the results in CSV text files, named after the given `OUTPUT_BASE_NAME`, but with distinguishing suffixes such as `_ics` for integrated cross-sections, `_dcs` for differential ones, `_irp` for integrated radiation pressure forces, etc. More details are given issuing:

```
$PATH_TO_SCRIPT/parse_output.py -help
```

2.7 Extract the dynamic range of a complex matrix

This script examines a complex matrix and extracts histograms of the orders of magnitude of the real parts, the imaginary parts and the absolute values. Optionally, if `matplotlib` is available, a plot of the dynamic range is drawn. To use this script, issue:

```
$PATH_TO_SCRIPT/pydynrange.py MATRIX_FILE
```

where `MATRIX_FILE` must be a CSV representation of the matrix. More details are given issuing:

```
$PATH_TO_SCRIPT/pydynrange.py -help
```

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Chapter 7

Namespace Documentation

7.1 inertia Namespace Reference

Compute the momentum of inertia for a model particle.

Functions

- [main](#) ()
Main execution code.
- [filling_factor](#) (stypes, i_tot_cm, config)
Compute the particle's filling factor.
- [get_centers](#) (config)
Create a dictionary of sphere types.
- [get_cm_inertia_tensor](#) (masses, stypes, centers)
Compute the tensor of inertia of an aggregate with respect to the center of mass.
- [get_inertia_tensor](#) (masses, stypes, centers)
Compute the tensor of inertia of an aggregate with respect to origin of coordinates.
- [get_particle_mass](#) (stypes, config)
Compute the total particle mass.
- [get_sphere_mass](#) (isphere, stypes, config)
Compute the mass of a given sphere.
- [get_sphere_type_mass](#) (itype, stypes, config)
Compute the mass of a given type of sphere.
- [get_total_volume](#) (stypes, config)
Compute the total volume of the spheres.
- [get_types](#) (config)
Create a dictionary of sphere types.
- [index_of](#) (element, array)
Get the index of an element in an array.
- [ingest_line](#) (line)
Transform a line in a sequence of string elements.
- [parse_arguments](#) ()
Parse the command line arguments.
- [print_help](#) ()
Print a command-line help summary.

7.1.1 Detailed Description

The complete description of a model particle requires knowledge of some properties, such as filling factor and symmetry. These can be estimated by comparing the model particle with its ellipsoid of inertia. The purpose of this script is to perform such comparison on NP_TMcode model particles.

7.1.2 Function Documentation

7.1.2.1 filling_factor()

```
inertia.filling_factor (
    stypes,
    i_tot_cm,
    config )
```

In this application the filling factor is defined as the ratio of the sum of the volumes of the spheres that compose the particle with respect to the volume of a homogeneous ellipsoid that has the same rotational properties as the particle.

Parameters

<i>stypes</i>	dict Dictionary of sphere types.
<i>i_tot_cm</i>	numpy.ndarray Inertia tensor of the particle with respect to the center of mass.
<i>config</i>	dict Configuration dictionary.

Returns

ff: float The particle's filling factor.

7.1.2.2 get_centers()

```
inertia.get_centers (
    config )
```

This function opens a NP_TMcode scatterer configuration file and creates sphere type dictionary from it.

Parameters

<i>config</i>	dict Configuration dictionary.
---------------	--------------------------------

Returns

centers: dict Sphere centers dictionary.

7.1.2.3 get_cm_inertia_tensor()

```
inertia.get_cm_inertia_tensor (
    masses,
```

```

    stypes,
    centers )

```

Parameters

<i>masses</i>	<code>numpy.ndarray</code> Array of masses in the aggregate.
<i>stypes</i>	<code>dict</code> Dictionary of sphere types.
<i>centers</i>	<code>dict</code> Dictionary containing the positions of the spheres.

Returns

`ltot: numpy.ndarray` The total tensor of inertia with respect to the center of mass.

7.1.2.4 get_inertia_tensor()

```

inertia.get_inertia_tensor (
    masses,
    stypes,
    centers )

```

Parameters

<i>masses</i>	list-like Array of masses in the aggregate.
<i>stypes</i>	<code>dict</code> Dictionary of sphere types.
<i>centers</i>	<code>dict</code> Dictionary containing the positions of the spheres.

Returns

`ltot: numpy.ndarray` The total tensor of inertia with respect to the reference frame.

7.1.2.5 get_particle_mass()

```

inertia.get_particle_mass (
    stypes,
    config )

```

Parameters

<i>stypes</i>	<code>dict</code> Sphere types dictionary.
<i>config</i>	<code>dict</code> Configuration dictionary.

Returns

`mass: float` The mass of the particle, expressed in kg.

7.1.2.6 get_sphere_mass()

```

inertia.get_sphere_mass (

```

```

    isphere,
    stypes,
    config )

```

Parameters

<i>isphere</i>	int ID of the sphere in the aggregate (base 1).
<i>stypes</i>	dict Sphere types dictionary.
<i>config</i>	dict Configuration dictionary.

Returns

mass: float The mass of the sphere, expressed in kg.

7.1.2.7 get_sphere_type_mass()

```

inertia.get_sphere_type_mass (
    itype,
    stypes,
    config )

```

Parameters

<i>itype</i>	int ID of the sphere type (base 1).
<i>stypes</i>	dict Sphere types dictionary.
<i>config</i>	dict Configuration dictionary.

Returns

type_mass: float The mass of the type of sphere, expressed in kg.

7.1.2.8 get_total_volume()

```

inertia.get_total_volume (
    stypes,
    config )

```

Parameters

<i>stypes</i>	dict Sphere types dictionary.
<i>config</i>	dict Configuration dictionary.

Returns

vtot: float The sum of the volumes of the spheres.

7.1.2.9 get_types()

```
inertia.get_types (
    config )
```

This function opens a NP_TMcode scatterer configuration file and creates sphere type dictionary from it.

Parameters

<i>config</i>	dict Configuration dictionary.
---------------	--------------------------------

Returns

stypes: dict Sphere types dictionary.

7.1.2.10 index_of()

```
inertia.index_of (
    element,
    array )
```

Parameters

<i>element</i>	Anything.
<i>array</i>	array-like Array with the same type of the element sought for.

Returns

index: int 0-based index of the element.

7.1.2.11 ingest_line()

```
inertia.ingest_line (
    line )
```

Parameters

<i>line</i>	string Input file line
-------------	------------------------

Returns

result: List of string elements.

7.1.2.12 main()

```
inertia.main ( )
```

`main()` is the function that handles the creation of the script configuration and the execution of the comparison. It returns an integer value corresponding to the script exit code.

Returns

errors: `int` Number of runtime errors (0 means successful run).

7.1.2.13 parse_arguments()

```
inertia.parse_arguments ( )
```

The script behaviour can be modified through a set of mandatory and optional arguments. Mandatory arguments are those required to execute a meaningful parsing and they are limited to the names of the files that need to be read and the operational mode ("edfb" for scattering model file, "geom" for geometry model file).

Returns

config: `dict` A dictionary containing the script configuration.

7.2 model_maker Namespace Reference

Script to build models from YAML configuration files.

Functions

- [main](#) ()
Main execution code.
- [interpolate_constants](#) (sconf)
Populate the dielectric constant data via interpolation.
- [load_model](#) (model_file)
Create the calculation configuration structure from YAML input.
- [match_grid](#) (sconf)
Populate the dielectric constant data matching a grid.
- [parse_arguments](#) ()
Parse the command line arguments.
- [print_help](#) ()
Print a command-line help summary.
- [print_model_summary](#) (scatterer, geometry)
Print a summary of model properties.
- [random_aggregate](#) (scatterer, geometry, seed, max_rad, max_attempts=100, bbox=(0.0, 0.0, 0.0))
Generate a random cluster aggregate from YAML configuration options.
- [random_compact](#) (scatterer, geometry, seed, max_rad)
Generate a random compact cluster from YAML configuration options.
- [recenter_aggregate](#) (vec_spheres)
Adjust a growing aggregate around its geometric center.
- [test_system_resources](#) (model, gconf, sconf)
Perform a preliminary test of the resources required by the model.
- [write_legacy_gconf](#) (conf)
Write the geometry configuration dictionary to legacy format.
- [write_legacy_sconf](#) (conf)
Write the scatterer configuration dictionary to legacy format.
- [write_obj](#) (scatterer, geometry, max_rad)
Export the model to a set of OBJ files for 3D visualization.

Variables

- `bool allow_3d = True`
3D software generation capability flag.
- `exit_code = main()`
Exit code (0 for success)

7.2.1 Detailed Description

Script to assist in the creation of model input files starting from a YAML descriptor.

The script requires python3.

7.2.2 Function Documentation

7.2.2.1 interpolate_constants()

```
model_maker.interpolate_constants (
    sconf )
```

Parameters

<code>sconf</code>	<code>dict</code> Scatterer configuration dictionary.
--------------------	---

Returns

result: `int` An exit code (0 if successful).

7.2.2.2 load_model()

```
model_maker.load_model (
    model_file )
```

Parameters

<code>model_file</code>	<code>str</code> Full path to the YAML input file.
-------------------------	--

Returns

sconf, gconf: `tuple` A dictionary tuple for scatterer and geometric configurations.

7.2.2.3 main()

```
model_maker.main ( )
```

`main()` is the function that handles the creation of the code configuration. It returns an integer value as exit code, using 0 to signal successful execution.

Returns

result: `int` Exit code (0 = SUCCESS).

7.2.2.4 match_grid()

```
model_maker.match_grid (
    sconf )
```

Important note: if the configuration requests that more than one optical constants file should be used, all the files must provide their constants for the same vector of wavelengths.

Parameters

<i>sconf</i>	<code>dict</code> Scatterer configuration dictionary.
--------------	---

Returns

result: `int` An exit code (0 if successful).

7.2.2.5 parse_arguments()

```
model_maker.parse_arguments ( )
```

The script behaviour can be modified through a set of optional arguments. The purpose of this function is to parse the command line in search for such arguments and prepare the execution accordingly.

Returns

config: `dict` A dictionary containing the script configuration.

7.2.2.6 print_model_summary()

```
model_maker.print_model_summary (
    scatterer,
    geometry )
```

This function provides a summary of useful information concerning the radii of the particle monomers and of the equivalent mass sphere, to assist in the selection of the proper starting orders.

Parameters

<i>scatterer</i>	<code>dict</code> A dictionary for the scatterer configuration.
<i>geometry</i>	<code>dict</code> A dictionary for the geometry configuration.

7.2.2.7 random_aggregate()

```
model_maker.random_aggregate (
    scatterer,
    geometry,
    seed,
    max_rad,
    max_attempts = 100,
    bbox = (0.0, 0.0, 0.0) )
```

This function generates a random aggregate of spheres using radial ejection in random directions of new spheres until they become tangent to the outermost sphere existing in that direction. The result of the generated model is directly saved in the parameters of the scatterer and geometry configuration dictionaries.

Parameters

in, out	<i>scatterer</i>	dict Scatterer configuration dictionary.
in, out	<i>geometry</i>	dict Geometry configuration dictionary.
in	<i>seed</i>	int Seed for the random sequence generation.
in	<i>max_rad</i>	float Maximum allowed radial extension of the aggregate.
in	<i>max_attempts</i>	int Maximum number of attempts to place a particle in any direction (optional, defaults to 100).
in	<i>bbox</i>	tuple(float, float, float) Particle's bounding box in meters (optional).

Returns

result: int Function exit code (0 for success, otherwise number of spheres that could not be placed)

7.2.2.8 random_compact()

```
model_maker.random_compact (
    scatterer,
    geometry,
    seed,
    max_rad )
```

This function generates a random aggregate of spheres using the maximum compactness packaging to fill a spherical volume with given maximum radius, then it proceeds by subtracting random spheres from the outer layers, until the aggregate is reduced to the desired number of spheres. The function can only be used if all sphere types have the same radius. The result of the generated model is directly saved in the parameters of the scatterer and geometry configuration dictionaries.

Parameters

<i>scatterer</i>	dict Scatterer configuration dictionary (gets modified)
<i>geometry</i>	dict Geometry configuration dictionary (gets modified)
<i>seed</i>	int Seed for the random sequence generation
<i>max_rad</i>	float Maximum allowed radial extension of the aggregate

Returns

result: `int` Function exit code (0 for success, otherwise error code)

7.2.2.9 recenter_aggregate()

```
model_maker.recenter_aggregate (
    vec_spheres )
```

To be written.

Parameters

<i>vec_spheres</i>	<code>list</code> A list of sphere position dictionaries.
--------------------	---

Returns

result: `int` Function exit code (0 for success)

7.2.2.10 test_system_resources()

```
model_maker.test_system_resources (
    model,
    gconf,
    sconf )
```

The resolution of models requires the availability of memory resources that increase as a function of the model complexity. This function aims at evaluating the complexity of the model and estimate whether the declared system resources are sufficient to run the calculation.

Parameters

<i>model</i>	<code>dict</code> Model description dictionary.
<i>gconf</i>	<code>dict</code> Geometry description dictionary.
<i>sconf</i>	<code>dict</code> Scattering description dictionary.

7.2.2.11 write_legacy_gconf()

```
model_maker.write_legacy_gconf (
    conf )
```

Parameters

<i>conf</i>	<code>dict</code> Geometry configuration dictionary.
-------------	--

Returns

result: `int` An exit code (0 if successful).

7.2.2.12 write_legacy_sconf()

```
model_maker.write_legacy_sconf (
    sconf )
```

Parameters

<i>sconf</i>	<code>dict</code> Scatterer configuration dictionary.
--------------	---

Returns

result: `int` An exit code (0 if successful).

7.2.2.13 write_obj()

```
model_maker.write_obj (
    scatterer,
    geometry,
    max_rad )
```

This function exports the model as a single OBJ file, containing the information to visualize the particle with 3D software tools. The model file is associated with a MTL material library file, used to assign colors to spheres of different type.

Parameters

<i>scatterer</i>	<code>dict</code> Scatterer configuration dictionary (gets modified)
<i>geometry</i>	<code>dict</code> Geometry configuration dictionary (gets modified)
<i>max_rad</i>	<code>float</code> Maximum allowed radial extension of the aggregate

7.3 parse_output Namespace Reference

Script to extract physical quantities from NPTMcode output files.

Functions

- [main](#) ()
Main execution code.
- [parse_arguments](#) ()
Parse the command line arguments.
- [parse_legacy_oclu](#) (config)

- Parse a legacy output file of np_cluster.*
- [parse_legacy_oinclu \(config\)](#)
Parse a legacy output file of np_inclusion.
- [parse_legacy_osph \(config\)](#)
Parse a legacy output file of np_sphere.
- [print_help \(\)](#)
Print a command-line help summary.

7.3.1 Detailed Description

This script is intended to assist users in the inspection of results. It can be used to extract physical quantities, such as radiation pressure forces and cross-sections from the standard code output files and save the data in machine readable CSV files.

The script execution requires python3.

7.3.2 Function Documentation

7.3.2.1 main()

```
parse_output.main ( )
```

[main\(\)](#) is the function that handles the creation of the script configuration and the execution of the comparison. It returns an integer value corresponding to the number of detected error-level inconsistencies.

Returns

errors: `int` Number of runtime errors (0 means successful run).

7.3.2.2 parse_arguments()

```
parse_output.parse_arguments ( )
```

The script behaviour can be modified through a set of mandatory and optional arguments. Mandatory arguments are those required to execute a meaningful parsing and they are limited to the names of the files that need to be read and written. The other arguments affect the format and the data fields that are sought for.

Returns

config: `dict` A dictionary containing the script configuration.

7.3.2.3 parse_legacy_oclu()

```
parse_output.parse_legacy_oclu (
    config )
```


Parameters

<code>config</code>	<code>dict</code> A dictionary containing the script configuration.
---------------------	---

Returns

`errors: int` The number of encountered errors.

7.3.2.4 parse_legacy_oinclu()

```
parse_output.parse_legacy_oinclu (  
    config )
```

Parameters

<code>config</code>	<code>dict</code> A dictionary containing the script configuration.
---------------------	---

Returns

`errors: int` The number of encountered errors.

7.3.2.5 parse_legacy_osph()

```
parse_output.parse_legacy_osph (  
    config )
```

Parameters

<code>config</code>	<code>dict</code> A dictionary containing the script configuration.
---------------------	---

Returns

`errors: int` The number of encountered errors.

7.4 pycompare Namespace Reference

Script to perform output consistency tests.

Functions

- `main ()`
Main execution code.
- `compare_files (config)`
Perform the comparison of two files.

- `compare_lines` (`f_line`, `c_line`, `config`, `line_num=0`, `num_len=4`, `log_file=None`)
Perform the comparison of two file lines.
- `mismatch_severities` (`str_f_values`, `str_c_values`, `config`)
Determine the severity of a numerical mismatch.
- `parse_arguments` ()
Parse the command line arguments.
- `print_help` ()
Print a command-line help summary.
- `reformat_log` (`config`, `errors`, `warnings`, `noisy`)
Add summary information to the HTML log file.

7.4.1 Detailed Description

Script to calculate the execution time of logged operations.

Comparing the numeric output can be rendered hard by the amount of information contained in a typical output file and the necessity to determine whether a difference is actually significant or just caused by numeric noise hitting negligible values. The task of `pycompare.py` is to compare two output files, in the assumption that they were written by the FORTRAN and the C++ versions of the code and to flag all the possible inconsistencies according to various severity levels (namely: NOISE, WARNING, and ERROR).

After execution, the script returns an exit code, which is set to 0, if no error-level inconsistencies were found, or 1 otherwise. This can be used by subsequent system calls to set up a testing suite checking whether the code is able to reproduce legacy results.

The script execution requires python3.

7.4.2 Function Documentation

7.4.2.1 `compare_files()`

```
pycompare.compare_files (
    config )
```

The comparison is executed as a line-by-line process. In order to process files correctly, it is required that the two input files have exactly the same format (with the exception of some number formatting subtleties that are handled by regular expressions). Therefore, the first comparison step is testing whether the input files have the same number of lines. If this condition is not met, a formatting problem is assumed and every line in the C++ output file is counted as an error. Otherwise, all the numeric values found in the result are compared for consistency within warning tolerance. Warnings and errors are issued if two values do not match by a fractional amount being, respectively, below or above the threshold for warning. A special case is the detection of suspect numeric noise. This arises on very small quantities as a consequence of differences in the level of approximation or in the hardware implementation of the numeric values, which typically has negligible impact on the overall results, even though it can have potentially large fractional mismatches, because it is caused by values that are close to 0.

Numeric noise is filtered by taking advantage from the fact that the output files are formatted in such a way that values with similar physical meaning are written to the same output line. If the comparison results in a large fractional mismatch on a value that is more than 5 orders of magnitude smaller than the highest order of magnitude that was read from the current row, the discrepancy is flagged as a potential noise effect.

Parameters

<i>config</i>	dict A dictionary containing the script configuration.
---------------	--

Returns

`mismatch_count: tuple(int, int, int)` A tuple that bundles together the numbers of detected errors, warnings and noisy values.

7.4.2.2 compare_lines()

```
pycompare.compare_lines (
    f_line,
    c_line,
    config,
    line_num = 0,
    num_len = 4,
    log_file = None )
```

This function handles the line-by-line comparison of coded result files. Depending on whether a HTML log report was requested, it also undertakes the task of formatting the HTML code to show the comparison results as highlighted entries, according to the severity degree of the mismatch.

Parameters

<i>f_line</i>	string A line extracted from the FORTRAN output file.
<i>c_line</i>	string A line extracted from the C++ output file.
<i>config</i>	dict A dictionary containing the script configuration.
<i>line_num</i>	int The number of the current line (0-indexed).
<i>num_len</i>	int The number digits to format the line number tag in the HTML log.
<i>log_file</i>	file A file where to write logging information, if required.

Returns

`mismatch_count: tuple(int, int, int)` A tuple that bundles together the numbers of detected errors, warnings and noisy values.

7.4.2.3 main()

```
pycompare.main ( )
```

`main()` is the function that handles the creation of the script configuration and the execution of the comparison. It returns an integer value corresponding to the number of detected error-level inconsistencies.

Returns

`errors: int` Number of detected error-level inconsistencies.

7.4.2.4 mismatch_severities()

```
pycompare.mismatch_severities (
    str_f_values,
    str_c_values,
    config )
```

The severity scale is currently designed with the following integer codes:

0 - the values are equal

1 - the values are subject to suspect numerical noise (green fonts)

2 - the values are different but below error threshold (blue fonts)

3 - the values differ more than error threshold (red fonts)

Parameters

<i>str_f_values</i>	array(string) The strings representing the numeric values read from the FORTRAN output file.
<i>str_c_values</i>	array(string) The strings representing the numeric values read from the C++ output file.
<i>config</i>	dict A dictionary containing the configuration options from which to read the warning and the error threshold.

Returns

result: array(int) An array of severity codes ordered as the input numeric values.

7.4.2.5 parse_arguments()

```
pycompare.parse_arguments ( )
```

The script behaviour can be modified through a set of mandatory and optional arguments. Mandatory arguments are those required to execute a meaningful comparison and they are limited to the names of the files that need to be compared. The other arguments affect whether the script should produce an HTML log file and what level of detail needs to be included in this log.

Returns

config: dict A dictionary containing the script configuration.

7.4.2.6 reformat_log()

```
pycompare.reformat_log (
    config,
    errors,
    warnings,
    noisy )
```

In the case when a HTML log is requested, it is useful to obtain an overview of the detected inconsistencies. This function undertakes the task of adding a summary of the error, warning and noise counts on top of the log.

Parameters

<i>config</i>	<code>dict</code> A dictionary containing the script configuration.
<i>errors</i>	<code>int</code> The number of errors detected by the comparison.
<i>warnings</i>	<code>int</code> The number of warnings detected by the comparison.
<i>noisy</i>	<code>int</code> The number of noisy values detected by the comparison.

7.5 pydynrange Namespace Reference

Script to calculate the dynamic range of a complex matrix.

Classes

- class [PlotData](#)
Class to represent the dynamic range memory structure.

Functions

- [main](#) ()
Main execution code.
- [get_dynamic_range](#) ([config](#))
Compute the dynamic range of a matrix.
- [parse_arguments](#) ()
Parse the command line arguments.
- [print_help](#) ()
Print a command-line help summary.

7.5.1 Detailed Description

The script execution requires python3.

7.5.2 Function Documentation

7.5.2.1 [get_dynamic_range\(\)](#)

```
pydynrange.get_dynamic_range (  
    config )
```

Parameters

<i>config</i>	<code>dict</code> A dictionary containing the script configuration.
---------------	---

Returns

result: `int` The exit code of the operation (0 for success).

7.5.2.2 main()

```
pydynrange.main ( )
```

`main()` is the function that handles the creation of the script configuration and the execution of the calculation. It returns 0 on successful completion.

Returns

exit_code: `int` 0 on successful completion.

7.5.2.3 parse_arguments()

```
pydynrange.parse_arguments ( )
```

The script behaviour can be modified through a set of mandatory and optional arguments. The only mandatory argument is the name of the log file to be parsed. Additional optional arguments are an operation filter, which should be the starting sequence of the log strings to be included in the timing calculation and the number of threads used during code execution.

Returns

config: `dict` A dictionary containing the script configuration.

7.6 pywiscombe Namespace Reference

Script to obtain a quick guess of the correct orders for a model.

Functions

- `main()`
Main execution code.
- `parse_arguments()`
Parse the command line arguments.
- `print_help()`
Print a command-line help summary.

7.6.1 Detailed Description

This script uses the Wiscombe criterion and its modifications to suggest the proper multipole truncation orders for a particle. It can be used to obtain an estimate of the proper internal order LI (used with the model components or with the model in far field approximation) and the proper external order LE (used for the model in near field approximation).

The script execution requires python3.

7.6.2 Function Documentation

7.6.2.1 `main()`

```
pywiscombe.main ( )
```

`main()` is the function that handles the creation of the script configuration and the execution of the comparison. It returns an integer value corresponding to the number of detected error-level inconsistencies.

Returns

errors: `int` Number of runtime errors (0 means successful run).

7.6.2.2 `parse_arguments()`

```
pywiscombe.parse_arguments ( )
```

The script behaviour can be modified through a set of mandatory and optional arguments. Mandatory arguments are those required to execute a meaningful parsing and they are limited to the names of the files that need to be read and written. The other arguments affect the format and the data fields that are sought for.

Returns

config: `dict` A dictionary containing the script configuration.

7.7 `scale_model` Namespace Reference

Script to re-scale an existing model by multiplicative factor.

Functions

- `main()`
Main execution code.
- `parse_arguments()`
Parse the command line arguments.
- `scale_legacy_edfb(config)`
Scale a model legacy [ScattererConfiguration](#) file.
- `scale_legacy_geom(config)`
Scale a model legacy [GeometryConfiguration](#) file.
- `print_help()`
Print a command-line help summary.

7.7.1 Detailed Description

This script is designed to create scaled versions of pre-existing models, to allow for quick production of sets of particle models characterized by the same structure, but with a different size.

The script execution requires python3.

7.7.2 Function Documentation

7.7.2.1 `main()`

```
scale_model.main ( )
```

`main()` is the function that handles the creation of the script configuration and the execution of the comparison. It returns an integer value corresponding to the script exit code.

Returns

`errors: int` Number of runtime errors (0 means successful run).

7.7.2.2 `parse_arguments()`

```
scale_model.parse_arguments ( )
```

The script behaviour can be modified through a set of mandatory and optional arguments. Mandatory arguments are those required to execute a meaningful parsing and they are limited to the names of the files that need to be read and the operational mode ("edfb" for scattering model file, "geom" for geometry model file).

Returns

`config: dict` A dictionary containing the script configuration.

7.7.2.3 `scale_legacy_edfb()`

```
scale_model.scale_legacy_edfb (
    config )
```

Parameters

<i>config</i>	<code>dict</code> A dictionary containing the script configuration.
---------------	---

Returns

`errors: int` The number of encountered errors.

7.7.2.4 `scale_legacy_geom()`

```
scale_model.scale_legacy_geom (
    config )
```

Parameters

<i>config</i>	<code>dict</code> A dictionary containing the script configuration.
---------------	---

Returns

`errors`: `int` The number of encountered errors.

Chapter 8

Class Documentation

8.1 CCR Struct Reference

CCR data structure.

```
#include <tra_subs.h>
```

Public Attributes

- double **cof**
First coefficient.
- double **cimu**
Second coefficient.

8.1.1 Detailed Description

A structure containing geometrical asymmetry parameter normalization coefficients.

The documentation for this struct was generated from the following file:

- np_tmcode/src/include/tra_subs.h

8.2 CIL Struct Reference

CIL data structure.

```
#include <tra_subs.h>
```

Public Attributes

- int **le**
Maximum external field expansion order.
- int **nlem**
 $NLEM = LE * (LE + 2)$
- int **nlemt**
 $NLEMT = 2 * NLEM.$
- int **mxmpo**
MXMPO is read from T-matrix with $IS < 0$ (not used here).
- int **mxim**
 $2 * mxmpo - 1.$

8.2.1 Detailed Description

A structure containing field expansion order configuration.

The documentation for this struct was generated from the following file:

- [np_tmcode/src/include/tra_subs.h](#)

8.3 ClusterIterationData Class Reference

A data structure representing the information used for a single scale of the CLUSTER case.

```
#include <IterationData.h>
```

Public Member Functions

- [ClusterIterationData](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf, const [mixMPI](#) *mpidata, const int device_count)
[ClusterIterationData](#) default instance constructor.
- [ClusterIterationData](#) (const [ClusterIterationData](#) &rhs)
[ClusterIterationData](#) copy constructor.
- [~ClusterIterationData](#) ()
[ClusterIterationData](#) instance destroyer.
- int [update_orders](#) (double **rcf, int inner_order, int outer_order)
Update field expansion orders.

Static Public Member Functions

- static long [get_size](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
Compute the memory requirements of an instance.

Public Attributes

- [ParticleDescriptor](#) * **c1**
Pointer to a [ParticleDescriptor](#) structure.
- double * **gaps**
Vector of geometric asymmetry factors.
- double ** **tqse**
Components of extinction contribution to radiation torque on a single sphere along k.
- [dcomplex](#) ** **tqspe**
Components of polarized extinction contribution to radiation torque on a single sphere along k.
- double ** **tqss**
Components of scattering contribution to radiation torque on a single sphere along k.
- [dcomplex](#) ** **tqsps**
Components of polarized scattering contribution to radiation torque on a single sphere along k.
- double **** **zpv**
L-dependent coefficients of the geometric asymmetry parameter.
- double ** **gapm**
Mean geometric asymmetry parameters.
- [dcomplex](#) ** **gappm**
Mean geometric asymmetry parameters referred to polarization plane.
- double * **argi**
Imaginary part of the harmonic functions argument.
- double * **args**
Argument of the harmonic functions referred to the scattering plane.
- double ** **gap**
Geometric asymmetry parameters.
- [dcomplex](#) ** **gapp**
Geometric asymmetry parameters referred to polarization plane.
- double ** **tqce**
Components of extinction contribution to radiation torque on the cluster along k.
- [dcomplex](#) ** **tqcpe**
Components of extinction contribution to radiation torque on the cluster along k referred to polarization plane.
- double ** **tqcs**
Components of scattering contribution to radiation torque on the cluster along k.
- [dcomplex](#) ** **tqcps**
Components of scattering contribution to radiation torque on the cluster along k referred to polarization plane.
- double * **duk**
Variation of unitary radiation vector.
- double ** **cextlr**
Cluster extinction cross-section components referred to scattering plane.
- double ** **cext**
Cluster extinction cross-section components referred to meridional plane.
- double ** **cmullr**
Cluster Mueller Transformation Matrix components referred to scattering plane.
- double ** **cmul**
Cluster Mueller Transformation Matrix components referred to meridional plane.
- double * **gapv**
Geometric asymmetry parameter components.
- double * **tqev**
Radiation extinction torque components.
- double * **tqsv**

- *Radiation scattering torque components.*
- double * **u**
 - *Incident unitary vector components.*
- double * **us**
 - *Scattered unitary vector components.*
- double * **un**
 - *Normal unitary vector components.*
- double * **uns**
 - *Normal scattered unitary vector components.*
- double * **up**
 - *Incident unitary vector components on polarization plane.*
- double * **ups**
 - *Scattered unitary vector components on polarization plane.*
- double * **unmp**
 - *Mean unitary vector components normal to polarization plane.*
- double * **unsmp**
 - *Mean scattered unitary vector components normal to polarization plane.*
- double * **upmp**
 - *Mean incident unitary vector components on polarization plane.*
- double * **upsmp**
 - *Mean scattered unitary vector components on polarization plane.*
- double **scan**
 - *Scattering angle.*
- double **cfmp**
 - *Control parameter on incidence direction referred to meridional plane.*
- double **sfmp**
 - *Control parameter on scattering direction referred to meridional plane.*
- double **cfsp**
 - *Control parameter on incidence direction referred to scattering plane.*
- double **sfsp**
 - *Control parameter on scattering direction referred to scattering plane.*
- double **sqsf**
 - $SQSFI = XI^{\wedge} -2.$
- dcomplex * **am_vector**
 - *Vectorized scattering coefficient matrix.*
- dcomplex ** **am**
 - *Scattering coefficient matrix.*
- dcomplex **arg**
 - *ANNOTATION: Argument of harmonic functions.*
- double **vk**
 - *Vacuum magnitude of wave vector.*
- double **wn**
 - *Wave number.*
- double **xip**
 - *ANNOTATION: Normalization scale.*
- int **number_of_scales**
 - *Number of scales (wavelengths) to be computed.*
- int **xiblock**
 - *Size of the block of scales handled by the current process.*
- int **firstxi**
 - *Index of the first scale handled by the current process.*

- int **lastxi**
Index of the last scale handled by the current process.
- int **proc_device**
ID of the GPU used by one MPI process.
- bool **is_first_scale**
flag defining a first iteration

8.3.1 Constructor & Destructor Documentation

8.3.1.1 ClusterIterationData() [1/2]

```
ClusterIterationData::ClusterIterationData (
    GeometryConfiguration * gconf,
    ScattererConfiguration * sconf,
    const mixMPI * mpidata,
    const int device_count )
```

Parameters

<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration object.
<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration object.
<i>mpidata</i>	mixMPI * Pointer to a mixMPI object.
<i>device_count</i>	const int Number of offload devices available on the system.

8.3.1.2 ClusterIterationData() [2/2]

```
ClusterIterationData::ClusterIterationData (
    const ClusterIterationData & rhs )
```

Parameters

<i>rhs</i>	const ClusterIterationData & Reference to the ClusterIterationData object to be copied.
------------	---

8.3.2 Member Function Documentation

8.3.2.1 get_size()

```
long ClusterIterationData::get_size (
    GeometryConfiguration * gconf,
    ScattererConfiguration * sconf ) [static]
```

Parameters

<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration object.
<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration object.

Returns

result: `long` Estimated size in bytes.

8.3.2.2 update_orders()

```
int ClusterIterationData::update_orders (
    double ** rcf,
    int inner_order,
    int outer_order )
```

Parameters

<i>rcf</i>	double ** Matrix of sphere fractional radii.
<i>inner_order</i>	int The new inner expansion order to be set.
<i>outer_order</i>	int The new outer expansion order to be set.

Returns

result: `int` An exit code (0 if successful).

The documentation for this class was generated from the following files:

- `np_tmcode/src/include/IterationData.h`
- `np_tmcode/src/cluster/cluster.cpp`

8.4 ClusterOutputInfo Class Reference

Class to collect output information for scattering from clusters.

```
#include <outputs.h>
```

Public Member Functions

- `ClusterOutputInfo` (`ScattererConfiguration` *sc, `GeometryConfiguration` *gc, const `mixMPI` *mpidata, int `first_xi`=1, int `xi_length`=0)
ClusterOutputInfo default instance constructor.
- `ClusterOutputInfo` (const std::string &hdf5_name)
ClusterOutputInfo constructor from HDF5 input.
- `ClusterOutputInfo` (const int `skip_flag`)
ClusterOutputInfo constructor for the dummy NULL case
- `~ClusterOutputInfo` ()
ClusterOutputInfo instance destroyer.
- long `compute_size` ()
Get the size of a *ClusterOutputInfo* instance in bytes.
- int `insert` (const `ClusterOutputInfo` &rhs)
Insert in the current output data the data of another block.
- int `write` (const std::string &output, const std::string &format)
Write the output to a file.

Static Public Member Functions

- static long `compute_size` (`ScattererConfiguration` *sc, `GeometryConfiguration` *gc, int `first_xi`=1, int `xi_`↔
length=0)

Estimate the size of the structure that would be built for given input.

Public Attributes

- const int & `skip_flag` = `_skip_flag`
Read-only view on skip_flag.
- const int & `first_xi` = `_first_xi`
Read-only view on the ID of the first scale.
- int `nsph`
Number of spheres in the aggregate.
- int `li`
Maximum internal field expansion order.
- int `le`
Maximum external field expansion order.
- int `lm`
Maximum field expansion order.
- `np_int` `mxndm`
Maximum coefficient matrix dimension.
- int `inpol`
Incident polarization flag.
- int `npnt`
Number of points for transition layer integration.
- int `npntts`
Number of points for non-transition layer integration.
- int `iavm`
Flag for intensity.
- int `isam`
Flag for reference to meridional plane.
- int `idfc`
Flag for dielectric function definition.
- double * `vec_x_coords`
Vector of spherical components X Cartesian coordinates.
- double * `vec_y_coords`
Vector of spherical components Y Cartesian coordinates.
- double * `vec_z_coords`
Vector of spherical components Z Cartesian coordinates.
- double `th`
First incident radiation azimuth angle.
- double `thstp`
Incident radiation azimuth angle step.
- double `thlst`
Last incident radiation azimuth angle.
- double `ths`
First scattered radiation azimuth angle.
- double `thsstp`
Scattered radiation azimuth angle step.

- double **thslst**
Last scattered radiation azimuth angle.
- double **ph**
First incident radiation elevation angle.
- double **phstp**
Incident radiation elevation angle step.
- double **phlst**
Last incident radiation elevation angle.
- double **phs**
First scattered radiation elevation angle.
- double **phsstp**
Scattered radiation elevation angle step.
- double **phslst**
Last scattered radiation elevation angle.
- int **ndirs**
Number of directions to be explicitly solved.
- double **exri**
Refractive index of external medium.
- int **nxi**
Number of scales (wavelengths)
- int **xi_block_size**
Number of scales handled by the current process.
- int **jwtm**
Index of the wavelength for T-matrix output.
- int * **vec_jxi**
Vector of scale (wavelength) indices.
- short * **vec_ier**
Vector of error severities (0 - success, 1 - HJV, 2 - DME).
- double * **vec_vk**
Vector of vacuum wave numbers.
- double * **vec_xi**
Vector of computed scales.
- int **configurations**
Number of sphere configurations.
- double * **vec_sphere_sizes**
Vector of sphere sizes (all configurations for every scale).
- **dcomplex** * **vec_sphere_ref_indices**
Vector of sphere refractive indices (all configurations for every scale).
- double * **vec_sphere_scs**
Vector of sphere scattering cross-sections (all configurations for every scale).
- double * **vec_sphere_abs**
Vector of sphere absorption cross-sections (all configurations for every scale).
- double * **vec_sphere_exs**
Vector of sphere extinction cross-sections (all configurations for every scale).
- double * **vec_sphere_albs**
Vector of sphere albedos (all configurations for every scale).
- double * **vec_sphere_sqscs**
Vector of sphere scattering cross-section to geometric section ratios (all configurations for every scale).
- double * **vec_sphere_sqabs**
Vector of sphere absorption cross-sections to geometric section ratios (all configurations for every scale).
- double * **vec_sphere_sqexs**

- Vector of sphere extinction cross-sections to geometric section ratios (all configurations for every scale).*

 - **dcomplex * vec_fsas**

Vector of sphere forward scattering amplitudes (all configurations for every scale).
 - **double * vec_qschus**

Vector of QSCHU = $4 \pi \text{IMAG}(\text{FSAS}) / \text{TOTAL_GEOM_SECTION}$ (all configurations for every scale).
 - **double * vec_pschus**

Vector of PSCHU = $4 \pi \text{REAL}(\text{FSAS}) / \text{TOTAL_GEOM_SECTION}$ (all configurations for every scale).
 - **double * vec_s0mags**

Vector of S0MAG = $\text{ABS}(\text{FSAS}) / (4 \pi k^3)$ (all configurations for every scale).
 - **double * vec_cosavs**

Vector of sphere asymmetry parameters (all configurations for every scale).
 - **double * vec_rapr**

Vector of sphere radiation pressure forces (all configurations for every scale).
 - **double * vec_tqek1**

Vector of extinction contributions to radiation torques along k for parallel linear polarization (all configurations for every scale).
 - **double * vec_tqsk1**

Vector of scattering contributions to radiation torques along k for parallel linear polarization (all configurations for every scale).
 - **double * vec_tqek2**

Vector of extinction contributions to radiation torques along k for perpendicular linear polarization (all configurations for every scale).
 - **double * vec_tqsk2**

Vector of scattering contributions to radiation torques along k for perpendicular linear polarization (all configurations for every scale).
 - **dcomplex * vec_fsat**

Vector of total forward scattering amplitudes (one for each scale).
 - **double * vec_qschut**

Vector of total QSCHU (one for each scale).
 - **double * vec_pschut**

Vector of total PSCHU (one for each scale).
 - **double * vec_s0magt**

Vector of total S0MAG (one for each scale).
 - **double tgs**

Total geometric section.
 - **double * vec_scc1**

Vector of cluster scattering cross-sections (parallel polarization).
 - **double * vec_scc2**

Vector of cluster scattering cross-sections (perpendicular polarization).
 - **double * vec_abc1**

Vector of cluster absorption cross-sections (parallel polarization).
 - **double * vec_abc2**

Vector of cluster absorption cross-sections (perpendicular polarization).
 - **double * vec_exc1**

Vector of cluster extinction cross-sections (parallel polarization).
 - **double * vec_exc2**

Vector of cluster extinction cross-sections (perpendicular polarization).
 - **double * vec_albedc1**

Vector of cluster albedos (parallel polarization).
 - **double * vec_albedc2**

Vector of cluster albedos (perpendicular polarization).

- double * **vec_qscamc1**
Vector of cluster scattering to geometric cross-section ratios (parallel polarization).
- double * **vec_qscamc2**
Vector of cluster scattering to geometric cross-section ratios (perpendicular polarization).
- double * **vec_qabsmc1**
Vector of cluster absorption to geometric cross-section ratios (parallel polarization).
- double * **vec_qabsmc2**
Vector of cluster absorption to geometric cross-section ratios (perpendicular polarization).
- double * **vec_qextmc1**
Vector of cluster extinction to geometric cross-section ratios (parallel polarization).
- double * **vec_qextmc2**
Vector of cluster extinction to geometric cross-section ratios (perpendicular polarization).
- double * **vec_sccrt1**
Vector of cluster-to-sum-of-spheres scattering cross-section ratios (parallel polarization).
- double * **vec_sccrt2**
Vector of cluster-to-sum-of-spheres scattering cross-section ratios (parallel polarization).
- double * **vec_abcr1**
Vector of cluster-to-sum-of-spheres absorption cross-section ratios (parallel polarization).
- double * **vec_abcr2**
Vector of cluster-to-sum-of-spheres absorption cross-section ratios (perpendicular polarization).
- double * **vec_excr1**
Vector of cluster-to-sum-of-spheres extinction cross-section ratios (parallel polarization).
- double * **vec_excr2**
Vector of cluster-to-sum-of-spheres extinction cross-section ratios (perpendicular polarization).
- **dcomplex** * **vec_fsac11**
Vector of forward scattering amplitudes for polarization parallel to incidence (one per scale).
- **dcomplex** * **vec_fsac21**
Vector of forward scattering amplitudes for polarization perpendicular to incidence (one per scale).
- **dcomplex** * **vec_fsac22**
Vector of forward scattering amplitudes for polarization parallel to incidence (one per scale).
- **dcomplex** * **vec_fsac12**
Vector of forward scattering amplitudes for polarization perpendicular to incidence (one per scale).
- double * **vec_qschuc1**
Vector of cluster QSCHU (parallel polarization).
- double * **vec_qschuc2**
Vector of cluster QSCHU (perpendicular polarization).
- double * **vec_pschuc1**
Vector of cluster PSCHU (parallel polarization).
- double * **vec_pschuc2**
Vector of cluster PSCHU (perpendicular polarization).
- double * **vec_s0magc1**
Vector of cluster S0MAG (parallel polarization).
- double * **vec_s0magc2**
Vector of cluster S0MAG (perpendicular polarization).
- double * **vec_cosavc1**
Vector of cluster asymmetry parameters (parallel polarization).
- double * **vec_cosavc2**
Vector of cluster asymmetry parameters (perpendicular polarization).
- double * **vec_raprc1**
Vector of cluster radiation pressure forces (parallel polarization).
- double * **vec_raprc2**

- Vector of cluster radiation pressure forces (perpendicular polarization).*
- double * **vec_fkc1**
Vector of optical forces along incidence direction [N] (parallel polarization).
- double * **vec_fkc2**
Vector of optical forces along incidence direction [N] (perpendicular polarization).
- double * **vec_dir_tidg**
Vector of incidence azimuth directions (one per incidence azimuth).
- double * **vec_dir_pidg**
Vector of incidence elevation directions (one per incidence elevation).
- double * **vec_dir_tsdg**
Vector of scattering azimuth directions (one per scattering azimuth).
- double * **vec_dir_psdg**
Vector of scattering elevation directions (one per scattering elevation).
- double * **vec_dir_scand**
Vector of scattering angles (one per direction).
- double * **vec_dir_cfmp**
Control parameter for incidence plane referred to meridional plane (one per direction).
- double * **vec_dir_sfmp**
Control parameter for scattering plane referred to meridional plane (one per direction).
- double * **vec_dir_cfsp**
Control parameter for incidence plane referred to scattering plane (one per direction).
- double * **vec_dir_sfsp**
Control parameter for scattering plane referred to scattering plane (one per direction).
- double * **vec_dir_un**
Components of the unitary vector perpendicular to incidence plane (three per direction).
- double * **vec_dir_uns**
Components of the unitary vector perpendicular to scattering plane (three per direction).
- dcomplex * **vec_dir_sas11**
Vector of sphere differential scattering amplitude with polarization parallel to parallel incidence field.
- dcomplex * **vec_dir_sas21**
Vector of sphere differential scattering amplitude with polarization perpendicular to the parallel incidence field.
- dcomplex * **vec_dir_sas12**
Vector of sphere differential scattering amplitude with polarization perpendicular to perpendicular incidence field.
- dcomplex * **vec_dir_sas22**
Vector of sphere differential scattering amplitude with polarization parallel the perpendicular incidence field.
- double * **vec_dir_muls**
Vector of sphere Mueller transformation matrices referred to meridional plane.
- double * **vec_dir_mulsr**
Vector of sphere Mueller transformation matrices referred to scattering plane.
- dcomplex * **vec_dir_sat11**
Vector of sphere total differential scattering amplitude with polarization parallel to parallel incidence field.
- dcomplex * **vec_dir_sat21**
Vector of sphere total differential scattering amplitude with polarization perpendicular to the parallel incidence field.
- dcomplex * **vec_dir_sat12**
Vector of sphere total differential scattering amplitude with polarization perpendicular to perpendicular incidence field.
- dcomplex * **vec_dir_sat22**
Vector of sphere total differential scattering amplitude with polarization parallel the perpendicular incidence field.
- double * **vec_dir_scc1**
Vector of cluster differential scattering cross-sections (parallel polarization).
- double * **vec_dir_scc2**
Vector of cluster differential scattering cross-sections (perpendicular polarization).

- double * **vec_dir_abc1**
Vector of cluster differential absorption cross-sections (parallel polarization).
- double * **vec_dir_abc2**
Vector of cluster differential absorption cross-sections (perpendicular polarization).
- double * **vec_dir_exc1**
Vector of cluster differential extinction cross-sections (parallel polarization).
- double * **vec_dir_exc2**
Vector of cluster differential extinction cross-sections (perpendicular polarization).
- double * **vec_dir_albedc1**
Vector of cluster differential albedos (parallel polarization).
- double * **vec_dir_albedc2**
Vector of cluster differential albedos (perpendicular polarization).
- double * **vec_dir_qsc1**
Vector of differential scattering to geometric cross-section ratios (parallel polarization).
- double * **vec_dir_qsc2**
Vector of differential scattering to geometric cross-section ratios (perpendicular polarization).
- double * **vec_dir_qabc1**
Vector of differential absorption to geometric cross-section ratios (parallel polarization).
- double * **vec_dir_qabc2**
Vector of differential absorption to geometric cross-section ratios (perpendicular polarization).
- double * **vec_dir_qexc1**
Vector of differential extinction to geometric cross-section ratios (parallel polarization).
- double * **vec_dir_qexc2**
Vector of differential extinction to geometric cross-section ratios (perpendicular polarization).
- double * **vec_dir_sccr1**
Vector of differential cluster-to-total scattering cross-section ratios (parallel polarization).
- double * **vec_dir_sccr2**
Vector of differential cluster-to-total scattering cross-section ratios (perpendicular polarization).
- double * **vec_dir_abcrt1**
Vector of differential cluster-to-total absorption cross-section ratios (parallel polarization).
- double * **vec_dir_abcrt2**
Vector of differential cluster-to-total absorption cross-section ratios (perpendicular polarization).
- double * **vec_dir_excrt1**
Vector of differential cluster-to-total extinction cross-section ratios (parallel polarization).
- double * **vec_dir_excrt2**
Vector of differential cluster-to-total extinction cross-section ratios (perpendicular polarization).
- **dcomplex** * **vec_dir_fsac11**
Vector of differential cluster forward scattering amplitude with polarization parallel to parallel incidence field (one per direction and scale).
- **dcomplex** * **vec_dir_fsac21**
Vector of differential cluster forward scattering amplitude with polarization perpendicular to the parallel incidence field (one per direction and scale).
- **dcomplex** * **vec_dir_fsac12**
Vector of differential cluster forward scattering amplitude with polarization perpendicular to perpendicular incidence field (one per direction and scale).
- **dcomplex** * **vec_dir_fsac22**
Vector of differential cluster forward scattering amplitude with polarization parallel the perpendicular incidence field (one per direction and scale).
- **dcomplex** * **vec_dir_sac11**
Vector of differential cluster scattering amplitude with polarization parallel to parallel incidence field (one per direction and scale).
- **dcomplex** * **vec_dir_sac21**

Vector of differential cluster scattering amplitude with polarization perpendicular to the parallel incidence field (one per direction and scale).

- **dcomplex * vec_dir_sac12**

Vector of differential cluster scattering amplitude with polarization perpendicular to perpendicular incidence field (one per direction and scale).

- **dcomplex * vec_dir_sac22**

Vector of differential cluster scattering amplitude with polarization parallel the perpendicular incidence field (one per direction and scale).

- **double * vec_dir_qschuc1**

Vector of differential cluster QSCHU (parallel polarization).

- **double * vec_dir_qschuc2**

Vector of differential cluster QSCHU (perpendicular polarization).

- **double * vec_dir_pschuc1**

Vector of differential cluster PSCHU (parallel polarization).

- **double * vec_dir_pschuc2**

Vector of differential cluster PSCHU (perpendicular polarization).

- **double * vec_dir_s0magc1**

Vector of cluster differential S0MAG (parallel polarization).

- **double * vec_dir_s0magc2**

Vector of cluster differential S0MAG (perpendicular polarization).

- **double * vec_dir_cosavc1**

Vector of differential cluster asymmetry parameters (parallel polarization).

- **double * vec_dir_cosavc2**

Vector of differential cluster asymmetry parameters (perpendicular polarization).

- **double * vec_dir_raprc1**

Vector of differential cluster radiation pressure forces (1).

- **double * vec_dir_raprc2**

Vector of differential cluster radiation pressure forces (1).

- **double * vec_dir_flg1**

Vector of differential radiation pressure force components along the polarization direction (parallel polarization).

- **double * vec_dir_flg2**

Vector of differential radiation pressure force components along the polarization direction (perpendicular polarization).

- **double * vec_dir_frc1**

Vector of differential radiation pressure force components perpendicular to the polarization direction (parallel polarization).

- **double * vec_dir_frc2**

Vector of differential radiation pressure force components perpendicular to the polarization direction (perpendicular polarization).

- **double * vec_dir_fkc1**

Vector of differential radiation pressure force components along the incidence direction (parallel polarization).

- **double * vec_dir_fkc2**

Vector of differential radiation pressure force components along the incidence direction (perpendicular polarization).

- **double * vec_dir_fxc1**

Vector of differential radiation pressure force components along the X axis (parallel polarization).

- **double * vec_dir_fxc2**

Vector of differential radiation pressure force components along the X axis (perpendicular polarization).

- **double * vec_dir_fyc1**

Vector of differential radiation pressure force components along the Y axis (parallel polarization).

- **double * vec_dir_fyc2**

Vector of differential radiation pressure force components along the Y axis (perpendicular polarization).

- **double * vec_dir_fzc1**

Vector of differential radiation pressure force components along the Z axis (parallel polarization).

- double * **vec_dir_fzc2**
Vector of differential radiation pressure force components along the Z axis (perpendicular polarization).
- double * **vec_dir_tqelc1**
Vector of differential extinction contribution to radiation torque components along the polarization direction (parallel polarization).
- double * **vec_dir_tqelc2**
Vector of differential extinction contribution to radiation torque components along the polarization direction (perpendicular polarization).
- double * **vec_dir_tqerc1**
Vector of differential extinction contribution to radiation torque components perpendicular to the polarization direction (parallel polarization).
- double * **vec_dir_tqerc2**
Vector of differential extinction contribution to radiation torque components perpendicular to the polarization direction (perpendicular polarization).
- double * **vec_dir_tqekc1**
Vector of differential extinction contribution to radiation torque components along the incidence direction (parallel polarization).
- double * **vec_dir_tqekc2**
Vector of differential extinction contribution to radiation torque components along the incidence direction (perpendicular polarization).
- double * **vec_dir_tqexc1**
Vector of differential extinction contribution to radiation torque components along the X axis (parallel polarization).
- double * **vec_dir_tqexc2**
Vector of differential extinction contribution to radiation torque components along the X axis (perpendicular polarization).
- double * **vec_dir_tqeyc1**
Vector of differential extinction contribution to radiation torque components along the Y axis (parallel polarization).
- double * **vec_dir_tqeyc2**
Vector of differential extinction contribution to radiation torque components along the Y axis (perpendicular polarization).
- double * **vec_dir_tqezc1**
Vector of differential extinction contribution to radiation torque components along the Z axis (parallel polarization).
- double * **vec_dir_tqezc2**
Vector of differential extinction contribution to radiation torque components along the Z axis (perpendicular polarization).
- double * **vec_dir_tqslc1**
Vector of differential scattering contribution to radiation torque components along the polarization direction (parallel polarization).
- double * **vec_dir_tqslc2**
Vector of differential scattering contribution to radiation torque components along the polarization direction (perpendicular polarization).
- double * **vec_dir_tqsrc1**
Vector of differential scattering contribution to radiation torque components perpendicular to the polarization direction (parallel polarization).
- double * **vec_dir_tqsrc2**
Vector of differential scattering contribution to radiation torque components perpendicular to the polarization direction (perpendicular polarization).
- double * **vec_dir_tqskc1**
Vector of differential scattering contribution to radiation torque components along the incidence direction (parallel polarization).
- double * **vec_dir_tqskc2**
Vector of differential scattering contribution to radiation torque components along the incidence direction (perpendicular polarization).
- double * **vec_dir_tqsxc1**

- *Vector of differential scattering contribution to radiation torque components along X axis (parallel polarization).*
- double * **vec_dir_tqsrc2**
- *Vector of differential scattering contribution to radiation torque components along X axis (perpendicular polarization).*
- double * **vec_dir_tqsrc1**
- *Vector of differential scattering contribution to radiation torque components along Y axis (parallel polarization).*
- double * **vec_dir_tqsrc2**
- *Vector of differential scattering contribution to radiation torque components along Y axis (perpendicular polarization).*
- double * **vec_dir_tqsrc1**
- *Vector of differential scattering contribution to radiation torque components along Z axis (parallel polarization).*
- double * **vec_dir_tqsrc2**
- *Vector of differential scattering contribution to radiation torque components along Z axis (perpendicular polarization).*
- double * **vec_dir_mulc**
- *Vector of cluster Mueller transformation matrices referred to meridional plane (16 per direction per scale).*
- double * **vec_dir_mulclr**
- *Vector of cluster Mueller transformation matrices referred to scattering plane (16 per direction per scale).*

Protected Member Functions

- int [write_hdf5](#) (const std::string &file_name)
Write the output to a HDF5 file.
- int [write_legacy](#) (const std::string &output)
Write the output to a legacy text file.

Protected Attributes

- int **_skip_flag**
Flag for skipping mpisend() and mpireceive()
- int **_num_theta**
Number of incident azimuth calculations.
- int **_num_thetas**
Number of scattered azimuth calculations.
- int **_num_phi**
Number of incident elevation calculations.
- int **_num_phis**
Number of scattered elevation calculations.
- int **_first_xi**
ID of the first computed wavelength.

8.4.1 Detailed Description

The results of the calculation can be saved in different formats. It is therefore convenient to have a proper memory structure that allows for storing the results and flushing them in any of the permitted formats with just one operation. The purpose of the [ClusterOutputInfo](#) class is to provide a wrapper for the output of the cluster scattering solver.

8.4.2 Constructor & Destructor Documentation

8.4.2.1 ClusterOutputInfo() [1/3]

```
ClusterOutputInfo::ClusterOutputInfo (
    ScattererConfiguration * sc,
    GeometryConfiguration * gc,
    const mixMPI * mpidata,
    int first_xi = 1,
    int xi_length = 0 )
```

Parameters

<i>sc</i>	ScattererConfiguration * Pointer to a ScattererConfiguration instance.
<i>gc</i>	GeometryConfiguration * Pointer to a GeometryConfiguration instance.
<i>mpidata</i>	const mixMPI* Pointer to a mixMPI instance.
<i>first_xi</i>	int Index of the first scale in output (optional, default is 1).
<i>xi_length</i>	int Number of scales to be included in output (optional, default is 0, meaning all).

8.4.2.2 ClusterOutputInfo() [2/3]

```
ClusterOutputInfo::ClusterOutputInfo (
    const std::string & hdf5_name )
```

Parameters

<i>hdf5_name</i>	const string & Path to the HDF5 file to be read.
------------------	--

8.4.2.3 ClusterOutputInfo() [3/3]

```
ClusterOutputInfo::ClusterOutputInfo (
    const int skip_flag )
```

Parameters

<i>skip_flag</i>	const int must be passed as the 1 constant.
------------------	---

8.4.3 Member Function Documentation

8.4.3.1 compute_size() [1/2]

```
long ClusterOutputInfo::compute_size ( )
```

Returns

size: long Estimated instance size in bytes.

8.4.3.2 compute_size() [2/2]

```
long ClusterOutputInfo::compute_size (
    ScattererConfiguration * sc,
    GeometryConfiguration * gc,
    int first_xi = 1,
    int xi_length = 0 ) [static]
```

Parameters

<i>sc</i>	ScattererConfiguration * Pointer to a ScattererConfiguration instance.
<i>gc</i>	GeometryConfiguration * Pointer to a GeometryConfiguration instance.
<i>first_xi</i>	int Index of the first scale in output (optional, default is 1).
<i>xi_length</i>	int Number of scales to be included in output (optional, default is all).

Returns

size: long Estimated instance size in bytes.

8.4.3.3 insert()

```
int ClusterOutputInfo::insert (
    const ClusterOutputInfo & rhs )
```

Parameters

<i>rhs</i>	const ClusterOutputInfo & Reference to the source data block.
------------	---

Returns

result: int Exit code (0 if successful).

8.4.3.4 write()

```
int ClusterOutputInfo::write (
    const std::string & output,
    const std::string & format )
```

Parameters

<i>output</i>	const string & Path to the output to be written.
<i>format</i>	const string & Output format (one of LEGACY or HDF5).

Returns

result: int Exit code (0 if successful).

8.4.3.5 write_hdf5()

```
int ClusterOutputInfo::write_hdf5 (
    const std::string & file_name ) [protected]
```

Parameters

<i>file_name</i>	const string & Path to the output to be written.
------------------	--

Returns

result: int Exit code (0 if successful).

8.4.3.6 write_legacy()

```
int ClusterOutputInfo::write_legacy (
    const std::string & output ) [protected]
```

This function takes care of writing the output using the legacy formatted ASCII structure. If the output file does not exist, it is created. If it exists, the new content overwritten.

Parameters

<i>output</i>	const string & Path to the output to be written.
---------------	--

Returns

result: int Exit code (0 if successful).

The documentation for this class was generated from the following files:

- [np_tmcode/src/include/outputs.h](#)
- [np_tmcode/src/libnptm/outputs.cpp](#)

8.5 List< T >::element Struct Reference

[List](#) element connector.

```
#include <List.h>
```

Public Attributes

- **T value**
Value of the list element.
- [element](#) * **p_prev**
Pointer to the previous element in the list.

The documentation for this struct was generated from the following file:

- [np_tmcode/src/include/List.h](#)

8.6 FileSchema Class Reference

File content descriptor.

```
#include <file_io.h>
```

Public Member Functions

- [FileSchema](#) (int num_rec, const std::string *rec_types, const std::string *rec_names=NULL)
FileSchema instance constructor.
- [~FileSchema](#) ()
FileSchema instance destroyer.
- int [get_record_number](#) ()
Get the number of records in file.
- std::string * [get_record_names](#) ()
Get a copy of the record names.
- std::string * [get_record_types](#) ()
Get a copy of the record types.

Protected Attributes

- int **num_records**
Number of records contained in the file.
- std::string * **record_names**
Array of record names.
- std::string * **record_types**
Array of record descriptors.

8.6.1 Detailed Description

Accessing binary files requires detailed knowledge of their contents. The [FileSchema](#) class is intended to encapsulate this information and use it as a wrapper to control I/O operations towards different file formats. Any file can be thought of as a sequence of records, which may further contain arbitrarily complex structures. By describing the structure of records, it is possible to support virtually any type of format.

8.6.2 Constructor & Destructor Documentation

8.6.2.1 FileSchema()

```
FileSchema::FileSchema (
    int num_rec,
    const std::string * rec_types,
    const std::string * rec_names = NULL )
```

Parameters

<i>num_rec</i>	int Number of records in the file.
<i>rec_types</i>	string * Description of the records in the file.
<i>rec_names</i>	string * Names of the records in the file.

8.6.3 Member Function Documentation

8.6.3.1 get_record_names()

```
string * FileSchema::get_record_names ( )
```

Returns

rec_types: string * A new vector of strings with description of records.

8.6.3.2 get_record_number()

```
int FileSchema::get_record_number ( ) [inline]
```

Returns

num_records: int The number of records contained in the file.

8.6.3.3 get_record_types()

```
string * FileSchema::get_record_types ( )
```

Returns

rec_names: string * A new vector of strings with record names.

The documentation for this class was generated from the following files:

- [np_tmcode/src/include/file_io.h](#)
- [np_tmcode/src/libnptm/file_io.cpp](#)

8.7 GeometryConfiguration Class Reference

A class to represent the configuration of the scattering geometry.

```
#include <Configuration.h>
```

Public Member Functions

- [GeometryConfiguration](#) (int nsph, int lm, int [in_pol](#), int [npnt](#), int [npntts](#), int meridional_type, int [li](#), int [le](#), [np_int](#) [mxndm](#), int [iavm](#), double *x, double *y, double *z, double in_th_start, double in_th_step, double in_th_end, double sc_th_start, double sc_th_step, double sc_th_end, double in_ph_start, double in_ph_step, double in_ph_end, double sc_ph_start, double sc_ph_step, double sc_ph_end, int [jwtm](#))
Build a scattering geometry configuration structure.
- [GeometryConfiguration](#) (const [GeometryConfiguration](#) &rhs)
Build a scattering geometry configuration structure copying it from an existing one.
- [~GeometryConfiguration](#) ()
Destroy a [GeometryConfiguration](#) instance.
- double [get_sph_x](#) (int index)
Get the X coordinate of a sphere by its index.
- double [get_sph_y](#) (int index)
Get the Y coordinate of a sphere by its index.
- double [get_sph_z](#) (int index)
Get the Z coordinate of a sphere by its index.

Static Public Member Functions

- static [GeometryConfiguration](#) * [from_legacy](#) (const std::string &file_name)
Build geometry configuration from legacy configuration input file.

Public Attributes

- const int & [number_of_spheres](#) = [_number_of_spheres](#)
Read-only view on number of spherical components.
- const int & [l_max](#) = [_l_max](#)
Read-only view on maximum field expansion order.
- const int & [li](#) = [_li](#)
Read-only view on maximum internal field expansion order.
- const int & [le](#) = [_le](#)
Read-only view on maximum external field expansion order.
- const [np_int](#) & [mxndm](#) = [_mxndm](#)
Read-only view on maximum dimension of allocated matrix allowance (deprecated).
- const int & [iavm](#) = [_iavm](#)
Read-only view on the intensity mode flag.
- const int & [in_pol](#) = [_in_pol](#)
Read-only view on incident field polarization status (0 - linear, 1 - circular).
- const int & [npnt](#) = [_npnt](#)
Read-only view on number of points for transition layer integration.
- const int & [npntts](#) = [_npntts](#)
Read-only view on number of points for non-transition layer integration.
- const int & [isam](#) = [_isam](#)
Read-only view on type of meridional plane definition.
- const int & [jwtn](#) = [_jwtn](#)
Read-only view on scale index for T-matrix output.
- const double & [in_theta_start](#) = [_in_theta_start](#)
Read-only view on incident field initial azimuth.
- const double & [in_theta_step](#) = [_in_theta_step](#)
Read-only view on incident field azimuth step.
- const double & [in_theta_end](#) = [_in_theta_end](#)
Read-only view on incident field final azimuth.
- const double & [sc_theta_start](#) = [_sc_theta_start](#)
Read-only view on scattered field initial azimuth.
- const double & [sc_theta_step](#) = [_sc_theta_step](#)
Read-only view on scattered field azimuth step.
- const double & [sc_theta_end](#) = [_sc_theta_end](#)
Read-only view on scattered field final azimuth.
- const double & [in_phi_start](#) = [_in_phi_start](#)
Read-only view on incident field initial elevation.
- const double & [in_phi_step](#) = [_in_phi_step](#)
Read-only view on incident field elevation step.
- const double & [in_phi_end](#) = [_in_phi_end](#)
Read-only view on incident field final elevation.
- const double & [sc_phi_start](#) = [_sc_phi_start](#)
Read-only view on scattered field initial elevation.
- const double & [sc_phi_step](#) = [_sc_phi_step](#)

- *Read-only view on scattered field elevation step.*
const double & **sc_phi_end** = [_sc_phi_end](#)
- *Read-only view on scattered field final elevation.*
const bool & **refine_flag** = [_refine_flag](#)
- *Read-only view on flag for matrix inversion refinement.*
const short & **dyn_order_flag** = [_dyn_order_flag](#)
- *Read-only view on flag for dynamic order management.*
const double & **host_ram_gb** = [_host_ram_gb](#)
- *Read-only view on host RAM in GB.*
const double & **gpu_ram_gb** = [_gpu_ram_gb](#)
- *Read-only view on GPU RAM in GB.*
const double & **tolerance** = [_tolerance](#)
- *Read-only view on sphere overlap tolerance.*
const short & **invert_mode** = [_invert_mode](#)
- *Read-only view on inversion algorithm.*
const double & **accuracy_goal** = [_accuracy_goal](#)
- *Read-only view on desired numerical accuracy for matrix inversion.*
const int & **ref_iters** = [_ref_iters](#)
- *Read-only view on maximum number of refinement iterations.*
const bool & **offload_flag** = [_offload_flag](#)
- *Read-only view on flag to use target offload, if available.*

Protected Attributes

- int **_number_of_spheres**
Number of spherical components.
- int **_l_max**
Maximum field expansion order.
- int **_li**
Maximum internal field expansion order.
- int **_le**
Maximum external field expansion order.
- [np_int](#) **_mxndm**
Maximum dimension of allocated matrix allowance (deprecated).
- int **_iavm**
Flag for intensity.
- int **_in_pol**
Incident field polarization status (0 - linear, 1 - circular).
- int **_npnt**
Number of points for transition layer integration.
- int **_npntts**
Number of points for non-transition layer integration.
- int **_isam**
Type of meridional plane definition.
- int **_jwtm**
Scale index of the T-matrix output.
- double **_in_theta_start**
Incident field initial azimuth.
- double **_in_theta_step**
Incident field azimuth step.

- **double _in_theta_end**
Incident field final azimuth.
- **double _sc_theta_start**
Scattered field initial azimuth.
- **double _sc_theta_step**
Scattered field azimuth step.
- **double _sc_theta_end**
Scattered field final azimuth.
- **double _in_phi_start**
Incident field initial elevation.
- **double _in_phi_step**
Incident field elevation step.
- **double _in_phi_end**
Incident field final elevation.
- **double _sc_phi_start**
Scattered field initial elevation.
- **double _sc_phi_step**
Scattered field elevation step.
- **double _sc_phi_end**
Scattered field final elevation.
- **double * _sph_x**
Vector of spherical components X coordinates.
- **double * _sph_y**
Vector of spherical components Y coordinates.
- **double * _sph_z**
Vector of spherical components Z coordinates.
- **bool _refine_flag**
Flag for matrix inversion refinement.
- **short _dyn_order_flag**
Flag for dynamic order management.
- **double _host_ram_gb**
Host RAM in GB.
- **double _gpu_ram_gb**
GPU RAM in GB.
- **double _tolerance**
Sphere overlap tolerance;.
- **short _invert_mode**
Inversion algorithm (0 - LU, 1 - SVD, 2 - RBT)
- **double _accuracy_goal**
Desired numerical accuracy for matrix inversion.
- **int _ref_iters**
Maximum number of refinement iterations.
- **bool _offload_flag**
Flag to use target offload, if available.

8.7.1 Detailed Description

[GeometryConfiguration](#) is a class designed to store the necessary configuration data to describe the scattering geometry, including the distribution of the particle components, the orientation of the incident and scattered radiation fields and their polarization properties.

8.7.2 Constructor & Destructor Documentation

8.7.2.1 GeometryConfiguration() [1/2]

```
GeometryConfiguration::GeometryConfiguration (
    int nsph,
    int lm,
    int in_pol,
    int npnt,
    int npntts,
    int meridional_type,
    int li,
    int le,
    np_int mxndm,
    int iavm,
    double * x,
    double * y,
    double * z,
    double in_th_start,
    double in_th_step,
    double in_th_end,
    double sc_th_start,
    double sc_th_step,
    double sc_th_end,
    double in_ph_start,
    double in_ph_step,
    double in_ph_end,
    double sc_ph_start,
    double sc_ph_step,
    double sc_ph_end,
    int jwtm )
```

Parameters

<i>nsph</i>	int Number of spheres to be used in calculation.
<i>lm</i>	int Maximum field angular momentum expansion order.
<i>in_pol</i>	int Incident field polarization status
<i>npnt</i>	int ANNOTATION: Number of points for non transition layer integration.
<i>npntts</i>	int ANNOTATION: Number of points for transition layer integration.
<i>meridional_type</i>	int Type of meridional plane definition (<0 for incident angles, 0 if determined by incidence and observation, =1 accross z-axis for incidence and observation, >1 across z-axis as a function of incidence angles for fixed scattering).
<i>li</i>	int
<i>le</i>	int
<i>mxndm</i>	int
<i>iavm</i>	int
<i>x</i>	double* Vector of spherical components X coordinates.
<i>y</i>	double* Vector of spherical components Y coordinates.
<i>z</i>	double* Vector of spherical components Z coordinates.
<i>in_th_start</i>	double Incident field starting azimuth angle.
<i>in_th_step</i>	double Incident field azimuth angle step.
<i>in_th_end</i>	double Incident field final azimuth angle.
<i>sc_th_start</i>	double Scattered field starting azimuth angle.
<i>sc_th_step</i>	double Scattered field azimuth angle step.

Parameters

<i>sc_th_end</i>	double Scattered field final azimuth angle.
<i>in_ph_start</i>	double Incident field starting elevation angle.
<i>in_ph_step</i>	double Incident field elevation angle step.
<i>in_ph_end</i>	double Incident field final elevation angle.
<i>sc_ph_start</i>	double Scattered field starting elevation angle.
<i>sc_ph_step</i>	double Scattered field elevation angle step.
<i>sc_ph_end</i>	double Scattered field final elevation angle.
<i>jwtm</i>	int Transition Matrix layer ID.

8.7.2.2 GeometryConfiguration() [2/2]

```
GeometryConfiguration::GeometryConfiguration (
    const GeometryConfiguration & rhs )
```

Parameters

<i>rhs</i>	GeometryConfiguration preexisting object to copy from.
------------	--

8.7.3 Member Function Documentation

8.7.3.1 from_legacy()

```
GeometryConfiguration * GeometryConfiguration::from_legacy (
    const std::string & file_name ) [static]
```

To allow for consistency tests and backward compatibility, geometry configurations can be built from legacy configuration files. This function replicates the approach implemented by the FORTRAN SPH and CLU codes, but using a C++ oriented work-flow.

Parameters

<i>file_name</i>	string Name of the legacy configuration data file.
------------------	--

Returns

config: GeometryConfiguration* Pointer to object containing the configuration data.

8.7.3.2 get_sph_x()

```
double GeometryConfiguration::get_sph_x (
    int index ) [inline]
```

This is a specialized function to access the X coordinate of a sphere through its index.

Parameters

<i>index</i>	<code>int</code> Index of the sphere (0-based).
--------------	---

Returns

`scale: double` The X coordinate of the requested sphere.

8.7.3.3 get_sph_y()

```
double GeometryConfiguration::get_sph_y (
    int index ) [inline]
```

This is a specialized function to access the Y coordinate of a sphere through its index.

Parameters

<i>index</i>	<code>int</code> Index of the sphere (0-based).
--------------	---

Returns

`scale: double` The Y coordinate of the requested sphere.

8.7.3.4 get_sph_z()

```
double GeometryConfiguration::get_sph_z (
    int index ) [inline]
```

This is a specialized function to access the Z coordinate of a sphere through its index.

Parameters

<i>index</i>	<code>int</code> Index of the sphere (0-based).
--------------	---

Returns

`scale: double` The Z coordinate of the requested sphere.

The documentation for this class was generated from the following files:

- [np_tmcode/src/include/Configuration.h](#)
- [np_tmcode/src/libnptm/Configuration.cpp](#)

8.8 H5File Class Reference

HDF5 I/O wrapper class.

```
#include <file_io.h>
```

Public Member Functions

- [HDFFile](#) (const std::string &name, unsigned int flags=H5F_ACC_EXCL, hid_t fcpl_id=H5P_DEFAULT, hid_t fapl_id=H5P_DEFAULT)
HDFFile instance constructor.
- [~HDFFile](#) ()
HDFFile instance destroyer.
- herr_t **close** ()
Close the current file.
- hid_t **get_file_id** ()
Get current status.
- herr_t **get_status** ()
Get current status.
- bool **is_open** ()
Check whether the attached file is currently open.
- herr_t **read** (const std::string &dataset_name, const std::string &data_type, void *buffer, hid_t mem_space_id=H5S_ALL, hid_t file_space_id=H5S_ALL, hid_t dapl_id=H5P_DEFAULT, hid_t dxpl_id=H5P_DEFAULT)
Read data from attached file.
- herr_t **write** (const std::string &dataset_name, const std::string &data_type, const void *buffer, hid_t mem_space_id=H5S_ALL, hid_t file_space_id=H5S_ALL, hid_t dapl_id=H5P_DEFAULT, hid_t dxpl_id=H5P_DEFAULT)
Write data to attached file.

Static Public Member Functions

- static [HDFFile](#) * **from_schema** ([FileSchema](#) &schema, const std::string &name, unsigned int flags=H5F_ACC_EXCL, hid_t fcpl_id=H5P_DEFAULT, hid_t fapl_id=H5P_DEFAULT)
Create an empty file from a [FileSchema](#) instance.

Protected Attributes

- [List](#)< hid_t > * **id_list**
Identifier list.
- std::string **file_name**
Name of the file.
- bool **file_open_flag**
Flag for the open file status.
- hid_t **file_id**
File identifier handle.
- herr_t **status**
Return status of the last operation.

8.8.1 Detailed Description

This class manages I/O operations toward HDF5 format files.

8.8.2 Constructor & Destructor Documentation

8.8.2.1 HDFFile()

```
HDFFile::HDFFile (
    const std::string & name,
    unsigned int flags = H5F_ACC_EXCL,
    hid_t fcpl_id = H5P_DEFAULT,
    hid_t fapl_id = H5P_DEFAULT )
```

Parameters

<i>name</i>	string Name of the file.
<i>flags</i>	unsigned int File access flags (default is H5F_ACC_EXCL).
<i>fcpl_id</i>	hid_t File creation property list identifier (default is H5P_DEFAULT).
<i>fcpl_id</i>	hid_t File access property list identifier (default is H5P_DEFAULT).

8.8.3 Member Function Documentation

8.8.3.1 from_schema()

```
HDFFile * HDFFile::from_schema (
    FileSchema & schema,
    const std::string & name,
    unsigned int flags = H5F_ACC_EXCL,
    hid_t fcpl_id = H5P_DEFAULT,
    hid_t fcpl_id = H5P_DEFAULT ) [static]
```

Parameters

<i>schema</i>	FileSchema & Reference to FileSchema instance.
<i>name</i>	string Name of the file.
<i>flags</i>	unsigned int File access flags (default is H5F_ACC_EXCL).
<i>fcpl_id</i>	hid_t File creation property list identifier (default is H5P_DEFAULT).
<i>fcpl_id</i>	hid_t File access property list identifier (default is H5P_DEFAULT).

Returns

hdf_file: HDFFile * Pointer to a new, open HDF5 file.

8.8.3.2 read()

```
herr_t HDFFile::read (
    const std::string & dataset_name,
    const std::string & data_type,
    void * buffer,
    hid_t mem_space_id = H5S_ALL,
    hid_t file_space_id = H5S_ALL,
    hid_t dapl_id = H5P_DEFAULT,
    hid_t dxpl_id = H5P_DEFAULT )
```

Parameters

<i>dataset_name</i>	string Name of the dataset to read from.
<i>data_type</i>	string Memory data type identifier.
<i>buffer</i>	hid_t Starting address of the memory sector to store the data.

Parameters

<i>mem_space↔ _id</i>	hid_t Memory data space identifier (defaults to H5S_ALL).
<i>file_space_id</i>	hid_t File space identifier (defaults to H5S_ALL).
<i>dapl_id</i>	hid_t Data access property list identifier (defaults to H5P_DEFAULT).
<i>dxml_id</i>	hid_t Data transfer property list identifier (defaults to H5P_DEFAULT).

Returns

status: herr_t Exit status of the operation.

8.8.3.3 write()

```
herr_t HDFFile::write (
    const std::string & dataset_name,
    const std::string & data_type,
    const void * buffer,
    hid_t mem_space_id = H5S_ALL,
    hid_t file_space_id = H5S_ALL,
    hid_t dapl_id = H5P_DEFAULT,
    hid_t dxml_id = H5P_DEFAULT )
```

Parameters

<i>dataset_name</i>	string Name of the dataset to write to.
<i>data_type</i>	string Memory data type identifier.
<i>buffer</i>	hid_t Starting address of the memory sector to be written.
<i>mem_space↔ _id</i>	hid_t Memory data space identifier (defaults to H5S_ALL).
<i>file_space_id</i>	hid_t File space identifier (defaults to H5S_ALL).
<i>dapl_id</i>	hid_t Data access property list identifier (defaults to H5P_DEFAULT).
<i>dxml_id</i>	hid_t Data transfer property list identifier (defaults to H5P_DEFAULT).

Returns

status: herr_t Exit status of the operation.

The documentation for this class was generated from the following files:

- [np_tmcode/src/include/file_io.h](#)
- [np_tmcode/src/libnptm/file_io.cpp](#)

8.9 InclusionIterationData Class Reference

A data structure representing the information used for a single scale of the INCLUSION case.

```
#include <IterationData.h>
```


Public Member Functions

- [InclusionIterationData](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf, const [mixMPI](#) *mpidata, const int device_count)
[InclusionIterationData](#) default instance constructor.
- [InclusionIterationData](#) (const [InclusionIterationData](#) &rhs)
[InclusionIterationData](#) copy constructor.
- [~InclusionIterationData](#) ()
[InclusionIterationData](#) instance destroyer.
- int [update_orders](#) (double **rcf, int inner_order, int outer_order)
Update field expansion orders.

Static Public Member Functions

- static long [get_size](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
Compute the memory requirements of an instance.

Public Attributes

- int **nimd**
External layer index.
- double **extr**
External layer radius.
- [ParticleDescriptor](#) * **c1**
Pointer to a [ParticleDescriptor](#) structure.
- double * **gaps**
Vector of geometric asymmetry factors.
- double ** **tqse**
Components of extinction contribution to radiation torque on a single sphere along k.
- [dcomplex](#) ** **tqspe**
Components of polarized extinction contribution to radiation torque on a single sphere along k.
- double ** **tqss**
Components of scattering contribution to radiation torque on a single sphere along k.
- [dcomplex](#) ** **tqsps**
Components of polarized scattering contribution to radiation torque on a single sphere along k.
- double **** **zpv**
L-dependent coefficients of the geometric asymmetry parameter.
- double ** **gapm**
Mean geometric asymmetry parameters.
- [dcomplex](#) ** **gappm**
Mean geometric asymmetry parameters referred to polarization plane.
- double * **argi**
Imaginary part of the harmonic functions argument.
- double * **args**
Argument of the harmonic functions referred to the scattering plane.
- double ** **gap**
Geometric asymmetry parameters.
- [dcomplex](#) ** **gapp**
Geometric asymmetry parameters referred to polarization plane.
- double ** **tqce**

- Components of extinction contribution to radiation torque on the cluster along k.*

 - **dcomplex ** tqcpe**

Components of extinction contribution to radiation torque on the cluster along k referred to polarization plane.
 - **double ** tqcs**

Components of scattering contribution to radiation torque on the cluster along k.
 - **dcomplex ** tqcps**

Components of scattering contribution to radiation torque on the cluster along k referred to polarization plane.
 - **double * duk**

Variation of unitary radiation vector.
 - **double ** cextlr**

Cluster extinction cross-section components referred to scattering plane.
 - **double ** cext**

Cluster extinction cross-section components referred to meridional plane.
 - **double ** cmullr**

Cluster Mueller Transformation Matrix components referred to scattering plane.
 - **double ** cmul**

Cluster Mueller Transformation Matrix components referred to meridional plane.
 - **double * gapv**

Geometric asymmetry parameter components.
 - **double * tqev**

Radiation extinction torque components.
 - **double * tqsv**

Radiation scattering torque components.
 - **double * u**

Incident unitary vector components.
 - **double * us**

Scattered unitary vector components.
 - **double * un**

Normal unitary vector components.
 - **double * uns**

Normal scattered unitary vector components.
 - **double * up**

Incident unitary vector components on polarization plane.
 - **double * ups**

Scattered unitary vector components on polarization plane.
 - **double * unmp**

Mean unitary vector components normal to polarization plane.
 - **double * unsmp**

Mean scattered unitary vector components normal to polarization plane.
 - **double * upmp**

Mean incident unitary vector components on polarization plane.
 - **double * upsmp**

Mean scattered unitary vector components on polarization plane.
 - **double scan**

Scattering angle.
 - **double cfmp**

Control parameter on incidence direction referred to meridional plane.
 - **double sfmp**

Control parameter on scattering direction referred to meridional plane.
 - **double cfsp**

Control parameter on incidence direction referred to scattering plane.

- double **sfsp**
Control parameter on scattering direction referred to scattering plane.
- double **sqsfi**
 $SQSFI = XI^{-2}$.
- [dcomplex](#) * **am_vector**
Vectorized scattering coefficient matrix.
- [dcomplex](#) ** **am**
Scattering coefficient matrix.
- [dcomplex](#) **arg**
Argument of harmonic functions.
- double **vk**
Vacuum magnitude of wave vector.
- double **wn**
Wave number.
- double **xip**
ANNOTATION: Normalization scale.
- int **number_of_scales**
Number of scales (wavelengths) to be computed.
- int **xiblock**
Size of the block of scales handled by the current process.
- int **firstxi**
Index of the first scale handled by the current process.
- int **lastxi**
Index of the last scale handled by the current process.
- int **proc_device**
ID of the GPU used by one MPI process.
- bool **is_first_scale**
flag defining a first iteration

Protected Attributes

- double * **vec_zpv**
Vectorized geometric asymmetry parameter components.

8.9.1 Constructor & Destructor Documentation

8.9.1.1 InclusionIterationData() [1/2]

```
InclusionIterationData::InclusionIterationData (
    GeometryConfiguration * gconf,
    ScattererConfiguration * sconf,
    const mixMPI * mpidata,
    const int device_count )
```

Parameters

<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration object.
<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration object.
<i>mpidata</i>	mixMPI * Pointer to a mixMPI object.
<i>device_count</i>	const int Number of offload devices available on the system.

8.9.1.2 InclusionIterationData() [2/2]

```
InclusionIterationData::InclusionIterationData (
    const InclusionIterationData & rhs )
```

Parameters

<i>rhs</i>	const InclusionIterationData & Reference to the InclusionIterationData object to be copied.
------------	---

8.9.2 Member Function Documentation

8.9.2.1 get_size()

```
long InclusionIterationData::get_size (
    GeometryConfiguration * gconf,
    ScattererConfiguration * sconf ) [static]
```

Parameters

<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration object.
<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration object.

Returns

result: long Estimated size in bytes.

8.9.2.2 update_orders()

```
int InclusionIterationData::update_orders (
    double ** rcf,
    int inner_order,
    int outer_order )
```

Parameters

<i>rcf</i>	double ** Matrix of sphere fractional radii.
<i>inner_order</i>	int The new inner expansion order to be set.
<i>outer_order</i>	int The new outer expansion order to be set.

Returns

result: int An exit code (0 if successful).

The documentation for this class was generated from the following files:

- [np_tmcode/src/include/IterationData.h](#)
- [np_tmcode/src/inclusion/inclusion.cpp](#)

8.10 InclusionOutputInfo Class Reference

Class to collect output information for scattering from particle with inclusions.

```
#include <outputs.h>
```

Public Member Functions

- [InclusionOutputInfo](#) ([ScattererConfiguration](#) *sc, [GeometryConfiguration](#) *gc, const [mixMPI](#) *mpidata, int first_xi=1, int xi_length=0)
InclusionOutputInfo default instance constructor.
- [InclusionOutputInfo](#) (const std::string &hdf5_name)
InclusionOutputInfo constructor from HDF5 input.
- [InclusionOutputInfo](#) (const int skip_flag)
InclusionOutputInfo constructor for the dummy NULL case
- [~InclusionOutputInfo](#) ()
InclusionOutputInfo instance destroyer.
- long [compute_size](#) ()
Get the size of an *InclusionOutputInfo* instance in bytes.
- int [insert](#) (const [InclusionOutputInfo](#) &rhs)
Insert in the current output data the data of another block.
- int [write](#) (const std::string &output, const std::string &format)
Write the output to a file.

Static Public Member Functions

- static long [compute_size](#) ([ScattererConfiguration](#) *sc, [GeometryConfiguration](#) *gc, int first_xi=1, int xi_length=0)
Estimate the size of the structure that would be built for given input.

Public Attributes

- const int & [skip_flag](#) = [_skip_flag](#)
Read-only view on skip_flag.
- const int & [first_xi](#) = [_first_xi](#)
Read-only view on the ID of the first scale.
- int [nsph](#)
Number of spheres in the aggregate.
- int [li](#)
Maximum internal field expansion order.
- int [le](#)
Maximum external field expansion order.
- int [lm](#)
Maximum field expansion order.
- [np_int](#) [mxndm](#)
Maximum coefficient matrix dimension.
- int [inpol](#)
Incident polarization flag.
- int [npnt](#)

- Number of points for transition layer integration.*

 - int **npntts**
- Number of points for non-transition layer integration.*

 - int **iavm**
- Flag for intensity.*

 - int **isam**
- Flag for reference to meridional plane.*

 - int **idfc**
- Flag for dielectric function definition.*

 - double * **vec_x_coords**
- Vector of spherical components X Cartesian coordinates.*

 - double * **vec_y_coords**
- Vector of spherical components Y Cartesian coordinates.*

 - double * **vec_z_coords**
- Vector of spherical components Z Cartesian coordinates.*

 - double **th**
- First incident radiation azimuth angle.*

 - double **thstp**
- Incident radiation azimuth angle step.*

 - double **thlst**
- Last incident radiation azimuth angle.*

 - double **ths**
- First scattered radiation azimuth angle.*

 - double **thsstp**
- Scattered radiation azimuth angle step.*

 - double **thslst**
- Last scattered radiation azimuth angle.*

 - double **ph**
- First incident radiation elevation angle.*

 - double **phstp**
- Incident radiation elevation angle step.*

 - double **phlst**
- Last incident radiation elevation angle.*

 - double **phs**
- First scattered radiation elevation angle.*

 - double **phsstp**
- Scattered radiation elevation angle step.*

 - double **phslst**
- Last scattered radiation elevation angle.*

 - int **ndirs**
- Number of directions to be explicitly solved.*

 - double **exri**
- Refractive index of external medium.*

 - int **nxi**
- Number of scales (wavelengths)*

 - int **xi_block_size**
- Number of scales handled by the current process.*

 - int **jwtm**
- Index of the wavelength for T-matrix output.*

 - int * **vec_jxi**
- Vector of scale (wavelength) indices.*

- short * **vec_ier**
Vector of error severities (0 - success, 1 - INDME, 2 - OSPV).
- double * **vec_vk**
Vector of vacuum wave numbers.
- double * **vec_xi**
Vector of computed scales.
- int **configurations**
Number of sphere configurations.
- double * **vec_sphere_sizes**
Vector of sphere sizes (all configurations for every scale).
- [dcomplex](#) * **vec_sphere_ref_indices**
Vector of sphere refractive indices (all configurations for every scale).
- double * **vec_scs1**
Vector of particle scattering cross-sections (parallel polarization).
- double * **vec_scs2**
Vector of particle scattering cross-sections (perpendicular polarization).
- double * **vec_abs1**
Vector of particle absorption cross-sections (parallel polarization).
- double * **vec_abs2**
Vector of particle absorption cross-sections (perpendicular polarization).
- double * **vec_exs1**
Vector of particle extinction cross-sections (parallel polarization).
- double * **vec_exs2**
Vector of particle extinction cross-sections (perpendicular polarization).
- double * **vec_albeds1**
Vector of particle albedos (parallel polarization).
- double * **vec_albeds2**
Vector of particle albedos (perpendicular polarization).
- double * **vec_scsrt1**
Vector of particle scattering-to-geometric cross-sections (parallel polarization).
- double * **vec_scsrt2**
Vector of particle scattering-to-geometric cross-sections (perpendicular polarization).
- double * **vec_absrt1**
Vector of particle absorption-to-geometric cross-sections (parallel polarization).
- double * **vec_absrt2**
Vector of particle absorption-to-geometric cross-sections (perpendicular polarization).
- double * **vec_exsrt1**
Vector of particle extinction-to-geometric cross-sections (parallel polarization).
- double * **vec_exsrt2**
Vector of particle extinction-to-geometric cross-sections (perpendicular polarization).
- double * **vec_qschu1**
Vector of particle QSCHU (parallel polarization).
- double * **vec_qschu2**
Vector of particle QSCHU (perpendicular polarization).
- double * **vec_pschu1**
Vector of particle PSCHU (parallel polarization).
- double * **vec_pschu2**
Vector of particle PSCHU (perpendicular polarization).
- double * **vec_s0mag1**
Vector of particle S0MAG (parallel polarization).
- double * **vec_s0mag2**

- Vector of particle S0MAG (perpendicular polarization).*
- double * **vec_cosav1**
Vector of particle average asymmetry parameter (parallel polarization).
- double * **vec_cosav2**
Vector of particle average asymmetry parameter (perpendicular polarization).
- double * **vec_raprs1**
Vector of particle average radiation pressure force (N - parallel polarization).
- double * **vec_raprs2**
Vector of particle average radiation pressure force (N - perpendicular polarization).
- double * **vec_fk1**
Vector of particle average radiation force along incidence direction (N - parallel polarization).
- double * **vec_fk2**
Vector of particle average radiation force along incidence direction (N - perpendicular polarization).
- **dcomplex** * **vec_fsas11**
Vector of forward scattering amplitudes for polarization parallel to incidence (one per scale).
- **dcomplex** * **vec_fsas21**
Vector of forward scattering amplitudes for polarization perpendicular to incidence (one per scale).
- **dcomplex** * **vec_fsas22**
Vector of forward scattering amplitudes for polarization parallel to incidence (one per scale).
- **dcomplex** * **vec_fsas12**
Vector of forward scattering amplitudes for polarization perpendicular to incidence (one per scale).
- double * **vec_dir_tidg**
Vector of incidence azimuth directions (one per incidence azimuth).
- double * **vec_dir_pidg**
Vector of incidence elevation directions (one per incidence elevation).
- double * **vec_dir_tsdg**
Vector of scattering azimuth directions (one per scattering azimuth).
- double * **vec_dir_psdg**
Vector of scattering elevation directions (one per scattering elevation).
- double * **vec_dir_scand**
Vector of scattering angles (one per direction).
- double * **vec_dir_cfmp**
Control parameter for incidence plane referred to meridional plane (one per direction).
- double * **vec_dir_sfmp**
Control parameter for scattering plane referred to meridional plane (one per direction).
- double * **vec_dir_cfsp**
Control parameter for incidence plane referred to scattering plane (one per direction).
- double * **vec_dir_sfsp**
Control parameter for scattering plane referred to scattering plane (one per direction).
- double * **vec_dir_un**
Components of the unitary vector perpendicular to incidence plane (three per direction).
- double * **vec_dir_uns**
Components of the unitary vector perpendicular to scattering plane (three per direction).
- double * **vec_dir_scs1**
Vector of particle differential scattering cross-sections (parallel polarization).
- double * **vec_dir_scs2**
Vector of particle differential scattering cross-sections (perpendicular polarization).
- double * **vec_dir_abs1**
Vector of particle differential absorption cross-sections (parallel polarization).
- double * **vec_dir_abs2**
Vector of particle differential absorption cross-sections (perpendicular polarization).

- double * **vec_dir_exs1**
Vector of particle differential extinction cross-sections (parallel polarization).
- double * **vec_dir_exs2**
Vector of particle differential extinction cross-sections (perpendicular polarization).
- double * **vec_dir_albeds1**
Vector of particle differential albedos (parallel polarization).
- double * **vec_dir_albeds2**
Vector of particle differential albedos (perpendicular polarization).
- double * **vec_dir_scsrt1**
Vector of particle differential scattering-to-geometric cross-sections (parallel polarization).
- double * **vec_dir_scsrt2**
Vector of particle differential scattering-to-geometric cross-sections (perpendicular polarization).
- double * **vec_dir_absrt1**
Vector of particle differential absorption-to-geometric cross-sections (parallel polarization).
- double * **vec_dir_absrt2**
Vector of particle differential absorption-to-geometric cross-sections (perpendicular polarization).
- double * **vec_dir_exsrt1**
Vector of particle differential extinction-to-geometric cross-sections (parallel polarization).
- double * **vec_dir_exsrt2**
Vector of particle differential extinction-to-geometric cross-sections (perpendicular polarization).
- dcomplex * **vec_dir_fsas11**
Vector of particle differential forward scattering amplitude with polarization parallel to parallel incidence field.
- dcomplex * **vec_dir_fsas21**
Vector of particle differential forward scattering amplitude with polarization perpendicular to the parallel incidence field.
- dcomplex * **vec_dir_fsas12**
Vector of particle differential forward scattering amplitude with polarization perpendicular to perpendicular incidence field.
- dcomplex * **vec_dir_fsas22**
Vector of particle differential forward scattering amplitude with polarization parallel the perpendicular incidence field.
- dcomplex * **vec_dir_sas11**
Vector of particle differential scattering amplitude with polarization parallel to parallel incidence field.
- dcomplex * **vec_dir_sas21**
Vector of particle differential scattering amplitude with polarization perpendicular to the parallel incidence field.
- dcomplex * **vec_dir_sas12**
Vector of particle differential scattering amplitude with polarization perpendicular to perpendicular incidence field.
- dcomplex * **vec_dir_sas22**
Vector of particle differential scattering amplitude with polarization parallel the perpendicular incidence field.
- double * **vec_dir_qschu1**
Vector of differential particle QSCHU (parallel polarization).
- double * **vec_dir_qschu2**
Vector of differential particle QSCHU (perpendicular polarization).
- double * **vec_dir_pschu1**
Vector of differential particle PSCHU (parallel polarization).
- double * **vec_dir_pschu2**
Vector of differential particle PSCHU (perpendicular polarization).
- double * **vec_dir_s0mag1**
Vector of particle differential S0MAG (parallel polarization).
- double * **vec_dir_s0mag2**
Vector of particle differential S0MAG (perpendicular polarization).
- double * **vec_dir_cosav1**

- Vector of differential particle asymmetry parameters (parallel polarization).*

 - double * **vec_dir_cosav2**
- Vector of differential particle asymmetry parameters (perpendicular polarization).*

 - double * **vec_dir_rapr1**
- Vector of differential particle radiation pressure forces (1).*

 - double * **vec_dir_rapr2**
- Vector of differential particle radiation pressure forces (1).*

 - double * **vec_dir_fl1**
- Vector of differential radiation pressure force components along the polarization direction (parallel polarization).*

 - double * **vec_dir_fl2**
- Vector of differential radiation pressure force components along the polarization direction (perpendicular polarization).*

 - double * **vec_dir_fr1**
- Vector of differential radiation pressure force components perpendicular to the polarization direction (parallel polarization).*

 - double * **vec_dir_fr2**
- Vector of differential radiation pressure force components perpendicular to the polarization direction (perpendicular polarization).*

 - double * **vec_dir_fk1**
- Vector of differential radiation pressure force components along the incidence direction (parallel polarization).*

 - double * **vec_dir_fk2**
- Vector of differential radiation pressure force components along the incidence direction (perpendicular polarization).*

 - double * **vec_dir_fx1**
- Vector of differential radiation pressure force components along the X axis (parallel polarization).*

 - double * **vec_dir_fx2**
- Vector of differential radiation pressure force components along the X axis (perpendicular polarization).*

 - double * **vec_dir_fy1**
- Vector of differential radiation pressure force components along the Y axis (parallel polarization).*

 - double * **vec_dir_fy2**
- Vector of differential radiation pressure force components along the Y axis (perpendicular polarization).*

 - double * **vec_dir_fz1**
- Vector of differential radiation pressure force components along the Z axis (parallel polarization).*

 - double * **vec_dir_fz2**
- Vector of differential radiation pressure force components along the Z axis (perpendicular polarization).*

 - double * **vec_dir_tqel1**
- Vector of differential extinction contribution to radiation torque components along the polarization direction (parallel polarization).*

 - double * **vec_dir_tqel2**
- Vector of differential extinction contribution to radiation torque components along the polarization direction (perpendicular polarization).*

 - double * **vec_dir_tqer1**
- Vector of differential extinction contribution to radiation torque components perpendicular to the polarization direction (parallel polarization).*

 - double * **vec_dir_tqer2**
- Vector of differential extinction contribution to radiation torque components perpendicular to the polarization direction (perpendicular polarization).*

 - double * **vec_dir_tqek1**
- Vector of differential extinction contribution to radiation torque components along the incidence direction (parallel polarization).*

 - double * **vec_dir_tqek2**
- Vector of differential extinction contribution to radiation torque components along the incidence direction (perpendicular polarization).*

 - double * **vec_dir_tqex1**

- Vector of differential extinction contribution to radiation torque components along the X axis (parallel polarization).*
- double * **vec_dir_tqex2**
Vector of differential extinction contribution to radiation torque components along the X axis (perpendicular polarization).
- double * **vec_dir_tqey1**
Vector of differential extinction contribution to radiation torque components along the Y axis (parallel polarization).
- double * **vec_dir_tqey2**
Vector of differential extinction contribution to radiation torque components along the Y axis (perpendicular polarization).
- double * **vec_dir_tqez1**
Vector of differential extinction contribution to radiation torque components along the Z axis (parallel polarization).
- double * **vec_dir_tqez2**
Vector of differential extinction contribution to radiation torque components along the Z axis (perpendicular polarization).
- double * **vec_dir_tqsl1**
Vector of differential scattering contribution to radiation torque components along the polarization direction (parallel polarization).
- double * **vec_dir_tqsl2**
Vector of differential scattering contribution to radiation torque components along the polarization direction (perpendicular polarization).
- double * **vec_dir_tqsr1**
Vector of differential scattering contribution to radiation torque components perpendicular to the polarization direction (parallel polarization).
- double * **vec_dir_tqsr2**
Vector of differential scattering contribution to radiation torque components perpendicular to the polarization direction (perpendicular polarization).
- double * **vec_dir_tqsk1**
Vector of differential scattering contribution to radiation torque components along the incidence direction (parallel polarization).
- double * **vec_dir_tqsk2**
Vector of differential scattering contribution to radiation torque components along the incidence direction (perpendicular polarization).
- double * **vec_dir_tqsx1**
Vector of differential scattering contribution to radiation torque components along X axis (parallel polarization).
- double * **vec_dir_tqsx2**
Vector of differential scattering contribution to radiation torque components along X axis (perpendicular polarization).
- double * **vec_dir_tqsy1**
Vector of differential scattering contribution to radiation torque components along Y axis (parallel polarization).
- double * **vec_dir_tqsy2**
Vector of differential scattering contribution to radiation torque components along Y axis (perpendicular polarization).
- double * **vec_dir_tqsx1**
Vector of differential scattering contribution to radiation torque components along Z axis (parallel polarization).
- double * **vec_dir_tqsx2**
Vector of differential scattering contribution to radiation torque components along Z axis (perpendicular polarization).
- double * **vec_dir_mull**
Vector of cluster Mueller transformation matrices referred to meridional plane (16 per direction per scale).
- double * **vec_dir_mulllr**
Vector of cluster Mueller transformation matrices referred to scattering plane (16 per direction per scale).

Protected Member Functions

- int **write_hdf5** (const std::string &file_name)
Write the output to a HDF5 file.
- int **write_legacy** (const std::string &output)
Write the output to a legacy text file.

Protected Attributes

- **int _skip_flag**
Flag for skipping mpisend() and mpireceive()
- **int _num_theta**
Number of incident azimuth calculations.
- **int _num_thetas**
Number of scattered azimuth calculations.
- **int _num_phi**
Number of incident elevation calculations.
- **int _num_phis**
Number of scattered elevation calculations.
- **int _first_xi**
ID of the first computed wavelength.

8.10.1 Detailed Description

The results of the calculation can be saved in different formats. It is therefore convenient to have a proper memory structure that allows for storing the results and flushing them in any of the permitted formats with just one operation. The purpose of the [InclusionOutputInfo](#) class is to provide a wrapper for the output of the particle with inclusions scattering solver.

8.10.2 Constructor & Destructor Documentation

8.10.2.1 InclusionOutputInfo() [1/3]

```
InclusionOutputInfo::InclusionOutputInfo (
    ScattererConfiguration * sc,
    GeometryConfiguration * gc,
    const mixMPI * mpidata,
    int first_xi = 1,
    int xi_length = 0 )
```

Parameters

<i>sc</i>	ScattererConfiguration * Pointer to a ScattererConfiguration instance.
<i>gc</i>	GeometryConfiguration * Pointer to a GeometryConfiguration instance.
<i>mpidata</i>	const mixMPI* Pointer to a mixMPI instance.
<i>first_xi</i>	int Index of the first scale in output (optional, default is 1).
<i>xi_length</i>	int Number of scales to be included in output (optional, default is 0, meaning all).

8.10.2.2 InclusionOutputInfo() [2/3]

```
InclusionOutputInfo::InclusionOutputInfo (
    const std::string & hdf5_name )
```

Parameters

<i>hdf5_name</i>	const string & Path to the HDF5 file to be read.
------------------	--

8.10.2.3 InclusionOutputInfo() [3/3]

```
InclusionOutputInfo::InclusionOutputInfo (
    const int skip_flag )
```

Parameters

<i>skip_flag</i>	const int must be passed as the 1 constant.
------------------	---

8.10.3 Member Function Documentation

8.10.3.1 compute_size() [1/2]

```
long InclusionOutputInfo::compute_size ( )
```

Returns

size: long Estimated instance size in bytes.

8.10.3.2 compute_size() [2/2]

```
long InclusionOutputInfo::compute_size (
    ScattererConfiguration * sc,
    GeometryConfiguration * gc,
    int first_xi = 1,
    int xi_length = 0 ) [static]
```

Parameters

<i>sc</i>	ScattererConfiguration * Pointer to a ScattererConfiguration instance.
<i>gc</i>	GeometryConfiguration * Pointer to a GeometryConfiguration instance.
<i>first_xi</i>	int Index of the first scale in output (optional, default is 1).
<i>xi_length</i>	int Number of scales to be included in output (optional, default is all).

Returns

size: long Estimated instance size in bytes.

8.10.3.3 insert()

```
int InclusionOutputInfo::insert (
    const InclusionOutputInfo & rhs )
```

Parameters

<i>rhs</i>	const InclusionOutputInfo & Reference to the source data block.
------------	---

Returns

result: int Exit code (0 if successful).

8.10.3.4 write()

```
int InclusionOutputInfo::write (
    const std::string & output,
    const std::string & format )
```

Parameters

<i>output</i>	const string & Path to the output to be written.
<i>format</i>	const string & Output format (one of LEGACY or HDF5).

Returns

result: int Exit code (0 if successful).

8.10.3.5 write_hdf5()

```
int InclusionOutputInfo::write_hdf5 (
    const std::string & file_name ) [protected]
```

Parameters

<i>file_name</i>	const string & Path to the output to be written.
------------------	--

Returns

result: int Exit code (0 if successful).

8.10.3.6 write_legacy()

```
int InclusionOutputInfo::write_legacy (
    const std::string & output ) [protected]
```

This function takes care of writing the output using the legacy formatted ASCII structure. If the output file does not exist, it is created. If it exists, the new content is overwritten.

Parameters

<i>output</i>	const string & Path to the output to be written.
---------------	--

Returns

result: int Exit code (0 if successful).

The documentation for this class was generated from the following files:

- np_tmcode/src/include/outputs.h
- np_tmcode/src/libnptm/outputs.cpp

8.11 List< T > Class Template Reference

A class to represent dynamic lists.

```
#include <List.h>
```

Classes

- struct [element](#)
List element connector.

Public Member Functions

- [List](#) (int [length](#)=1)
List constructor.
- [~List](#) ()
Destroy a List instance.
- void [append](#) (T value)
Append an element at the end of the List.
- T [get](#) (int index)
Get the element at given index.
- int [length](#) ()
Get the number of elements in the List.
- void [set](#) (int index, T value)
Set an element by index and value.
- T * [to_array](#) ()
Convert the list to a C array.

Protected Attributes

- int [size](#)
Size of the List.
- [element](#) * [current](#)
Pointer to element affected by last operation.
- [element](#) * [first](#)
Pointer to the first element in the List.
- [element](#) * [last](#)
Pointer to the last element in the List.

8.11.1 Detailed Description

```
template<class T>
class List< T >
```

This class helps in the creation and management of dynamic lists of objects, whose size is not known in advance. [List](#) offers the advantage of saving memory, since only the necessary space will be allocated, but it has the disadvantage of creating an object which is not contiguous in memory and, therefore, is very inefficient for subsequent manipulation.

For this reason, the best use of [List](#) objects is to collect all the desired members and then, once the element number is known, to convert the [List](#) to C array, by calling [List.to_array\(\)](#). This function returns a contiguous array of type T[SIZE] that can be used for indexed access.

8.11.2 Constructor & Destructor Documentation

8.11.2.1 List()

```
template<class T >
List< T >::List (
    int length = 1 ) [inline]
```

Use the constructor [List](#)<T>([int length]) to create a new list with a given size. If the required size is not known in advance, it is recommended to create a [List](#) with SIZE=1 (this is the default behavior) and then to append the elements dynamically, using [List.append\(ELEMENT\)](#) (where ELEMENT needs to be a value of type T, corresponding to the class template specialization). Note that, due to the default behavior, the following calls are equivalent and they both produce an integer [List](#) with size equal to 1:

```
List<int> a = List<int>(1);
List<int> b = List<int>();
```

Parameters

<i>length</i>	int The size of the list to be constructed [OPTIONAL, default=1].
---------------	---

8.11.3 Member Function Documentation

8.11.3.1 append()

```
template<class T >
void List< T >::append (
    T value ) [inline]
```

To dynamically create a list whose size is not known in advance, elements can be appended in an iterative way. Note that element manipulation is much more effective in a C array than in a [List](#) object. For this reason, after the [List](#) has been created, it is strongly advised to convert it to a C array by calling the function [List.to_array\(\)](#).

Parameters

<i>value</i>	T The value of the element to be appended.
--------------	--

8.11.3.2 get()

```
template<class T >
T List< T >::get (
    int index ) [inline]
```

Get the element specified by the index argument. The first element has index 0 and the last one has index [size - 1].

Parameters

<i>index</i>	int The index of the element to be retrieved. 0 for first.
--------------	--

Returns

value T The value of the element at the requested position.

Exceptions

ListOutOfBoundsException	Raised if the index is out of bounds.
--	---------------------------------------

8.11.3.3 length()

```
template<class T >
int List< T >::length ( ) [inline]
```

Get the number of elements currently stored in the [List](#).

Returns

size int The size of the [List](#).

8.11.3.4 set()

```
template<class T >
void List< T >::set (
    int index,
    T value ) [inline]
```

Set the element at the position specified by the index to the value specified by the value argument.

Parameters

<i>index</i>	int The index of the element to be set. 0 for first.
<i>value</i>	int The value to store in the pointed element.

Exceptions

ListOutOfBoundsException	Raised if the index is out of bounds.
--	---------------------------------------

8.11.3.5 to_array()

```
template<class T >
T * List< T >::to_array ( ) [inline]
```

The [List](#) object is useful to dynamically manage a set of objects when the number of elements is not known in advance. However, the resulting object is not contiguously stored in memory. As a result, access to specific elements in the middle of the list is not very effective, because the list needs to be walked every time up to the desired position. In order to avoid this, [List.to_array\(\)](#) makes a conversion from [List](#) to C array, returning a contiguous object, where indexed access can be used.

Returns

array T* A C array of type T and size equal to the [List](#) size.

The documentation for this class was generated from the following file:

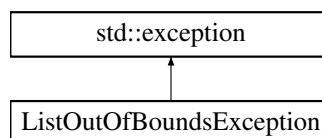
- [np_tmcode/src/include/List.h](#)

8.12 ListOutOfBoundsException Class Reference

Exception for out of bounds [List](#) requests.

```
#include <errors.h>
```

Inheritance diagram for ListOutOfBoundsException:



Public Member Functions

- [ListOutOfBoundsException](#) (int requested, int min, int max)
Exception instance constructor.
- [virtual const char * what \(\) const throw \(\)](#)
Exception message.

Protected Attributes

- `std::string message`
Description of the problem.

8.12.1 Constructor & Destructor Documentation

8.12.1.1 ListOutOfBounds Exception()

```
ListOutOfBounds Exception::ListOutOfBounds Exception (
    int requested,
    int min,
    int max ) [inline]
```

Parameters

<i>requested</i>	<code>int</code> The index that was requested.
<i>min</i>	<code>int</code> The minimum index allowed by the list.
<i>max</i>	<code>int</code> The maximum index allowed by the list.

The documentation for this class was generated from the following file:

- `np_tmcode/src/include/errors.h`

8.13 Logger Class Reference

`Logger` class.

```
#include <logging.h>
```

Public Member Functions

- `Logger (int threshold, FILE *logging_output=stdout, FILE *error_output=stderr)`
Logger instance constructor.
- `~Logger ()`
Logger instance destroyer.
- `void err (const std::string &message) const`
Print a message to the error output.
- `void flush (int level=LOG_DEBUG)`
Print a summary of recurrent messages and clear the stack.
- `void log (const std::string &message, int level=LOG_INFO) const`
Print a message, depending on its logging level.
- `void push (const std::string &message)`
Push a recurrent message to the message stack.

Protected Attributes

- `FILE * err_output`
Pointer to error stream.
- `FILE * log_output`
Pointer to logging stream.
- `std::string * last_message`
Last logged message.
- `int log_threshold`
Threshold of logging level.
- `long repetitions`
Number of identical message repetitions.

8.13.1 Detailed Description

Loggers are objects used to track the execution of a code, reporting activities such as function calls and parameter settings. They can be used to inform the user about the execution of operations at runtime (e.g. by printing messages to the terminal), as well as to record the execution history in appropriate log files. The `Logger` class offers an implementation of logging system complying with the requirements of the NP_TMcode project.

The `Logger` class is designed to work with open files. It is a user responsibility to check that the required log files are properly opened before use, and closed thereafter, if they are not the standard `stdout` and `stderr` streams.

8.13.2 Constructor & Destructor Documentation

8.13.2.1 `Logger()`

```
Logger::Logger (
    int threshold,
    FILE * logging_output = stdout,
    FILE * error_output = stderr )
```

Parameters

<i>threshold</i>	int Threshold of the messages to be included in log. Can be <code>LOG_DEBUG</code> (log everything), <code>LOG_INFO</code> (give detailed information), <code>LOG_WARN</code> (log odd looking effects), or <code>LOG_ERROR</code> (print error messages, always active). The default behaviour is <code>LOG_WARN</code> .
<i>logging_output</i>	FILE * Pointer to an open output file for common messages (optional, default is <code>stdout</code>).
<i>error_output</i>	FILE * Pointer to an open output file for error messages (optional, default is <code>stderr</code>).

8.13.3 Member Function Documentation

8.13.3.1 `err()`

```
void Logger::err (
    const std::string & message ) const
```

Parameters

<i>message</i>	string The message to be printed.
----------------	-----------------------------------

8.13.3.2 `flush()`

```
void Logger::flush (
    int level = LOG_DEBUG )
```

Parameters

<i>level</i>	int The priority level (default is <code>LOG_DEBUG</code> = 0).
--------------	---

8.13.3.3 log()

```
void Logger::log (
    const std::string & message,
    int level = LOG_INFO ) const
```

Parameters

<i>message</i>	string The message to be printed.
<i>level</i>	int The priority level (default is LOG_INFO = 1).

8.13.3.4 push()

```
void Logger::push (
    const std::string & message )
```

When a long stream of identical messages is expected, it may be more convenient to put them in a stack. The stack is the error output stream, so that no memory is consumed. After the call stack is over, or is the message changes, the stack is flushed, meaning that a summary message is written to the logging output and the stack counter is reset to 0.

Parameters

<i>message</i>	string The message to be stacked.
----------------	-----------------------------------

The documentation for this class was generated from the following files:

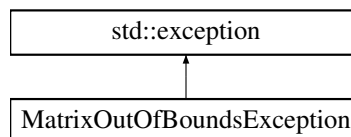
- [np_tmcode/src/include/logging.h](#)
- [np_tmcode/src/libnptm/logging.cpp](#)

8.14 MatrixOutOfBoundsException Class Reference

Exception for access requests out of matrix bounds.

```
#include <errors.h>
```

Inheritance diagram for MatrixOutOfBoundsException:



Public Member Functions

- [MatrixOutOfBoundsException](#) ([const std::string &problem](#))
Exception instance constructor.
- [virtual const char * what \(\) const throw \(\)](#)
Exception message.

Protected Attributes

- `std::string message`
Description of the problem.

8.14.1 Constructor & Destructor Documentation**8.14.1.1 MatrixOutOfBoundsException()**

```
MatrixOutOfBoundsException::MatrixOutOfBoundsException (
    const std::string & problem ) [inline]
```

Parameters

<i>problem</i>	string Description of the problem that occurred.
----------------	--

The documentation for this class was generated from the following file:

- `np_tmcode/src/include/errors.h`

8.15 mixMPI Class Reference

Structure with essential MPI data.

```
#include <Commons.h>
```

Public Member Functions

- `mixMPI ()`
empty [mixMPI](#) instance constructor.
- `mixMPI (const mixMPI &rhs)`
[mixMPI](#) instance constructor from an actual MPI communicator.
- `~mixMPI ()`
[mixMPI](#) instance destroyer.

Public Attributes

- `bool mpirunning`
was MPI initialised?
- `int rank`
MPI rank.
- `int nprocs`
MPI nprocs.

8.15.1 Constructor & Destructor Documentation

8.15.1.1 mixMPI()

```
mixMPI::mixMPI (
    const mixMPI & rhs )
```

`mixMPI` instance constructor copying its contents from a preexisting object.

The documentation for this class was generated from the following files:

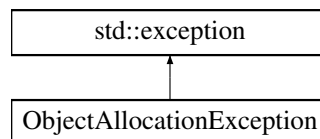
- `np_tmcode/src/include/Commons.h`
- `np_tmcode/src/libnptm/Commons.cpp`

8.16 ObjectAllocationException Class Reference

Exception for object allocation error handlers.

```
#include <errors.h>
```

Inheritance diagram for `ObjectAllocationException`:



Public Member Functions

- `ObjectAllocationException (const std::string &name)`
Exception instance constructor.
- `virtual const char * what () const throw ()`
Exception message.

Protected Attributes

- `std::string file_name`
Name of the file that was accessed.

8.16.1 Constructor & Destructor Documentation

8.16.1.1 ObjectAllocationException()

```
ObjectAllocationException::ObjectAllocationException (
    const std::string & name ) [inline]
```


Parameters

<i>name</i>	string Name of the file that was accessed.
-------------	--

The documentation for this class was generated from the following file:

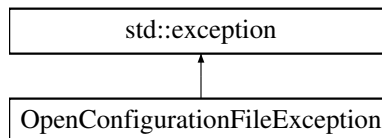
- np_tmcode/src/include/errors.h

8.17 OpenConfigurationFileException Class Reference

Exception for open file error handlers.

```
#include <errors.h>
```

Inheritance diagram for OpenConfigurationFileException:



Public Member Functions

- [OpenConfigurationFileException](#) ([const](#) std::string &[name](#))
Exception instance constructor.
- [virtual const char * what](#) () [const throw](#) ()
Exception message.

Protected Attributes

- std::string **file_name**
Name of the file that was accessed.

8.17.1 Constructor & Destructor Documentation

8.17.1.1 OpenConfigurationFileException()

```
OpenConfigurationFileException::OpenConfigurationFileException (
    const std::string & name ) [inline]
```

Parameters

<i>name</i>	string Name of the file that was accessed.
-------------	--

The documentation for this class was generated from the following file:

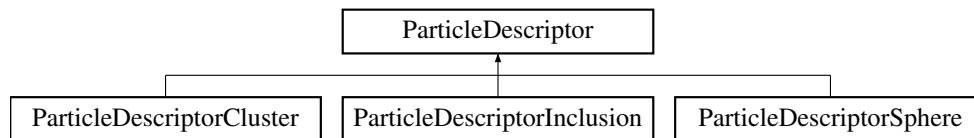
- [np_tmcode/src/include/errors.h](#)

8.18 ParticleDescriptor Class Reference

Basic data structure describing the particle model and its interaction with fields.

```
#include <Commons.h>
```

Inheritance diagram for ParticleDescriptor:



Public Member Functions

- [ParticleDescriptor](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
ParticleDescriptor instance constructor.
- [ParticleDescriptor](#) (const [ParticleDescriptor](#) &rhs)
ParticleDescriptor copy constructor.
- [~ParticleDescriptor](#) ()
ParticleDescriptor instance destroyer.
- [virtual](#) std::string [get_descriptor_type](#) ()
Interface function to return the descriptor type as string.

Static Public Member Functions

- [static long](#) [get_size](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
Compute the memory requirements of an instance.

Public Attributes

- [const short](#) & [class_type](#) = [_class_type](#)
Sub-class identification code.
- [const int](#) & [nsph](#) = [_nsph](#)
Read-only view of number of spheres composing the model.
- [const int](#) & [li](#) = [_li](#)
Read-only view of maximum internal field expansion order.
- [const int](#) & [max_layers](#) = [_max_layers](#)
Read-only view of the maximum number of layers in types.
- [const int](#) & [num_configurations](#) = [_num_configurations](#)
Read-only view of number of different sphere types.
- [const int](#) & [num_layers](#) = [_num_layers](#)
Read-only view of total number of layers from all sphere types.

- `const int & nhspo = _nhspo`
Read-only view of NHSPQ.
- `const int & npnt = _npnt`
Read-only view on number of points for numerical integration in layered spheres.
- `const int & npntts = _npntts`
Read-only view on number of points for numerical integration in transition layer.
- `dcomplex ** rmi`
Matrix of Mie scattering b-coefficients for different orders and spheres.
- `dcomplex ** rei`
Matrix of Matrix of Mie scattering a-coefficients for different orders and spheres.
- `dcomplex ** w`
Matrix of multipole amplitudes for incident and scattered fields.
- `dcomplex * vint`
Vector of intensity components.
- `double * rxx`
Vector of sphere Cartesian X coordinates.
- `double * ryy`
Vector of sphere Cartesian Y coordinates.
- `double * rzz`
Vector of sphere Cartesian Z coordinates.
- `double * ros`
Vector of sphere radii.
- `double ** rc`
Matrix of sphere fractional transition radii.
- `int * iog`
Vector of sphere type identifiers.
- `int * nshl`
Vector of number of layers in sphere type.
- `dcomplex * ris`
ANNOTATION: Square inverted coefficients.
- `dcomplex * dlri`
ANNOTATION: Refractive index variation.
- `dcomplex * dc0`
Vector of dielectric constants.
- `dcomplex * vkt`
Vector of complex refractive indices.
- `double * vsz`
Vector of sizes in units of $2\pi/\text{LAMBDA}$.
- `double gcs`
Total geometric cross-section.
- `dcomplex *** sas`
Tensor of sphere scattering amplitudes.
- `dcomplex ** vints`
Array of vectors of intensity components for single spheres.
- `dcomplex * fsas`
Vector of sphere forward scattering amplitudes.
- `double * sscs`
Vector of scattering cross-sections for spheres.
- `double * sexs`
Vector of extinction cross-sections for spheres.
- `double * sabs`

- Vector of absorption cross-sections for spheres.*

 - **double * sqscs**

Vector of scattering efficiencies for spheres.
- **double * sqexs**

Vector of extinction efficiencies for spheres.
- **double * sqabs**

Vector of absorption efficiencies for spheres.
- **double * gcsv**

Vector of geometric cross-sections for spheres.
- **dcomplex * vintt**

Vector of field intensity components.
- **dcomplex tfsas**

Total forward scattering amplitude of the spheres.
- **dcomplex ** tsas**

Total scattering amplitude of the spheres.
- **double scs**

Sphere scattering cross-section.
- **double ecs**

Sphere extinction cross-section.
- **double acs**

Sphere absorption cross-section.
- **dcomplex ** gis**

ANNOTATION: Geometric tensor.
- **dcomplex ** gls**

ANNOTATION: Geometric tensor.
- **dcomplex ** sam**

Mean scattering amplitude components.
- **const int & le = _le**

Read-only view of maximum external field expansion order.
- **const int & lm = _lm**

Read-only view of maximum field expansion order.
- **const int & nlim = _nlim**

Read-only view of NLIM.
- **const int & nlem = _nlem**

Read-only view of NLEM.
- **const int & nlemt = _nlemt**

Read-only view of NLEMT.
- **const int & ncou = _ncou**

Read-only view of NCOU.
- **const int & litpo = _litpo**

Read-only view of LITPO.
- **const int & litpos = _litpos**

Read-only view of LITPOS.
- **const int & lmpo = _lmpo**

Read-only view of LMPO.
- **const int & lmtpo = _lmtpo**

Read-only view of LMTPO.
- **const int & lmtpos = _lmtpos**

Read-only view of LMTPOS.
- **const int & nv3j = _nv3j**

Read-only view of NV3J.

- `const int & ndi = _ndi`
Read-only view of NDI.
- `const int & ndit = _ndit`
Read-only view of NDIT.
- `const int & ndm = _ndm`
Read-only view of NDM.
- `dcomplex * vh`
ANNOTATION: Hankel vector.
- `dcomplex * vj0`
ANNOTATION: J0 vector.
- `dcomplex * vyhj`
ANNOTATION: Translation vector.
- `dcomplex * vyj0`
ANNOTATION: J0 translation vector.
- `dcomplex vj`
ANNOTATION: J vector.
- `dcomplex ** am0m`
Transition matrix.
- `dcomplex ** fsac`
Cluster forward scattering amplitude.
- `dcomplex ** sac`
Cluster scattering amplitude.
- `dcomplex ** fsacm`
Mean cluster forward scattering amplitude.
- `dcomplex * vintm`
Mean intensity components vector.
- `dcomplex * scscp`
Cluster polarized scattering cross-sections.
- `dcomplex * ecscp`
Cluster polarized extinction cross-sections.
- `dcomplex * scscpm`
Mean cluster polarized scattering cross-sections.
- `dcomplex * ecscpm`
Mean cluster polarized extinction cross-sections.
- `double * v3j0`
ANNOTATION: J0 vector.
- `double * scsc`
Cluster scattering cross-sections.
- `double * ecsc`
Cluster extinction cross-sections.
- `double * scscm`
Mean cluster scattering cross-sections.
- `double * ecscm`
Mean cluster extinction cross-sections.
- `int ** ind3j`
J-vector components index matrix.
- `double * rac3j`
ANNOTATION: J-vector boundary conditions.
- `dcomplex * rm0`
ANNOTATION: M coefficients.
- `dcomplex * re0`

- *ANNOTATION: E coefficients.*
- **dcomplex * rmw**
ANNOTATION: M amplitude coefficients.
- **dcomplex * rew**
ANNOTATION: E amplitude coefficients.
- **dcomplex * tm**
ANNOTATION: Transition M coefficients.
- **dcomplex * te**
ANNOTATION: Transition E coefficients.
- **dcomplex * tm0**
ANNOTATION: Initial transition M coefficients.
- **dcomplex * te0**
ANNOTATION: Initial transition E coefficients.
- **dcomplex ** at**
ANNOTATION: Included particle T-matrix.

Static Public Attributes

- **static const short BASE_TYPE = 0**
Base sub-class identification code.
- **static const short SPHERE_TYPE = 1**
Sphere sub-class identification code.
- **static const short CLUSTER_TYPE = 2**
Cluster sub-class identification code.
- **static const short INCLUSION_TYPE = 3**
Inclusion sub-class identification code.

Protected Attributes

- **short _class_type**
Sub-class identification code.
- **int _nsph**
Number of spheres composing the model.
- **int _li**
Maximum internal field expansion order.
- **int _max_layers**
Maximum number of layers in known sphere types.
- **int _num_configurations**
Number of different sphere types.
- **int _num_layers**
Total number of layers from all sphere types.
- **int _nhspo**
 *$NHSPO = 2 * MAX(NPNT, NPNTTS) - 1$.*
- **int _npnt**
Number of points for numerical integration in layered spheres.
- **int _npntts**
Number of points for numerical integration in transition layer.
- **dcomplex * vec_rmi**
Contiguous space for RMI.
- **dcomplex * vec_rei**

- Contiguous space for REI.
- **dcomplex * vec_w**
 - Contiguous space for W.
- **double * vec_rc**
 - Contiguous space for RC.
- **dcomplex * vec_sas**
 - Contiguous space for SAS.
- **dcomplex * vec_vints**
 - Contiguous space for VINTS.
- **dcomplex * vec_tsas**
 - Contiguous space for TSAS.
- **dcomplex * vec_gis**
 - Contiguous space for GIS.
- **dcomplex * vec_gls**
 - Contiguous space for GLS.
- **dcomplex * vec_sam**
 - Contiguous space for SAM.
- **int _le**
 - Maximum external field expansion order.
- **int _lm**
 - Maximum field expansion order.
- **int _nlim**
 - $NLIM = LI * (LI + 2)$
- **int _nlem**
 - $NLEM = LE * (LE + 2)$
- **int _nlemt**
 - $NLEMT = 2 * NLEM.$
- **int _ncou**
 - $NCOU = NSPH * NSPH - 1.$
- **int _litpo**
 - $LITPO = 2 * LI + 1.$
- **int _litpos**
 - $LITPOS = LITPO * LITPO.$
- **int _lmpo**
 - $LMPO = LM + 1.$
- **int _lmtpo**
 - $LMTPO = LI + LE + 1.$
- **int _lmtpos**
 - $LMTPOS = LMTPO * LMTPO.$
- **int _nv3j**
 - $NV3J = (LM * (LM + 1) * (2 * LM + 7)) / 6.$
- **int _ndi**
 - $NDI = NSPH * NLIM.$
- **int _ndit**
 - $NDIT = 2 * NSPH * NLIM.$
- **dcomplex * vec_am0m**
 - Contiguous space for AM0M.
- **dcomplex * vec_fsac**
 - Contiguous space for FSAC.
- **dcomplex * vec_sac**
 - Contiguous space for SAC.

- `dcomplex * vec_fsacm`
Contiguous space for FSACM.
- `int * vec_ind3j`
Contiguous space for IND3J.
- `int _ndm`
 $NDM = NDIT + NLEMT$.
- `dcomplex * vec_at`
Contiguous space for AT.

8.18.1 Detailed Description

This class forms a base of the data structure collections that are used by the the code to handle different scattering problems. The class implements real members that are used by all code sections.

8.18.2 Constructor & Destructor Documentation

8.18.2.1 ParticleDescriptor() [1/2]

```
ParticleDescriptor::ParticleDescriptor (
    GeometryConfiguration * gconf,
    ScattererConfiguration * sconf )
```

Parameters

<i>gconf</i>	const GeometryConfiguration * Pointer to GeometryConfiguration instance.
<i>sconf</i>	const ScattererConfiguration * Pointer to ScattererConfiguration instance.

8.18.2.2 ParticleDescriptor() [2/2]

```
ParticleDescriptor::ParticleDescriptor (
    const ParticleDescriptor & rhs )
```

Parameters

<i>rhs</i>	const ParticleDescriptor & Reference to ParticleDescriptor object to be copied.
------------	---

8.18.3 Member Function Documentation

8.18.3.1 get_descriptor_type()

```
virtual std::string ParticleDescriptor::get_descriptor_type ( ) [inline], [virtual]
```

Returns

`descriptor_type`: `string` The descriptor type name.

Reimplemented in [ParticleDescriptorCluster](#), [ParticleDescriptorInclusion](#), and [ParticleDescriptorSphere](#).

8.18.3.2 get_size()

```
long ParticleDescriptor::get_size (
    GeometryConfiguration * gconf,
    ScattererConfiguration * sconf ) [static]
```

Parameters

<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration object.
<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration object.

Returns

result: long Estimated size in bytes.

The documentation for this class was generated from the following files:

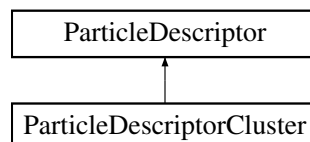
- [np_tmcode/src/include/Commons.h](#)
- [np_tmcode/src/libnptm/Commons.cpp](#)

8.19 ParticleDescriptorCluster Class Reference

The data structure describing a particle model made by a cluster of spheres.

```
#include <Commons.h>
```

Inheritance diagram for ParticleDescriptorCluster:



Public Member Functions

- [ParticleDescriptorCluster](#) ([GeometryConfiguration](#) **gconf*, [ScattererConfiguration](#) **sconf*)
ParticleDescriptorCluster instance constructor.
- [ParticleDescriptorCluster](#) (const [ParticleDescriptorCluster](#) &*rhs*)
ParticleDescriptorCluster copy constructor.
- `std::string` [get_descriptor_type](#) () *override*
Interface function to return the descriptor type as string.
- `int` [update_orders](#) (`int` *inner_order*, `int` *outer_order*)
Update the field expansion orders.

Public Member Functions inherited from [ParticleDescriptor](#)

- [ParticleDescriptor](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
ParticleDescriptor instance constructor.
- [ParticleDescriptor](#) (const [ParticleDescriptor](#) &rhs)
ParticleDescriptor copy constructor.
- [~ParticleDescriptor](#) ()
ParticleDescriptor instance destroyer.

Static Public Member Functions

- [static long get_size](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
Compute the memory requirements of an instance.

Static Public Member Functions inherited from [ParticleDescriptor](#)

- [static long get_size](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
Compute the memory requirements of an instance.

Additional Inherited Members

Public Attributes inherited from [ParticleDescriptor](#)

- [const short & class_type](#) = [_class_type](#)
Sub-class identification code.
- [const int & nsph](#) = [_nsph](#)
Read-only view of number of spheres composing the model.
- [const int & li](#) = [_li](#)
Read-only view of maximum internal field expansion order.
- [const int & max_layers](#) = [_max_layers](#)
Read-only view of the maximum number of layers in types.
- [const int & num_configurations](#) = [_num_configurations](#)
Read-only view of number of different sphere types.
- [const int & num_layers](#) = [_num_layers](#)
Read-only view of total number of layers from all sphere types.
- [const int & nhspo](#) = [_nhspo](#)
Read-only view of NHSP0.
- [const int & npnt](#) = [_npnt](#)
Read-only view on number of points for numerical integration in layered spheres.
- [const int & npntts](#) = [_npntts](#)
Read-only view on number of points for numerical integration in transition layer.
- [dcomplex ** rmi](#)
Matrix of Mie scattering b-coefficients for different orders and spheres.
- [dcomplex ** rei](#)
Matrix of Matrix of Mie scattering a-coefficients for different orders and spheres.
- [dcomplex ** w](#)
Matrix of multipole amplitudes for incident and scattered fields.
- [dcomplex * vint](#)
Vector of intensity components.

- **double * rxx**
Vector of sphere Cartesian X coordinates.
- **double * ryy**
Vector of sphere Cartesian Y coordinates.
- **double * rzz**
Vector of sphere Cartesian Z coordinates.
- **double * ros**
Vector of sphere radii.
- **double ** rc**
Matrix of sphere fractional transition radii.
- **int * iog**
Vector of sphere type identifiers.
- **int * nshl**
Vector of number of layers in sphere type.
- **dcomplex * ris**
ANNOTATION: Square inverted coefficients.
- **dcomplex * dlri**
ANNOTATION: Refractive index variation.
- **dcomplex * dc0**
Vector of dielectric constants.
- **dcomplex * vkt**
Vector of complex refractive indices.
- **double * vsz**
Vector of sizes in units of $2\pi/\text{LAMBDA}$.
- **double gcs**
Total geometric cross-section.
- **dcomplex *** sas**
Tensor of sphere scattering amplitudes.
- **dcomplex ** vints**
Array of vectors of intensity components for single spheres.
- **dcomplex * fsas**
Vector of sphere forward scattering amplitudes.
- **double * sscs**
Vector of scattering cross-sections for spheres.
- **double * sexs**
Vector of extinction cross-sections for spheres.
- **double * sabs**
Vector of absorption cross-sections for spheres.
- **double * sqscs**
Vector of scattering efficiencies for spheres.
- **double * sqexs**
Vector of extinction efficiencies for spheres.
- **double * sqabs**
Vector of absorption efficiencies for spheres.
- **double * gcsv**
Vector of geometric cross-sections for spheres.
- **dcomplex * vintt**
Vector of field intensity components.
- **dcomplex tfsas**
Total forward scattering amplitude of the spheres.
- **dcomplex ** tsas**

- Total scattering amplitude of the spheres.*
- **double** **scs**
Sphere scattering cross-section.
- **double** **ecs**
Sphere extinction cross-section.
- **double** **acs**
Sphere absorption cross-section.
- **dcomplex** ** **gis**
ANNOTATION: Geometric tensor.
- **dcomplex** ** **gls**
ANNOTATION: Geometric tensor.
- **dcomplex** ** **sam**
Mean scattering amplitude components.
- **const int** & **le** = **_le**
Read-only view of maximum external field expansion order.
- **const int** & **lm** = **_lm**
Read-only view of maximum field expansion order.
- **const int** & **nlim** = **_nlim**
Read-only view of NLIM.
- **const int** & **nlem** = **_nlem**
Read-only view of NLEM.
- **const int** & **nlemt** = **_nlemt**
Read-only view of NLEMT.
- **const int** & **ncou** = **_ncou**
Read-only view of NCOU.
- **const int** & **litpo** = **_litpo**
Read-only view of LITPO.
- **const int** & **litpos** = **_litpos**
Read-only view of LITPOS.
- **const int** & **lmpo** = **_lmpo**
Read-only view of LMPO.
- **const int** & **lmtpo** = **_lmtpo**
Read-only view of LMTPO.
- **const int** & **lmtpos** = **_lmtpos**
Read-only view of LMTPOS.
- **const int** & **nv3j** = **_nv3j**
Read-only view of NV3J.
- **const int** & **ndi** = **_ndi**
Read-only view of NDI.
- **const int** & **ndit** = **_ndit**
Read-only view of NDIT.
- **const int** & **ndm** = **_ndm**
Read-only view of NDM.
- **dcomplex** * **vh**
ANNOTATION: Hankel vector.
- **dcomplex** * **vj0**
ANNOTATION: J0 vector.
- **dcomplex** * **vyhj**
ANNOTATION: Translation vector.
- **dcomplex** * **vyj0**
ANNOTATION: J0 translation vector.

- **dcomplex** **vj**
ANNOTATION: J vector.
- **dcomplex** ** **am0m**
Transition matrix.
- **dcomplex** ** **fsac**
Cluster forward scattering amplitude.
- **dcomplex** ** **sac**
Cluster scattering amplitude.
- **dcomplex** ** **fsacm**
Mean cluster forward scattering amplitude.
- **dcomplex** * **vintm**
Mean intensity components vector.
- **dcomplex** * **scscp**
Cluster polarized scattering cross-sections.
- **dcomplex** * **ecscp**
Cluster polarized extinction cross-sections.
- **dcomplex** * **scscpm**
Mean cluster polarized scattering cross-sections.
- **dcomplex** * **ecscpm**
Mean cluster polarized extinction cross-sections.
- **double** * **v3j0**
ANNOTATION: J0 vector.
- **double** * **scsc**
Cluster scattering cross-sections.
- **double** * **ecsc**
Cluster extinction cross-sections.
- **double** * **scscm**
Mean cluster scattering cross-sections.
- **double** * **ecscm**
Mean cluster extinction cross-sections.
- **int** ** **ind3j**
J-vector components index matrix.
- **double** * **rac3j**
ANNOTATION: J-vector boundary conditions.
- **dcomplex** * **rm0**
ANNOTATION: M coefficients.
- **dcomplex** * **re0**
ANNOTATION: E coefficients.
- **dcomplex** * **rmw**
ANNOTATION: M amplitude coefficients.
- **dcomplex** * **rew**
ANNOTATION: E amplitude coefficients.
- **dcomplex** * **tm**
ANNOTATION: Transition M coefficients.
- **dcomplex** * **te**
ANNOTATION: Transition E coefficients.
- **dcomplex** * **tm0**
ANNOTATION: Initial transition M coefficients.
- **dcomplex** * **te0**
ANNOTATION: Initial transition E coefficients.
- **dcomplex** ** **at**
ANNOTATION: Included particle T-matrix.

Static Public Attributes inherited from [ParticleDescriptor](#)

- **static const short** **BASE_TYPE** = 0
Base sub-class identification code.
- **static const short** **SPHERE_TYPE** = 1
Sphere sub-class identification code.
- **static const short** **CLUSTER_TYPE** = 2
Cluster sub-class identification code.
- **static const short** **INCLUSION_TYPE** = 3
Inclusion sub-class identification code.

Protected Attributes inherited from [ParticleDescriptor](#)

- **short** **_class_type**
Sub-class identification code.
- **int** **_nsph**
Number of spheres composing the model.
- **int** **_li**
Maximum internal field expansion order.
- **int** **_max_layers**
Maximum number of layers in known sphere types.
- **int** **_num_configurations**
Number of different sphere types.
- **int** **_num_layers**
Total number of layers from all sphere types.
- **int** **_nhspo**
 $NHSPO = 2 * MAX(NPNT, NPNTTS) - 1.$
- **int** **_npnt**
Number of points for numerical integration in layered spheres.
- **int** **_npntts**
Number of points for numerical integration in transition layer.
- **dcomplex** * **vec_rmi**
Contiguous space for RMI.
- **dcomplex** * **vec_rei**
Contiguous space for REI.
- **dcomplex** * **vec_w**
Contiguous space for W.
- **double** * **vec_rc**
Contiguous space for RC.
- **dcomplex** * **vec_sas**
Contiguous space for SAS.
- **dcomplex** * **vec_vints**
Contiguous space for VINTS.
- **dcomplex** * **vec_tsas**
Contiguous space for TSAS.
- **dcomplex** * **vec_gis**
Contiguous space for GIS.
- **dcomplex** * **vec_gls**
Contiguous space for GLS.
- **dcomplex** * **vec_sam**

- Contiguous space for SAM.*

 - **int_le**

Maximum external field expansion order.
 - **int_lm**

Maximum field expansion order.
 - **int_nlim**

$NLIM = LI * (LI + 2)$
 - **int_nlem**

$NLEM = LE * (LE + 2)$
 - **int_nlemt**

$NLEMT = 2 * NLEM.$
 - **int_ncou**

$NCOU = NSPH * NSPH - 1.$
 - **int_litpo**

$LITPO = 2 * LI + 1.$
 - **int_litpos**

$LITPOS = LITPO * LITPO.$
 - **int_lmipo**

$LMPO = LM + 1.$
 - **int_lmtpo**

$LMTPO = LI + LE + 1.$
 - **int_lmtpos**

$LMTPOS = LMTPO * LMTPO.$
 - **int_nv3j**

$NV3J = (LM * (LM + 1) * (2 * LM + 7)) / 6.$
 - **int_ndi**

$NDI = NSPH * NLIM.$
 - **int_ndit**

$NDIT = 2 * NSPH * NLIM.$
 - **dcomplex * vec_am0m**

Contiguous space for AM0M.
 - **dcomplex * vec_fsac**

Contiguous space for FSAC.
 - **dcomplex * vec_sac**

Contiguous space for SAC.
 - **dcomplex * vec_fsacm**

Contiguous space for FSACM.
 - **int * vec_ind3j**

Contiguous space for IND3J.
 - **int_ndm**

$NDM = NDIT + NLEMT.$
 - **dcomplex * vec_at**

Contiguous space for AT.

8.19.1 Detailed Description

This class is used to solve the problem of a cluster of spherical particles. It replaces the C1 and C1_AddOns class implementation of the C1 FORTRAN common block.

8.19.2 Constructor & Destructor Documentation

8.19.2.1 ParticleDescriptorCluster() [1/2]

```
ParticleDescriptorCluster::ParticleDescriptorCluster (
    GeometryConfiguration * gconf,
    ScattererConfiguration * sconf )
```

Parameters

<i>gconf</i>	const GeometryConfiguration * Pointer to GeometryConfiguration instance.
<i>sconf</i>	const ScattererConfiguration * Pointer to ScattererConfiguration instance.

8.19.2.2 ParticleDescriptorCluster() [2/2]

```
ParticleDescriptorCluster::ParticleDescriptorCluster (
    const ParticleDescriptorCluster & rhs )
```

Parameters

<i>rhs</i>	const ParticleDescriptorCluster & Reference to ParticleDescriptorCluster object to be copied.
------------	---

8.19.3 Member Function Documentation

8.19.3.1 get_descriptor_type()

```
std::string ParticleDescriptorCluster::get_descriptor_type ( ) [inline], [override], [virtual]
```

Returns

descriptor_type: string The descriptor type name.

Reimplemented from [ParticleDescriptor](#).

8.19.3.2 get_size()

```
long ParticleDescriptorCluster::get_size (
    GeometryConfiguration * gconf,
    ScattererConfiguration * sconf ) [static]
```

Parameters

<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration object.
<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration object.

Returns

result: `long` Estimated size in bytes.

8.19.3.3 update_orders()

```
int ParticleDescriptorCluster::update_orders (
    int inner_order,
    int outer_order )
```

Parameters

<i>inner_order</i>	<code>int</code> The new inner expansion order to be set.
<i>outer_order</i>	<code>int</code> The new outer expansion order to be set.

Returns

result: `int` An exit code (0 if successful).

The documentation for this class was generated from the following files:

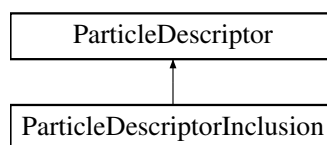
- `np_tmcode/src/include/`[Commons.h](#)
- `np_tmcode/src/libnptm/`[Commons.cpp](#)

8.20 ParticleDescriptorInclusion Class Reference

The data structure describing a particle model for a sphere with inclusions.

```
#include <Commons.h>
```

Inheritance diagram for ParticleDescriptorInclusion:

**Public Member Functions**

- [ParticleDescriptorInclusion](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
ParticleDescriptorInclusion instance constructor.
- [ParticleDescriptorInclusion](#) (const [ParticleDescriptorInclusion](#) &rhs)
ParticleDescriptorInclusion copy constructor.
- `std::string` [get_descriptor_type](#) () **override**
Interface function to return the descriptor type as string.
- `int` [update_orders](#) (`int` inner_order, `int` outer_order)
Update the field expansion orders.

Public Member Functions inherited from [ParticleDescriptor](#)

- [ParticleDescriptor](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
ParticleDescriptor instance constructor.
- [ParticleDescriptor](#) (const [ParticleDescriptor](#) &rhs)
ParticleDescriptor copy constructor.
- [~ParticleDescriptor](#) ()
ParticleDescriptor instance destroyer.

Static Public Member Functions

- [static long get_size](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
Compute the memory requirements of an instance.

Static Public Member Functions inherited from [ParticleDescriptor](#)

- [static long get_size](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
Compute the memory requirements of an instance.

Additional Inherited Members

Public Attributes inherited from [ParticleDescriptor](#)

- [const short & class_type](#) = [_class_type](#)
Sub-class identification code.
- [const int & nsph](#) = [_nsph](#)
Read-only view of number of spheres composing the model.
- [const int & li](#) = [_li](#)
Read-only view of maximum internal field expansion order.
- [const int & max_layers](#) = [_max_layers](#)
Read-only view of the maximum number of layers in types.
- [const int & num_configurations](#) = [_num_configurations](#)
Read-only view of number of different sphere types.
- [const int & num_layers](#) = [_num_layers](#)
Read-only view of total number of layers from all sphere types.
- [const int & nhspo](#) = [_nhspo](#)
Read-only view of NHSP0.
- [const int & npnt](#) = [_npnt](#)
Read-only view on number of points for numerical integration in layered spheres.
- [const int & npntts](#) = [_npntts](#)
Read-only view on number of points for numerical integration in transition layer.
- [dcomplex ** rmi](#)
Matrix of Mie scattering b-coefficients for different orders and spheres.
- [dcomplex ** rei](#)
Matrix of Matrix of Mie scattering a-coefficients for different orders and spheres.
- [dcomplex ** w](#)
Matrix of multipole amplitudes for incident and scattered fields.
- [dcomplex * vint](#)
Vector of intensity components.

- **double * rxx**
Vector of sphere Cartesian X coordinates.
- **double * ryy**
Vector of sphere Cartesian Y coordinates.
- **double * rzz**
Vector of sphere Cartesian Z coordinates.
- **double * ros**
Vector of sphere radii.
- **double ** rc**
Matrix of sphere fractional transition radii.
- **int * iog**
Vector of sphere type identifiers.
- **int * nshl**
Vector of number of layers in sphere type.
- **dcomplex * ris**
ANNOTATION: Square inverted coefficients.
- **dcomplex * dlri**
ANNOTATION: Refractive index variation.
- **dcomplex * dc0**
Vector of dielectric constants.
- **dcomplex * vkt**
Vector of complex refractive indices.
- **double * vsz**
Vector of sizes in units of $2\pi/\text{LAMBDA}$.
- **double gcs**
Total geometric cross-section.
- **dcomplex *** sas**
Tensor of sphere scattering amplitudes.
- **dcomplex ** vints**
Array of vectors of intensity components for single spheres.
- **dcomplex * fsas**
Vector of sphere forward scattering amplitudes.
- **double * sscs**
Vector of scattering cross-sections for spheres.
- **double * sexs**
Vector of extinction cross-sections for spheres.
- **double * sabs**
Vector of absorption cross-sections for spheres.
- **double * sqscs**
Vector of scattering efficiencies for spheres.
- **double * sqexs**
Vector of extinction efficiencies for spheres.
- **double * sqabs**
Vector of absorption efficiencies for spheres.
- **double * gcsv**
Vector of geometric cross-sections for spheres.
- **dcomplex * vintt**
Vector of field intensity components.
- **dcomplex tfsas**
Total forward scattering amplitude of the spheres.
- **dcomplex ** tsas**

- Total scattering amplitude of the spheres.*
- **double** **scs**
Sphere scattering cross-section.
- **double** **ecs**
Sphere extinction cross-section.
- **double** **acs**
Sphere absorption cross-section.
- **dcomplex** ** **gis**
ANNOTATION: Geometric tensor.
- **dcomplex** ** **gls**
ANNOTATION: Geometric tensor.
- **dcomplex** ** **sam**
Mean scattering amplitude components.
- **const int** & **le** = **_le**
Read-only view of maximum external field expansion order.
- **const int** & **lm** = **_lm**
Read-only view of maximum field expansion order.
- **const int** & **nlim** = **_nlim**
Read-only view of NLIM.
- **const int** & **nlem** = **_nlem**
Read-only view of NLEM.
- **const int** & **nlemt** = **_nlemt**
Read-only view of NLEMT.
- **const int** & **ncou** = **_ncou**
Read-only view of NCOU.
- **const int** & **litpo** = **_litpo**
Read-only view of LITPO.
- **const int** & **litpos** = **_litpos**
Read-only view of LITPOS.
- **const int** & **lmpo** = **_lmpo**
Read-only view of LMPO.
- **const int** & **lmtpo** = **_lmtpo**
Read-only view of LMTPO.
- **const int** & **lmtpos** = **_lmtpos**
Read-only view of LMTPOS.
- **const int** & **nv3j** = **_nv3j**
Read-only view of NV3J.
- **const int** & **ndi** = **_ndi**
Read-only view of NDI.
- **const int** & **ndit** = **_ndit**
Read-only view of NDIT.
- **const int** & **ndm** = **_ndm**
Read-only view of NDM.
- **dcomplex** * **vh**
ANNOTATION: Hankel vector.
- **dcomplex** * **vj0**
ANNOTATION: J0 vector.
- **dcomplex** * **vyhj**
ANNOTATION: Translation vector.
- **dcomplex** * **vyj0**
ANNOTATION: J0 translation vector.

- **dcomplex** **vj**
ANNOTATION: J vector.
- **dcomplex** ** **am0m**
Transition matrix.
- **dcomplex** ** **fsac**
Cluster forward scattering amplitude.
- **dcomplex** ** **sac**
Cluster scattering amplitude.
- **dcomplex** ** **fsacm**
Mean cluster forward scattering amplitude.
- **dcomplex** * **vintm**
Mean intensity components vector.
- **dcomplex** * **scscp**
Cluster polarized scattering cross-sections.
- **dcomplex** * **ecscp**
Cluster polarized extinction cross-sections.
- **dcomplex** * **scscpm**
Mean cluster polarized scattering cross-sections.
- **dcomplex** * **ecscpm**
Mean cluster polarized extinction cross-sections.
- **double** * **v3j0**
ANNOTATION: J0 vector.
- **double** * **scsc**
Cluster scattering cross-sections.
- **double** * **ecsc**
Cluster extinction cross-sections.
- **double** * **scscm**
Mean cluster scattering cross-sections.
- **double** * **ecscm**
Mean cluster extinction cross-sections.
- **int** ** **ind3j**
J-vector components index matrix.
- **double** * **rac3j**
ANNOTATION: J-vector boundary conditions.
- **dcomplex** * **rm0**
ANNOTATION: M coefficients.
- **dcomplex** * **re0**
ANNOTATION: E coefficients.
- **dcomplex** * **rmw**
ANNOTATION: M amplitude coefficients.
- **dcomplex** * **rew**
ANNOTATION: E amplitude coefficients.
- **dcomplex** * **tm**
ANNOTATION: Transition M coefficients.
- **dcomplex** * **te**
ANNOTATION: Transition E coefficients.
- **dcomplex** * **tm0**
ANNOTATION: Initial transition M coefficients.
- **dcomplex** * **te0**
ANNOTATION: Initial transition E coefficients.
- **dcomplex** ** **at**
ANNOTATION: Included particle T-matrix.

Static Public Attributes inherited from [ParticleDescriptor](#)

- **static const short** **BASE_TYPE** = 0
Base sub-class identification code.
- **static const short** **SPHERE_TYPE** = 1
Sphere sub-class identification code.
- **static const short** **CLUSTER_TYPE** = 2
Cluster sub-class identification code.
- **static const short** **INCLUSION_TYPE** = 3
Inclusion sub-class identification code.

Protected Attributes inherited from [ParticleDescriptor](#)

- **short** **_class_type**
Sub-class identification code.
- **int** **_nsph**
Number of spheres composing the model.
- **int** **_li**
Maximum internal field expansion order.
- **int** **_max_layers**
Maximum number of layers in known sphere types.
- **int** **_num_configurations**
Number of different sphere types.
- **int** **_num_layers**
Total number of layers from all sphere types.
- **int** **_nhspo**
 $NHSPO = 2 * MAX(NPNT, NPNTTS) - 1.$
- **int** **_npnt**
Number of points for numerical integration in layered spheres.
- **int** **_npntts**
Number of points for numerical integration in transition layer.
- **dcomplex** * **vec_rmi**
Contiguous space for RMI.
- **dcomplex** * **vec_rei**
Contiguous space for REI.
- **dcomplex** * **vec_w**
Contiguous space for W.
- **double** * **vec_rc**
Contiguous space for RC.
- **dcomplex** * **vec_sas**
Contiguous space for SAS.
- **dcomplex** * **vec_vints**
Contiguous space for VINTS.
- **dcomplex** * **vec_tsas**
Contiguous space for TSAS.
- **dcomplex** * **vec_gis**
Contiguous space for GIS.
- **dcomplex** * **vec_gls**
Contiguous space for GLS.
- **dcomplex** * **vec_sam**

- Contiguous space for SAM.*

 - **int_le**

Maximum external field expansion order.
 - **int_lm**

Maximum field expansion order.
 - **int_nlim**

$NLIM = LI * (LI + 2)$
 - **int_nlem**

$NLEM = LE * (LE + 2)$
 - **int_nlemt**

$NLEMT = 2 * NLEM.$
 - **int_ncou**

$NCOU = NSPH * NSPH - 1.$
 - **int_litpo**

$LITPO = 2 * LI + 1.$
 - **int_litpos**

$LITPOS = LITPO * LITPO.$
 - **int_lmipo**

$LMPO = LM + 1.$
 - **int_lmtpo**

$LMTPO = LI + LE + 1.$
 - **int_lmtpos**

$LMTPOS = LMTPO * LMTPO.$
 - **int_nv3j**

$NV3J = (LM * (LM + 1) * (2 * LM + 7)) / 6.$
 - **int_ndi**

$NDI = NSPH * NLIM.$
 - **int_ndit**

$NDIT = 2 * NSPH * NLIM.$
 - **dcomplex * vec_am0m**

Contiguous space for AM0M.
 - **dcomplex * vec_fsac**

Contiguous space for FSAC.
 - **dcomplex * vec_sac**

Contiguous space for SAC.
 - **dcomplex * vec_fsacm**

Contiguous space for FSACM.
 - **int * vec_ind3j**

Contiguous space for IND3J.
 - **int_ndm**

$NDM = NDIT + NLEMT.$
 - **dcomplex * vec_at**

Contiguous space for AT.

8.20.1 Detailed Description

This class is used to solve the problem of a spherical particle with a cluster of inclusions. It replaces the C1 and C1_AddOns class implementation of the C1 FORTRAN common block.

8.20.2 Constructor & Destructor Documentation

8.20.2.1 ParticleDescriptorInclusion() [1/2]

```
ParticleDescriptorInclusion::ParticleDescriptorInclusion (
    GeometryConfiguration * gconf,
    ScattererConfiguration * sconf )
```

Parameters

<i>gconf</i>	const GeometryConfiguration * Pointer to GeometryConfiguration instance.
<i>sconf</i>	const ScattererConfiguration * Pointer to ScattererConfiguration instance.

8.20.2.2 ParticleDescriptorInclusion() [2/2]

```
ParticleDescriptorInclusion::ParticleDescriptorInclusion (
    const ParticleDescriptorInclusion & rhs )
```

Parameters

<i>rhs</i>	const ParticleDescriptorInclusion & Reference to ParticleDescriptorInclusion object to be copied.
------------	---

8.20.3 Member Function Documentation

8.20.3.1 get_descriptor_type()

```
std::string ParticleDescriptorInclusion::get_descriptor_type ( ) [inline], [override], [virtual]
```

Returns

descriptor_type: string The descriptor type name.

Reimplemented from [ParticleDescriptor](#).

8.20.3.2 get_size()

```
long ParticleDescriptorInclusion::get_size (
    GeometryConfiguration * gconf,
    ScattererConfiguration * sconf ) [static]
```

Parameters

<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration object.
<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration object.

Returns

result: `long` Estimated size in bytes.

8.20.3.3 update_orders()

```
int ParticleDescriptorInclusion::update_orders (
    int inner_order,
    int outer_order )
```

Parameters

<i>inner_order</i>	<code>int</code> The new inner expansion order to be set.
<i>outer_order</i>	<code>int</code> The new outer expansion order to be set.

Returns

result: `int` An exit code (0 if successful).

The documentation for this class was generated from the following files:

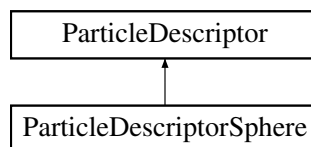
- `np_tmcode/src/include/`[Commons.h](#)
- `np_tmcode/src/libnptm/`[Commons.cpp](#)

8.21 ParticleDescriptorSphere Class Reference

The data structure describing a spherical particle model.

```
#include <Commons.h>
```

Inheritance diagram for ParticleDescriptorSphere:

**Public Member Functions**

- [ParticleDescriptorSphere](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
ParticleDescriptorSphere instance constructor.
- [ParticleDescriptorSphere](#) ([const ParticleDescriptorSphere](#) &rhs)
ParticleDescriptorSphere copy constructor.
- `std::string` [get_descriptor_type](#) () **override**
Interface function to return the descriptor type as string.
- `int` [update_order](#) (`int` order)
Update the field expansion order.

Public Member Functions inherited from ParticleDescriptor

- [ParticleDescriptor](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
ParticleDescriptor instance constructor.
- [ParticleDescriptor](#) (const [ParticleDescriptor](#) &rhs)
ParticleDescriptor copy constructor.
- [~ParticleDescriptor](#) ()
ParticleDescriptor instance destroyer.

Static Public Member Functions

- [static long get_size](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
Compute the memory requirements of an instance.

Static Public Member Functions inherited from ParticleDescriptor

- [static long get_size](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf)
Compute the memory requirements of an instance.

Additional Inherited Members

Public Attributes inherited from ParticleDescriptor

- [const short & class_type](#) = [_class_type](#)
Sub-class identification code.
- [const int & nsph](#) = [_nsph](#)
Read-only view of number of spheres composing the model.
- [const int & li](#) = [_li](#)
Read-only view of maximum internal field expansion order.
- [const int & max_layers](#) = [_max_layers](#)
Read-only view of the maximum number of layers in types.
- [const int & num_configurations](#) = [_num_configurations](#)
Read-only view of number of different sphere types.
- [const int & num_layers](#) = [_num_layers](#)
Read-only view of total number of layers from all sphere types.
- [const int & nhspo](#) = [_nhspo](#)
Read-only view of NHSP0.
- [const int & npnt](#) = [_npnt](#)
Read-only view on number of points for numerical integration in layered spheres.
- [const int & npntts](#) = [_npntts](#)
Read-only view on number of points for numerical integration in transition layer.
- [dcomplex ** rmi](#)
Matrix of Mie scattering b-coefficients for different orders and spheres.
- [dcomplex ** rei](#)
Matrix of Matrix of Mie scattering a-coefficients for different orders and spheres.
- [dcomplex ** w](#)
Matrix of multipole amplitudes for incident and scattered fields.
- [dcomplex * vint](#)
Vector of intensity components.

- **double * rxx**
Vector of sphere Cartesian X coordinates.
- **double * ryy**
Vector of sphere Cartesian Y coordinates.
- **double * rzz**
Vector of sphere Cartesian Z coordinates.
- **double * ros**
Vector of sphere radii.
- **double ** rc**
Matrix of sphere fractional transition radii.
- **int * iog**
Vector of sphere type identifiers.
- **int * nshl**
Vector of number of layers in sphere type.
- **dcomplex * ris**
ANNOTATION: Square inverted coefficients.
- **dcomplex * dlri**
ANNOTATION: Refractive index variation.
- **dcomplex * dc0**
Vector of dielectric constants.
- **dcomplex * vkt**
Vector of complex refractive indices.
- **double * vsz**
Vector of sizes in units of $2\pi/\text{LAMBDA}$.
- **double gcs**
Total geometric cross-section.
- **dcomplex *** sas**
Tensor of sphere scattering amplitudes.
- **dcomplex ** vints**
Array of vectors of intensity components for single spheres.
- **dcomplex * fsas**
Vector of sphere forward scattering amplitudes.
- **double * sscs**
Vector of scattering cross-sections for spheres.
- **double * sexs**
Vector of extinction cross-sections for spheres.
- **double * sabs**
Vector of absorption cross-sections for spheres.
- **double * sqscs**
Vector of scattering efficiencies for spheres.
- **double * sqexs**
Vector of extinction efficiencies for spheres.
- **double * sqabs**
Vector of absorption efficiencies for spheres.
- **double * gcsv**
Vector of geometric cross-sections for spheres.
- **dcomplex * vintt**
Vector of field intensity components.
- **dcomplex tfsas**
Total forward scattering amplitude of the spheres.
- **dcomplex ** tsas**

- Total scattering amplitude of the spheres.*
- **double** **scs**
Sphere scattering cross-section.
- **double** **ecs**
Sphere extinction cross-section.
- **double** **acs**
Sphere absorption cross-section.
- **dcomplex** ** **gis**
ANNOTATION: Geometric tensor.
- **dcomplex** ** **gls**
ANNOTATION: Geometric tensor.
- **dcomplex** ** **sam**
Mean scattering amplitude components.
- **const int** & **le** = **_le**
Read-only view of maximum external field expansion order.
- **const int** & **lm** = **_lm**
Read-only view of maximum field expansion order.
- **const int** & **nlim** = **_nlim**
Read-only view of NLIM.
- **const int** & **nlem** = **_nlem**
Read-only view of NLEM.
- **const int** & **nlemt** = **_nlemt**
Read-only view of NLEMT.
- **const int** & **ncou** = **_ncou**
Read-only view of NCOU.
- **const int** & **litpo** = **_litpo**
Read-only view of LITPO.
- **const int** & **litpos** = **_litpos**
Read-only view of LITPOS.
- **const int** & **lmpo** = **_lmpo**
Read-only view of LMPO.
- **const int** & **lmtpo** = **_lmtpo**
Read-only view of LMTPO.
- **const int** & **lmtpos** = **_lmtpos**
Read-only view of LMTPOS.
- **const int** & **nv3j** = **_nv3j**
Read-only view of NV3J.
- **const int** & **ndi** = **_ndi**
Read-only view of NDI.
- **const int** & **ndit** = **_ndit**
Read-only view of NDIT.
- **const int** & **ndm** = **_ndm**
Read-only view of NDM.
- **dcomplex** * **vh**
ANNOTATION: Hankel vector.
- **dcomplex** * **vj0**
ANNOTATION: J0 vector.
- **dcomplex** * **vyhj**
ANNOTATION: Translation vector.
- **dcomplex** * **vyj0**
ANNOTATION: J0 translation vector.

- **dcomplex** **vj**
ANNOTATION: J vector.
- **dcomplex** ** **am0m**
Transition matrix.
- **dcomplex** ** **fsac**
Cluster forward scattering amplitude.
- **dcomplex** ** **sac**
Cluster scattering amplitude.
- **dcomplex** ** **fsacm**
Mean cluster forward scattering amplitude.
- **dcomplex** * **vintm**
Mean intensity components vector.
- **dcomplex** * **scscp**
Cluster polarized scattering cross-sections.
- **dcomplex** * **ecscp**
Cluster polarized extinction cross-sections.
- **dcomplex** * **scscpm**
Mean cluster polarized scattering cross-sections.
- **dcomplex** * **ecscpm**
Mean cluster polarized extinction cross-sections.
- **double** * **v3j0**
ANNOTATION: J0 vector.
- **double** * **scsc**
Cluster scattering cross-sections.
- **double** * **ecsc**
Cluster extinction cross-sections.
- **double** * **scscm**
Mean cluster scattering cross-sections.
- **double** * **ecscm**
Mean cluster extinction cross-sections.
- **int** ** **ind3j**
J-vector components index matrix.
- **double** * **rac3j**
ANNOTATION: J-vector boundary conditions.
- **dcomplex** * **rm0**
ANNOTATION: M coefficients.
- **dcomplex** * **re0**
ANNOTATION: E coefficients.
- **dcomplex** * **rmw**
ANNOTATION: M amplitude coefficients.
- **dcomplex** * **rew**
ANNOTATION: E amplitude coefficients.
- **dcomplex** * **tm**
ANNOTATION: Transition M coefficients.
- **dcomplex** * **te**
ANNOTATION: Transition E coefficients.
- **dcomplex** * **tm0**
ANNOTATION: Initial transition M coefficients.
- **dcomplex** * **te0**
ANNOTATION: Initial transition E coefficients.
- **dcomplex** ** **at**
ANNOTATION: Included particle T-matrix.

Static Public Attributes inherited from [ParticleDescriptor](#)

- **static const short** **BASE_TYPE** = 0
Base sub-class identification code.
- **static const short** **SPHERE_TYPE** = 1
Sphere sub-class identification code.
- **static const short** **CLUSTER_TYPE** = 2
Cluster sub-class identification code.
- **static const short** **INCLUSION_TYPE** = 3
Inclusion sub-class identification code.

Protected Attributes inherited from [ParticleDescriptor](#)

- **short** **_class_type**
Sub-class identification code.
- **int** **_nsph**
Number of spheres composing the model.
- **int** **_li**
Maximum internal field expansion order.
- **int** **_max_layers**
Maximum number of layers in known sphere types.
- **int** **_num_configurations**
Number of different sphere types.
- **int** **_num_layers**
Total number of layers from all sphere types.
- **int** **_nhspo**
 $NHSPO = 2 * MAX(NPNT, NPNTTS) - 1.$
- **int** **_npnt**
Number of points for numerical integration in layered spheres.
- **int** **_npntts**
Number of points for numerical integration in transition layer.
- **dcomplex** * **vec_rmi**
Contiguous space for RMI.
- **dcomplex** * **vec_rei**
Contiguous space for REI.
- **dcomplex** * **vec_w**
Contiguous space for W.
- **double** * **vec_rc**
Contiguous space for RC.
- **dcomplex** * **vec_sas**
Contiguous space for SAS.
- **dcomplex** * **vec_vints**
Contiguous space for VINTS.
- **dcomplex** * **vec_tsas**
Contiguous space for TSAS.
- **dcomplex** * **vec_gis**
Contiguous space for GIS.
- **dcomplex** * **vec_gls**
Contiguous space for GLS.
- **dcomplex** * **vec_sam**

- Contiguous space for SAM.*

 - **int_le**

Maximum external field expansion order.
 - **int_lm**

Maximum field expansion order.
 - **int_nlim**

$NLIM = LI * (LI + 2)$
 - **int_nlem**

$NLEM = LE * (LE + 2)$
 - **int_nlemt**

$NLEMT = 2 * NLEM.$
 - **int_ncou**

$NCOU = NSPH * NSPH - 1.$
 - **int_litpo**

$LITPO = 2 * LI + 1.$
 - **int_litpos**

$LITPOS = LITPO * LITPO.$
 - **int_lmipo**

$LMPO = LM + 1.$
 - **int_lmtpo**

$LMTPO = LI + LE + 1.$
 - **int_lmtpos**

$LMTPOS = LMTPO * LMTPO.$
 - **int_nv3j**

$NV3J = (LM * (LM + 1) * (2 * LM + 7)) / 6.$
 - **int_ndi**

$NDI = NSPH * NLIM.$
 - **int_ndit**

$NDIT = 2 * NSPH * NLIM.$
 - **dcomplex * vec_am0m**

Contiguous space for AM0M.
 - **dcomplex * vec_fsac**

Contiguous space for FSAC.
 - **dcomplex * vec_sac**

Contiguous space for SAC.
 - **dcomplex * vec_fsacm**

Contiguous space for FSACM.
 - **int * vec_ind3j**

Contiguous space for IND3J.
 - **int_ndm**

$NDM = NDIT + NLEMT.$
 - **dcomplex * vec_at**

Contiguous space for AT.

8.21.1 Detailed Description

This class is used to solve the problem of a single spherical particle. It replaces the old C1 class implementation of the corresponding FORTRAN common block.

8.21.2 Constructor & Destructor Documentation

8.21.2.1 ParticleDescriptorSphere() [1/2]

```
ParticleDescriptorSphere::ParticleDescriptorSphere (
    GeometryConfiguration * gconf,
    ScattererConfiguration * sconf )
```

Parameters

<i>gconf</i>	const GeometryConfiguration * Pointer to GeometryConfiguration instance.
<i>sconf</i>	const ScattererConfiguration * Pointer to ScattererConfiguration instance.

8.21.2.2 ParticleDescriptorSphere() [2/2]

```
ParticleDescriptorSphere::ParticleDescriptorSphere (
    const ParticleDescriptorSphere & rhs )
```

Parameters

<i>rhs</i>	const ParticleDescriptorSphere & Reference to ParticleDescriptorSphere object to be copied.
------------	---

8.21.3 Member Function Documentation

8.21.3.1 get_descriptor_type()

```
std::string ParticleDescriptorSphere::get_descriptor_type ( ) [inline], [override], [virtual]
```

Returns

descriptor_type: string The descriptor type name.

Reimplemented from [ParticleDescriptor](#).

8.21.3.2 get_size()

```
long ParticleDescriptorSphere::get_size (
    GeometryConfiguration * gconf,
    ScattererConfiguration * sconf ) [static]
```

Parameters

<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration object.
<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration object.

Returns

result: `long` Estimated size in bytes.

8.21.3.3 update_order()

```
int ParticleDescriptorSphere::update_order (
    int order )
```

Parameters

<code>order</code>	<code>int</code> The new field expansion order to be set.
--------------------	---

Returns

result: `int` An exit code (0 if successful).

The documentation for this class was generated from the following files:

- `np_tmcode/src/include/Commons.h`
- `np_tmcode/src/libnptm/Commons.cpp`

8.22 pydynrange.PlotData Class Reference

Class to represent the dynamic range memory structure.

Public Member Functions

- `__init__` (`self`)
PlotData instance constructor.
- `log_dynamic_range` (`self`)
Print a text log of the dynamic range.
- `plot_dynamic_range` (`self`, `config`)
Make histograms of dynamic range with matplotlib.

Public Attributes

- `max_real`
Maximum real part value.
- `max_imag`
Maximum imaginary part value.
- `max_absl`
Maximum absolute value.
- `min_real`
Minimum (most negative) real part value.
- `min_imag`

- Minimum (most negative) imaginary part value.*
- **sml_absl**
Smallest absolute value.
- **sml_real**
Smallest absolute real part value.
- **sml_imag**
Smallest absolute imaginary part value.
- **absl_hist**
Histogram of absolute values.
- **real_hist**
Histogram of real values.
- **imag_hist**
Histogram of imaginary values.
- **max_real_mag**
Maximum real order of magnitude.
- **min_real_mag**
Minimum real order of magnitude.
- **max_imag_mag**
Maximum imaginary order of magnitude.
- **min_imag_mag**
Minimum imaginary order of magnitude.
- **max_absl_mag**
Maximum absolute order of magnitude.
- **min_absl_mag**
Minimum absolute order of magnitude.

8.22.1 Detailed Description

In order to implement a light-weight test that is able to choose whether to produce a text-only output or a text + diagnostic plot output, the `PlotData` class stores the memory structures to produce the text log and the plots. The data for the plots are filled with dummy place-holders if the required output is text only, while they are filled with actual data, otherwise.

Returns

`exit_code`: `int` 0 on successful completion.

8.22.2 Member Function Documentation

8.22.2.1 `plot_dynamic_range()`

```
pydynrange.PlotData.plot_dynamic_range (
    self,
    config )
```

Parameters

<code>config</code>	<code>dict</code> Dictionary of configuration options.
---------------------	--

The documentation for this class was generated from the following file:

- np_tmcode/src/scripts/pydynrange.py

8.23 RuntimeSettings Class Reference

A class to represent runtime settings.

```
#include <Configuration.h>
```

Public Member Functions

- **RuntimeSettings** ()
Default [RuntimeSettings](#) instance constructor.
- **RuntimeSettings** ([GeometryConfiguration](#) *gconf, [Logger](#) *ptr_logger=NULL)
[RuntimeSettings](#) instance constructor.
- **void set_accuracy_goal** (const double goal)
Update matrix inversion accuracy to new value.

Public Attributes

- **const short & invert_mode** = [_invert_mode](#)
Read-only view on inversion mode flag.
- **const double & accuracy_goal** = [_accuracy_goal](#)
Read-only view on target matrix inversion accuracy.
- **const double & gpu_ram_gb** = [_gpu_ram_gb](#)
Read-only view on GPU memory in GiB.
- **const double & host_ram_gb** = [_host_ram_gb](#)
Read-only view on host system memory in GiB.
- **const int & max_ref_iters** = [_max_ref_iters](#)
Read-only view on maximum number of refinement iterations.
- **const bool & use_refinement** = [_use_refinement](#)
Read-only view on refinement flag.
- **const bool & use_offload** = [_use_offload](#)
Read-only view on offload flag.
- **const [Logger](#) * logger**
Pointer to a [Logger](#) instance.

Static Public Attributes

- **static const short INV_MODE_LU** = 0
Flag for LU factorization inversion.
- **static const short INV_MODE_GESV** = 1
Flag for linear system solver.
- **static const short INV_MODE_RBT** = 2
Flag for random butterfly transformation inversion.
- **static const short INV_MODE_SVD** = 3
Flag for singular value decomposition.

Protected Attributes

- [short _invert_mode](#)
Inversion mode flag.
- [double _accuracy_goal](#)
Target matrix inversion accuracy.
- [double _gpu_ram_gb](#)
GPU memory in GiB.
- [double _host_ram_gb](#)
Host system memory in GiB.
- [int _max_ref_iters](#)
Maximum number of refinement iterations.
- [bool _use_refinement](#)
Refinement flag.
- [bool _use_offload](#)
Offload flag.

8.23.1 Detailed Description

DEVELOPMENT NOTE: this class does not need MPI / OpenMP interface methods, because it is designed to be a shared memory read-only structure, independently constructed by each MPI process, immediately after getting its [GeometryConfiguration](#) instance.

8.23.2 Constructor & Destructor Documentation

8.23.2.1 RuntimeSettings()

```
RuntimeSettings::RuntimeSettings (
    GeometryConfiguration * gconf,
    Logger * ptr_logger = NULL )
```

Parameters

<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration instance.
<i>ptr_logger</i>	Logger * Pointer to a Logger instance (optional).

8.23.3 Member Function Documentation

8.23.3.1 set_accuracy_goal()

```
void RuntimeSettings::set_accuracy_goal (
    const double goal ) [inline]
```

Parameters

<i>goal</i>	const double Desired new accuracy value.
-------------	--

The documentation for this class was generated from the following files:

- np_tmcode/src/include/Configuration.h
- np_tmcode/src/libnptm/Configuration.cpp

8.24 ScattererConfiguration Class Reference

A class to represent scatterer configuration objects.

```
#include <Configuration.h>
```

Public Member Functions

- [ScattererConfiguration](#) (int nsph, int configs, double *scale_vector, int nxi, const std::string &variable_name, int *iog_vector, double *ros_vector, int *nshl_vector, double **rcf_vector, int dielectric_func_type, dcomplex ***dc_matrix, bool has_external, double exdc, double wp, double xip)
Build a scatterer configuration structure.
- [ScattererConfiguration](#) (const ScattererConfiguration &rhs)
Build a scatterer configuration structure copying its contents from a preexisting one.
- [~ScattererConfiguration](#) ()
Destroy a scatterer configuration instance.
- [dcomplex get_dielectric_constant](#) (int i, int j, int k)
Get the dielectric constant of a material for a specific wavelength.
- [int get_iog](#) (int index)
Get the ID of a configuration from the index of the sphere.
- [double get_max_radius](#) ()
Get the maximum radius of the sphere components.
- [double get_min_radius](#) ()
Get the minimum radius of the sphere components.
- [int get_nshl](#) (int index)
Get the number of layers for a given configuration.
- [double get_particle_radius](#) (GeometryConfiguration *gc)
Get the radius of the smallest sphere containing the particle.
- [double get_radius](#) (int index)
Get the radius of a sphere by its index.
- [double get_rcf](#) (int row, int column)
Get the value of a scale by its index.
- [double get_scale](#) (int index)
Get the value of a scale by its index.
- [double get_type_radius](#) (int index)
Get the radius of a sphere type by its index.
- [void print](#) ()
Print the contents of the configuration object to terminal.
- [void write_binary](#) (const std::string &file_name, const std::string &mode="LEGACY")
Write the scatterer configuration data to binary output.
- [void write_formatted](#) (const std::string &file_name)
Write the scatterer configuration data to formatted text output.
- [bool operator==](#) (const ScattererConfiguration &other)
Test whether two instances of [ScattererConfiguration](#) are equal.

Static Public Member Functions

- `static ScattererConfiguration * from_binary (const std::string &file_name, const std::string &mode="LEGACY")`
Build configuration from binary configuration input file.
- `static ScattererConfiguration * from_dedfb (const std::string &file_name)`
Build scatterer configuration from legacy configuration input file.

Public Attributes

- `const std::string & reference_variable_name = _reference_variable_name`
Read-only view on name of the reference variable type.
- `const int & number_of_spheres = _number_of_spheres`
Read-only view on number of spherical components.
- `const int & configurations = _configurations`
Read-only view on number of configurations.
- `const int & number_of_scales = _number_of_scales`
Read-only view on number of scales to use in calculation.
- `const int & idfc = _idfc`
Read-only view on type of dielectric functions.
- `const double & exdc = _exdc`
Read-only view on external medium dielectric constant.
- `const double & wp = _wp`
Read-only view on WP.
- `const double & xip = _xip`
Read-only view on peak XI.
- `const int & max_layers = _max_layers`
Read-only view on the maximum number of layers for the particle components.
- `const bool & use_external_sphere = _use_external_sphere`
Read-only view on flag to control whether to add an external layer.
- `double ** _rcf`
Matrix of fractional transition radii with size [CONFIGURATIONS x LAYERS].

Protected Member Functions

- `void write_hdf5 (const std::string &file_name)`
Write the scatterer configuration data to HDF5 binary output.
- `void write_legacy (const std::string &file_name)`
Write the scatterer configuration data to legacy binary output.

Static Protected Member Functions

- `static ScattererConfiguration * from_hdf5 (const std::string &file_name)`
Build configuration from a HDF5 binary input file.
- `static ScattererConfiguration * from_legacy (const std::string &file_name)`
Build configuration from legacy binary input file.

Protected Attributes

- **dcomplex *** _dc0_matrix**
Matrix of dielectric parameters with size [NON_TRANS_LAYERS x N_SPHERES x N_SCALES].
- **double * _radii_of_spheres**
Vector of sphere radii expressed in m, with size [N_SPHERES].
- **int * _iog_vec**
Vector of sphere ID numbers, with size [N_SPHERES].
- **int * _nshl_vec**
Vector of layer numbers for every sphere, with size [CONFIGURATIONS].
- **double * _scale_vec**
Vector of scale parameters, with size [N_SCALES].
- **std::string _reference_variable_name**
Name of the reference variable type (one of XIV, WNS, WLS, PUS, EVS).
- **int _number_of_spheres**
Number of spherical components.
- **int _configurations**
Number of configurations.
- **int _number_of_scales**
Number of scales to use in calculation.
- **int _idfc**
Type of dielectric functions (<0 at XIP, =0 as function of XI, >0 constants).
- **double _exdc**
External medium dielectric constant.
- **double _wp**
WP. Peak pulsation.
- **double _xip**
Peak scale.
- **int _max_layers**
Maximum number of layers for the particle components.
- **bool _use_external_sphere**
Flag to control whether to add an external layer.

8.24.1 Detailed Description

[ScattererConfiguration](#) is a class designed to store the necessary configuration data to describe the scatterer properties.

8.24.2 Constructor & Destructor Documentation

8.24.2.1 ScattererConfiguration() [1/2]

```
ScattererConfiguration::ScattererConfiguration (
    int nsph,
    int configs,
    double * scale_vector,
    int nxi,
    const std::string & variable_name,
    int * iog_vector,
```

```

double * ros_vector,
int * nshl_vector,
double ** rcf_vector,
int dielectric_func_type,
dcomplex *** dc_matrix,
bool has_external,
double exdc,
double wp,
double xip )

```

Prepare a default configuration structure by allocating the necessary memory structures.

Parameters

<i>nsph</i>	int The number of spheres in the simulation.
<i>configs</i>	int Number of spherical monometer configuration types.
<i>scale_vector</i>	double* The radiation-particle scale vector.
<i>nxi</i>	int The number of radiation-particle scalings.
<i>variable_name</i>	string The name of the radiation-particle scaling type.
<i>iog_vector</i>	int* Array of sphere identification numbers.
<i>ros_vector</i>	double* Sphere radius array.
<i>nshl_vector</i>	int* Array of layer numbers.
<i>rcf_vector</i>	double** Array of fractional break radii.
<i>dielectric_func_type</i>	int Type of dielectric function definition (=0 for constant, >0 as function of scale parameter, <0 for functions at XIP value and XI is scale factor for dimensions).
<i>dc_matrix</i>	complex double *** Matrix of reference dielectric constants.
<i>has_external</i>	bool Flag to set whether to add an external spherical layer.
<i>exdc</i>	double External medium dielectric constant.
<i>wp</i>	double wp
<i>xip</i>	double xip

8.24.2.2 ScattererConfiguration() [2/2]

```

ScattererConfiguration::ScattererConfiguration (
    const ScattererConfiguration & rhs )

```

Prepare a default configuration structure by allocating the necessary memory structures.

Parameters

<i>rhs</i>	ScattererConfiguration & Reference to the ScattererConfiguration object to be copied.
------------	---

8.24.3 Member Function Documentation

8.24.3.1 from_binary()

```

ScattererConfiguration * ScattererConfiguration::from_binary (
    const std::string & file_name,
    const std::string & mode = "LEGACY" ) [static]

```


The configuration step can save configuration data as a binary file. The original FORTRAN code used this possibility to manage communication between the configuring code and the calculation program. This possibility is maintained, in case the configuration step needs to be separated from the calculation execution. In this case, `from_binary()` is the class method that restores a `ScattererConfiguration` object from a previously saved binary file.

Parameters

<i>file_name</i>	string Name of the binary configuration data file.
<i>mode</i>	string Binary encoding. Can be one of "LEGACY", Optional (default is "LEGACY").

Returns

config: `ScattererConfiguration*` Pointer to object containing the scatterer configuration data.

8.24.3.2 from_dedfb()

```
ScattererConfiguration * ScattererConfiguration::from_dedfb (
    const std::string & file_name ) [static]
```

To allow for consistency tests and backward compatibility, `ScattererConfiguration` objects can be built from legacy configuration files. This function replicates the approach implemented by the FORTRAN EDFB code, but using a C++ oriented work-flow.

Parameters

<i>file_name</i>	string Name of the legacy configuration data file.
------------------	--

Returns

config: `ScattererConfiguration*` Pointer to object containing the scatterer configuration data.

8.24.3.3 from_hdf5()

```
ScattererConfiguration * ScattererConfiguration::from_hdf5 (
    const std::string & file_name ) [static], [protected]
```

This is the function called by the public method `from_binary()` in case of HDF5 mode selection. This function creates a configuration structure from a binary file written according to the HDF5 format standard.

Parameters

<i>file_name</i>	string Name of the binary configuration data file.
------------------	--

Returns

config: `ScattererConfiguration*` Pointer to object containing the scatterer configuration data.

8.24.3.4 from_legacy()

```
ScattererConfiguration * ScattererConfiguration::from_legacy (
    const std::string & file_name ) [static], [protected]
```

This is the function called by the public method `from_binary()` in case of legacy mode selection. This function creates a configuration structure from a binary file written according to the proprietary mode used by the original FORTRAN code.

Parameters

<i>file_name</i>	string Name of the binary configuration data file.
------------------	--

Returns

config: ScattererConfiguration* Pointer to object containing the scatterer configuration data.

8.24.3.5 get_dielectric_constant()

```
dcomplex ScattererConfiguration::get_dielectric_constant (
    int i,
    int j,
    int k ) [inline]
```

Dielectric constants are stored in a 3D complex matrix, whose dimensions map to [NUMBER_OF_CONFIGURATIONS x NUMBER_OF_SPHERES x NUMBER_OF_SCALES]. This function extracts such values from the matrix through their indices.

Parameters

<i>i</i>	int Index of the configuration.
<i>j</i>	int Index of the sphere.
<i>k</i>	int Index of the current scale.

Returns

radius: dcomplex The requested dielectric constant.

8.24.3.6 get_iog()

```
int ScattererConfiguration::get_iog (
    int index ) [inline]
```

This is a specialized function to get a configuration ID through the index of the sphere it applies to.

Parameters

<i>index</i>	int Index of the sphere.
--------------	--------------------------

Returns

ID: `int` ID of the configuration to be applied.

8.24.3.7 get_max_radius()

```
double ScattererConfiguration::get_max_radius ( )
```

Returns

radius: `double` The radius of the largest sphere.

8.24.3.8 get_min_radius()

```
double ScattererConfiguration::get_min_radius ( )
```

Returns

radius: `double` The radius of the smallest sphere.

8.24.3.9 get_nshl()

```
int ScattererConfiguration::get_nshl (
    int index ) [inline]
```

This is a specialized function to get the number of layers in a specific configuration.

Parameters

<i>index</i>	<code>int</code> Index of the configuration.
--------------	--

Returns

nl: `int` The number of layers for the given configuration.

8.24.3.10 get_particle_radius()

```
double ScattererConfiguration::get_particle_radius (
    GeometryConfiguration * gc )
```

Parameters

<i>gc</i>	<code>GeometryConfiguration *</code> Pointer to a <code>GeometryConfiguration</code> instance.
-----------	--

Returns

radius: double The radius of the sphere containing the particle.

8.24.3.11 get_radius()

```
double ScattererConfiguration::get_radius (
    int index )
```

This is a specialized function to get the radius of a sphere through its index.

Parameters

<i>index</i>	int Index of the ID to be retrieved.
--------------	--------------------------------------

Returns

radius: double The requested sphere radius.

8.24.3.12 get_rcf()

```
double ScattererConfiguration::get_rcf (
    int row,
    int column ) [inline]
```

This is a specialized function to access a scale (generally a wavelength), through its index.

Parameters

<i>row</i>	int Row index of the element to be retrieved.
<i>column</i>	int Column index of the element to be retrieved.

Returns

scale: double The desired scale.

8.24.3.13 get_scale()

```
double ScattererConfiguration::get_scale (
    int index ) [inline]
```

This is a specialized function to access a scale (generally a wavelength), through its index.

Parameters

<i>index</i>	int Index of the scale to be retrieved.
--------------	---

Returns

scale: double The desired scale.

8.24.3.14 get_type_radius()

```
double ScattererConfiguration::get_type_radius (
    int index ) [inline]
```

This is a specialized function to get the radius of a sphere type through its index. Sphere types range from 0 to NUM_CONFIGURATIONS - 1.

Parameters

index	int Index of the ID to be retrieved.
-------	--------------------------------------

Returns

radius: double The requested sphere radius.

8.24.3.15 operator==()

```
bool ScattererConfiguration::operator== (
    const ScattererConfiguration & other )
```

[ScattererConfiguration](#) objects can be obtained in a variety of manner. They can be constructed manually, through the class constructor, they can be read from formatted configuration files, or they can be loaded from binary files. The == operator tests for the equivalence of two configuration instances, returning `true` if they are equivalent, `false` otherwise.

Parameters

other	ScattererConfiguration & Reference to the instance to be compared with.
-------	---

Returns

result: bool True, if the two instances are equal, false otherwise.

8.24.3.16 print()

```
void ScattererConfiguration::print ( )
```

In case of quick debug testing, [ScattererConfiguration.print\(\)](#) allows printing a formatted summary of the configuration data to terminal.

8.24.3.17 write_binary()

```
void ScattererConfiguration::write_binary (
    const std::string & file_name,
    const std::string & mode = "LEGACY" )
```

The execution work-flow may be split in a configuration step and one or more calculation steps. In case the calculation is not being run all-in-one, it can be useful to save the configuration data. [ScattererConfiguration.write_binary\(\)](#) performs the operation of saving the configuration in binary format. This function can work in legacy mode, to write backward compatible configuration files, as well as by wrapping the data into common scientific formats.

Parameters

<i>file_name</i>	string Name of the file to be written.
<i>mode</i>	string Binary encoding. Can be one of ["LEGACY", "HDF5"] . Optional (default is "LEGACY").

8.24.3.18 write_formatted()

```
void ScattererConfiguration::write_formatted (
    const std::string & file_name )
```

Writing configuration to formatted text is an optional operation, which may turn out to be useful for consistency checks. As a matter of fact, formatted configuration output is not read back by the FORTRAN code work-flow and it can be safely omitted, unless there is a specific interest in assessing that the legacy code and the updated one are doing the same thing.

Parameters

<i>file_name</i>	string Name of the file to be written.
------------------	--

8.24.3.19 write_hdf5()

```
void ScattererConfiguration::write_hdf5 (
    const std::string & file_name ) [protected]
```

This function is invoked by the public method [write_binary\(\)](#) with the "HDF5" format mode. It undertakes the task of writing the configuration information to a binary file using the standard HDF5 format.

Parameters

<i>file_name</i>	string Name of the binary configuration data file.
------------------	--

8.24.3.20 write_legacy()

```
void ScattererConfiguration::write_legacy (
    const std::string & file_name ) [protected]
```

This function is invoked by the public method `write_binary()` with the "LEGACY" format mode. It undertakes the task of writing the configuration information to a binary file using a proprietary format, as it was done originally in the FORTRAN code.

Parameters

<code>file_name</code>	string Name of the binary configuration data file.
------------------------	--

The documentation for this class was generated from the following files:

- `np_tmcode/src/include/Configuration.h`
- `np_tmcode/src/libnptm/Configuration.cpp`

8.25 ScatteringAngles Class Reference

A data structure representing the angles to be evaluated in the problem.

```
#include <Commons.h>
```

Public Member Functions

- [ScatteringAngles](#) ([GeometryConfiguration](#) *gconf)
ScatteringAngles instance constructor.
- [ScatteringAngles](#) (const [ScatteringAngles](#) &rhs)
ScatteringAngles copy constructor.

Public Attributes

- [const int & nth](#) = [_nth](#)
Read only view of _nth.
- [const int & nth_s](#) = [_nth_s](#)
Read only view of _nth_s.
- [const int & nph](#) = [_nph](#)
Read only view of _nph.
- [const int & nph_s](#) = [_nph_s](#)
Read only view of _nph_s.
- [const int & nk](#) = [_nk](#)
Read only view of _nk.
- [const int & nks](#) = [_nks](#)
Read only view of _nks.
- [const int & nkks](#) = [_nkks](#)
Read only view of _nkks.
- [const double & th](#) = [_th](#)
Read only view of _th.
- [const double & thstp](#) = [_thstp](#)
Read only view of _thstp.
- [const double & thlst](#) = [_thlst](#)
Read only view of _thlst.

- `const double & ths = _ths`
Read only view of `_ths`.
- `const double & thsstp = _thsstp`
Read only view of `_thsstp`.
- `const double & thslst = _thslst`
Read only view of `_thslst`.
- `const double & ph = _ph`
Read only view of `_ph`.
- `const double & phstp = _phstp`
Read only view of `_phstp`.
- `const double & phlst = _phlst`
Read only view of `_phlst`.
- `const double & phs = _phs`
Read only view of `_phs`.
- `const double & phsstp = _phsstp`
Read only view of `_phsstp`.
- `const double & phslst = _phslst`
Read only view of `_phslst`.
- `const double & thsca = _thsca`
Read only view of `_thsca`.

Protected Attributes

- `int _nth`
Number of incident field azimuth angles.
- `int _nths`
Number of scattered field azimuth angles.
- `int _nph`
Number of incident field elevation angles.
- `int _nphs`
Number of scattered field elevation angles.
- `int _nk`
Number of incident field propagation angles.
- `int _nks`
Number of scattered field propagation angles.
- `int _nkks`
Total number of field propagation angles.
- `double _th`
First incident field azimuth angle.
- `double _thstp`
Incident field azimuth angle increment.
- `double _thlst`
Last incident field azimuth angle.
- `double _ths`
First scattered field azimuth angle.
- `double _thsstp`
Scattered field azimuth angle increment.
- `double _thslst`
Last scattered field azimuth angle.
- `double _ph`

- `double _phstp`
First incident field elevation angle.
- `double _phlst`
Incident field elevation angle increment.
- `double _phs`
Last incident field elevation angle.
- `double _phsstp`
First scattered field elevation angle.
- `double _phsstp`
Scattered field elevation angle increment.
- `double _phslst`
Last scattered field elevation angle.
- `double _thsca`
Azimuth scattering deflection.

8.25.1 Constructor & Destructor Documentation

8.25.1.1 ScatteringAngles() [1/2]

```
ScatteringAngles::ScatteringAngles (
    GeometryConfiguration * gconf )
```

Parameters

<code>gconf</code>	GeometryConfiguration* Pointer to a GeometryConfiguration object.
--------------------	---

8.25.1.2 ScatteringAngles() [2/2]

```
ScatteringAngles::ScatteringAngles (
    const ScatteringAngles & rhs )
```

Parameters

<code>rhs</code>	ScatteringAngles & Reference to the ScatteringAngles object to be copied.
------------------	---

The documentation for this class was generated from the following files:

- `np_tmcode/src/include/`[Commons.h](#)
- `np_tmcode/src/libnptm/`[Commons.cpp](#)

8.26 SphereliterationData Class Reference

A data structure representing the information used for a single scale of the SPHERE case.

```
#include <IterationData.h>
```

Public Member Functions

- [SphereIterationData](#) ([GeometryConfiguration](#) *gconf, [ScattererConfiguration](#) *sconf, const [mixMPI](#) *mpidata, const int device_count)
SphereIterationData default instance constructor.
- [SphereIterationData](#) (const [SphereIterationData](#) &rhs)
SphereIterationData copy constructor.
- [~SphereIterationData](#) ()
SphereIterationData instance destroyer.
- [int](#) update_order ([int](#) order)
Update field expansion order.

Public Attributes

- [double](#) cost
Cosine of incident radiation azimuth.
- [double](#) sint
Sine of incident radiation azimuth.
- [double](#) cosp
Cosine of incident radiation elevation.
- [double](#) sinp
Sine of incident radiation elevation.
- [double](#) costs
Cosine of scattered radiation azimuth.
- [double](#) sints
Sine of scattered radiation azimuth.
- [double](#) cosps
Cosine of scattered radiation elevation.
- [double](#) sinps
Sine of scattered radiation elevation.
- [double](#) vk
Vacuum magnitude of wave vector.
- [double](#) wn
Wave number.
- [double](#) xip
ANNOTATION: Normalization scale.
- [int](#) number_of_scales
Number of scales (wavelengths) to be computed.
- [int](#) xiblock
Size of the block of scales handled by the current process.
- [int](#) firstxi
Index of the first scale handled by the current process.
- [int](#) lastxi
Index of the last scale handled by the current process.
- [dcomplex](#) arg
Argument of harmonic functions.
- [dcomplex](#) s0
 $S0 = FSAS / (4 \pi K^3)$.
- [dcomplex](#) tfsas
Total forward scattering amplitude of the spheres.

- **ParticleDescriptor * c1**
Pointer to a sphere particle descriptor.
- **double * argi**
Imaginary part of arg .
- **double * args**
 arg squared.
- **double scan**
Scattering angle.
- **double cfmp**
Control parameter on incidence direction referred to meridional plane.
- **double sfmp**
Control parameter on scattering direction referred to meridional plane.
- **double cfsp**
Control parameter on incidence direction referred to scattering plane.
- **double sfsp**
Control parameter on scattering direction referred to scattering plane.
- **double * gaps**
Geometry asymmetry parameter for spheres.
- **double * duk**
Variation of unitary wave vector.
- **double * u**
Incidence direction unitary vector.
- **double * us**
Scattering direction unitary vector.
- **double * un**
Normal direction unitary vector.
- **double * uns**
Scattering normal direction unitary vector.
- **double * up**
Polarization direction unitary vector.
- **double * ups**
Scattered polarization direction unitary vector.
- **double * upmp**
Polarization direction unitary vector referred to meridional plane.
- **double * upsmp**
Scattered polarization direction unitary vector referred to meridional plane.
- **double * unmp**
Normal direction unitary vector referred to meridional plane.
- **double * unsmp**
Scattering normal direction unitary vector referred to meridional plane.
- **double ** cmul**
Mueller matrix components.
- **double ** cmullr**
Mueller matrix components referred to meridional plane.
- **dcomplex ** tqspe**
Polarization-dependent extinction contribution to torque for each sphere.
- **dcomplex ** tqsp**
Polarization-dependent scattering contribution to torque for each sphere.
- **double ** tqse**
Extinction contribution to torque for each sphere.
- **double ** tqss**

Scattering contribution to torque for each sphere.

- **double **** zpv**

Scattering coefficients tensor.

- **bool is_first_scale**

flag for first time initialisation

Protected Attributes

- **int _nsph**

Number of spheres.

- **int _lm**

Maximum field expansion order.

- **double * vec_cmul**

Vector of Mueller matrix components.

- **double * vec_cmullr**

Vector of Mueller matrix components referred to meridional plane.

- **dcomplex * vec_tqspe**

Vectorized TQSPE.

- **dcomplex * vec_tqsps**

Vectorized TQSPS.

- **double * vec_tqse**

Vectorized TQSE.

- **double * vec_tqss**

Vectorized TQSS.

- **double * vec_zpv**

Vectorized ZPV.

8.26.1 Constructor & Destructor Documentation

8.26.1.1 SphereIterationData() [1/2]

```
SphereIterationData::SphereIterationData (
    GeometryConfiguration * gconf,
    ScattererConfiguration * sconf,
    const mixMPI * mpidata,
    const int device_count )
```

Parameters

<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration object.
<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration object.
<i>mpidata</i>	mixMPI * Pointer to a mixMPI object.
<i>device_count</i>	const int Number of offload devices available on the system.

8.26.1.2 SphereIterationData() [2/2]

```
SphereIterationData::SphereIterationData (
    const SphereIterationData & rhs )
```

Parameters

<i>rhs</i>	const SphereIterationData & Reference to the object to be copied.
------------	---

8.26.2 Member Function Documentation

8.26.2.1 update_order()

```
int SphereIterationData::update_order (
    int order )
```

Parameters

<i>order</i>	int The new expansion order to be set.
--------------	--

Returns

result: int An exit code (0 if successful).

The documentation for this class was generated from the following files:

- np_tmcode/src/include/[IterationData.h](#)
- np_tmcode/src/sphere/[sphere.cpp](#)

8.27 SphereOutputInfo Class Reference

Class to collect output information for scattering from a single sphere.

```
#include <outputs.h>
```

Public Member Functions

- [SphereOutputInfo](#) ([ScattererConfiguration](#) *sc, [GeometryConfiguration](#) *gc, const [mixMPI](#) *mpidata, int first_xi=1, int xi_length=0)
SphereOutputInfo default instance constructor.
- [SphereOutputInfo](#) (const std::string &hdf5_name)
SphereOutputInfo constructor from HDF5 input.
- [SphereOutputInfo](#) (const int skip_flag)
SphereOutputInfo constructor for the dummy NULL case
- [~SphereOutputInfo](#) ()
SphereOutputInfo instance destroyer.
- [long compute_size](#) ()
Get the size of a *SphereOutputInfo* instance in bytes.
- [int insert](#) (const [SphereOutputInfo](#) &rhs)
Insert in the current output data the data of another block.
- [int write](#) (const std::string &output, const std::string &format)
Write the output to a file.

Static Public Member Functions

- `static long compute_size (ScattererConfiguration *sc, GeometryConfiguration *gc, int first_xi=1, int xi_length=0)`

Estimate the size of the structure that would be built for given input.

Public Attributes

- `const int & skip_flag = _skip_flag`
Read-only view on skip_flag.
- `const int & first_xi = _first_xi`
Read-only view on the ID of the first scale.
- `int nsph`
Number of spheres.
- `int lm`
Maximum field expansion order.
- `int inpol`
Incident polarization flag.
- `int npnt`
Number of points for transition layer integration.
- `int npntts`
Number of points for non-transition layer integration.
- `int isam`
Flag for reference to meridional plane.
- `int idfc`
Flag for dielectric function definition.
- `double th`
First incident radiation azimuth angle.
- `double thstp`
Incident radiation azimuth angle step.
- `double thlst`
Last incident radiation azimuth angle.
- `double ths`
First scattered radiation azimuth angle.
- `double thsstp`
Scattered radiation azimuth angle step.
- `double thslst`
Last scattered radiation azimuth angle.
- `double ph`
First incident radiation elevation angle.
- `double phstp`
Incident radiation elevation angle step.
- `double phlst`
Last incident radiation elevation angle.
- `double phs`
First scattered radiation elevation angle.
- `double phsstp`
Scattered radiation elevation angle step.
- `double phslst`
Last scattered radiation elevation angle.

- **int ndirs**
Number of directions to be explicitly solved.
- **double exri**
Refractive index of external medium.
- **int nxi**
Number of scales (wavelengths)
- **int xi_block_size**
Number of scales handled by the current process.
- **int jwtm**
Index of the wavelength for T-matrix output.
- **int configurations**
Number of sphere types.
- **int lcalc**
Highest expansion order achieved in calculations.
- **dcomplex arg**
Harmonic functions argument.
- **int * vec_jxi**
Vector of scale (wavelength) indices.
- **short * vec_ier**
Vector of error severities (0 - success, 1 - DME).
- **double * vec_vk**
Vector of vacuum wave numbers.
- **double * vec_xi**
Vector of computed scales.
- **double * vec_sphere_sizes**
Vector of sphere sizes (one for every configuration and scale).
- **dcomplex * vec_sphere_ref_indices**
Vector of sphere refractive indices (one for every configuration and scale).
- **double * vec_scs**
Vector of sphere scattering cross-sections.
- **double * vec_abs**
Vector of sphere absorption cross-sections.
- **double * vec_exs**
Vector of sphere extinction cross-sections.
- **double * vec_albeds**
Vector of sphere albedos.
- **double * vec_scsrt**
Vector of sphere scattering-to-geometric cross-sections.
- **double * vec_absrt**
Vector of sphere absorption-to-geometric cross-sections.
- **double * vec_exsrt**
Vector of sphere extinction-to-geometric cross-sections.
- **dcomplex * vec_fsas**
Vector of sphere forward scattering amplitudes.
- **double * vec_qschu**
Vector of sphere QSCHU.
- **double * vec_pschu**
Vector of sphere PSCHU.
- **double * vec_s0mag**
Vector of sphere S0MAG.
- **double * vec_cosav**

- Vector of sphere average asymmetry parameter.*
- **double * vec_raprs**
Vector of sphere average radiation pressure force (N).
- **double * vec_tqek1**
Vector of sphere average extinction torque along incidence direction (parallel polarization).
- **double * vec_tqek2**
Vector of sphere average extinction torque along incidence direction (perpendicular polarization).
- **double * vec_tqsk1**
Vector of sphere average scattering torque along incidence direction (parallel polarization).
- **double * vec_tqsk2**
Vector of sphere average scattering torque along incidence direction (perpendicular polarization).
- **dcomplex * vec_fsat**
Vector of total forward scattering amplitudes.
- **double * vec_qschut**
Vector of total QSCHU.
- **double * vec_pschut**
Vector of total PSCHU.
- **double * vec_s0magt**
Vector of total S0MAG.
- **double * vec_dir_tidg**
Vector of incidence azimuth directions (one per incidence azimuth).
- **double * vec_dir_pidg**
Vector of incidence elevation directions (one per incidence elevation).
- **double * vec_dir_tsdg**
Vector of scattering azimuth directions (one per scattering azimuth).
- **double * vec_dir_psdg**
Vector of scattering elevation directions (one per scattering elevation).
- **double * vec_dir_scand**
Vector of scattering angles (one per direction).
- **double * vec_dir_cfmp**
Control parameter for incidence plane referred to meridional plane (one per direction).
- **double * vec_dir_sfmp**
Control parameter for scattering plane referred to meridional plane (one per direction).
- **double * vec_dir_cfsp**
Control parameter for incidence plane referred to scattering plane (one per direction).
- **double * vec_dir_sfsp**
Control parameter for scattering plane referred to scattering plane (one per direction).
- **double * vec_dir_un**
Components of the unitary vector perpendicular to incidence plane (three per direction).
- **double * vec_dir_uns**
Components of the unitary vector perpendicular to scattering plane (three per direction).
- **dcomplex * vec_dir_sas11**
Vector of sphere differential scattering amplitude with polarization parallel to parallel incidence field.
- **dcomplex * vec_dir_sas21**
Vector of sphere differential scattering amplitude with polarization perpendicular to the parallel incidence field.
- **dcomplex * vec_dir_sas12**
Vector of sphere differential scattering amplitude with polarization perpendicular to perpendicular incidence field.
- **dcomplex * vec_dir_sas22**
Vector of sphere differential scattering amplitude with polarization parallel the perpendicular incidence field.
- **double * vec_dir_fx**
Vector of differential radiation pressure force components along the X axis.

- `double * vec_dir_fy`
Vector of differential radiation pressure force components along the Y axis.
- `double * vec_dir_fz`
Vector of differential radiation pressure force components along the Z axis.
- `double * vec_dir_muls`
Vector of sphere Mueller transformation matrices referred to meridional plane.
- `double * vec_dir_mulsr`
Vector of sphere Mueller transformation matrices referred to scattering plane.

Protected Member Functions

- `int write_hdf5 (const std::string &file_name)`
Write the output to a HDF5 file.
- `int write_legacy (const std::string &output)`
Write the output to a legacy text file.

Protected Attributes

- `int _skip_flag`
Flag for skipping `mpisend()` and `mpireceive()`
- `int _num_theta`
Number of incident azimuth calculations.
- `int _num_thetas`
Number of scattered azimuth calculations.
- `int _num_phi`
Number of incident elevation calculations.
- `int _num_phis`
Number of scattered elevation calculations.
- `int _first_xi`
ID of the first computed wavelength.

8.27.1 Detailed Description

The results of the calculation can be saved in different formats. It is therefore convenient to have a proper memory structure that allows for storing the results and flushing them in any of the permitted formats with just one operation. The purpose of the `SphereOutputInfo` class is to provide a wrapper for the output of the spherical particle scattering solver.

8.27.2 Constructor & Destructor Documentation

8.27.2.1 SphereOutputInfo() [1/3]

```
SphereOutputInfo::SphereOutputInfo (
    ScattererConfiguration * sc,
    GeometryConfiguration * gc,
    const mixMPI * mpidata,
    int first_xi = 1,
    int xi_length = 0 )
```

Parameters

<i>sc</i>	ScattererConfiguration * Pointer to a ScattererConfiguration instance.
<i>gc</i>	GeometryConfiguration * Pointer to a GeometryConfiguration instance.
<i>mpidata</i>	const mixMPI* Pointer to a mixMPI instance.
<i>first_xi</i>	int Index of the first scale in output (optional, default is 1).
<i>xi_length</i>	int Number of scales to be included in output (optional, default is 0, meaning all).

8.27.2.2 SphereOutputInfo() [2/3]

```
SphereOutputInfo::SphereOutputInfo (
    const std::string & hdf5_name )
```

Parameters

<i>hdf5_name</i>	const string & Path to the HDF5 file to be read.
------------------	--

8.27.2.3 SphereOutputInfo() [3/3]

```
SphereOutputInfo::SphereOutputInfo (
    const int skip_flag )
```

Parameters

<i>skip_flag</i>	const int must be passed as the 1 constant.
------------------	---

8.27.3 Member Function Documentation**8.27.3.1 compute_size() [1/2]**

```
long SphereOutputInfo::compute_size ( )
```

Returns

size: long Estimated instance size in bytes.

8.27.3.2 compute_size() [2/2]

```
long SphereOutputInfo::compute_size (
    ScattererConfiguration * sc,
    GeometryConfiguration * gc,
    int first_xi = 1,
    int xi_length = 0 ) [static]
```

Parameters

<i>sc</i>	ScattererConfiguration * Pointer to a ScattererConfiguration instance.
<i>gc</i>	GeometryConfiguration * Pointer to a GeometryConfiguration instance.
<i>first_xi</i>	int Index of the first scale in output (optional, default is 1).
<i>xi_length</i>	int Number of scales to be included in output (optional, default is all).

Returns

size: long Estimated instance size in bytes.

8.27.3.3 insert()

```
int SphereOutputInfo::insert (
    const SphereOutputInfo & rhs )
```

Parameters

<i>rhs</i>	const SphereOutputInfo & Reference to the source data block.
------------	--

Returns

result: int Exit code (0 if successful).

8.27.3.4 write()

```
int SphereOutputInfo::write (
    const std::string & output,
    const std::string & format )
```

Parameters

<i>output</i>	const string & Path to the output to be written.
<i>format</i>	const string & Output format (one of LEGACY or HDF5).

Returns

result: int Exit code (0 if successful).

8.27.3.5 write_hdf5()

```
int SphereOutputInfo::write_hdf5 (
    const std::string & file_name ) [protected]
```

Parameters

<i>file_name</i>	<code>const string & Path to the output to be written.</code>
------------------	---

Returns

result: `int` Exit code (0 if successful).

8.27.3.6 write_legacy()

```
int SphereOutputInfo::write_legacy (
    const std::string & output ) [protected]
```

This function takes care of writing the output using the legacy formatted ASCII structure. If the output file does not exist, it is created. If it exists, the new content is overwritten.

Parameters

<i>output</i>	<code>const string & Path to the output to be written.</code>
---------------	---

Returns

result: `int` Exit code (0 if successful).

The documentation for this class was generated from the following files:

- [np_tmcode/src/include/outputs.h](#)
- [np_tmcode/src/libnptm/outputs.cpp](#)

8.28 Swap1 Class Reference

Class to represent the first group of trapping swap data.

```
#include <tfrfme.h>
```

Public Member Functions

- [Swap1](#) (`int` lm, `int` nk)
 - Swap1 instance constructor.*
- [~Swap1](#) ()
 - Swap1 instance destroyer.*
- [void append](#) (`dcomplex` value)
 - Append an element at the end of the vector.*
- [void reset](#) ()
 - Bring the pointer to the next element at the start of vector.*
- [void write_binary](#) (`const` `std::string` &file_name, `const` `std::string` &mode="LEGACY")
 - Write a [Swap1](#) instance to binary file.*
- [bool operator==](#) ([Swap1](#) &other)
 - Test whether two instances of [Swap1](#) are equal.*

Static Public Member Functions

- `static Swap1 * from_binary (const std::string &file_name, const std::string &mode="LEGACY")`
Load a *Swap1* instance from binary file.
- `static long get_size (int lm, int nkV)`
Calculate the necessary amount of memory to create a new instance.

Public Attributes

- `const dcomplex * wk`
Read only view on WK.

Protected Member Functions

- `void write_hdf5 (const std::string &file_name)`
Save a *Swap1* instance to a HDF5 binary file.
- `void write_legacy (const std::string &file_name)`
Save a *Swap1* instance to a legacy binary file.

Static Protected Member Functions

- `static Swap1 * from_hdf5 (const std::string &file_name)`
Load a *Swap1* instance from a HDF5 binary file.
- `static Swap1 * from_legacy (const std::string &file_name)`
Load a *Swap1* instance from a legacy binary file.

Protected Attributes

- `int _last_index`
Index of the last element to be filled.
- `int _nkV`
Number of beam description wave-numbers.
- `int _nlmmt`
 $NLMMT = 2 * LM * (LM + 2)$
- `dcomplex * _wk`
ANNOTATION: incident amplitudes.

8.28.1 Constructor & Destructor Documentation

8.28.1.1 Swap1()

```
Swap1::Swap1 (
    int lm,
    int nkV )
```

Parameters

<i>lm</i>	int Maximum field expansion order.
<i>nkx</i>	int Number of beam description wave numbers.

8.28.2 Member Function Documentation

8.28.2.1 append()

```
void Swap1::append (
    dcomplex value ) [inline]
```

Parameters

<i>value</i>	complex double The value to be added to the vector.
--------------	---

8.28.2.2 from_binary()

```
Swap1 * Swap1::from_binary (
    const std::string & file_name,
    const std::string & mode = "LEGACY" ) [static]
```

Parameters

<i>file_name</i>	string Name of the file.
<i>mode</i>	string Format of the file (can be either "HDF5" or "LEGACY". Default is "LEGACY").

Returns

instance: `Swap1` * Pointer to a newly created `Swap1` instance.

8.28.2.3 from_hdf5()

```
Swap1 * Swap1::from_hdf5 (
    const std::string & file_name ) [static], [protected]
```

Parameters

<i>file_name</i>	string Name of the file to be loaded.
------------------	---------------------------------------

Returns

instance: `Swap1` * Pointer to a new `Swap1` instance.

8.28.2.4 from_legacy()

```
Swap1 * Swap1::from_legacy (
    const std::string & file_name ) [static], [protected]
```

Parameters

<i>file_name</i>	string Name of the file to be loaded.
------------------	---------------------------------------

Returns

instance: [Swap1](#) * Pointer to a new [Swap1](#) instance.

8.28.2.5 get_size()

```
long Swap1::get_size (
    int lm,
    int nkx ) [static]
```

Parameters

<i>lm</i>	int Maximum field expansion order.
<i>nkx</i>	int ANNOTATION: Number of wave vectors.

Returns

size: long The necessary memory size in bytes.

8.28.2.6 operator==()

```
bool Swap1::operator== (
    Swap1 & other )
```

Parameters

<i>other</i>	Swap1 & Reference to the instance to be compared with.
--------------	--

Returns

result: bool True, if the two instances are equal, false otherwise.

8.28.2.7 write_binary()

```
void Swap1::write_binary (
    const std::string & file_name,
    const std::string & mode = "LEGACY" )
```

Parameters

<i>file_name</i>	string Name of the file.
<i>mode</i>	string Format of the file (can be either "HDF5" or "LEGACY". Default is "LEGACY").

8.28.2.8 write_hdf5()

```
void Swap1::write_hdf5 (
    const std::string & file_name ) [protected]
```

Parameters

<i>file_name</i>	string Name of the file to be written.
------------------	--

8.28.2.9 write_legacy()

```
void Swap1::write_legacy (
    const std::string & file_name ) [protected]
```

Parameters

<i>file_name</i>	string Name of the file to be written.
------------------	--

The documentation for this class was generated from the following files:

- [np_tmcode/src/include/tfrfme.h](#)
- [np_tmcode/src/libnptm/tfrfme.cpp](#)

8.29 Swap2 Class Reference

Class to represent the second group of trapping swap data.

```
#include <tfrfme.h>
```

Public Member Functions

- [Swap2 \(int nk\)](#)
Swap2 instance constructor.
- [~Swap2 \(\)](#)
Swap2 instance destroyer.
- [double * get_vector \(\)](#)
Get the pointer to the VKV vector.
- [void push_matrix \(double value\)](#)
Append an element at the end of the matrix.
- [void push_vector \(double value\)](#)

- Append an element at the end of the vector.*

 - **void reset_matrix ()**
Bring the matrix pointer to the start of the array.
 - **void reset_vector ()**
Bring the vector pointer to the start of the array.
 - **void set_param** (const std::string ¶m_name, double value)
Set a parameter by its name and value.
 - **void write_binary** (const std::string &file_name, const std::string &mode="LEGACY")
*Write a *Swap2* instance to binary file.*
 - **bool operator==** (Swap2 &other)
*Test whether two instances of *Swap2* are equal.*

Static Public Member Functions

- **static Swap2 * from_binary** (const std::string &file_name, const std::string &mode="LEGACY")
*Load a *Swap2* instance from binary file.*
- **static long get_size** (int nkV)
Calculate the necessary amount of memory to create a new instance.

Public Attributes

- **const int & last_vector** = _last_vector
Read-only view on the index of the last vector element to be filled.
- **const int & last_matrix** = _last_matrix
Read-only view on the index of the last matrix element to be filled.
- **const int & nkV** = _nkV
Read-only view on the number of beam description wave numbers.
- **double * vkV**
ANNOTATION: wave number vector.
- **double * vec_vkzm**
ANNOTATION: Vectorized VKZ matrix.
- **const double & apfafa** = _apfafa
ANNOTATION: Normalized aperture.
- **const double & pmf** = _pmf
ANNOTATION: Aperture correction.
- **const double & spd** = _spd
ANNOTATION: Cover slip thickness.
- **const double & rir** = _rir
ANNOTATION: External over oil refractive index.
- **const double & ftcn** = _ftcn
ANNOTATION: Refraction factor.
- **const double & fshmx** = _fshmx
ANNOTATION: Cover slip correction factor.
- **const double & vxyzmx** = _vxyzmx
ANNOTATION: Maximum Cartesian extension of wave.
- **const double & delxyz** = _delxyz
ANNOTATION: Cartesian displacement.
- **const double & vknmx** = _vknmx
ANNOTATION: Maximum wave number.
- **const double & delk** = _delk

ANNOTATION: Wave number variation.

- `const double & delks = _delks`

ANNOTATION: Squared wave number variation.

- `const int & nlmmt = _nlmmt`

$NLMMT = LM * (LM + 2) * 2.$

- `const int & nrvc = _nrvc`

Read-only view on the number of radial vector coordinates.

Protected Member Functions

- `void write_hdf5 (const std::string &file_name)`
Save a [Swap2](#) instance to a HDF5 binary file.
- `void write_legacy (const std::string &file_name)`
Save a [Swap2](#) instance to a legacy binary file.

Static Protected Member Functions

- `static Swap2 * from_hdf5 (const std::string &file_name)`
Load a [Swap2](#) instance from a HDF5 binary file.
- `static Swap2 * from_legacy (const std::string &file_name)`
Load a [Swap2](#) instance from a legacy binary file.

Protected Attributes

- `int _last_vector`
Index of the last vector element to be filled.
- `int _last_matrix`
Index of the last matrix element to be filled.
- `int _nkx`
Number of beam description wave numbers.
- `double _apfafa`
ANNOTATION: normalized aperture.
- `double _pmf`
ANNOTATION: aperture correction.
- `double _spd`
ANNOTATION: cover slip thickness.
- `double _rir`
ANNOTATION: external over oil refractive index ratio.
- `double _ftcn`
ANNOTATION: refraction factor.
- `double _fshmx`
ANNOTATION: cover correction factor.
- `double _vxyzmx`
ANNOTATION: maximum Cartesian extension of wave.
- `double _delxyz`
ANNOTATION: Cartesian displacement.
- `double _vknmx`
ANNOTATION: maximum wave number.
- `double _delk`

Wave number grid spacing.

- `double _delks`

Square of wave number grid spacing.

- `int _nlmmt`

$NLMMT = LM * (LM + 2) * 2.$

- `int _nrvc`

Number of radial vector coordinates.

8.29.1 Constructor & Destructor Documentation

8.29.1.1 Swap2()

```
Swap2::Swap2 (
    int nkx )
```

Parameters

<code>nkx</code>	int Number of beam description wave numbers.
------------------	--

8.29.2 Member Function Documentation

8.29.2.1 from_binary()

```
Swap2 * Swap2::from_binary (
    const std::string & file_name,
    const std::string & mode = "LEGACY" ) [static]
```

Parameters

<code>file_name</code>	string Name of the file.
<code>mode</code>	string Format of the file (can be either "HDF5" or "LEGACY". Default is "LEGACY").

Returns

instance: `Swap2` * Pointer to a newly created `Swap2` instance.

8.29.2.2 from_hdf5()

```
Swap2 * Swap2::from_hdf5 (
    const std::string & file_name ) [static], [protected]
```

Parameters

<code>file_name</code>	string Name of the file to be loaded.
------------------------	---------------------------------------

Returns

instance: [Swap2](#) * Pointer to a new [Swap2](#) instance.

8.29.2.3 from_legacy()

```
Swap2 * Swap2::from_legacy (
    const std::string & file_name ) [static], [protected]
```

Parameters

<i>file_name</i>	string Name of the file to be loaded.
------------------	---------------------------------------

Returns

instance: [Swap2](#) * Pointer to a new [Swap2](#) instance.

8.29.2.4 get_size()

```
long Swap2::get_size (
    int nkV ) [static]
```

Parameters

<i>nkV</i>	int Number of beam description wave numbers.
------------	--

Returns

size: long The necessary memory size in bytes.

8.29.2.5 get_vector()

```
double * Swap2::get_vector ( ) [inline]
```

Returns

value: double * Pointer to the VKV vector.

8.29.2.6 operator==()

```
bool Swap2::operator== (
    Swap2 & other )
```

Parameters

<i>other</i>	Swap1 & Reference to the instance to be compared with.
--------------	--

Returns

result: `bool` True, if the two instances are equal, false otherwise.

8.29.2.7 push_matrix()

```
void Swap2::push_matrix (
    double value )
```

Parameters

<i>value</i>	double The value to be pushed in the matrix.
--------------	--

8.29.2.8 push_vector()

```
void Swap2::push_vector (
    double value ) [inline]
```

Parameters

<i>value</i>	double The value to be pushed in the vector.
--------------	--

8.29.2.9 set_param()

```
void Swap2::set_param (
    const std::string & param_name,
    double value )
```

Parameters

<i>param_name</i>	string Name of the parameter.
<i>value</i>	double The value of the parameter.

8.29.2.10 write_binary()

```
void Swap2::write_binary (
    const std::string & file_name,
    const std::string & mode = "LEGACY" )
```

Parameters

<i>file_name</i>	string Name of the file.
<i>mode</i>	string Format of the file (can be either "HDF5" or "LEGACY". Default is "LEGACY").

8.29.2.11 write_hdf5()

```
void Swap2::write_hdf5 (
    const std::string & file_name ) [protected]
```

Parameters

<i>file_name</i>	string Name of the file to be written.
------------------	--

8.29.2.12 write_legacy()

```
void Swap2::write_legacy (
    const std::string & file_name ) [protected]
```

Parameters

<i>file_name</i>	string Name of the file to be written.
------------------	--

The documentation for this class was generated from the following files:

- np_tmcode/src/include/tfrfme.h
- np_tmcode/src/libnptm/tfrfme.cpp

8.30 TFRFME Class Reference

Class to represent the trapping configuration.

```
#include <tfrfme.h>
```

Public Member Functions

- **TFRFME** (int lmode, int lm, int nkx, int nxv, int nyv, int nzv)
Trapping configuration instance constructor.
- **~TFRFME** ()
Trapping configuration instance destroyer.
- **double * get_x** ()
Get the pointer to the X coordinates vector.
- **double * get_y** ()
Get the pointer to the Y coordinates vector.
- **double * get_z** ()
Get the pointer to the Z coordinates vector.
- **void set_param** (const std::string ¶m_name, double value)
Set a configuration parameter.
- **void write_binary** (const std::string &file_name, const std::string &mode="LEGACY")
Write a trapping configuration instance to binary file.
- **bool operator==** (const TFRFME &other)
Test whether two instances of configuration are equal.

Static Public Member Functions

- `static TFRFME * from_binary (const std::string &file_name, const std::string &mode="LEGACY")`
Load a trapping configuration instance from binary file.
- `static long get_size (int lm, int nkx, int nxv, int nyv, int nzv)`
Calculate the necessary amount of memory to create a new instance.

Public Attributes

- `const int & nlmmt = _nlmmt`
Read-only view on NLMMT.
- `const int & nrvc = _nrvc`
Read-only view on NRVC.
- `const int & lmode = _lmode`
Read-only view on field expansion mode identifier.
- `const int & lm = _lm`
Read-only view on maximum field expansion order.
- `const int & nkx = _nkx`
ANNOTATION: Number of wave vectors.
- `const int & nxv = _nxv`
Read-only view on number of computed X coordinates.
- `const int & nyv = _nyv`
Read-only view on number of computed Y coordinates.
- `const int & nzv = _nzv`
Read-only view on number of computed Z coordinates.
- `const double & vk = _vk`
Read-only view on vacuum wave number.
- `const double & exri = _exri`
Read-only view on external medium refractive index.
- `const double & an = _an`
Read-only view on numeric aperture.
- `const double & ff = _ff`
Read-only view on filling factor.
- `const double & tra = _tra`
Read-only view on lens transmission.
- `const double & spd = _spd`
ANNOTATION: Cover slip thickness.
- `const double & frsh = _frsh`
ANNOTATION: Cover slip correction.
- `const double & exril = _exril`
ANNOTATION: Oil refractive index.
- `dcomplex * vec_wsum`
ANNOTATION: Vectorised WSUM matrix.

Protected Member Functions

- `void write_hdf5 (const std::string &file_name)`
Save a configuration instance to a HDF5 binary file.
- `void write_legacy (const std::string &file_name)`
Save a configuration instance to a legacy binary file.

Static Protected Member Functions

- `static TFRFME * from_hdf5 (const std::string &file_name)`
Load a configuration instance from a HDF5 binary file.
- `static TFRFME * from_legacy (const std::string &file_name)`
Load a configuration instance from a legacy binary file.

Protected Attributes

- `int _nlmmt`
 $NLM MT = 2 * LM * (LM + 2)$
- `int _nrvc`
 $NRVC = NXV * NYV * NZV$.
- `int _lmode`
Beam description mode.
- `int _lm`
Maximum field expansion order.
- `int _nkx`
Number of beam description wave numbers.
- `int _nxv`
Number of computed X coordinates.
- `int _nyv`
Number of computed Y coordinates.
- `int _nzv`
Number of computed Z coordinates.
- `double _vk`
Vacuum wave number.
- `double _exri`
External medium refractive index.
- `double _an`
Numerical aperture.
- `double _ff`
Filling factor.
- `double _tra`
Lens transmission.
- `double _spd`
ANNOTATION: Cover slip thickness.
- `double _frsh`
ANNOTATION: Cover slip correction factor.
- `double _exril`
ANNOTATION: Oil refractive index.
- `double * xv`
Vector of computed x positions.
- `double * yv`
Vector of computed y positions.
- `double * zv`
Vector of computed z positions.

8.30.1 Constructor & Destructor Documentation

8.30.1.1 TFRFME()

```
TFRFME::TFRFME (
    int lmode,
    int lm,
    int nkx,
    int nxv,
    int nyv,
    int nzv )
```

Parameters

<i>lmode</i>	int Order expansion mode flag.
<i>lm</i>	int Maximum field expansion order.
<i>nkx</i>	int ANNOTATION: Number of wave vectors.
<i>nxv</i>	int Number of computed X coordinates.
<i>nyv</i>	int Number of computed Y coordinates.
<i>nzv</i>	int Number of computed Z coordinates.

8.30.2 Member Function Documentation

8.30.2.1 from_binary()

```
TFRFME * TFRFME::from_binary (
    const std::string & file_name,
    const std::string & mode = "LEGACY" ) [static]
```

Parameters

<i>file_name</i>	string Name of the file.
<i>mode</i>	string Format of the file (can be either "HDF5" or "LEGACY". Default is "LEGACY").

Returns

instance: [TFRFME](#) * Pointer to a newly created configuration instance.

8.30.2.2 from_hdf5()

```
TFRFME * TFRFME::from_hdf5 (
    const std::string & file_name ) [static], [protected]
```

Parameters

<i>file_name</i>	string Name of the file to be loaded.
------------------	---------------------------------------

Returns

instance: [TFRFME](#) * Pointer to a new trapping configuration instance.

8.30.2.3 from_legacy()

```
TFRFME * TFRFME::from\_legacy (
    const std::string & file_name ) [static], [protected]
```

Parameters

<i>file_name</i>	string Name of the file to be loaded.
------------------	---

Returns

instance: [TFRFME](#) * Pointer to a new trapping configuration instance.

8.30.2.4 get_size()

```
long TFRFME::get\_size (
    int lm,
    int nkv,
    int nxv,
    int nyv,
    int nzv ) [static]
```

Parameters

<i>lm</i>	int Maximum field expansion order.
<i>nk_v</i>	int ANNOTATION: Number of wave vectors.
<i>nx_v</i>	int Number of computed X coordinates.
<i>ny_v</i>	int Number of computed Y coordinates.
<i>nz_v</i>	int Number of computed Z coordinates.

Returns

size: [long](#) The necessary memory size in bytes.

8.30.2.5 get_x()

```
double * TFRFME::get\_x ( ) [inline]
```

Returns

x: [double](#) * Pointer to X coordinates vector.

8.30.2.6 get_y()

```
double * TFRFME::get_y ( ) [inline]
```

Returns

y: double * Pointer to Y coordinates vector.

8.30.2.7 get_z()

```
double * TFRFME::get_z ( ) [inline]
```

Returns

z: double * Pointer to Z coordinates vector.

8.30.2.8 operator==()

```
bool TFRFME::operator== (
    const TFRFME & other )
```

Parameters

<i>other</i>	<code>TFRFME</code> & Reference to the instance to be compared with.
--------------	--

Returns

result: bool True, if the two instances are equal, false otherwise.

8.30.2.9 set_param()

```
void TFRFME::set_param (
    const std::string & param_name,
    double value )
```

Parameters

<i>param_name</i>	string Name of the parameter.
<i>value</i>	double Value to be stored as parameter.

8.30.2.10 write_binary()

```
void TFRFME::write_binary (
    const std::string & file_name,
    const std::string & mode = "LEGACY" )
```

Parameters

<i>file_name</i>	string Name of the file.
<i>mode</i>	string Format of the file (can be either "HDF5" or "LEGACY". Default is "LEGACY").

8.30.2.11 write_hdf5()

```
void TFRFME::write_hdf5 (
    const std::string & file_name ) [protected]
```

Parameters

<i>file_name</i>	string Name of the file to be written.
------------------	--

8.30.2.12 write_legacy()

```
void TFRFME::write_legacy (
    const std::string & file_name ) [protected]
```

Parameters

<i>file_name</i>	string Name of the file to be written.
------------------	--

The documentation for this class was generated from the following files:

- [np_tmcode/src/include/tfrfme.h](#)
- [np_tmcode/src/libnptm/tfrfme.cpp](#)

8.31 TransitionMatrix Class Reference

Class to represent the Transition Matrix.

```
#include <TransitionMatrix.h>
```

Public Member Functions

- [TransitionMatrix](#) (int is, int lm, double vk, double exri, dcomplex *_elems, double radius=0.0)
Default Transition Matrix instance constructor.
- [TransitionMatrix](#) (int lm, double vk, double exri, dcomplex **rmi, dcomplex **rei, double radius)
Transition Matrix instance constructor for single sphere.
- [TransitionMatrix](#) (int nlemt, int lm, double vk, double exri, dcomplex **am0m)
Transition Matrix instance constructor for a cluster of spheres.
- [~TransitionMatrix](#) ()
Transition Matrix instance destroyer.
- [void write_binary](#) (const std::string &file_name, const std::string &mode="LEGACY")
Write the Transition Matrix to a binary file.
- [bool operator==](#) ([TransitionMatrix](#) &other)
Test whether two instances of [TransitionMatrix](#) are equal.

Static Public Member Functions

- `static TransitionMatrix * from_binary (const std::string &file_name, const std::string &mode="LEGACY")`
Build transition matrix from binary input file.
- `static void write_binary (const std::string &file_name, np_int nlemt, int lm, double vk, double exri, dcomplex **am0m, const std::string &mode="LEGACY")`
Write a cluster Transition Matrix to a binary file without instanciating it.
- `static void write_binary (const std::string &file_name, int lm, double vk, double exri, dcomplex **rmi, dcomplex **rei, double sphere_radius, const std::string &mode="LEGACY")`
Write a single sphere Transition Matrix to a binary file without instanciating it.

Public Attributes

- `const int & is = _is`
Read-only view on matrix type identifier.
- `const int & l_max = _l_max`
Read-only view on maximum field expansion order.
- `const double & vk = _vk`
Read-only view on wave number in scale units.
- `const double & exri = _exri`
Read-only view on external medium refractive index.
- `dcomplex * elements`
Vectorized matrix elements.
- `const double & sphere_radius = _sphere_radius`
Read-only view on sphere radius.
- `int * shape`
Matrix shape.

Protected Member Functions

- `void write_hdf5 (const std::string &file_name)`
Write the Transition Matrix to HDF5 binary output.
- `void write_legacy (const std::string &file_name)`
Write the Transition Matrix to legacy binary output.

Static Protected Member Functions

- `static TransitionMatrix * from_hdf5 (const std::string &file_name)`
Build transition matrix from a HDF5 binary input file.
- `static TransitionMatrix * from_legacy (const std::string &file_name)`
Build transition matrix from a legacy binary input file.
- `static void write_hdf5 (const std::string &file_name, np_int nlemt, int lm, double vk, double exri, dcomplex **am0m)`
Write transition matrix data to HDF5 binary output.
- `static void write_hdf5 (const std::string &file_name, int lm, double vk, double exri, dcomplex **rmi, dcomplex **rei, double sphere_radius)`
Write transition matrix data to HDF5 binary output.
- `static void write_legacy (const std::string &file_name, np_int nlemt, int lm, double vk, double exri, dcomplex **am0m)`
Write transition matrix data to binary output using legacy format.
- `static void write_legacy (const std::string &file_name, int lm, double vk, double exri, dcomplex **rmi, dcomplex **rei, double radius)`
Write transition matrix data to binary output using legacy format.

Protected Attributes

- `int _is`
Matrix type identifier.
- `int _l_max`
Maximum field expansion order.
- `double _vk`
Wave number in scale units.
- `double _exri`
External medium refractive index.
- `double _sphere_radius`
Sphere radius.

8.31.1 Constructor & Destructor Documentation

8.31.1.1 TransitionMatrix() [1/3]

```
TransitionMatrix::TransitionMatrix (
    int is,
    int lm,
    double vk,
    double exri,
    dcomplex * _elems,
    double radius = 0.0 )
```

Parameters

<i>is</i>	int Matrix type identifier
<i>lm</i>	int Maximum field expansion order.
<i>vk</i>	double Wave number in scale units.
<i>exri</i>	double External medium refractive index.
<i>_elems</i>	complex double * Vectorized elements of the matrix.
<i>radius</i>	double Radius for the single sphere case (defaults to 0.0).

8.31.1.2 TransitionMatrix() [2/3]

```
TransitionMatrix::TransitionMatrix (
    int lm,
    double vk,
    double exri,
    dcomplex ** rmi,
    dcomplex ** rei,
    double radius )
```

This constructor allocates the memory structure needed to represent the transition matrix for the case of a single sphere.

Parameters

<i>lm</i>	int Maximum field expansion order.
-----------	------------------------------------

Parameters

<i>vk</i>	double Wave number in scale units.
<i>exri</i>	double External medium refractive index.
<i>rmi</i>	complex double **
<i>rei</i>	complex double **
<i>radius</i>	double Radius of the sphere.

8.31.1.3 TransitionMatrix() [3/3]

```
TransitionMatrix::TransitionMatrix (
    int nlemt,
    int lm,
    double vk,
    double exri,
    dcomplex ** am0m )
```

This constructor allocates the memory structure needed to represent the transition matrix for the case of a cluster of spheres.

Parameters

<i>nlemt</i>	int Size of the matrix (2 * LE * (LE + 2)).
<i>lm</i>	int Maximum field expansion order.
<i>vk</i>	double Wave number in scale units.
<i>exri</i>	double External medium refractive index.
<i>am0m</i>	complex double **

8.31.2 Member Function Documentation

8.31.2.1 from_binary()

```
TransitionMatrix * TransitionMatrix::from_binary (
    const std::string & file_name,
    const std::string & mode = "LEGACY" ) [static]
```

In some cases, it is necessary to perform calculations starting from a pre-computed transition matrix. If this matrix is not available in memory ([e.g.](#) because it was calculated by a different process), but it was saved in a binary file, this function can be used to load it back in memory. The function operates in two modes: "LEGACY", which reads the matrix data from a proprietary binary file, having the same structure as the one used by the original FORTRAN code, and "HDF5", which, instead, reads the data from an input file complying with the HDF5 format.

Parameters

<i>file_name</i>	string Name of the binary configuration data file.
<i>mode</i>	string Binary encoding. Can be one of ["LEGACY", "HDF5"]. Optional (default is "LEGACY").

Returns

config: [TransitionMatrix](#) * Pointer to object containing the transition matrix data.

8.31.2.2 from_hdf5()

```
TransitionMatrix * TransitionMatrix::from_hdf5 (
    const std::string & file_name ) [static], [protected]
```

This function takes care of the specific task of building a transition matrix memory data structure from a HDF5 binary input file.

Parameters

<i>file_name</i>	string Name of the binary configuration data file.
------------------	--

Returns

config: [TransitionMatrix](#) * Pointer to object containing the transition matrix data.

8.31.2.3 from_legacy()

```
TransitionMatrix * TransitionMatrix::from_legacy (
    const std::string & file_name ) [static], [protected]
```

Parameters

<i>file_name</i>	string Name of the binary configuration data file.
------------------	--

Returns

config: [TransitionMatrix](#) * Pointer to object containing the transition matrix data.

8.31.2.4 operator==()

```
bool TransitionMatrix::operator== (
    TransitionMatrix & other )
```

Transition matrices can be the result of a calculation or of a file input operation, reading from a previously computed object. The == operator tests for the equivalence of two transition matrices, returning `true` if they are equivalent, `false` otherwise.

Parameters

<i>other</i>	TransitionMatrix & Reference to the instance to be compared with.
--------------	---

Returns

result: `bool` True, if the two instances are equal, false otherwise.

8.31.2.5 write_binary() [1/3]

```
void TransitionMatrix::write_binary (
    const std::string & file_name,
    const std::string & mode = "LEGACY" )
```

This function writes a hard-copy of the transition matrix to an output file, making it available for subsequent processes to reload. The function operates in two modes: "LEGACY", which writes a proprietary binary file, using the same structure of the original FORTRAN code, and "HDF5", which, instead, writes the output to a file using the HDF5 format, thus leaving it available for inspection with external tools.

Parameters

<i>file_name</i>	string Name of the file to be written.
<i>mode</i>	string Binary encoding. Can be one of ["LEGACY", "HDF5"] . Optional (default is "LEGACY").

8.31.2.6 write_binary() [2/3]

```
void TransitionMatrix::write_binary (
    const std::string & file_name,
    int lm,
    double vk,
    double exri,
    dcomplex ** rmi,
    dcomplex ** rei,
    double sphere_radius,
    const std::string & mode = "LEGACY" ) [static]
```

Transition Matrix data can take a large amount of memory. For such reason, attempts to create [TransitionMatrix](#) instances only for writing purposes can create unnecessary resource consumption and computing time to duplicate the data into the output buffer. This function offers output to file as a static method. It takes the arguments of a constructor together with the usual arguments to specify the output file name and format, to write the required data directly to a file, without creating a new [TransitionMatrix](#) instance. The implementation works for [TransitionMatrix](#) objects built for the single sphere case. It belongs to the public class interface and it calls the proper versions of [write_legacy\(\)](#) and [write_hdf5\(\)](#), depending on the requested output format.

Parameters

<i>file_name</i>	string Name of the file to be written.
<i>lm</i>	int Maximum field expansion order.
<i>vk</i>	double Wave number in scale units.
<i>exri</i>	double External medium refractive index.
<i>rmi</i>	complex double **
<i>rei</i>	complex double **
<i>sphere_radius</i>	double Radius of the sphere.
<i>mode</i>	string Binary encoding. Can be one of ["LEGACY", "HDF5"] . Optional (default is "LEGACY").

8.31.2.7 write_binary() [3/3]

```
void TransitionMatrix::write_binary (
    const std::string & file_name,
    np_int nlemt,
    int lm,
    double vk,
    double exri,
    dcomplex ** am0m,
    const std::string & mode = "LEGACY" ) [static]
```

Transition Matrix data can take a large amount of memory. For such reason, attempts to create [TransitionMatrix](#) instances only for writing purposes can create unnecessary resource consumption and computing time to duplicate the data into the output buffer. This function offers output to file as a static method. It takes the arguments of a constructor together with the usual arguments to specify the output file name and format, to write the required data directly to a file, without creating a new [TransitionMatrix](#) instance. The implementation works for [TransitionMatrix](#) objects built for the CLUSTER case. It belongs to the public class interface and it calls the proper versions of [write_legacy\(\)](#) and [write_hdf5\(\)](#), depending on the requested output format.

Parameters

<i>file_name</i>	string Name of the file to be written.
<i>nlemt</i>	np_int Size of the matrix (2 * LE * (LE + 2)).
<i>lm</i>	int Maximum field expansion order.
<i>vk</i>	double Wave number in scale units.
<i>exri</i>	double External medium refractive index.
<i>am0m</i>	complex double **
<i>mode</i>	string Binary encoding. Can be one of ["LEGACY", "HDF5"] . Optional (default is "LEGACY").

8.31.2.8 write_hdf5() [1/3]

```
void TransitionMatrix::write_hdf5 (
    const std::string & file_name ) [protected]
```

This function takes care of the specific task of writing the transition matrix memory data structure to a binary output file formatted according to the HDF5 standard.

Parameters

<i>file_name</i>	string Name of the binary configuration data file.
------------------	--

8.31.2.9 write_hdf5() [2/3]

```
void TransitionMatrix::write_hdf5 (
    const std::string & file_name,
    int lm,
    double vk,
    double exri,
    dcomplex ** rmi,
```

```

dcomplex ** rei,
double sphere_radius ) [static], [protected]

```

This function takes care of the specific task of writing the transition matrix memory data structure to a binary output file formatted according to the HDF5 standard without a pre-existing instance. It is designed to work for the case of a single sphere.

Parameters

<i>file_name</i>	string Name of the binary configuration data file.
<i>lm</i>	int Maximum field expansion order.
<i>vk</i>	double Wave number in scale units.
<i>exri</i>	double External medium refractive index.
<i>rmi</i>	complex double **
<i>rei</i>	complex double **
<i>sphere_radius</i>	double Radius of the sphere.

8.31.2.10 write_hdf5() [3/3]

```

void TransitionMatrix::write_hdf5 (
    const std::string & file_name,
    np_int nlemt,
    int lm,
    double vk,
    double exri,
    dcomplex ** am0m ) [static], [protected]

```

This function takes care of the specific task of writing the transition matrix memory data structure to a binary output file formatted according to the HDF5 standard without a pre-existing instance. It is designed to work for the case of a cluster of spheres.

Parameters

<i>file_name</i>	string Name of the binary configuration data file.
<i>nlemt</i>	np_int Size of the matrix ($2 * LE * (LE + 2)$).
<i>lm</i>	int Maximum field expansion order.
<i>vk</i>	double Wave number in scale units.
<i>exri</i>	double External medium refractive index.
<i>am0m</i>	complex double **

8.31.2.11 write_legacy() [1/3]

```

void TransitionMatrix::write_legacy (
    const std::string & file_name ) [protected]

```

Parameters

<i>file_name</i>	string Name of the binary configuration data file.
------------------	--

8.31.2.12 write_legacy() [2/3]

```
void TransitionMatrix::write_legacy (
    const std::string & file_name,
    int lm,
    double vk,
    double exri,
    dcomplex ** rmi,
    dcomplex ** rei,
    double radius ) [static], [protected]
```

This function takes care of the specific task of writing the transition matrix memory data structure to a binary output file formatted according to the original code structure without a pre-existing instance. It is designed to work for the case of a single sphere.

Parameters

<i>file_name</i>	string Name of the binary configuration data file.
<i>lm</i>	int Maximum field expansion order.
<i>vk</i>	double Wave number in scale units.
<i>exri</i>	double External medium refractive index.
<i>rmi</i>	complex double **
<i>rei</i>	complex double **
<i>radius</i>	double Radius of the sphere.

8.31.2.13 write_legacy() [3/3]

```
void TransitionMatrix::write_legacy (
    const std::string & file_name,
    np_int nlemt,
    int lm,
    double vk,
    double exri,
    dcomplex ** am0m ) [static], [protected]
```

This function takes care of the specific task of writing the transition matrix memory data structure to a binary output file formatted according to the format used by the legacy FORTRAN code. It is designed to work for the case of clusters of spheres.

Parameters

<i>file_name</i>	string Name of the binary configuration data file.
<i>nlemt</i>	np_int Size of the matrix ($2 * LE * (LE + 2)$).
<i>lm</i>	int Maximum field expansion order.
<i>vk</i>	double Wave number in scale units.
<i>exri</i>	double External medium refractive index.
<i>am0m</i>	complex double **

The documentation for this class was generated from the following files:

- np_tmcode/src/include/[TransitionMatrix.h](#)
- np_tmcode/src/libnptm/[TransitionMatrix.cpp](#)

8.32 TrappingOutputInfo Class Reference

Class to collect output information for particle trapping.

```
#include <outputs.h>
```

Public Member Functions

- `TrappingOutputInfo (int ift, int iss, int itw, int nx, int ny, int nz)`
TrappingOutputInfo default instance constructor.
- `~TrappingOutputInfo ()`
TrappingOutputInfo instance destroyer.
- `long get_size ()`
Get the size of a TrappingOutputInfo instance in bytes.
- `void set_param (const std::string &pname, double pvalue)`
Set the value of a parameter by its name.
- `int write (const std::string &output, const std::string &format)`
Write the output to a file.

Static Public Member Functions

- `static long get_size (int ift, int iss, int nx, int ny, int nz)`
Estimate the size of the structure that would be built for given input.

Public Attributes

- `const int & jft = _jft`
Read-only view on force / torque mode.
- `const int & jss = _jss`
TWS / TFRFME switch.
- `const int & jtw = _jtw`
Flag for formatted output.
- `const int & nkx = _nkx`
Read-only view on number of wave vectors used to describe the radiation field.
- `const int & nxv = _nxv`
Read-only view on number of vertices along X-axis.
- `const int & nyv = _nyv`
Read-only view on number of vertices along Y-axis.
- `const int & nzv = _nzv`
Read-only view on number of vertices along Z-axis.
- `const int & lmode = _lmode`
Laser mode flag.
- `const int & le = _le`
Maximum field expansion order for the T-matrix.
- `const double & vk = _vk`
Vacuum wave number of laser radiation field.
- `const double & exri = _exri`
Refractive index of the external medium.
- `const double & an = _an`

- *Numeric aperture of focusing lens.*
- `const double & ff = _ff`
Lens filling factor.
- `const double & tra = _tra`
Lens transmittance.
- `const double & spd = _spd`
Offset of cover slip.
- `const double & frsh = _frsh`
Cover slip offset normalization.
- `const double & exril = _exril`
Refractive index of cover lens.
- `double * vec_x`
Vector of X-coordinates grid spacings.
- `double * vec_y`
Vector of Y-coordinates grid spacings.
- `double * vec_z`
Vector of Z-coordinates grid spacings.
- `double * vec_csf1`
Vector of first components for force cross-sections.
- `double * vec_csf2`
Vector of second components for force cross-sections.
- `double * vec_csf3`
Vector of third components for force cross-sections.
- `double * vec_cst1`
Vector of first components for torque cross-sections.
- `double * vec_cst2`
Vector of second components for torque cross-sections.
- `double * vec_cst3`
Vector of third components for torque cross-sections.

Protected Member Functions

- `int write_hdf5 (const std::string &file_name)`
Write the output to a HDF5 file.
- `int write_legacy (const std::string &output)`
Write the output to a legacy text file.

Protected Attributes

- `int _jft`
Force / torque mode.
- `int _jss`
TWS / TFRFME switch.
- `int _jtw`
Flag for formatted output.
- `int _nkx`
Number of wave vectors used to describe the radiation field.
- `int _nxv`
Number of vertices along X-axis.
- `int _nyv`

- `int _nzv`
Number of vertices along Y-axis.
- `int _lmode`
Laser mode flag.
- `int _le`
Maximum field expansion order for the T-matrix.
- `double _vk`
Vacuum wave number of laser radiation field.
- `double _exri`
Refractive index of the external medium.
- `double _an`
Numeric aperture of focusing lens.
- `double _ff`
Lens filling factor.
- `double _tra`
Lens transmittance.
- `double _spd`
Offset of cover slip.
- `double _frsh`
Cover slip offset normalization.
- `double _exril`
Refractive index of cover lens.

8.32.1 Detailed Description

The results of the calculation can be saved in different formats. It is therefore convenient to have a proper memory structure that allows for storing the results and flushing them in any of the permitted formats with just one operation. The purpose of the `TrappingOutputInfo` class is to provide a wrapper for the output of the particle trapping solver.

8.32.2 Constructor & Destructor Documentation

8.32.2.1 TrappingOutputInfo()

```
TrappingOutputInfo::TrappingOutputInfo (
    int ift,
    int iss,
    int itw,
    int nx,
    int ny,
    int nz )
```

Parameters

<i>ift</i>	'int' Force / torque toggle mode.
<i>iss</i>	'int' TWS / TFRFME switch.
<i>itw</i>	'int' Formatted output switch (0 - disabled, 1 - enabled).
<i>nx</i>	'int' Number of vertices along the X-axis.
<i>ny</i>	'int' Number of vertices along the Y-axis.
<i>nz</i>	'int' Number of vertices along the Z-axis.

8.32.3 Member Function Documentation

8.32.3.1 `get_size()` [1/2]

```
long TrappingOutputInfo::get_size ( )
```

Returns

size: long Estimated instance size in bytes.

8.32.3.2 `get_size()` [2/2]

```
long TrappingOutputInfo::get_size (
    int ift,
    int iss,
    int nx,
    int ny,
    int nz ) [static]
```

Parameters

<i>ift</i>	'int' Force / torque toggle mode.
<i>iss</i>	'int' TWS / TFRFME switch.
<i>nx</i>	'int' Number of vertices along the X-axis.
<i>ny</i>	'int' Number of vertices along the Y-axis.
<i>nz</i>	'int' Number of vertices along the Z-axis.

Returns

size: long Estimated instance size in bytes.

8.32.3.3 `set_param()`

```
void TrappingOutputInfo::set_param (
    const std::string & pname,
    double pvalue )
```

Parameters

<i>pname</i>	'const string&' Name of the parameter to be set.
<i>pvalue</i>	'double' Value of the parameter.

8.32.3.4 `write()`

```
int TrappingOutputInfo::write (
    const std::string & output,
    const std::string & format )
```


Parameters

<i>output</i>	<code>const string &</code> Path to the output to be written.
<i>format</i>	<code>const string &</code> Output format (one of LEGACY or HDF5).

Returns

result: `int` Exit code (0 if successful).

8.32.3.5 write_hdf5()

```
int TrappingOutputInfo::write_hdf5 (
    const std::string & file_name ) [protected]
```

Parameters

<i>file_name</i>	<code>const string &</code> Path to the output to be written.
------------------	---

Returns

result: `int` Exit code (0 if successful).

8.32.3.6 write_legacy()

```
int TrappingOutputInfo::write_legacy (
    const std::string & output ) [protected]
```

This function takes care of writing the output using the legacy formatted ASCII structure. If the output file does not exist, it is created. If it exists, the new content is overwritten.

Parameters

<i>output</i>	<code>const string &</code> Path to the output to be written.
---------------	---

Returns

result: `int` Exit code (0 if successful).

The documentation for this class was generated from the following files:

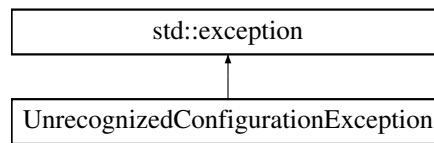
- [np_tmcode/src/include/outputs.h](#)
- [np_tmcode/src/libnptm/outputs.cpp](#)

8.33 UnrecognizedConfigurationException Class Reference

Exception for unrecognized configuration data sets.

```
#include <errors.h>
```

Inheritance diagram for UnrecognizedConfigurationException:



Public Member Functions

- [UnrecognizedConfigurationException](#) (`const` `std::string` &`problem`)
Exception instance constructor.
- `virtual const char *` `what` (`() const throw` ())
Exception message.

Protected Attributes

- `std::string` `message`
Description of the problem.

8.33.1 Constructor & Destructor Documentation

8.33.1.1 UnrecognizedConfigurationException()

```
UnrecognizedConfigurationException::UnrecognizedConfigurationException (
    const std::string & problem ) [inline]
```

Parameters

<i>problem</i>	<code>string</code> Description of the problem that occurred.
----------------	---

The documentation for this class was generated from the following file:

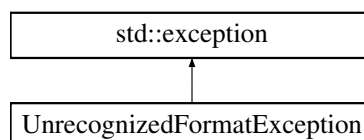
- `np_tmcode/src/include/`[errors.h](#)

8.34 UnrecognizedFormatException Class Reference

Exception for unrecognized file formats.

```
#include <errors.h>
```

Inheritance diagram for UnrecognizedFormatException:



Public Member Functions

- [UnrecognizedFormatException](#) (`const std::string &problem`)
Exception instance constructor.
- `virtual const char * what () const throw ()`
Exception message.

Protected Attributes

- `std::string message`
Description of the problem.

8.34.1 Constructor & Destructor Documentation

8.34.1.1 UnrecognizedFormatException()

```
UnrecognizedFormatException::UnrecognizedFormatException (
    const std::string & problem ) [inline]
```

Parameters

<i>problem</i>	string Description of the problem that occurred.
----------------	--

The documentation for this class was generated from the following file:

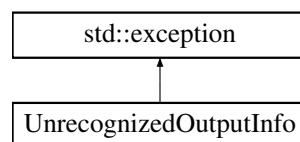
- `np_tmcode/src/include/errors.h`

8.35 UnrecognizedOutputInfo Class Reference

Exception for wrong OutputInfo NULL calls.

```
#include <errors.h>
```

Inheritance diagram for UnrecognizedOutputInfo:



Public Member Functions

- [UnrecognizedOutputInfo](#) (`int requested`)
Exception instance constructor.
- `virtual const char * what () const throw ()`
Exception message.

Protected Attributes

- `std::string message`
Description of the problem.

8.35.1 Constructor & Destructor Documentation

8.35.1.1 UnrecognizedOutputInfo()

```
UnrecognizedOutputInfo::UnrecognizedOutputInfo (
    int requested ) [inline]
```

Parameters

<i>requested</i>	<code>int</code> The index that was requested.
------------------	--

The documentation for this class was generated from the following file:

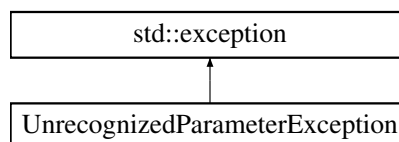
- `np_tmcode/src/include/errors.h`

8.36 UnrecognizedParameterException Class Reference

Exception for unrecognized parameters.

```
#include <errors.h>
```

Inheritance diagram for UnrecognizedParameterException:



Public Member Functions

- `UnrecognizedParameterException (const std::string &problem)`
Exception instance constructor.
- `virtual const char * what () const throw ()`
Exception message.

Protected Attributes

- `std::string message`
Description of the problem.

8.36.1 Constructor & Destructor Documentation

8.36.1.1 UnrecognizedParameterException()

```
UnrecognizedParameterException::UnrecognizedParameterException (
    const std::string & problem ) [inline]
```

Parameters

<i>problem</i>	string Description of the problem that occurred.
----------------	--

The documentation for this class was generated from the following file:

- [np_tmcode/src/include/errors.h](#)

8.37 VirtualAsciiFile Class Reference

Virtual representation of an ASCII file.

```
#include <file_io.h>
```

Public Member Functions

- [VirtualAsciiFile](#) (int32_t lines=0)
VirtualAsciiFile instance constructor.
- [VirtualAsciiFile](#) (const VirtualAsciiFile &rhs)
VirtualAsciiFile copy constructor.
- [VirtualAsciiFile](#) (const mixMPI *mpidata, int rr)
VirtualAsciiFile instance constructor copying all contents off MPISend() calls from MPI process rr.
- [~VirtualAsciiFile](#) ()
VirtualAsciiFile instance destroyer.
- [void append](#) (VirtualAsciiFile &rhs)
Append another VirtualAsciiFile at the end of the current instance.
- [void append_line](#) (const std::string &line)
Append a line at the end of the file.
- [int append_to_disk](#) (const std::string &file_name)
Append the contents of the VirtualAsciiFile to a physical file on disk.
- [int insert](#) (int32_t position, VirtualAsciiFile &rhs, int32_t start=0, int32_t end=0)
Insert another VirtualAsciiFile at a given position.
- [int32_t number_of_lines](#) ()
Get the number of lines in the current instance.
- [int write_to_disk](#) (const std::string &file_name)
Write virtual file contents to a real file on disk.
- [void mpisend](#) (const mixMPI *mpidata)
Send VirtualAsciiFile instance to MPI process 0 via MPISend() calls.

Protected Attributes

- `std::vector< std::string > * _file_lines`
The number of lines.

8.37.1 Constructor & Destructor Documentation

8.37.1.1 VirtualAsciiFile() [1/3]

```
VirtualAsciiFile::VirtualAsciiFile (
    int32_t lines = 0 )
```

Parameters

<i>lines</i>	<code>int32_t</code> Number of lines, if known in advance (optional, default is 0).
--------------	---

8.37.1.2 VirtualAsciiFile() [2/3]

```
VirtualAsciiFile::VirtualAsciiFile (
    const VirtualAsciiFile & rhs )
```

Parameters

<i>rhs</i>	const <code>VirtualAsciiFile</code> & Reference to a <code>VirtualAsciiFile</code> instance.
------------	--

8.37.1.3 VirtualAsciiFile() [3/3]

```
VirtualAsciiFile::VirtualAsciiFile (
    const mixMPI * mpidata,
    int rr )
```

Parameters

<i>mpidata</i>	<code>mixMPI</code> * pointer to MPI data structure.
<i>rr</i>	<code>int</code> rank of the MPI process sending the data.

8.37.2 Member Function Documentation**8.37.2.1 append()**

```
void VirtualAsciiFile::append (
    VirtualAsciiFile & rhs )
```

Parameters

<i>rhs</i>	const <code>VirtualAsciiFile</code> & Reference to the <code>VirtualAsciiFile</code> to be appended.
------------	--

8.37.2.2 append_line()

```
void VirtualAsciiFile::append_line (
    const std::string & line )
```

Parameters

<i>line</i>	const <code>string</code> & Reference to a string representing the line.
-------------	--

8.37.2.3 append_to_disk()

```
int VirtualAsciiFile::append_to_disk (
    const std::string & file_name )
```

Parameters

<i>file_name</i>	const string& Name of the file to append contents to.
------------------	---

Returns

result: int A result code (0 if successful).

8.37.2.4 insert()

```
int VirtualAsciiFile::insert (
    int32_t position,
    VirtualAsciiFile & rhs,
    int32_t start = 0,
    int32_t end = 0 )
```

This function inserts a target [VirtualAsciiFile](#) in the current one at the given position. Optionally, a range of lines to be inserted can be specified, otherwise the full content of the target file is inserted. This function DOES NOT increase the size of the inner storage and it can only be used if the inner storage has already been adjusted to contain the insertion target.

Parameters

<i>position</i>	int32_t The position at which the other file is inserted in this one.
<i>rhs</i>	const VirtualAsciiFile & The reference to the VirtualAsciiFile to be inserted.
<i>start</i>	int32_t The first line to be inserted (optional, default is 0).
<i>end</i>	int32_t The last line to be inserted (optional, default is 0 to read all).

Returns

result: int A result code (0 if successful).

8.37.2.5 mpisend()

```
void VirtualAsciiFile::mpisend (
    const mixMPI * mpidata )
```

Parameters

<i>mpidata</i>	mixMPI * pointer to MPI data structure.
----------------	---

8.37.2.6 number_of_lines()

```
int32_t VirtualAsciiFile::number_of_lines ( ) [inline]
```

Returns

size: int32_t The number of lines in the [VirtualAsciiFile](#) instance.

8.37.2.7 write_to_disk()

```
int VirtualAsciiFile::write_to_disk (
    const std::string & file_name )
```

Parameters

<i>file_name</i>	const string& Name of the file to append contents to.
------------------	---

Returns

result: int A result code (0 if successful).

8.37.3 Member Data Documentation

8.37.3.1 _file_lines

```
std::vector<std::string>* VirtualAsciiFile::_file_lines [protected]
```

A vector of strings representing the file lines.

The documentation for this class was generated from the following files:

- np_tmcode/src/include/[file_io.h](#)
- np_tmcode/src/libnptm/[file_io.cpp](#)

8.38 VirtualBinaryFile Class Reference

Virtual representation of a binary file.

```
#include <file_io.h>
```


Public Member Functions

- **VirtualBinaryFile** ()
VirtualBinaryFile empty instance constructor.
- **VirtualBinaryFile** (const **VirtualBinaryFile** &rhs)
VirtualBinaryFile copy constructor.
- **VirtualBinaryFile** (const mixMPI *mpidata, int rr)
VirtualBinaryFile instance constructor copying all contents off MPISend() calls from MPI process rr.
- **~VirtualBinaryFile** ()
VirtualBinaryFile instance destroyer.
- **void append** (**VirtualBinaryFile** &rhs)
Append another *VirtualBinaryFile* at the end of the current instance.
- **void append_line** (const **VirtualBinaryLine** &line)
Append a line at the end of the file.
- **int append_to_disk** (const std::string &file_name)
Append the contents of the *VirtualBinaryFile* to a physical file on disk.
- **int32_t number_of_lines** ()
Get the number of lines in the current instance.
- **int write_to_disk** (const std::string &file_name)
Write virtual file contents to a real file on disk.
- **void mpisend** (const mixMPI *mpidata)
Send *VirtualBinaryFile* instance to MPI process 0 via MPISend() calls.

Public Attributes

- std::vector< **VirtualBinaryLine** > * **_file_lines**
The number of lines.

8.38.1 Constructor & Destructor Documentation

8.38.1.1 VirtualBinaryFile() [1/2]

```
VirtualBinaryFile::VirtualBinaryFile (
    const VirtualBinaryFile & rhs )
```

Parameters

<i>rhs</i>	const VirtualBinaryFile & Reference to a VirtualBinaryFile instance.
------------	--

8.38.1.2 VirtualBinaryFile() [2/2]

```
VirtualBinaryFile::VirtualBinaryFile (
    const mixMPI * mpidata,
    int rr )
```

Parameters

<i>mpidata</i>	mixMPI * pointer to MPI data structure.
<i>rr</i>	int rank of the MPI process sending the data.

8.38.2 Member Function Documentation

8.38.2.1 append()

```
void VirtualBinaryFile::append (
    VirtualBinaryFile & rhs )
```

Parameters

<i>rhs</i>	const VirtualBinaryFile& Reference to the VirtualBinaryFile to be appended.
------------	---

8.38.2.2 append_line()

```
void VirtualBinaryFile::append_line (
    const VirtualBinaryLine & line )
```

Parameters

<i>line</i>	const string& Reference to a string representing the line.
-------------	--

8.38.2.3 append_to_disk()

```
int VirtualBinaryFile::append_to_disk (
    const std::string & file_name )
```

Parameters

<i>file_name</i>	const string& Name of the file to append contents to.
------------------	---

Returns

result: int A result code (0 if successful).

8.38.2.4 mpisend()

```
void VirtualBinaryFile::mpisend (
    const mixMPI * mpidata )
```

Parameters

<i>mpidata</i>	mixMPI * pointer to MPI data structure.
----------------	---

8.38.2.5 number_of_lines()

```
int32_t VirtualBinaryFile::number_of_lines ( ) [inline]
```

Returns

size: `int32_t` The number of lines in the [VirtualBinaryFile](#) instance.

8.38.2.6 write_to_disk()

```
int VirtualBinaryFile::write_to_disk (
    const std::string & file_name )
```

Parameters

<i>file_name</i>	<code>const string&</code> Name of the file to append contents to.
------------------	--

Returns

result: `int` A result code (0 if successful).

8.38.3 Member Data Documentation**8.38.3.1 _file_lines**

```
std::vector<VirtualBinaryLine>* VirtualBinaryFile::_file_lines
```

A vector of strings representing the file lines.

The documentation for this class was generated from the following files:

- [np_tmcode/src/include/file_io.h](#)
- [np_tmcode/src/libnptm/file_io.cpp](#)

8.39 VirtualBinaryLine Class Reference

Virtual representation of a binary file line.

```
#include <file_io.h>
```

Public Member Functions

- [VirtualBinaryLine \(int mydata\)](#)
VirtualBinaryLine instance constructor for int data.
- [VirtualBinaryLine \(long mydata\)](#)
VirtualBinaryLine instance constructor for long data.
- [VirtualBinaryLine \(float mydata\)](#)
VirtualBinaryLine instance constructor for single-precision floating point data.
- [VirtualBinaryLine \(double mydata\)](#)
VirtualBinaryLine instance constructor for double-precision floating point data.
- [VirtualBinaryLine \(dcomplex mydata\)](#)

- *VirtualBinaryLine* instance constructor for *dcomplex* data.
- *VirtualBinaryLine* (const *VirtualBinaryLine* &rhs)
VirtualBinaryLine copy constructor.
- *VirtualBinaryLine* (const mixMPI *mpidata, int rr)
VirtualBinaryLine instance constructor copying all contents off MPISend() calls from MPI process rr.
- ~*VirtualBinaryLine* ()
VirtualBinaryLine instance destroyer.
- void mpisend (const mixMPI *mpidata)
Send *VirtualBinaryLine* instance to MPI process 0 via MPISend() calls.

Public Attributes

- char * _data_pointer
The pointer to the piece of data to be written, cast to char *.
- size_t _data_size
the size of the data block.
- const char * data_pointer = _data_pointer
Read only view of _data_pointer.
- const size_t & data_size = _data_size
Read only view of _data_size.

8.39.1 Constructor & Destructor Documentation

8.39.1.1 VirtualBinaryLine() [1/7]

```
VirtualBinaryLine::VirtualBinaryLine (
    int mydata )
```

Parameters

<i>mydata</i>	int The piece of data to put in the line.
---------------	---

8.39.1.2 VirtualBinaryLine() [2/7]

```
VirtualBinaryLine::VirtualBinaryLine (
    long mydata )
```

Parameters

<i>mydata</i>	long The piece of data to put in the line.
---------------	--

8.39.1.3 VirtualBinaryLine() [3/7]

```
VirtualBinaryLine::VirtualBinaryLine (
    float mydata )
```

Parameters

<i>mydata</i>	float The piece of data to put in the line.
---------------	---

8.39.1.4 VirtualBinaryLine() [4/7]

```
VirtualBinaryLine::VirtualBinaryLine (
    double mydata )
```

Parameters

<i>mydata</i>	double The piece of data to put in the line.
---------------	--

8.39.1.5 VirtualBinaryLine() [5/7]

```
VirtualBinaryLine::VirtualBinaryLine (
    dcomplex mydata )
```

Parameters

<i>mydata</i>	dcomplex The piece of data to put in the line.
---------------	--

8.39.1.6 VirtualBinaryLine() [6/7]

```
VirtualBinaryLine::VirtualBinaryLine (
    const VirtualBinaryLine & rhs )
```

Parameters

<i>rhs</i>	const VirtualBinaryLine & Reference to a VirtualBinaryLine instance.
------------	--

8.39.1.7 VirtualBinaryLine() [7/7]

```
VirtualBinaryLine::VirtualBinaryLine (
    const mixMPI * mpidata,
    int rr )
```

Parameters

<i>mpidata</i>	mixMPI * pointer to MPI data structure.
<i>rr</i>	int rank of the MPI process sending the data.

8.39.2 Member Function Documentation

8.39.2.1 mpisend()

```
void VirtualBinaryLine::mpisend (
    const mixMPI * mpidata )
```

Parameters

<i>mpidata</i>	<code>mixMPI</code> * pointer to MPI data structure.
----------------	--

The documentation for this class was generated from the following files:

- [np_tmcode/src/include/file_io.h](#)
- [np_tmcode/src/libnptm/file_io.cpp](#)

Chapter 9

File Documentation

9.1 np_tmcode/src/cluster/cluster.cpp File Reference

Implementation of the calculation for a cluster of spheres.

```
#include <chrono>
#include <cmath>
#include <cstdio>
#include <exception>
#include <fstream>
#include <hdf5.h>
#include <string>
#include "../include/types.h"
#include "../include/errors.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/Commons.h"
#include "../include/sph_subs.h"
#include "../include/clu_subs.h"
#include "../include/TransitionMatrix.h"
#include "../include/algebraic.h"
#include "../include/List.h"
#include "../include/file_io.h"
#include "../include/utils.h"
#include "../include/outputs.h"
#include "../include/IterationData.h"
```

Functions

- int [cluster_jxi488_cycle](#) (int jxi488, [ScattererConfiguration](#) *sconf, [GeometryConfiguration](#) *gconf, [ScatteringAngles](#) *sa, [ClusterIterationData](#) *cid, [ClusterOutputInfo](#) *oi, const string &output_path)
Main calculation loop.
- void [cluster](#) (const string &config_file, const string &data_file, const string &output_path, const [mixMPI](#) *mpidata)
C++ implementation of CLU.

9.1.1 Function Documentation

9.1.1.1 cluster()

```
void cluster (
    const string & config_file,
    const string & data_file,
    const string & output_path,
    const mixMPI * mpidata )
```

Parameters

<i>config_file</i>	const string & Name of the configuration file.
<i>data_file</i>	const string & Name of the input data file.
<i>output_path</i>	const string & Name of the directory to write the output files in.
<i>mpidata</i>	const mixMPI * Pointer to a mixMPI data structure.

9.1.1.2 cluster_jxi488_cycle()

```
int cluster_jxi488_cycle (
    int jxi488,
    ScattererConfiguration * sconf,
    GeometryConfiguration * gconf,
    ScatteringAngles * sa,
    ClusterIterationData * cid,
    ClusterOutputInfo * oi,
    const string & output_path )
```

The solution of the scattering problem for different wavelengths is an embarrassingly parallel task. This function, therefore, collects all the operations that can be independently executed by different processes, after the configuration stage and the first calculation loop have been executed.

Parameters

<i>jxi488</i>	int Wavelength loop index.
<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration object.
<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration object.
<i>sa</i>	ScatteringAngles * Pointer to a ScatteringAngles object.
<i>cid</i>	ClusterIterationData * Pointer to a ClusterIterationData object.
<i>oi</i>	ClusterOutputInfo * Pointer to a ClusterOutputInfo object.
<i>output_path</i>	const string & Path to the output directory.

9.2 np_tmcode/src/cluster/np_cluster.cpp File Reference

Cluster of spheres scattering problem handler.

```
#include <stdio>
#include <string>
```



```
#include "../include/types.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/Commons.h"
```

Functions

- void [cluster](#) (const string &config_file, const string &data_file, const string &output_path, const [mixMPI](#) *mpidata)
C++ implementation of CLU.
- int [main](#) (int argc, char **argv)
Main program entry point.

9.2.1 Detailed Description

This program emulates the execution work-flow originally handled by the FORTRAN EDFB and CLU codes, which undertook the task of solving the scattering calculation for the case of a cluster of spheres, with one or more materials. The program is designed to work in two modes: emulation and enhanced. The emulation mode is activated by invoking the program without arguments. In this case, the code looks for hard-coded default input and writes output in the execution folder, replicating the behaviour of the original FORTRAN code. Advanced mode, instead, is activated by passing command line arguments that locate the desired input files and a valid folder to write the output into. The emulation mode is useful for testing, while the advanced mode implements the possibility to change input and output options, without having to modify the code.

9.2.2 Function Documentation

9.2.2.1 cluster()

```
void cluster (
    const string & config_file,
    const string & data_file,
    const string & output_path,
    const mixMPI * mpidata ) [extern]
```

Parameters

<i>config_file</i>	const string & Name of the configuration file.
<i>data_file</i>	const string & Name of the input data file.
<i>output_path</i>	const string & Name of the directory to write the output files in.
<i>mpidata</i>	const mixMPI * Pointer to a mixMPI data structure.

9.2.2.2 main()

```
int main (
    int argc,
    char ** argv )
```

This is the starting point of the execution flow. Here we may choose how to configure the code, e.g. by loading a legacy configuration file or some otherwise formatted configuration data set. The code can be linked to a launcher script or to a GUI oriented application that performs the configuration and runs the main program.

Parameters

<i>argc</i>	int The number of arguments given in command-line.
<i>argv</i>	char ** The vector of command-line arguments.

Returns

result: int An exit code passed to the OS (0 for succesful execution).

9.3 np_tmcode/src/include/algebraic.h File Reference

Declaration of algebraic functions with different call-backs.

Functions

- void [invert_matrix](#) (dcomplex **mat, np_int size, int &ier, const string &output_path, int jxi488, np_int max_size=0, int target_device=0, const [RuntimeSettings](#) &rs=[RuntimeSettings](#)())
Perform in-place matrix inversion.

9.3.1 Detailed Description

In principle, the system that runs NP_TMcode may offer various types of optimized features, such as multi-core or multi-node scaling, GPU offload, or external libraries. This header collects a set of functions that can perform standard algebraic operations choosing the most optimized available system as a call-back. If no optimization is detected, eventually the legacy serial function implementation is used as a fall-back.

9.3.2 Function Documentation

9.3.2.1 invert_matrix()

```
void invert_matrix (
    dcomplex ** mat,
    np_int size,
    int & ier,
    const string & output_path,
    int jxi488,
    np_int max_size = 0,
    int target_device = 0,
    const RuntimeSettings & rs = RuntimeSettings() )
```

Parameters

<i>mat</i>	complex double ** The matrix to be inverted (must be a square matrix).
<i>size</i>	np_int The size of the matrix (i.e. the number of its rows or columns).
<i>ier</i>	int & Reference to an integer variable for returning a result flag.
<i>output_path</i>	const string & Path where the output needs to be placed.
<i>jxi488</i>	int Index of the current wavelength calculation.
<i>max_size</i>	np_int The maximum expected size (required by some call-backs, optional, defaults to 0).
<i>target_device</i>	int ID of target GPU, if available (defaults to 0).

9.4 algebraic.h

[Go to the documentation of this file.](#)

```
00001 /* Copyright (C) 2025 INAF - Osservatorio Astronomico di Cagliari
00002
00003 This program is free software: you can redistribute it and/or modify
00004 it under the terms of the GNU General Public License as published by
00005 the Free Software Foundation, either version 3 of the License, or
00006 (at your option) any later version.
00007
00008 This program is distributed in the hope that it will be useful,
00009 but WITHOUT ANY WARRANTY; without even the implied warranty of
00010 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
00011 GNU General Public License for more details.
00012
00013 A copy of the GNU General Public License is distributed along with
00014 this program in the COPYING file. If not, see: <https://www.gnu.org/licenses/>.
00015 */
00016
00017 #ifndef INCLUDE_ALGEBRAIC_H_
00018 #define INCLUDE_ALGEBRAIC_H_
00019
00020 using namespace std;
00021
00022 void invert_matrix(dcomplex **mat, np_int size, int &ier, const string& output_path, int jxi488,
00023 np_int max_size=0, int target_device=0, const RuntimeSettings& rs=RuntimeSettings());
00024
00025 #endif
```

9.5 np_tmcode/src/include/clu_subs.h File Reference

C++ porting of CLU functions and subroutines.

Functions

- void **apc** (double ****zpv, int le, **dcomplex** **am0m, **dcomplex** **w, double sqk, double **gapr, **dcomplex** **gapp)

Compute the asymmetry-corrected scattering cross-section of a cluster.
- void **apcra** (double ****zpv, const int le, **dcomplex** **am0m, int inpol, double sqk, double **gaprm, **dcomplex** **gappm)

Compute the asymmetry-corrected scattering cross-section under random average conditions.
- **dcomplex** **cdtp** (**dcomplex** z, **dcomplex** *vec_am, int i, int jf, int k, int nj, **np_int** istep)

Complex inner product.
- double **cgev** (int ipamo, int mu, int l, int m)

C++ porting of CGEV. ANNOTATION: Get weight of T-matrix element.
- void **cms** (**dcomplex** *am, **ParticleDescriptor** *c1)

Build the multi-centered M-matrix of the cluster.
- void **crsm1** (double vk, double exri, **ParticleDescriptor** *c1)

Compute orientation-averaged scattered field intensity.
- **dcomplex** **ghit** (int ihi, int ipamo, int nbl, int l1, int m1, int l2, int m2, **ParticleDescriptor** *c1, double rac3j[])

Compute the transfer vector from N2 to N1.
- void **hvj** (double exri, double vk, int &jer, int &lcalc, **dcomplex** &arg, **ParticleDescriptor** *c1)

Compute Hankel funtion and Bessel functions.
- void **lucin** (**dcomplex** **am, const **np_int** nddmst, **np_int** n, int &ier)

Invert the multi-centered M-matrix.
- void **mextc** (double vk, double exri, **dcomplex** **fsac, double **cextlr, double **cext)

Compute the average extinction cross-section.
- void **pcros** (double vk, double exri, **ParticleDescriptor** *c1)

Compute cross-sections and forward scattering amplitude for the cluster.

- void [pcrsm0](#) (double vk, double exri, int inpol, [ParticleDescriptor](#) *c1)
Compute orientation-averaged cross-sections and forward scattering amplitude.
- void [polar](#) (double x, double y, double z, double &r, double &cth, double &sth, double &cph, double &sph)
Transform Cartesian quantities to spherical coordinates ones.
- void [r3j000](#) (int j2, int j3, double *rac3j)
Compute the 3j symbol for Clebsch-Gordan coefficients towards J=0.
- void [r3jjr](#) (int j2, int j3, int m2, int m3, double rac3j[])
Compute the 3j symbol for Clebsch-Gordan coefficients for JJ transitions.
- void [r3jmr](#) (int j1, int j2, int j3, int m1, double *rac3j)
Compute the 3j symbol for Clebsch-Gordan coefficients for JM transitions.
- void [raba](#) (int le, [dcomplex](#) **am0m, [dcomplex](#) **w, double **tqce, [dcomplex](#) **tqcpe, double **tqcs, [dcomplex](#) **tqcps)
Compute radiation torques on a particle in Cartesian coordinates.
- void [rfr](#) (double *u, double *up, double *un, double *gapv, double extins, double scatts, double &rapr, double &cosav, double &fp, double &fn, double &fk, double &fx, double &fy, double &fz)
Compute the radiation force Cartesian components.
- void [scr0](#) (double vk, double exri, [ParticleDescriptor](#) *c1)
Compute Mie cross-sections for the sphere units in the cluster.
- void [scr2](#) (double vk, double vkarg, double exri, double *duk, [ParticleDescriptor](#) *c1)
Compute the scattering amplitude for a single sphere in an aggregate.
- void [str](#) (double **rcf, [ParticleDescriptor](#) *c1)
Transform sphere Cartesian coordinates to spherical coordinates.
- void [tqr](#) (double *u, double *up, double *un, double *tqev, double *tqsv, double &tep, double &ten, double &tek, double &tsp, double &tsn, double &tsk)
Compute radiation torques on particles on a k-vector oriented system.
- void [ztm](#) ([dcomplex](#) **am, [ParticleDescriptor](#) *c1)
Calculate the single-centered inversion of the M-matrix.

9.5.1 Detailed Description

This library includes a collection of functions that are used to solve the scattering problem in the case of a cluster of spheres. The functions that were generalized from the case of the single sphere are imported the [sph_subs.h](#) library. As it occurs with the single sphere case functions, in most cases, the results of calculations do not fall back to fundamental data types. They are rather multi-component structures. In order to manage access to such variety of return values, most functions are declared as `void` and they operate on output arguments passed by reference.

9.5.2 Function Documentation

9.5.2.1 [apc\(\)](#)

```
void apc (
    double **** zpv,
    int le,
    dcomplex ** am0m,
    dcomplex ** w,
    double sqk,
    double ** gapr,
    dcomplex ** gapp )
```

This function computes the product between the geometrical asymmetry parameter and the scattering cross-section, like [aps\(\)](#), but for a cluster of spheres. See Eq. (3.16) in Borghese, Denti & Saija (2007).

Parameters

<i>zpv</i>	double ****
<i>le</i>	int
<i>am0m</i>	complex double **
<i>w</i>	complex double **
<i>sqk</i>	double
<i>gapr</i>	double **
<i>gapp</i>	complex double **

9.5.2.2 apcra()

```
void apcra (
    double **** zpv,
    const int le,
    dcomplex ** am0m,
    int inpol,
    double sqk,
    double ** gaprm,
    dcomplex ** gappm )
```

This function computes the product between the geometrical asymmetry parameter and the scattering cross-section of a cluster using the random average directions.

Parameters

<i>zpv</i>	double ****
<i>le</i>	int
<i>am0m</i>	complex double **
<i>inpol</i>	int Polarization type.
<i>sqk</i>	double
<i>gaprm</i>	double **
<i>gappm</i>	complex double **

9.5.2.3 cdtp()

```
dcomplex cdtp (
    dcomplex z,
    dcomplex * vec_am,
    int i,
    int jf,
    int k,
    int nj,
    np_int istep )
```

This function performs the complex inner product. It is used by `lucin()`.

Parameters

<i>z</i>	complex double Starting element.
----------	----------------------------------

Parameters

<i>vec_am</i>	complex double * Vectorized matrix.
<i>i</i>	int Index of first sub-matrix row.
<i>jf</i>	int Index of first submatrix column.
<i>k</i>	int Index of second sub-matrix row.
<i>nj</i>	int Range of first sub-matrix columns.
<i>istep</i>	np_int Size of rows in the matrix.

9.5.2.4 cgev()

```
double cgev (
    int ipamo,
    int mu,
    int l,
    int m )
```

Parameters

<i>ipamo</i>	int
<i>mu</i>	int
<i>l</i>	int
<i>m</i>	int

Returns

result: double

9.5.2.5 cms()

```
void cms (
    dcomplex * am,
    ParticleDescriptor * c1 )
```

This function constructs the multi-centered M-matrix of the cluster, according to Eq. (5.28) of Borghese, Denti & Saija (2007).

Parameters

in, out	<i>am</i>	complex double *
in	<i>c1</i>	ParticleDescriptor *

9.5.2.6 crsm1()

```
void crsm1 (
    double vk,
```

```
double exri,
ParticleDescriptor * c1 )
```

This function computes the intensity of the scattered field for the cluster, averaged on the orientations. It is invoked for IAVM=1 (ANNOTATION: geometry referred to the meridional plane).

Parameters

<i>vk</i>	double Wave number.
<i>exri</i>	double External medium refractive index.
<i>c1</i>	ParticleDescriptor *

9.5.2.7 ghit()

```
dcomplex ghit (
    int ihi,
    int ipamo,
    int nbl,
    int l1,
    int m1,
    int l2,
    int m2,
    ParticleDescriptor * c1,
    double rac3j[] )
```

This function computes the transfer vector going from N2 to N1, using either Hankel, Bessel or Bessel from origin functions.

Parameters

in	<i>ihi</i>	int Function mode.
in	<i>ipamo</i>	int
in	<i>nbl</i>	int Block identifier.
in	<i>l1</i>	int First L quantum number.
in	<i>m1</i>	int First M quantum number.
in	<i>l2</i>	int Second L quantum number.
in	<i>m2</i>	int Second M quantum number.
in	<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.
in, out	<i>rac3j</i>	double[] J connection vector.

Returns

result: dcomplex Matrix element.

9.5.2.8 hjv()

```
void hjv (
    double exri,
    double vk,
    int & jer,
```

```

    int & lcalc,
    dcomplex & arg,
    ParticleDescriptor * c1 )

```

This function constructs the Hankel function and the Bessel functions vectors. See page 331 in Borghese, Denti & Saija (2007).

Parameters

<i>exri</i>	double External medium refractive index.
<i>vk</i>	double Wave number.
<i>jer</i>	int & Reference to error code flag.
<i>lcalc</i>	int & Reference to the highest order accounted for in calculation.
<i>arg</i>	complex\<double\> &
<i>c1</i>	ParticleDescriptor *

9.5.2.9 lucin()

```

void lucin (
    dcomplex ** am,
    const np_int nddmst,
    np_int n,
    int & ier )

```

This function performs the inversion of the multi-centered M-matrix through LU decomposition. See Eq. (5.29) in Borghese, Denti & Saija (2007).

Parameters

<i>am</i>	complex double **
<i>nddmst</i>	const int64_t
<i>n</i>	int64_t
<i>ier</i>	int &

9.5.2.10 mextc()

```

void mextc (
    double vk,
    double exri,
    dcomplex ** fsac,
    double ** cextlr,
    double ** cext )

```

This function computes the average extinction cross-section starting from the definition of the scattering amplitude. See Sec. 3.2.1 of Borghese, Denti & Saija (2007).

Parameters

<i>vk</i>	double Wave number.
<i>exri</i>	double External medium refractive index.
<i>fsac</i>	Matrix of complex
<i>cextlr</i>	double **
<i>cext</i>	double **

9.5.2.11 pcros()

```
void pcros (
    double vk,
    double exri,
    ParticleDescriptor * c1 )
```

This function computes the scattering, absorption and extinction cross-sections of the cluster, together with the Forward Scattering Amplitude.

This function is intended to evaluate the particle cross-section. QUESTIUON: correct?

Parameters

<i>vk</i>	double Wave number.
<i>exri</i>	double External medium refractive index.
<i>c1</i>	ParticleDescriptor *

9.5.2.12 pcrsm0()

```
void pcrsm0 (
    double vk,
    double exri,
    int inpol,
    ParticleDescriptor * c1 )
```

This function computes the orientation-averaged scattering, absorption and extinction cross-sections of the cluster, together with the averaged Forward Scattering Amplitude.

Parameters

<i>vk</i>	double Wave number.
<i>exri</i>	double External medium refractive index.
<i>inpol</i>	int Incident field polarization type.
<i>c1</i>	ParticleDescriptor *

9.5.2.13 polar()

```
void polar (
    double x,
    double y,
    double z,
    double & r,
    double & cth,
    double & sth,
    double & cph,
    double & sph )
```

This function performs a conversion from the Cartesian coordinates system to the spherical one. It is used by `sphar()`.

Parameters

<i>x</i>	double X-axis Cartesian coordinate.
<i>y</i>	double Y-axis Cartesian coordinate.
<i>z</i>	double Z-axis Cartesian coordinate.
<i>r</i>	double & Reference to radial vector (output value).
<i>cth</i>	double & Reference to the cosine of the azimuth coordinate (output value).
<i>sth</i>	double & Reference to the sine of the azimuth coordinate (output value).
<i>cph</i>	double & Reference to the cosine of the elevation coordinate (output value).
<i>sph</i>	double & Reference to the sine of the elevation coordinate (output value).

9.5.2.14 r3j000()

```
void r3j000 (
    int j2,
    int j3,
    double * rac3j )
```

This function calculates the $3j(J,J2,J3;0,0,0)$ symbol for the Clebsch-Gordan coefficients. See Appendix a.3.1 in Borghese, Denti & Saija (2007).

Parameters

<i>j2</i>	int
<i>j3</i>	int
<i>rac3j</i>	double * Vector of 3j symbols.

9.5.2.15 r3jjr()

```
void r3jjr (
    int j2,
    int j3,
    int m2,
    int m3,
    double rac3j[] )
```

This function calculates the $3j(J,J2,J3;-M2-M3,M2,M3)$ symbol for the Clebsch-Gordan coefficients. See Appendix a.3.1 in Borghese, Denti & Saija (2007).

Parameters

in	<i>j2</i>	int Value of J2.
in	<i>j3</i>	int Value of J3.
in	<i>m2</i>	int Value of M2.
in	<i>m3</i>	int Value of M3.
out	<i>rac3j</i>	double[] Vector of 3j symbols.

9.5.2.16 r3jmr()

```
void r3jmr (
    int j1,
    int j2,
    int j3,
    int m1,
    double * rac3j )
```

This function calculates the $3j(J,J2,J3;M1,M,-M1-M)$ symbol for the Clebsch-Gordan coefficients. See Appendix a.3.1 in Borghese, Denti & Saija (2007).

Parameters

<i>j1</i>	int
<i>j2</i>	int
<i>j3</i>	int
<i>m1</i>	int
<i>rac3j</i>	double * Vector of 3j symbols.

9.5.2.17 raba()

```
void raba (
    int le,
    dcomplex ** am0m,
    dcomplex ** w,
    double ** tqce,
    dcomplex ** tqcpe,
    double ** tqcs,
    dcomplex ** tqcps )
```

This function computes radiation torque on on a cluster of spheres as the result of the difference between the extinction and the scattering contributions for a Cartesian coordinate system, as `rabas()`. See Sec. 4.9 in Borghese, Denti & Saija (2007).

Parameters

<i>le</i>	int
<i>am0m</i>	complex double **
<i>w</i>	complex double **
<i>tqce</i>	double **
<i>tqcpe</i>	complex double **
<i>tqcs</i>	double **
<i>tqcps</i>	complex double **

9.5.2.18 rftr()

```
void rftr (
    double * u,
```

```

double * up,
double * un,
double * gapv,
double extins,
double scatts,
double & rapr,
double & cosav,
double & fp,
double & fn,
double & fk,
double & fx,
double & fy,
double & fz )

```

This function computes the Cartesian components of the radiation force exerted on a particle. See Sec. 3.2.1 in Borghese, Denti & Saija (2007).

Parameters

<i>u</i>	double *
<i>up</i>	double *
<i>un</i>	double *
<i>gapv</i>	double *
<i>extins</i>	double
<i>scatts</i>	double
<i>rapr</i>	double &
<i>cosav</i>	double &
<i>fp</i>	double &
<i>fn</i>	double &
<i>fk</i>	double &
<i>fx</i>	double &
<i>fy</i>	double &
<i>fz</i>	double &

9.5.2.19 scr0()

```

void scr0 (
    double vk,
    double exri,
    ParticleDescriptor * c1 )

```

This function computes the scattering, absorption and extinction cross-sections for the spheres composing the cluster, in terms of Mie coefficients, together with the Forward Scattering Amplitude. See Sec. 4.2.1 in Borghese, Denti & Saija (2007).

Parameters

<i>vk</i>	double Wave number
<i>exri</i>	double External medium refractive index.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.5.2.20 scr2()

```
void scr2 (
    double vk,
    double vkarg,
    double exri,
    double * duk,
    ParticleDescriptor * c1 )
```

This function computes the scattering amplitude for single spheres constituting an aggregate. See Sec. 4.2.1 in Borghese, Denti & Saija (2007).

Parameters

<i>vk</i>	double Wave number.
<i>vkarg</i>	double ANNOTATION: Argument of wave number.
<i>exri</i>	double External medium refractive index.
<i>duk</i>	double * ANNOTATION: variation of unit wave vector.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.5.2.21 str()

```
void str (
    double ** rcf,
    ParticleDescriptor * c1 )
```

This function transforms the Cartesian coordinates of the spheres in an aggregate to radial coordinates, then it calls `sphar()` to calculate the vector of spherical harmonics of the incident field.

Parameters

<i>rcf</i>	double ** Matrix of sphere configuration fractional radii.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.5.2.22 tqr()

```
void tqr (
    double * u,
    double * up,
    double * un,
    double * tqev,
    double * tqsv,
    double & tep,
    double & ten,
    double & tek,
    double & tsp,
    double & tsn,
    double & tsk )
```

This function computes the radiation torques resulting from the difference between absorption and scattering contributions, like `rabas()`, but for a coordinate system oriented along the wave vector and its orthogonal axis. See Sec. 4.9 in Borghese, Denti & Saija (2007).

Parameters

<i>u</i>	double *
<i>up</i>	double *
<i>un</i>	double *
<i>tqev</i>	double *
<i>tqsv</i>	double *
<i>tep</i>	double &
<i>ten</i>	double &
<i>tek</i>	double &
<i>tsp</i>	double &
<i>tsn</i>	double &
<i>tsk</i>	double &

9.5.2.23 ztm()

```
void ztm (
    dcomplex ** am,
    ParticleDescriptor * c1 )
```

This function computes the single-centered inverted M-matrix appearing in Eq. (5.28) of Borghese, Denti & Saija (2007).

Parameters

<i>am</i>	complex double **
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.6 clu_subsh

[Go to the documentation of this file.](#)

```
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00012
00013 A copy of the GNU General Public License is distributed along with
00014 this program in the COPYING file. If not, see: <https://www.gnu.org/licenses/>.
00015 */
00016
00032 #ifndef INCLUDE_CLU_SUBS_H_
00033 #define INCLUDE_CLU_SUBS_H_
00034
00050 void apc(
00051     double ****zpv, int le, dcomplex **am0m, dcomplex **w,
00052     double sqk, double **gapr, dcomplex **gapp
00053 );
00054
00070 void apcra(
00071     double ****zpv, const int le, dcomplex **am0m, int inpol, double sqk,
00072     double **gaprm, dcomplex **gappm
00073 );
```

```

00074
00088 dcomplex cdtp(dcomplex z, dcomplex *vec_am, int i, int jf, int k, int nj, np_int istep);
00089
00099 double cgev(int ipamo, int mu, int l, int m);
00100
00110 void cms(dcomplex *am, ParticleDescriptor *cl);
00111
00112 #ifdef USE_TARGET_OFFLOAD
00122 void cms_flat(dcomplex *am, ParticleDescriptor *cl);
00123 #endif // USE_TARGET_OFFLOAD
00124
00136 void crsml(double vk, double exri, ParticleDescriptor *cl);
00137
00155 dcomplex ghit(
00156     int ihi, int ipamo, int nbl, int l1, int m1, int l2, int m2, ParticleDescriptor *cl,
00157     double rac3j[]
00158 );
00159
00173 void hjv(
00174     double exri, double vk, int &jer, int &lcalc, dcomplex &arg,
00175     ParticleDescriptor *cl
00176 );
00177
00189 void lucin(dcomplex **am, const np_int nddmst, np_int n, int &ier);
00190
00204 void mextc(double vk, double exri, dcomplex **fsac, double **cextlr, double **cext);
00205
00217 void pcros(double vk, double exri, ParticleDescriptor *cl);
00218
00230 void pcasm0(double vk, double exri, int inpol, ParticleDescriptor *cl);
00231
00247 void polar(
00248     double x, double y, double z, double &r, double &cth, double &sth,
00249     double &cph, double &sph
00250 );
00251
00262 void r3j000(int j2, int j3, double *rac3j);
00263
00276 void r3jjr(int j2, int j3, int m2, int m3, double rac3j[]);
00277
00290 void r3jmr(int j1, int j2, int j3, int m1, double *rac3j);
00291
00308 void raba(
00309     int le, dcomplex **am0m, dcomplex **w, double **tqce,
00310     dcomplex **tqcpe, double **tqcs, dcomplex **tqcps
00311 );
00312
00334 void rftr(
00335     double *u, double *up, double *un, double *gapv, double extins, double scatts,
00336     double &rapr, double &cosav, double &fp, double &fn, double &fk, double &fx,
00337     double &fy, double &fz
00338 );
00339
00352 void scr0(double vk, double exri, ParticleDescriptor *cl);
00353
00366 void scr2(
00367     double vk, double vkarg, double exri, double *duk, ParticleDescriptor *cl
00368 );
00369
00380 void str(double **rcf, ParticleDescriptor *cl);
00381
00402 void tqr(
00403     double *u, double *up, double *un, double *tqev, double *tqsv, double &tep,
00404     double &ten, double &tek, double &tsp, double &ttn, double &tsk
00405 );
00406
00416 void ztm(dcomplex **am, ParticleDescriptor *cl);
00417
00418 #endif

```

9.7 np_tmcode/src/include/Commons.h File Reference

C++ porting of common data structures.

Classes

- class `mixMPI`

- *Structure with essential MPI data.*
- class [ParticleDescriptor](#)
 - *Basic data structure describing the particle model and its interaction with fields.*
- class [ParticleDescriptorCluster](#)
 - *The data structure describing a particle model made by a cluster of spheres.*
- class [ParticleDescriptorInclusion](#)
 - *The data structure describing a particle model for a sphere with inclusions.*
- class [ParticleDescriptorSphere](#)
 - *The data structure describing a spherical particle model.*
- class [ScatteringAngles](#)
 - *A data structure representing the angles to be evaluated in the problem.*

9.7.1 Detailed Description

Many functions of the original FORTRAN code share complex data blocks in form of COMMON blocks. This poses the limit of freezing the structure of the data blocks in the code, therefore implying the necessity to modify the code to adapt it to the input and to recompile before running. C++, on the contrary, offers the possibility to represent the necessary data structures as classes that can dynamically instantiate the shared information in the most convenient format for the current configuration. This approach adds an abstraction layer that lifts the need to modify and recompile the code depending on the input.

9.8 Commons.h

[Go to the documentation of this file.](#)

```

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00002
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00007
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00011 GNU General Public License for more details.
00012
00013 A copy of the GNU General Public License is distributed along with
00014 this program in the COPYING file. If not, see: <https://www.gnu.org/licenses/>.
00015 */
00016
00033 #ifndef INCLUDE_COMMONS_H_
00034 #define INCLUDE_COMMONS_H_
00035
00036 class ParticleDescriptor;
00037
00040 class mixMPI {
00041 public:
00043     bool mpirunning;
00045     int rank;
00047     int nprocs;
00048
00051     mixMPI();
00052
00055 #ifdef MPI_VERSION
00056     mixMPI(MPI_Comm comm);
00057 #endif
00058
00061     mixMPI(const mixMPI& rhs);
00062
00065     ~mixMPI();
00066 };
00067
00074 class ParticleDescriptor {
00075 protected:
00076     // >> COMMON TO ALL DESCRIPTOR TYPES << //
00078     short _class_type;

```



```

00080 int _nsph;
00082 int _li;
00084 int _max_layers;
00086 int _num_configurations;
00088 int _num_layers;
00090 int _nhspo;
00092 int _npnt;
00094 int _npntts;
00096 dcomplex *vec_rmi;
00098 dcomplex *vec_rei;
00100 dcomplex *vec_w;
00102 double *vec_rc;
00103 // >> END OF SECTION COMMON TO ALL DESCRIPTOR TYPES << //
00104
00105 // >> NEEDED BY SPHERE AND CLUSTER << //
00107 dcomplex *vec_sas;
00109 dcomplex *vec_vints;
00110 // >> END OF SECTION NEEDED BY SPHERE AND CLUSTER << //
00111
00112 // >> NEEDED BY CLUSTER << //
00114 dcomplex *vec_tsas;
00116 dcomplex *vec_gis;
00118 dcomplex *vec_gls;
00120 dcomplex *vec_sam;
00121 // >> END OF SECTION NEEDED BY CLUSTER << //
00122
00123 // >> NEEDED BY CLUSTER AND INCLU <<
00125 int _le;
00127 int _lm;
00129 int _nlim;
00131 int _nlem;
00133 int _nlemt;
00135 int _ncou;
00137 int _litpo;
00139 int _litpos;
00141 int _lmpo;
00143 int _lmtpo;
00145 int _lmtpos;
00147 int _nv3j;
00149 int _ndi;
00151 int _ndit;
00153 dcomplex *vec_am0m;
00155 dcomplex *vec_fsac;
00157 dcomplex *vec_sac;
00159 dcomplex *vec_fsacm;
00161 int *vec_ind3j;
00162 // >> END OF SECTION NEEDED BY CLUSTER AND INCLU << //
00163
00164 // >> NEEDED BY INCLU << //
00166 int _ndm;
00168 dcomplex *vec_at;
00169 // >> END OF SECTION NEEDED BY INCLU << //
00170 public:
00171 // >> COMMON TO ALL DESCRIPTOR TYPES << //
00173 const static short BASE_TYPE = 0;
00175 const static short SPHERE_TYPE = 1;
00177 const static short CLUSTER_TYPE = 2;
00179 const static short INCLUSION_TYPE = 3;
00180
00182 const short& class_type = _class_type;
00184 const int &nsph = _nsph;
00186 const int &li = _li;
00188 const int &max_layers = _max_layers;
00190 const int &num_configurations = _num_configurations;
00192 const int &num_layers = _num_layers;
00194 const int &nhspo = _nhspo;
00196 const int &npnt = _npnt;
00198 const int &npntts = _npntts;
00200 dcomplex **rmi;
00202 dcomplex **rei;
00204 dcomplex **w;
00206 dcomplex *vint;
00208 double *rxx;
00210 double *ryy;
00212 double *rzz;
00214 double *ros;
00216 double **rc;
00218 int *iog;
00220 int *nshl;
00222 dcomplex *ris;
00224 dcomplex *dlri;
00226 dcomplex *dc0;
00228 dcomplex *vkt;
00230 double *vsz;
00232 double *gcs;
00233 // >> END OF SECTION COMMON TO ALL DESCRIPTOR TYPES << //
00234

```

```

00235 // >> NEEDED BY SPHERE AND CLUSTER << //
00237 dcomplex ****sas;
00239 dcomplex **vints;
00241 dcomplex *fsas;
00243 double **sscs;
00245 double *sexs;
00247 double *sabs;
00249 double *sqscs;
00251 double *sqexs;
00253 double *sqabs;
00255 double *gcsv;
00256 // >> END OF SECTION NEEDED BY SPHERE AND CLUSTER << //
00257
00258 // >> NEEDED BY CLUSTER <<
00260 dcomplex *vintt;
00262 dcomplex tfsas;
00264 dcomplex **tsas;
00266 double scs;
00268 double ecs;
00270 double acs;
00272 dcomplex **gis;
00274 dcomplex **gls;
00276 dcomplex **sam;
00277 // >> END OF SECTION NEEDED BY CLUSTER << //
00278
00279 // >> NEEDED BY CLUSTER AND INCLU <<
00281 const int& le = _le;
00283 const int& lm = _lm;
00285 const int& nlim = _nlim;
00287 const int& nlem = _nlem;
00289 const int& nlemt = _nlemt;
00291 const int& ncou = _ncou;
00293 const int& litpo = _litpo;
00295 const int& litpos = _litpos;
00297 const int& lmpo = _lmpo;
00299 const int& lmtpo = _lmtpo;
00301 const int& lmtpos = _lmtpos;
00303 const int& nv3j = _nv3j;
00305 const int& ndi = _ndi;
00307 const int& ndit = _ndit;
00309 const int& ndm = _ndm;
00310
00312 dcomplex *vh;
00314 dcomplex *vj0;
00316 dcomplex *vyhj;
00318 dcomplex *vyj0;
00320 dcomplex vj;
00322 dcomplex **am0m;
00324 dcomplex **fsac;
00326 dcomplex **sac;
00328 dcomplex **fsacm;
00330 dcomplex *vintm;
00332 dcomplex *scscp;
00334 dcomplex *ecscp;
00336 dcomplex *scscpm;
00338 dcomplex *ecscpm;
00340 double *v3j0;
00342 double *scsc;
00344 double *ecsc;
00346 double *scscm;
00348 double *ecscm;
00350 int **ind3j;
00352 double *rac3j;
00353 // >> END OF SECTION NEEDED BY CLUSTER AND INCLU << //
00354
00355 // >> NEEDED BY INCLU << //
00357 dcomplex *rm0;
00359 dcomplex *re0;
00361 dcomplex *rmw;
00363 dcomplex *rew;
00365 dcomplex *tm;
00367 dcomplex *te;
00369 dcomplex *tm0;
00371 dcomplex *te0;
00373 dcomplex **at;
00374 // >> END OF SECTION NEEDED BY INCLU << //
00375
00381 ParticleDescriptor(GeometryConfiguration *gconf, ScattererConfiguration *sconf);
00382
00387 ParticleDescriptor(const ParticleDescriptor &rhs);
00388
00389 #ifdef MPI_VERSION
00394 ParticleDescriptor(const mixMPI *mpidata);
00395
00400 void mpibcast(const mixMPI *mpidata);
00401 #endif // MPI_VERSION
00402

```

```

00405 ~ParticleDescriptor();
00406
00411 virtual std::string get_descriptor_type() { return "base descriptor"; }
00412
00419 static long get_size(GeometryConfiguration *gconf, ScattererConfiguration *sconf);
00420 };
00421
00428 class ParticleDescriptorCluster: public ParticleDescriptor {
00429 public:
00435 ParticleDescriptorCluster(GeometryConfiguration *gconf, ScattererConfiguration *sconf);
00436
00441 ParticleDescriptorCluster(const ParticleDescriptorCluster &rhs);
00442
00443 #ifndef MPI_VERSION
00448 ParticleDescriptorCluster(const mixMPI *mpidata);
00449
00454 void mpibcast(const mixMPI *mpidata);
00455 #endif // MPI_VERSION
00456
00461 std::string get_descriptor_type() override { return "cluster descriptor"; }
00462
00469 static long get_size(GeometryConfiguration *gconf, ScattererConfiguration *sconf);
00470
00477 int update_orders(int inner_order, int outer_order);
00478 };
00479
00486 class ParticleDescriptorInclusion: public ParticleDescriptor {
00487 public:
00493 ParticleDescriptorInclusion(GeometryConfiguration *gconf, ScattererConfiguration *sconf);
00494
00499 ParticleDescriptorInclusion(const ParticleDescriptorInclusion &rhs);
00500
00501 #ifndef MPI_VERSION
00506 ParticleDescriptorInclusion(const mixMPI *mpidata);
00507
00512 void mpibcast(const mixMPI *mpidata);
00513 #endif // MPI_VERSION
00514
00519 std::string get_descriptor_type() override { return "inclusion descriptor"; }
00520
00527 static long get_size(GeometryConfiguration *gconf, ScattererConfiguration *sconf);
00528
00535 int update_orders(int inner_order, int outer_order);
00536 };
00537
00544 class ParticleDescriptorSphere: public ParticleDescriptor {
00545 public:
00551 ParticleDescriptorSphere(GeometryConfiguration *gconf, ScattererConfiguration *sconf);
00552
00557 ParticleDescriptorSphere(const ParticleDescriptorSphere &rhs);
00558
00559 #ifndef MPI_VERSION
00564 ParticleDescriptorSphere(const mixMPI *mpidata);
00565
00570 void mpibcast(const mixMPI *mpidata);
00571 #endif // MPI_VERSION
00572
00577 std::string get_descriptor_type() override { return "sphere descriptor"; }
00578
00585 static long get_size(GeometryConfiguration *gconf, ScattererConfiguration *sconf);
00586
00592 int update_order(int order);
00593 };
00594
00598 class ScatteringAngles {
00599 protected:
00601 int _nth;
00603 int _nths;
00605 int _nph;
00607 int _nphs;
00609 int _nk;
00611 int _nks;
00613 int _nkks;
00615 double _th;
00617 double _thstp;
00619 double _thlst;
00621 double _ths;
00623 double _thsstp;
00625 double _thslst;
00627 double _ph;
00629 double _phstp;
00631 double _phlst;
00633 double _phs;
00635 double _phsstp;
00637 double _phslst;
00639 double _thsca;
00640

```

```

00641 public:
00643     const int& nth = _nth;
00645     const int& nthS = _nthS;
00647     const int& nph = _nph;
00649     const int& nphS = _nphS;
00651     const int& nk = _nk;
00653     const int& nks = _nks;
00655     const int& nkks = _nkks;
00657     const double& th = _th;
00659     const double& thstp = _thstp;
00661     const double& thlst = _thlst;
00663     const double& ths = _ths;
00665     const double& thsstp = _thsstp;
00667     const double& thslst = _thslst;
00669     const double& ph = _ph;
00671     const double& phstp = _phstp;
00673     const double& phlst = _phlst;
00675     const double& phs = _phs;
00677     const double& phsstp = _phsstp;
00679     const double& phslst = _phslst;
00681     const double& thsca = _thsca;
00682
00687     ScatteringAngles(GeometryConfiguration *gconf);
00688
00693     ScatteringAngles(const ScatteringAngles &rhs);
00694
00695 #ifdef MPI_VERSION
00700     ScatteringAngles(const mixMPI *mpidata);
00701
00711     void mpibcast(const mixMPI *mpidata);
00712 #endif
00713
00714 };
00715
00716
00717 #endif

```

9.9 np_tmcode/src/include/Configuration.h File Reference

Configuration data structures.

Classes

- class [GeometryConfiguration](#)
A class to represent the configuration of the scattering geometry.
- class [RuntimeSettings](#)
A class to represent runtime settings.
- class [ScattererConfiguration](#)
A class to represent scatterer configuration objects.

Functions

- int * [check_overlaps](#) ([ScattererConfiguration](#) *sconf, [GeometryConfiguration](#) *gconf, const double tolerance=0.0)
Check the presence of overlapping spheres in an aggregate.
- double [get_overlap](#) ([ScattererConfiguration](#) *sconf, [GeometryConfiguration](#) *gconf, const int index_0, const int index_1)
Get the overlap among two spheres.

9.9.1 Detailed Description

To handle the calculation of a scattering problem, the code needs a set of configuration parameters that define the properties of the scattering particle, of the incident radiation field and of the geometry of the problem. The necessary information is managed through the use of two classes: [ScattererConfiguration](#) and [GeometryConfiguration](#). The first class, [ScattererConfiguration](#), undertakes the role of describing the scattering particle properties, by defining its structure in terms of a distribution of spherical sub-particles, and the optical properties of the constituting materials. It also defines the scaling vector, which tracks the ratio of particle size and radiation wavelength, eventually allowing for the iteration of the calculation on different wavelengths. Multiple materials and layered structures are allowed, while sphere compenetration is not handled, due to the implied discontinuity on the spherical symmetry of the elementary sub-particles. The second class, [GeometryConfiguration](#), on the contrary, describes the properties of the radiation field impinging on the particle. It defines the incoming radiation polarization state, the incident direction and the scattering directions that need to be accounted for. It also stores information concerning the locations of the single spheres and a set of runtime options read from the model input file. Both classes expose methods to read the required configuration data from input files formatted according to the same rules expected by the original code. In addition, they offer perform I/O operations towards proprietary and standard binary formats.

Additional functions and auxiliary classes are defined to manage the runtime settings, the communication with MPI, and the sanity checks on input and system resources.

9.9.2 Function Documentation

9.9.2.1 `check_overlaps()`

```
int * check_overlaps (
    ScattererConfiguration * sconf,
    GeometryConfiguration * gconf,
    const double tolerance = 0.0 )
```

The theoretical implementation of the code is based on the assumption that the spherical monomers that compose the aggregate do not overlap in space. Small violations of this assumption do not critically affect the results, but they need to be reported. This function performs a scan of the aggregate and returns a summary of any potential overlap, if detected.

Parameters

<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration instance.
<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration instance.
<i>tolerance</i>	const double A tolerance value below which overlap is ignored.

Returns

overlaps: int * A vector containing the number of overlaps and a list of overlapping sphere indices.

9.9.2.2 `get_overlap()`

```
double get_overlap (
    ScattererConfiguration * sconf,
    GeometryConfiguration * gconf,
```

```
const int index_0,
const int index_1 )
```

This function computes the thickness of the compenetration between two spheres, in order to compare it with the current tolerance settings.

Parameters

<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration instance.
<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration instance.
<i>index_0</i>	const int Index of the first sphere.
<i>index_1</i>	const int Index of the second sphere.

Returns

overlap: double The thickness of the overlapping layer in meters.

9.10 Configuration.h

[Go to the documentation of this file.](#)

```
00001 /* Copyright (C) 2025   INAF - Osservatorio Astronomico di Cagliari
00002
00003     This program is free software: you can redistribute it and/or modify
00004     it under the terms of the GNU General Public License as published by
00005     the Free Software Foundation, either version 3 of the License, or
00006     (at your option) any later version.
00007
00008     This program is distributed in the hope that it will be useful,
00009     but WITHOUT ANY WARRANTY; without even the implied warranty of
00010     MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
00011     GNU General Public License for more details.
00012
00013     A copy of the GNU General Public License is distributed along with
00014     this program in the COPYING file. If not, see: <https://www.gnu.org/licenses/>.
00015 */
00016
00051 #ifndef INCLUDE_CONFIGURATION_H_
00052 #define INCLUDE_CONFIGURATION_H_
00053
00054 class mixMPI;
00055
00064 class GeometryConfiguration {
00065
00066 protected:
00068     int _number_of_spheres;
00070     int _l_max;
00072     int _li;
00074     int _le;
00076     np_int _mxndm;
00078     int _iavm;
00080     int _in_pol;
00082     int _npnt;
00084     int _npntts;
00086     int _isam;
00088     int _jwrm;
00090     double _in_theta_start;
00092     double _in_theta_step;
00094     double _in_theta_end;
00096     double _sc_theta_start;
00098     double _sc_theta_step;
00100     double _sc_theta_end;
00102     double _in_phi_start;
00104     double _in_phi_step;
00106     double _in_phi_end;
00108     double _sc_phi_start;
00110     double _sc_phi_step;
00112     double _sc_phi_end;
00114     double *_sph_x;
```

```

00116     double *_sph_y;
00118     double *_sph_z;
00120     bool _refine_flag;
00122     short _dyn_order_flag;
00124     double _host_ram_gb;
00126     double _gpu_ram_gb;
00128     double _tolerance;
00130     short _invert_mode;
00132     double _accuracy_goal;
00134     int _ref_iters;
00136     bool _offload_flag;
00137
00138 public:
00140     const int& number_of_spheres = _number_of_spheres;
00142     const int& l_max = _l_max;
00144     const int& li = _li;
00146     const int& le = _le;
00148     const np_int& mxndm = _mxndm;
00150     const int& iavm = _iavm;
00152     const int& in_pol = _in_pol;
00154     const int& npnt = _npnt;
00156     const int& npntts = _npntts;
00158     const int& isam = _isam;
00160     const int& jwtn = _jwtn;
00162     const double& in_theta_start = _in_theta_start;
00164     const double& in_theta_step = _in_theta_step;
00166     const double& in_theta_end = _in_theta_end;
00168     const double& sc_theta_start = _sc_theta_start;
00170     const double& sc_theta_step = _sc_theta_step;
00172     const double& sc_theta_end = _sc_theta_end;
00174     const double& in_phi_start = _in_phi_start;
00176     const double& in_phi_step = _in_phi_step;
00178     const double& in_phi_end = _in_phi_end;
00180     const double& sc_phi_start = _sc_phi_start;
00182     const double& sc_phi_step = _sc_phi_step;
00184     const double& sc_phi_end = _sc_phi_end;
00186     const bool& refine_flag = _refine_flag;
00188     const short& dyn_order_flag = _dyn_order_flag;
00190     const double& host_ram_gb = _host_ram_gb;
00192     const double& gpu_ram_gb = _gpu_ram_gb;
00194     const double& tolerance = _tolerance;
00196     const short& invert_mode = _invert_mode;
00198     const double& accuracy_goal = _accuracy_goal;
00200     const int& ref_iters = _ref_iters;
00202     const bool& offload_flag = _offload_flag;
00203
00237     GeometryConfiguration(
00238         int nsph, int lm, int in_pol, int npnt, int npntts, int meridional_type,
00239         int li, int le, np_int mxndm, int iavm,
00240         double *x, double *y, double *z,
00241         double in_th_start, double in_th_step, double in_th_end,
00242         double sc_th_start, double sc_th_step, double sc_th_end,
00243         double in_ph_start, double in_ph_step, double in_ph_end,
00244         double sc_ph_start, double sc_ph_step, double sc_ph_end,
00245         int jwtn
00246     );
00247
00253     GeometryConfiguration(const GeometryConfiguration& rhs);
00254
00255 #ifdef MPI_VERSION
00261     GeometryConfiguration(const mixMPI *mpidata);
00262
00273     void mpibcast(const mixMPI *mpidata);
00274 #endif
00275
00279     ~GeometryConfiguration();
00280
00293     static GeometryConfiguration *from_legacy(const std::string& file_name);
00294
00304     double get_sph_x(int index) { return _sph_x[index]; }
00305
00315     double get_sph_y(int index) { return _sph_y[index]; }
00316
00326     double get_sph_z(int index) { return _sph_z[index]; }
00327
00328 };
00329
00338 class RuntimeSettings {
00339 protected:
00341     short _invert_mode;
00343     double _accuracy_goal;
00345     double _gpu_ram_gb;
00347     double _host_ram_gb;
00349     int _max_ref_iters;
00351     bool _use_refinement;
00353     bool _use_offload;
00354

```

```

00355 public:
00357     const static short INV_MODE_LU = 0;
00359     const static short INV_MODE_GESV = 1;
00361     const static short INV_MODE_RBT = 2;
00363     const static short INV_MODE_SVD = 3;
00364
00366     const short& invert_mode = _invert_mode;
00368     const double& accuracy_goal = _accuracy_goal;
00370     const double& gpu_ram_gb = _gpu_ram_gb;
00372     const double& host_ram_gb = _host_ram_gb;
00374     const int& max_ref_iters = _max_ref_iters;
00376     const bool& use_refinement = _use_refinement;
00378     const bool& use_offload = _use_offload;
00380     const Logger *logger;
00381
00385     RuntimeSettings();
00386
00393     RuntimeSettings(GeometryConfiguration *gconf, Logger *ptr_logger=NULL);
00394
00400     void set_accuracy_goal(const double goal) { _accuracy_goal = goal; }
00401 };
00402
00409 class ScattererConfiguration {
00410
00411 protected:
00413     dcomplex ***dc0_matrix;
00415     double *_radii_of_spheres;
00417     int *_iog_vec;
00419     int *_nshl_vec;
00421     double *_scale_vec;
00423     std::string _reference_variable_name;
00425     int _number_of_spheres;
00427     int _configurations;
00429     int _number_of_scales;
00431     int _idfc;
00433     double _exdc;
00435     double _wp;
00437     double _xip;
00439     int _max_layers;
00441     bool _use_external_sphere;
00442
00454     static ScattererConfiguration *from_hdf5(const std::string& file_name);
00455
00468     static ScattererConfiguration *from_legacy(const std::string& file_name);
00469
00479     void write_hdf5(const std::string& file_name);
00480
00491     void write_legacy(const std::string& file_name);
00492 public:
00494     const std::string& reference_variable_name = _reference_variable_name;
00496     const int& number_of_spheres = _number_of_spheres;
00498     const int& configurations = _configurations;
00500     const int& number_of_scales = _number_of_scales;
00502     const int& idfc = _idfc;
00504     const double& exdc = _exdc;
00506     const double& wp = _wp;
00508     const double& xip = _xip;
00510     const int& max_layers = _max_layers;
00512     const bool& use_external_sphere = _use_external_sphere;
00514     double **rcf;
00515
00540     ScattererConfiguration(
00541         int nsph,
00542         int configs,
00543         double *scale_vector,
00544         int nxi,
00545         const std::string& variable_name,
00546         int *iog_vector,
00547         double *ros_vector,
00548         int *nshl_vector,
00549         double **rcf_vector,
00550         int dielectric_func_type,
00551         dcomplex ***dc_matrix,
00552         bool has_external,
00553         double exdc,
00554         double wp,
00555         double xip
00556     );
00557
00567     ScattererConfiguration(const ScattererConfiguration& rhs);
00568
00569 #ifdef MPI_VERSION
00575     ScattererConfiguration(const mixMPI *mpidata);
00576
00587     void mpibcast(const mixMPI *mpidata);
00588 #endif
00589

```



```

00593 ~ScattererConfiguration();
00594
00611 static ScattererConfiguration* from_binary(const std::string& file_name, const std::string& mode =
"LEGACY");
00612
00625 static ScattererConfiguration* from_dedfb(const std::string& file_name);
00626
00639 dcomplex get_dielectric_constant(int i, int j, int k) { return _dc0_matrix[i][j][k]; }
00640
00650 int get_iog(int index) { return _iog_vec[index]; }
00651
00657 double get_max_radius();
00658
00664 double get_min_radius();
00665
00675 int get_nshl(int index) { return _nshl_vec[index]; }
00676
00683 double get_particle_radius(GeometryConfiguration *gc);
00684
00694 double get_radius(int index);
00695
00706 double get_rcf(int row, int column) { return _rcf[row][column]; }
00707
00717 double get_scale(int index) { return _scale_vec[index]; }
00718
00728 double get_type_radius(int index) { return _radii_of_spheres[index]; }
00729
00736 void print();
00737
00752 void write_binary(const std::string& file_name, const std::string& mode="LEGACY");
00753
00765 void write_formatted(const std::string& file_name);
00766
00779 bool operator ==(const ScattererConfiguration &other);
00780
00781 };
00782
00798 int *check_overlaps(
00799 ScattererConfiguration *sconf, GeometryConfiguration *gconf, const double tolerance=0.0
00800 );
00801
00814 double get_overlap(
00815 ScattererConfiguration *sconf, GeometryConfiguration *gconf,
00816 const int index_0, const int index_1
00817 );
00818
00819 #endif // INCLUDE_CONFIGURATION_H_

```

9.11 np_tmcode/src/include/cublas_calls.h File Reference

C++ interface to CUBLAS calls.

Functions

- void `cublas_zinvert` (`dcomplex` **mat, `np_int` n, int device_id, const `RuntimeSettings` &rs=`RuntimeSettings()`)
Invert a complex matrix with double precision elements.

9.11.1 Function Documentation

9.11.1.1 cublas_zinvert()

```

void cublas_zinvert (
    dcomplex ** mat,
    np_int n,
    int device_id,
    const RuntimeSettings & rs = RuntimeSettings() )

```

Use cuBLAS to perform an in-place matrix inversion for a complex matrix with double precision elements.

Parameters

<i>mat</i>	Matrix of complex. The matrix to be inverted.
<i>n</i>	np_int The number of rows and columns of the [n x n] matrix.
<i>device_id</i>	int ID of the device for matrix inversion offloading.
<i>rs</i>	const RuntimeSettings & Runtime settings instance.

9.12 cublas_calls.h

[Go to the documentation of this file.](#)

```

00001 /* Copyright (C) 2025 INAF - Osservatorio Astronomico di Cagliari
00002
00003 This program is free software: you can redistribute it and/or modify
00004 it under the terms of the GNU General Public License as published by
00005 the Free Software Foundation, either version 3 of the License, or
00006 (at your option) any later version.
00007
00008 This program is distributed in the hope that it will be useful,
00009 but WITHOUT ANY WARRANTY; without even the implied warranty of
00010 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
00011 GNU General Public License for more details.
00012
00013 A copy of the GNU General Public License is distributed along with
00014 this program in the COPYING file. If not, see: <https://www.gnu.org/licenses/>.
00015 */
00016
00024 #ifndef INCLUDE_CUBLAS_CALLS_H_
00025 #define INCLUDE_CUBLAS_CALLS_H_
00026
00038 void cublas_zinvert(
00039     dcomplex **mat, np_int n, int device_id, const RuntimeSettings& rs=RuntimeSettings()
00040 );
00041
00042 #endif

```

9.13 np_tmcode/src/include/errors.h File Reference

Collection of proprietary code exceptions.

Classes

- class [UnrecognizedOutputInfo](#)
Exception for wrong OutputInfo NULL calls.
- class [ListOutOfBoundsException](#)
Exception for out of bounds List requests.
- class [ObjectAllocationException](#)
Exception for object allocation error handlers.
- class [OpenConfigurationFileException](#)
Exception for open file error handlers.
- class [MatrixOutOfBoundsException](#)
Exception for access requests out of matrix bounds.
- class [UnrecognizedConfigurationException](#)
Exception for unrecognized configuration data sets.
- class [UnrecognizedFormatException](#)
Exception for unrecognized file formats.
- class [UnrecognizedParameterException](#)
Exception for unrecognized parameters.

9.13.1 Detailed Description

There are many circumstances that can prevent the correct execution of a code. These range from user mistakes, to improper configuration, to unsupported hardware and all the way up to various system failures. Although it is not possible to grant proper execution in all cases, it is often possible to design a code in such a way that the program detects unexpected conditions, informs the user and takes the proper actions, eventually stopping without crash, if no other options are available. C++ handles such unexpected circumstances by means of `exceptions`. These are special procedures that can be launched whenever an unexpected situation occurs and they allow to restore the code work-flow and attempt recovery. Exceptions can be divided in different categories, which respond to various types of problems. This library contains a set of exceptions designed to the most common problems that may occur while executing an application of the NP_TMcode suite.

9.14 errors.h

[Go to the documentation of this file.](#)

```

00001
00022 #ifndef INCLUDE_ERRORS_H_
00023 #define INCLUDE_ERRORS_H_
00024
00027 class UnrecognizedOutputInfo: public std::exception {
00028 protected:
00030     std::string message;
00031
00032 public:
00038     UnrecognizedOutputInfo(int requested) {
00039         message = "Error: passed parameter " + std::to_string(requested)
00040             + ", but only 1 is allowed";
00041     }
00042
00046     virtual const char* what() const throw() {
00047         return message.c_str();
00048     }
00049 };
00050
00053 class ListOutOfBoundsException: public std::exception {
00054 protected:
00056     std::string message;
00057
00058 public:
00066     ListOutOfBoundsException(int requested, int min, int max) {
00067         message = "Error: requested index " + std::to_string(requested)
00068             + " out of list allowed bounds [" + std::to_string(min) + ", "
00069             + std::to_string(max - 1) + "]";
00070     }
00071
00075     virtual const char* what() const throw() {
00076         return message.c_str();
00077     }
00078 };
00079
00082 class ObjectAllocationException: public std::exception {
00083 protected:
00085     std::string file_name;
00086
00087 public:
00093     ObjectAllocationException(const std::string& name) { file_name = name; }
00094
00098     virtual const char* what() const throw() {
00099         return file_name.c_str();
00100     }
00101 };
00102
00105 class OpenConfigurationFileException: public std::exception {
00106 protected:
00108     std::string file_name;
00109
00110 public:
00116     OpenConfigurationFileException(const std::string& name) { file_name = name; }
00117
00121     virtual const char* what() const throw() {
00122         return file_name.c_str();
00123     }
00124 };
00125
00128 class MatrixOutOfBoundsException: public std::exception {

```

```

00129 protected:
00131     std::string message;
00132 public:
00138     MatrixOutOfBoundsException(const std::string& problem) { message = problem; }
00142     virtual const char* what() const throw() {
00143         return message.c_str();
00144     }
00145 };
00146
00149 class UnrecognizedConfigurationException: public std::exception {
00150 protected:
00152     std::string message;
00153 public:
00159     UnrecognizedConfigurationException(const std::string& problem) { message = problem; }
00163     virtual const char* what() const throw() {
00164         return message.c_str();
00165     }
00166 };
00167
00170 class UnrecognizedFormatException: public std::exception {
00171 protected:
00173     std::string message;
00174 public:
00180     UnrecognizedFormatException(const std::string& problem) { message = problem; }
00184     virtual const char* what() const throw() {
00185         return message.c_str();
00186     }
00187 };
00188
00191 class UnrecognizedParameterException: public std::exception {
00192 protected:
00194     std::string message;
00195 public:
00201     UnrecognizedParameterException(const std::string& problem) { message = problem; }
00205     virtual const char* what() const throw() {
00206         return message.c_str();
00207     }
00208 };
00209
00210 #endif

```

9.15 np_tmcode/src/include/file_io.h File Reference

Library to handle I/O operations with files.

```
#include <vector>
```

Classes

- class [FileSchema](#)
File content descriptor.
- class [HDFFile](#)
HDF5 I/O wrapper class.
- class [VirtualAsciiFile](#)
Virtual representation of an ASCII file.
- class [VirtualBinaryLine](#)
Virtual representation of a binary file line.
- class [VirtualBinaryFile](#)
Virtual representation of a binary file.

9.16 file_io.h

[Go to the documentation of this file.](#)

```

00001
00005 #ifndef INCLUDE_FILE_IO_H_
00006 #define INCLUDE_FILE_IO_H_
00007
00008 #include <vector>
00009
00010 class mixMPI;
00011
00022 class FileSchema {
00023 protected:
00025     int num_records;
00027     std::string *record_names;
00029     std::string *record_types;
00030
00031 public:
00038     FileSchema(int num_rec, const std::string *rec_types, const std::string *rec_names=NULL);
00039
00042     ~FileSchema();
00043
00048     int get_record_number() { return num_records; }
00049
00054     std::string *get_record_names();
00055
00060     std::string *get_record_types();
00061 };
00062
00069 class HDFFile {
00070 protected:
00072     List<hid_t> *id_list;
00074     std::string file_name;
00076     bool file_open_flag;
00078     hid_t file_id;
00080     herr_t status;
00081
00082 public:
00090     HDFFile(
00091         const std::string& name, unsigned int flags=H5F_ACC_EXCL,
00092         hid_t fcpl_id=H5P_DEFAULT, hid_t fapl_id=H5P_DEFAULT
00093     );
00094
00097     ~HDFFile();
00098
00101     herr_t close();
00102
00112     static HDFFile* from_schema(
00113         FileSchema &schema, const std::string& name, unsigned int flags=H5F_ACC_EXCL,
00114         hid_t fcpl_id=H5P_DEFAULT, hid_t fapl_id=H5P_DEFAULT
00115     );
00116
00119     hid_t get_file_id() { return file_id; }
00120
00123     herr_t get_status() { return status; }
00124
00127     bool is_open() { return file_open_flag; }
00128
00140     herr_t read(
00141         const std::string& dataset_name, const std::string& data_type, void *buffer,
00142         hid_t mem_space_id=H5S_ALL, hid_t file_space_id=H5S_ALL,
00143         hid_t dapl_id=H5P_DEFAULT, hid_t dxpl_id=H5P_DEFAULT
00144     );
00145
00157     herr_t write(
00158         const std::string& dataset_name, const std::string& data_type, const void *buffer,
00159         hid_t mem_space_id=H5S_ALL, hid_t file_space_id=H5S_ALL,
00160         hid_t dapl_id=H5P_DEFAULT, hid_t dxpl_id=H5P_DEFAULT
00161     );
00162 };
00163
00168 class VirtualAsciiFile {
00169 protected:
00171     // int32_t _num_lines;
00173     std::vector<std::string> *_file_lines;
00174
00175 public:
00176     // const int32_t &num_lines = _num_lines;
00181     VirtualAsciiFile(int32_t lines = 0);
00182
00187     VirtualAsciiFile(const VirtualAsciiFile& rhs);
00188
00194     VirtualAsciiFile(const mixMPI *mpidata, int rr);
00195
00198     ~VirtualAsciiFile();

```

```

00199
00204 void append(VirtualAsciiFile& rhs);
00205
00210 void append_line(const std::string& line);
00211
00217 int append_to_disk(const std::string& file_name);
00218
00233 int insert(int32_t position, VirtualAsciiFile& rhs, int32_t start = 0, int32_t end = 0);
00234
00239 int32_t number_of_lines() { return _file_lines->size(); }
00240
00246 int write_to_disk(const std::string& file_name);
00247
00252 void mpisend(const mixMPI *mpidata);
00253 };
00254
00255
00260 class VirtualBinaryLine {
00261 // protected:
00262 //     /// \brief The pointer to the piece of data to be written, cast to char *
00263 //     char *_data_pointer;
00264 //     /// \brief the size of the data block.
00265 //     size_t _data_size;
00266
00267 public:
00269     char *_data_pointer;
00271     size_t _data_size;
00273     const char* data_pointer = _data_pointer;
00275     const size_t & data_size = _data_size;
00276
00281     VirtualBinaryLine(int mydata);
00282
00287     VirtualBinaryLine(long mydata);
00288
00293     VirtualBinaryLine(float mydata);
00294
00299     VirtualBinaryLine(double mydata);
00300
00305     VirtualBinaryLine(dcomplex mydata);
00306
00311     VirtualBinaryLine(const VirtualBinaryLine& rhs);
00312
00318     VirtualBinaryLine(const mixMPI *mpidata, int rr);
00319
00322     ~VirtualBinaryLine();
00323
00328     void mpisend(const mixMPI *mpidata);
00329 };
00330
00331
00336 class VirtualBinaryFile {
00337 protected:
00339     // int32_t _num_lines;
00340     // /// \brief A vector of strings representing the file lines.
00341     // std::vector<VirtualBinaryLine> *_file_lines;
00342
00343 public:
00345     std::vector<VirtualBinaryLine> *_file_lines;
00346     // const int32_t &num_lines = _num_lines;
00350     VirtualBinaryFile();
00351
00356     VirtualBinaryFile(const VirtualBinaryFile& rhs);
00357
00363     VirtualBinaryFile(const mixMPI *mpidata, int rr);
00364
00367     ~VirtualBinaryFile();
00368
00373     void append(VirtualBinaryFile& rhs);
00374
00379     void append_line(const VirtualBinaryLine& line);
00380
00386     int append_to_disk(const std::string& file_name);
00387
00392     int32_t number_of_lines() { return _file_lines->size(); }
00393
00399     int write_to_disk(const std::string& file_name);
00400
00405     void mpisend(const mixMPI *mpidata);
00406 };
00407 #endif
00408
00409

```

9.17 np_tmcode/src/include/inclu_subs.h File Reference

C++ porting of INCLU functions and subroutines.

Functions

- void `cnf` (int `n`, `dcomplex` `z`, int `nm`, `dcomplex` *`csj`, `dcomplex` *`csy`)
C++ porting of CNF.
- void `exma` (`dcomplex` **`am`, `ParticleDescriptor` *`c1`)
C++ porting of EXMA.
- void `incms` (`dcomplex` **`am`, double `enti`, `ParticleDescriptor` *`c1`)
C++ porting of INCMS.
- void `indme` (int `i`, int `npnt`, int `npntts`, double `vk`, `dcomplex` `ent`, double `enti`, `dcomplex` `entn`, int &`jer`, int &`lcalc`, `dcomplex` &`arg`, `ParticleDescriptor` *`c1`)
C++ porting of INDME.
- void `instr` (double **`rcf`, `ParticleDescriptor` *`c1`)
C++ porting of INSTR.
- void `ospv` (`ParticleDescriptor` *`c1`, double `vk`, double `sze`, double `exri`, `dcomplex` `entn`, double `enti`, int &`jer`, int &`lcalc`, `dcomplex` &`arg`)
C++ porting of OSPV.

9.17.1 Detailed Description

This library includes a collection of functions that are used to solve the scattering problem in the case of a sphere with a cluster of inclusions. Like the other use cases, many of the functions declared here execute various calculations on different data structures. In order to manage access to such variety of calculations, most functions are declared as `void` and they operate on output arguments passed by reference.

9.17.2 Function Documentation

9.17.2.1 `cnf()`

```
void cnf (
    int n,
    dcomplex z,
    int nm,
    dcomplex * csj,
    dcomplex * csy )
```

Parameters

<code>n</code>	int Bessel y function order.
<code>z</code>	dcomplex Argument of Bessel y function.
<code>nm</code>	int Maximum computed order.
<code>csj</code>	dcomplex * ANNOTATION: Complex spherical J vector.
<code>csy</code>	dcomplex * Complex spherical Bessel functions up to desired order.

9.17.2.2 exma()

```
void exma (
    dcomplex ** am,
    ParticleDescriptor * c1 )
```

Parameters

<i>am</i>	dcomplex ** Field transition coefficients matrix.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.17.2.3 incms()

```
void incms (
    dcomplex ** am,
    double enti,
    ParticleDescriptor * c1 )
```

Parameters

<i>am</i>	dcomplex ** Field transition coefficients matrix.
<i>enti</i>	double ANNOTATION: imaginary part of coating dielectric function.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.17.2.4 indme()

```
void indme (
    int i,
    int npnt,
    int npntts,
    double vk,
    dcomplex ent,
    double enti,
    dcomplex entn,
    int & jer,
    int & lcalc,
    dcomplex & arg,
    ParticleDescriptor * c1 )
```

Parameters

<i>i</i>	int 1-based sphere configuration index.
<i>npnt</i>	int ANNOTATION: Number of non transition layer integration points.
<i>npntts</i>	int ANNOTATION: Number of transition layer integrtion points.
<i>vk</i>	double Vacuum wave vector magnitude.
<i>ent</i>	dcomplex ANNOTATION: coating dielectric function.
<i>enti</i>	double ANNOTATION: imaginary part of coating dielectric function.
<i>entn</i>	dcomplex ANNOTATION: coating refractive index.
<i>jer</i>	int & Error code flag.

Parameters

<i>lcalc</i>	int & Maximum order achieved in calculation.
<i>arg</i>	dcomplex & ANNOTATION: argument of calculation.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.17.2.5 instr()

```
void instr (
    double ** rcf,
    ParticleDescriptor * c1 )
```

Parameters

<i>rcf</i>	double ** Pointer to matrix of fractional radii.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.17.2.6 ospv()

```
void ospv (
    ParticleDescriptor * c1,
    double vk,
    double sze,
    double exri,
    dcomplex entn,
    double enti,
    int & jer,
    int & lcalc,
    dcomplex & arg )
```

Parameters

<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.
<i>vk</i>	double ANNOTATION: vacuum wave number.
<i>sze</i>	double ANNOTATION: size.
<i>exri</i>	double External medium refractive index.
<i>entn</i>	dcomplex Outer sphere refractive index.
<i>enti</i>	double Imaginary part of the outer medium refractive index.
<i>jer</i>	int & Reference to an integer error flag.
<i>lcalc</i>	int & Maximum order achieved in calculation.
<i>arg</i>	dcomplex Complex Bessel function argument.

9.18 inclu_subs.h

[Go to the documentation of this file.](#)

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```

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00014     this program in the COPYING file. If not, see: <https://www.gnu.org/licenses/>.
00015 */
00016
00029 #ifndef INCLUDE_INCLU_SUBS_H_
00030 #define INCLUDE_INCLU_SUBS_H_
00031
00040 void cnf(int n, dcomplex z, int nm, dcomplex *csj, dcomplex *csy);
00041
00047 void exma(dcomplex **am, ParticleDescriptor *cl);
00048
00055 void incms(dcomplex **am, double enti, ParticleDescriptor *cl);
00056
00071 void indme(
00072     int i, int npnt, int npntts, double vk, dcomplex ent, double enti,
00073     dcomplex entn, int &jer, int &lcalc, dcomplex &arg, ParticleDescriptor *cl
00074 );
00075
00081 void instr(double **rcf, ParticleDescriptor *cl);
00082
00095 void ospv(ParticleDescriptor *cl, double vk, double sze, double exri, dcomplex entn, double enti, int
00096     &jer, int &lcalc, dcomplex &arg);
00096
00097 #endif // INCLUDE_INCLU_SUBS_H_

```

9.19 np_tmcode/src/include/IterationData.h File Reference

Multi-process communication data structures.

Classes

- class [ClusterIterationData](#)
A data structure representing the information used for a single scale of the CLUSTER case.
- class [InclusionIterationData](#)
A data structure representing the information used for a single scale of the INCLUSION case.
- class [SphereIterationData](#)
A data structure representing the information used for a single scale of the SPHERE case.

9.20 IterationData.h

[Go to the documentation of this file.](#)

```

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00012
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00014     this program in the COPYING file. If not, see: <https://www.gnu.org/licenses/>.
00015 */
00016

```

```

00023 #ifndef INCLUDE_ITERATION_DATA_H_
00024 #define INCLUDE_ITERATION_DATA_H_
00025
00026 // >> DEFINITION OF ClusterIterationData CLASS <<
00030 class ClusterIterationData {
00031 public:
00033     ParticleDescriptor *cl;
00035     double *gaps;
00037     double **tqse;
00039     dcomplex **tqspe;
00041     double **tqss;
00043     dcomplex **tqsps;
00045     double ****zpv;
00047     double **gapm;
00049     dcomplex **gappm;
00051     double *argi;
00053     double *args;
00055     double **gap;
00057     dcomplex **gapp;
00059     double **tqce;
00061     dcomplex **tqcpe;
00063     double **tqcs;
00065     dcomplex **tqcps;
00067     double *duk;
00069     double **cextlr;
00071     double **cext;
00073     double **cmullr;
00075     double **cmul;
00077     double *gapv;
00079     double *tgev;
00081     double *tqsv;
00083     double *u;
00085     double *us;
00087     double *un;
00089     double *uns;
00091     double *up;
00093     double *ups;
00095     double *unmp;
00097     double *unsmp;
00099     double *upmp;
00101     double *upsm;
00103     double scan;
00105     double cfmp;
00107     double sfmp;
00109     double cfsp;
00111     double sfsp;
00113     double sqsfi;
00115     dcomplex *am_vector;
00117     dcomplex **am;
00119     dcomplex arg;
00121     double vk;
00123     double wn;
00125     double xip;
00127     int number_of_scales;
00129     int xiblock;
00131     int firstxi;
00133     int lastxi;
00135     int proc_device;
00137     bool is_first_scale;
00138
00146     ClusterIterationData(GeometryConfiguration *gconf, ScattererConfiguration *sconf, const mixMPI
    *mpidata, const int device_count);
00147
00152     ClusterIterationData(const ClusterIterationData& rhs);
00153
00154 #ifdef MPI_VERSION
00160     ClusterIterationData(const mixMPI *mpidata, const int device_count);
00161
00171     void mpibcast(const mixMPI *mpidata);
00172 #endif // MPI_VERSION
00173
00176     ~ClusterIterationData();
00177
00184     static long get_size(GeometryConfiguration *gconf, ScattererConfiguration *sconf);
00185
00193     int update_orders(double** rcf, int inner_order, int outer_order);
00194 };
00195 // >> END OF ClusterIterationData CLASS DEFINITION <<
00196
00197 // >> DEFINITION OF InclusionIterationData CLASS <<
00201 class InclusionIterationData {
00202 protected:
00204     double *vec_zpv;
00205
00206 public:
00208     int nimd;
00210     double extr;

```

```

00211
00213 ParticleDescriptor *cl;
00215 double *gaps;
00217 double **tqse;
00219 dcomplex **tqspe;
00221 double **tqss;
00223 dcomplex **tqsps;
00225 double ***zpv;
00227 double **gapm;
00229 dcomplex **gappm;
00231 double *argi;
00233 double *args;
00235 double **gap;
00237 dcomplex **gapp;
00239 double **tqce;
00241 dcomplex **tqcpe;
00243 double **tqcs;
00245 dcomplex **tqcps;
00247 double *duk;
00249 double **cextlr;
00251 double **cext;
00253 double **cmullr;
00255 double **cmul;
00257 double *gapv;
00259 double *tqev;
00261 double *tqsv;
00263 double *u;
00265 double *us;
00267 double *un;
00269 double *uns;
00271 double *up;
00273 double *ups;
00275 double *unmp;
00277 double *unsmp;
00279 double *upmp;
00281 double *upsmp;
00283 double scan;
00285 double cfmp;
00287 double sfmp;
00289 double cfsp;
00291 double sfsp;
00293 double sqsfi;
00295 dcomplex *am_vector;
00297 dcomplex **am;
00299 dcomplex arg;
00301 double vk;
00303 double wn;
00305 double xip;
00307 int number_of_scales;
00309 int xiblock;
00311 int firstxi;
00313 int lastxi;
00315 int proc_device;
00317 bool is_first_scale;
00318
00326 InclusionIterationData(GeometryConfiguration *gconf, ScattererConfiguration *sconf, const mixMPI
*mpidata, const int device_count);
00327
00332 InclusionIterationData(const InclusionIterationData& rhs);
00333
00334 #ifdef MPI_VERSION
00340 InclusionIterationData(const mixMPI *mpidata, const int device_count);
00341
00351 void mpibcast(const mixMPI *mpidata);
00352 #endif
00353
00356 ~InclusionIterationData();
00357
00364 static long get_size(GeometryConfiguration *gconf, ScattererConfiguration *sconf);
00365
00373 int update_orders(double** rcf, int inner_order, int outer_order);
00374 };
00375 // >> END OF InclusionIterationData CLASS DEFINITION << //
00376
00377 // >> DEFINITION OF SphereIterationData CLASS <<
00381 class SphereIterationData {
00382 protected:
00384 int _nsph;
00386 int _lm;
00388 double *vec_cmul;
00390 double *vec_cmullr;
00392 dcomplex *vec_tqspe;
00394 dcomplex *vec_tqsps;
00396 double *vec_tqse;
00398 double *vec_tqss;
0400 double *vec_zpv;
0401

```

```

00402 public:
00404     double cost;
00406     double sint;
00408     double cosp;
00410     double sinp;
00412     double costs;
00414     double sints;
00416     double cosps;
00418     double sinps;
00420     double vk;
00422     double wn;
00424     double xip;
00426     int number_of_scales;
00428     int xiblock;
00430     int firstxi;
00432     int lastxi;
00434     dcomplex arg;
00436     dcomplex s0;
00438     dcomplex tfas;
00440     ParticleDescriptor *cl;
00442     double *argi;
00444     double *args;
00446     double scan;
00448     double cfmp;
00450     double sfmp;
00452     double cfsp;
00454     double sfsp;
00456     double *gaps;
00458     double *duk;
00460     double *u;
00462     double *us;
00464     double *un;
00466     double *uns;
00468     double *up;
00470     double *ups;
00472     double *upmp;
00474     double *upsmp;
00476     double *unmp;
00478     double *unsmp;
00480     double **cmul;
00482     double **cmullr;
00484     dcomplex **tqspe;
00486     dcomplex **tqsps;
00488     double **tqse;
00490     double **tqss;
00492     double ****zpv;
00494     bool is_first_scale;
00495
00503     SphereIterationData(GeometryConfiguration *gconf, ScattererConfiguration *sconf, const mixMPI
*mpidata, const int device_count);
00504
00509     SphereIterationData(const SphereIterationData& rhs);
00510
00511 #ifdef MPI_VERSION
00517     SphereIterationData(const mixMPI *mpidata, const int device_count);
00518
00528     int mpibcast(const mixMPI *mpidata);
00529 #endif // MPI_VERSION
00530
00533     ~SphereIterationData();
00534
00540     int update_order(int order);
00541 };
00542 // >> END OF SphereIterationData CLASS DEFINITION <<
00543
00544 #endif // INCLUDE_ITERATION_DATA_H_

```

9.21 np_tmcode/src/include/lapack_calls.h File Reference

C++ interface to LAPACK calls.

Macros

- **#define LAPACK_SUCCESS 0**
Success code macro for LAPACK functions.

Typedefs

- typedef `dcomplex` `lcomplex`
Double precision complex for LAPACK implementation.

Functions

- void `zinvert` (`dcomplex` **mat, `np_int` n, int &jer, const `RuntimeSettings` &rs=`RuntimeSettings`())
Invert a complex matrix with double precision elements.
- `lapack_int` `lapack_newton` (const `RuntimeSettings` &rs, `lcomplex` *a_orig, const `lapack_int` m, `lcomplex` *a)
Perform Newton-Schulz iterative refinement of matrix inversion.

9.21.1 Function Documentation

9.21.1.1 lapack_newton()

```
lapack_int lapack_newton (
    const RuntimeSettings & rs,
    lcomplex * a_orig,
    const lapack_int m,
    lcomplex * a )
```

In this function the residual of the inversion of a matrix A is evaluated as:

$$R = A^{-1} A - I$$

and the convergence of refinement is estimated through the largest element modulus left in R.

Parameters

<i>rs</i>	const <code>RuntimeSettings</code> & Runtime settings instance. [IN]
<i>a_orig</i>	<code>lcomplex</code> * Pointer to the first element of the non-inverted matrix. [IN]
<i>m</i>	const <code>lapack_int</code> Number of rows / columns in a. [IN]
<i>a</i>	<code>lcomplex</code> * Pointer to the inverted matrix. [IN/OUT]

Returns

err: `lapack_int` An error code (LAPACK_SUCCESS, if everything was fine).

9.21.1.2 zinvert()

```
void zinvert (
    dcomplex ** mat,
    np_int n,
    int & jer,
    const RuntimeSettings & rs = RuntimeSettings() )
```

Use LAPACKE64 to perform an in-place matrix inversion for a complex matrix with double precision elements.

Parameters

<i>mat</i>	Matrix of complex. The matrix to be inverted.
<i>n</i>	np_int The number of rows and columns of the [n x n] matrix.
<i>jer</i>	int & Reference to an integer return flag.
<i>rs</i>	const RuntimeSettings & Runtime settings instance.

9.22 lapack_calls.h

[Go to the documentation of this file.](#)

```

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00014     this program in the COPYING file. If not, see: <https://www.gnu.org/licenses/>.
00015 */
00016
00023 #ifndef INCLUDE_LAPACK_CALLS_H_
00024 #define INCLUDE_LAPACK_CALLS_H_
00025
00026 #ifndef LAPACK_SUCCESS
00028 #define LAPACK_SUCCESS 0
00029 #endif
00030
00031 #ifndef USE_MKL
00033 typedef MKL_Complex16 lcomplex;
00034 #else
00036 typedef dcomplex lcomplex;
00037 #endif // USE_MKL
00038
00050 void zinvert(dcomplex **mat, np_int n, int &jer, const RuntimeSettings& rs=RuntimeSettings());
00051
00068 lapack_int lapack_newton(
00069     const RuntimeSettings& rs, lcomplex* a_orig, const lapack_int m, lcomplex* a
00070 );
00071
00072 #endif

```

9.23 np_tmcode/src/include/List.h File Reference

A library of classes used to manage dynamic lists.

Classes

- class [List< T >](#)
A class to represent dynamic lists.
- struct [List< T >::element](#)
[List](#) element connector.

9.24 List.h

[Go to the documentation of this file.](#)

```

00001
00006 #ifndef INCLUDE_LIST_H_
00007 #define INCLUDE_LIST_H_
00008
00022 template<class T> class List {
00023 protected:
00024     int size;
00025     struct element {
00026         T value;
00027         element* p_prev;
00028     };
00029     element *current,
00030     *first,
00031     *last;
00032
00033 public:
00052     List(int length = 1) {
00053         size = length;
00054         first = new element;
00055         first->p_prev = NULL;
00056         current = first;
00057         element *p_prev = first;
00058         for (int i = 1; i < size; i++) {
00059             current = new element;
00060             current->p_prev = p_prev;
00061             p_prev = current;
00062         }
00063         last = current;
00064     }
00065
00068     ~List() {
00069         current = last;
00070         element *old;
00071         while (current->p_prev) {
00072             old = current;
00073             current = old->p_prev;
00074             delete old;
00075         }
00076         delete current;
00077     }
00078
00090     void append(T value) {
00091         element *p_prev = last;
00092         current = new element;
00093         current->value = value;
00094         current->p_prev = p_prev;
00095         last = current;
00096         size++;
00097     }
00098
00108     T get(int index) {
00109         if (index < 0 || index > size - 1) {
00110             throw ListOutOfBoundsExpection(index, 0, size - 1);
00111         }
00112         current = last;
00113         for (int i = size - 1; i > index; i--) current = current->p_prev;
00114         return current->value;
00115     }
00116
00123     int length() {
00124         return size;
00125     }
00126
00136     void set(int index, T value) {
00137         if (index < 0 || index > size - 1) {
00138             throw ListOutOfBoundsExpection(index, 0, size - 1);
00139         }
00140         current = last;
00141         for (int i = size - 1; i > index; i--) current = current->p_prev;
00142         current->value = value;
00143     }
00144
00158     T* to_array() {
00159         T *array = new T[size];
00160         current = last;
00161         for (int i = size - 1; i > -1; i--) {
00162             array[i] = current->value;
00163             current = current->p_prev;
00164         }
00165         return array;
00166     }
00167 };

```



```
00168
00169 #endif
```

9.25 np_tmcode/src/include/logging.h File Reference

Definition of the logging system.

Classes

- class [Logger](#)
Logger class.

Macros

- `#define LOG_DEBG 0`
Debug level logging (maximum verbosity).
- `#define LOG_INFO 1`
Standard information level logging (default).
- `#define LOG_WARN 2`
Warning level logging (almost quiet).
- `#define LOG_ERRO 3`
Error level logging (silent, unless in pain).
- `#define TOSTRING(ARG) string(#ARG)`
Macro to stringize code lines.

9.26 logging.h

[Go to the documentation of this file.](#)

```
00001 /* Copyright (C) 2025   INAF - Osservatorio Astronomico di Cagliari
00002
00003     This program is free software: you can redistribute it and/or modify
00004     it under the terms of the GNU General Public License as published by
00005     the Free Software Foundation, either version 3 of the License, or
00006     (at your option) any later version.
00007
00008     This program is distributed in the hope that it will be useful,
00009     but WITHOUT ANY WARRANTY; without even the implied warranty of
00010     MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
00011     GNU General Public License for more details.
00012
00013     A copy of the GNU General Public License is distributed along with
00014     this program in the COPYING file. If not, see: <https://www.gnu.org/licenses/>.
00015 */
00016
00021 #ifndef INCLUDE_LOGGING_H_
00022 #define INCLUDE_LOGGING_H_
00023
00025 #define LOG_DEBG 0
00027 #define LOG_INFO 1
00029 #define LOG_WARN 2
00031 #define LOG_ERRO 3
00032
00034 #define TOSTRING(ARG) string(#ARG)
00035
00049 class Logger {
00050 protected:
00052     FILE *err_output;
00054     FILE *log_output;
00056     std::string *last_message;
00058     int log_threshold;
```

```

00060     long repetitions;
00061
00062 public:
00074     Logger(int threshold, FILE *logging_output = stdout, FILE *error_output = stderr);
00075
00078     ~Logger();
00079
00084     void err(const std::string& message) const;
00085
00090     void flush(int level=LOG_DEBUG);
00091
00097     void log(const std::string& message, int level=LOG_INFO) const;
00098
00109     void push(const std::string& message);
00110 };
00111
00112 #endif

```

9.27 np_tmcode/src/include/magma_calls.h File Reference

C++ interface to MAGMA calls.

Macros

- #define **MAX_GPU_STACK_DOUBLE** 128
Maximum number of elements on GPU static vectors.

Functions

- void **magma_zinvert** (dcomplex **mat, np_int n, int &jer, int device_id=0, const [RuntimeSettings](#) &rs=[RuntimeSettings](#)())
Invert a complex matrix with double precision elements.
- magma_int_t **magma_newton** (const [RuntimeSettings](#) &rs, magmaDoubleComplex *a, const magma_int_t m, magmaDoubleComplex *d_a, magma_queue_t queue)
Perform Newton-Schulz iterative refinement of matrix inversion.

9.27.1 Function Documentation

9.27.1.1 magma_newton()

```

magma_int_t magma_newton (
    const RuntimeSettings & rs,
    magmaDoubleComplex * a,
    const magma_int_t m,
    magmaDoubleComplex * d_a,
    magma_queue_t queue )

```

In this function the residual of the inversion of a matrix A is evaluated as:

$$R = A^{-1} A - I$$

and the convergence of refinement is estimated through the largest element modulus left in R. Since Newton-Schulz iterative refinement is dangerous for unstable matrices, the function first creates a host copy of the unrefined inverted matrix and then attempts refinement. If the residue generated by refinement does not improve over the original residue, the function returns an error code that informs the calling function about corruption danger and instructs it to use the back-up. In case of successful refinement, instead, the calling function will be responsible of getting the refined matrix from device back to the host.

Parameters

in	<i>rs</i>	const <code>RuntimeSettings</code> & Runtime settings instance.
in	<i>a</i>	<code>magmaDoubleComplex *</code> Pointer to the first element of the non-inverted matrix on host.
in	<i>m</i>	const <code>magma_int_t</code> Number of rows / columns in a.
in, out	<i>d_a</i>	<code>magmaDoubleComplex *</code> Pointer to the matrix on the GPU.
in	<i>queue</i>	<code>magma_queue_t</code> GPU communication queue.

Returns

`err: magma_int_t` An error code (MAGMA_SUCCESS, if everything was fine).

9.27.1.2 magma_zinvert()

```
void magma_zinvert (
    dcomplex ** mat,
    np_int n,
    int & jer,
    int device_id = 0,
    const RuntimeSettings & rs = RuntimeSettings() )
```

This function is a first implementation of GPU use. The data transfer steps are handled internally, via standard MAGMA calls, so that the matrix to be inverted is copied from the host memory to the GPU, inverted, possibly refined and finally taken back to the host, leaving the GPU free.

Parameters

in, out	<i>mat</i>	Matrix of complex. The matrix to be inverted.
in	<i>n</i>	<code>np_int</code> The number of rows and columns of the [n x n] matrix.
out	<i>jer</i>	<code>int &</code> Reference to an integer return flag.
in	<i>device_id</i>	<code>int</code> ID of the device for matrix inversion offloading (optional, default 0).
in	<i>rs</i>	const <code>RuntimeSettings</code> & Runtime settings instance (optional).

9.28 magma_calls.h

[Go to the documentation of this file.](#)

```
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00002
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00006 (at your option) any later version.
00007
00008 This program is distributed in the hope that it will be useful,
00009 but WITHOUT ANY WARRANTY; without even the implied warranty of
00010 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
00011 GNU General Public License for more details.
00012
00013 A copy of the GNU General Public License is distributed along with
00014 this program in the COPYING file. If not, see: <https://www.gnu.org/licenses/>.
00015 */
00016
00024 #ifndef INCLUDE_MAGMA_CALLS_H_
```

```

00025 #define INCLUDE_MAGMA_CALLS_H_
00026
00028 #define MAX_GPU_STACK_DOUBLE 128
00029
00030 #ifndef USE_TARGET_OFFLOAD
00031 #pragma omp begin declare target device_type(any)
00041 double magma_cgev(int ipamo, int mu, int l, int m);
00042 #pragma omp end declare target
00043
00063 magma_int_t magma_cms(
00064     magmaDoubleComplex *vec_am, ParticleDescriptor *c1, double *rxx, double *ryy, double *rzz,
00065     int *ind3j, double *v3j0, magmaDoubleComplex *vh, magmaDoubleComplex *vyhj,
00066     magmaDoubleComplex *vj0, magmaDoubleComplex *vyj0, magmaDoubleComplex *rmi,
00067     magmaDoubleComplex* rei, int device_id
00068 );
00069
00098 #pragma omp begin declare target device_type(any)
00099 template<int IHI> magmaDoubleComplex magma_ghit(
00100     int ipamo, int nbl, int l1, int m1, int l2, int m2,
00101     double *rxx, double *ryy, double *rzz, int *ind3j,
00102     double *v3j0, magmaDoubleComplex *vh, magmaDoubleComplex *vyhj,
00103     magmaDoubleComplex *vj0, magmaDoubleComplex *vyj0,
00104     magmaDoubleComplex vj, int li, int le, double rac3j[])
00105 );
00106 #pragma omp end declare target
00107
00108 #pragma omp begin declare target device_type(any)
00122 void magma_r3jrr(int j2, int j3, int m2, int m3, double rac3j[]);
00123 #pragma omp end declare target
00124
00135 void magma_zinvert_resident(
00136     magmaDoubleComplex *mat, magma_int_t n, int &jer, magma_queue_t queue, int device_id=0,
00137     const RuntimeSettings& rs=RuntimeSettings()
00138 );
00139
00160 magma_int_t magma_ztm(
00161     magmaDoubleComplex *vec_am, ParticleDescriptor *c1, double *rxx, double *ryy,
00162     double *rzz, int *ind3j, double *v3j0, magmaDoubleComplex *vh, magmaDoubleComplex *vyhj,
00163     magmaDoubleComplex *vj0, magmaDoubleComplex *vyj0, magmaDoubleComplex *sam_v,
00164     magmaDoubleComplex *gis_v, magmaDoubleComplex *gls_v, magmaDoubleComplex *vec_am0m,
00165     int device_id
00166 );
00167 #endif // USE_TARGET_OFFLOAD
00168
00183 void magma_zinvert(
00184     dcomplex **mat, np_int n, int &jer, int device_id=0,
00185     const RuntimeSettings& rs=RuntimeSettings()
00186 );
00187
00212 magma_int_t magma_newton(
00213     const RuntimeSettings& rs, magmaDoubleComplex* a, const magma_int_t m,
00214     magmaDoubleComplex* d_a, magma_queue_t queue
00215 );
00216
00217 #endif

```

9.29 np_tmcode/src/include/outputs.h File Reference

Definition of the output format system.

Classes

- class [ClusterOutputInfo](#)
Class to collect output information for scattering from clusters.
- class [InclusionOutputInfo](#)
Class to collect output information for scattering from particle with inclusions.
- class [SphereOutputInfo](#)
Class to collect output information for scattering from a single sphere.
- class [TrappingOutputInfo](#)
Class to collect output information for particle trapping.

9.30 outputs.h

[Go to the documentation of this file.](#)

```

00001  /* Copyright (C) 2025   INAF - Osservatorio Astronomico di Cagliari
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00006   (at your option) any later version.
00007
00008   This program is distributed in the hope that it will be useful,
00009   but WITHOUT ANY WARRANTY; without even the implied warranty of
00010   MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
00011   GNU General Public License for more details.
00012
00013   A copy of the GNU General Public License is distributed along with
00014   this program in the COPYING file. If not, see: <https://www.gnu.org/licenses/>.
00015  */
00016
00021  #ifndef INCLUDE_OUTPUTS_H_
00022  #define INCLUDE_OUTPUTS_H_
00023
00024  // >> OUTPUT FOR CLUSTER <<
00034  class ClusterOutputInfo {
00035  protected:
00037      int _skip_flag;
00039      int _num_theta;
00041      int _num_thetas;
00043      int _num_phi;
00045      int _num_phis;
00047      int _first_xi;
00048
00054      int write_hdf5(const std::string &file_name);
00055
00065      int write_legacy(const std::string &output);
00066
00067  public:
00069      const int &skip_flag = _skip_flag;
00071      const int &first_xi = _first_xi;
00073      int nsph;
00075      int li;
00077      int le;
00079      int lm;
00081      np_int mxndm;
00083      int inpol;
00085      int npnt;
00087      int npntts;
00089      int iavm;
00091      int isam;
00093      int idfc;
00095      double *vec_x_coords;
00097      double *vec_y_coords;
00099      double *vec_z_coords;
00101      double th;
00103      double thstp;
00105      double thlst;
00107      double ths;
00109      double thsstp;
00111      double thslst;
00113      double ph;
00115      double phstp;
00117      double phlst;
00119      double phs;
00121      double phsstp;
00123      double phslst;
00125      int ndirs;
00127      double exri;
00129      int nxi;
00131      int xi_block_size;
00133      int jwtm;
00135      int *vec_jxi;
00137      short *vec_ier;
00139      double *vec_vk;
00141      double *vec_xi;
00143      int configurations;
00145      double *vec_sphere_sizes;
00147      dcomplex *vec_sphere_ref_indices;
00149      double *vec_sphere_scs;
00151      double *vec_sphere_abs;
00153      double *vec_sphere_exs;
00155      double *vec_sphere_albs;
00157      double *vec_sphere_sqscs;
00159      double *vec_sphere_sqabs;
00161      double *vec_sphere_sqexs;
00163      dcomplex *vec_fsas;

```

```
00165 double *vec_qschus;
00167 double *vec_pschus;
00169 double *vec_s0mags;
00171 double *vec_cosavs;
00173 double *vec_raprs;
00175 double *vec_tqek1;
00177 double *vec_tqsk1;
00179 double *vec_tqek2;
00181 double *vec_tqsk2;
00183 dcomplex *vec_fsac;
00185 double *vec_qschut;
00187 double *vec_pschut;
00189 double *vec_s0magt;
00191 double tgs;
00193 double *vec_scc1;
00195 double *vec_scc2;
00197 double *vec_abc1;
00199 double *vec_abc2;
00201 double *vec_exc1;
00203 double *vec_exc2;
00205 double *vec_albedc1;
00207 double *vec_albedc2;
00209 double *vec_qscamc1;
00211 double *vec_qscamc2;
00213 double *vec_qabsmc1;
00215 double *vec_qabsmc2;
00217 double *vec_qextmc1;
00219 double *vec_qextmc2;
00221 double *vec_sccrt1;
00223 double *vec_sccrt2;
00225 double *vec_abcrt1;
00227 double *vec_abcrt2;
00229 double *vec_excrt1;
00231 double *vec_excrt2;
00233 dcomplex *vec_fsac11;
00235 dcomplex *vec_fsac21;
00237 dcomplex *vec_fsac22;
00239 dcomplex *vec_fsac12;
00241 double *vec_qschuc1;
00243 double *vec_qschuc2;
00245 double *vec_pschuc1;
00247 double *vec_pschuc2;
00249 double *vec_s0magc1;
00251 double *vec_s0magc2;
00253 double *vec_cosavc1;
00255 double *vec_cosavc2;
00257 double *vec_raprc1;
00259 double *vec_raprc2;
00261 double *vec_fk1;
00263 double *vec_fk2;
00265 double *vec_dir_tidg;
00267 double *vec_dir_pidg;
00269 double *vec_dir_tsdg;
00271 double *vec_dir_psdg;
00273 double *vec_dir_scand;
00275 double *vec_dir_cfmp;
00277 double *vec_dir_sfmp;
00279 double *vec_dir_cfsp;
00281 double *vec_dir_sfsp;
00283 double *vec_dir_un;
00285 double *vec_dir_uns;
00287 dcomplex *vec_dir_sas11;
00289 dcomplex *vec_dir_sas21;
00291 dcomplex *vec_dir_sas12;
00293 dcomplex *vec_dir_sas22;
00295 double *vec_dir_muls;
00297 double *vec_dir_muls1r;
00299 dcomplex *vec_dir_sat11;
00301 dcomplex *vec_dir_sat21;
00303 dcomplex *vec_dir_sat12;
00305 dcomplex *vec_dir_sat22;
00307 double *vec_dir_scc1;
00309 double *vec_dir_scc2;
00311 double *vec_dir_abc1;
00313 double *vec_dir_abc2;
00315 double *vec_dir_exc1;
00317 double *vec_dir_exc2;
00319 double *vec_dir_albedc1;
00321 double *vec_dir_albedc2;
00323 double *vec_dir_qsc1;
00325 double *vec_dir_qsc2;
00327 double *vec_dir_qabc1;
00329 double *vec_dir_qabc2;
00331 double *vec_dir_qexc1;
00333 double *vec_dir_qexc2;
00335 double *vec_dir_sccrt1;
00337 double *vec_dir_sccrt2;
```

```

00339 double *vec_dir_abcrt1;
00341 double *vec_dir_abcrt2;
00343 double *vec_dir_excrt1;
00345 double *vec_dir_excrt2;
00347 dcomplex *vec_dir_fsac11;
00349 dcomplex *vec_dir_fsac21;
00351 dcomplex *vec_dir_fsac12;
00353 dcomplex *vec_dir_fsac22;
00355 dcomplex *vec_dir_sac11;
00357 dcomplex *vec_dir_sac21;
00359 dcomplex *vec_dir_sac12;
00361 dcomplex *vec_dir_sac22;
00363 double *vec_dir_qschuc1;
00365 double *vec_dir_qschuc2;
00367 double *vec_dir_pschuc1;
00369 double *vec_dir_pschuc2;
00371 double *vec_dir_s0magc1;
00373 double *vec_dir_s0magc2;
00375 double *vec_dir_cosavc1;
00377 double *vec_dir_cosavc2;
00379 double *vec_dir_raprc1;
00381 double *vec_dir_raprc2;
00383 double *vec_dir_flc1;
00385 double *vec_dir_flc2;
00387 double *vec_dir_frc1;
00389 double *vec_dir_frc2;
00391 double *vec_dir_fkcl;
00393 double *vec_dir_fkc2;
00395 double *vec_dir_fxcl;
00397 double *vec_dir_fxc2;
00399 double *vec_dir_fycl;
00401 double *vec_dir_fyc2;
00403 double *vec_dir_fzcl;
00405 double *vec_dir_fzc2;
00407 double *vec_dir_tqelc1;
00409 double *vec_dir_tqelc2;
00411 double *vec_dir_tqerc1;
00413 double *vec_dir_tqerc2;
00415 double *vec_dir_tqekc1;
00417 double *vec_dir_tqekc2;
00419 double *vec_dir_tqexcl;
00421 double *vec_dir_tqexc2;
00423 double *vec_dir_tqeyc1;
00425 double *vec_dir_tqeyc2;
00427 double *vec_dir_tqezc1;
00429 double *vec_dir_tqezc2;
00431 double *vec_dir_tqslc1;
00433 double *vec_dir_tqslc2;
00435 double *vec_dir_tqsrc1;
00437 double *vec_dir_tqsrc2;
00439 double *vec_dir_tqskc1;
00441 double *vec_dir_tqskc2;
00443 double *vec_dir_tqsxc1;
00445 double *vec_dir_tqsxc2;
00447 double *vec_dir_tqsyc1;
00449 double *vec_dir_tqsyc2;
00451 double *vec_dir_tqszc1;
00453 double *vec_dir_tqszc2;
00455 double *vec_dir_mulc;
00457 double *vec_dir_mulc1r;
00458
00467 ClusterOutputInfo(
00468     ScattererConfiguration *sc, GeometryConfiguration *gc,
00469     const mixMPI *mpidata, int first_xi = 1, int xi_length = 0
00470 );
00471
00476 ClusterOutputInfo(const std::string &hdf5_name);
00477
00482 ClusterOutputInfo(const int skip_flag);
00483
00486 ~ClusterOutputInfo();
00487
00496 static long compute_size(
00497     ScattererConfiguration *sc, GeometryConfiguration *gc,
00498     int first_xi = 1, int xi_length = 0
00499 );
00500
00505 long compute_size();
00506
00512 int insert(const ClusterOutputInfo &rhs);
00513
00520 int write(const std::string &output, const std::string &format);
00521
00522 #ifdef MPI_VERSION
00534 int mpireceive(const mixMPI* mpidata, int pid);
00535
00547 int mpisend(const mixMPI *mpidata);

```

```

00548 #endif // MPI_VERSION
00549 };
00550 // >> END OF OUTPUT FOR CLUSTER <<
00551
00552 // >> OUTPUT FOR INCLUSION <<
00562 class InclusionOutputInfo {
00563 protected:
00565     int _skip_flag;
00567     int _num_theta;
00569     int _num_thetas;
00571     int _num_phi;
00573     int _num_phis;
00575     int _first_xi;
00576
00582     int write_hdf5(const std::string &file_name);
00583
00593     int write_legacy(const std::string &output);
00594
00595 public:
00597     const int &skip_flag = _skip_flag;
00599     const int &first_xi = _first_xi;
06001     int nsph;
06003     int li;
06005     int le;
06007     int lm;
06009     np_int mxndm;
06011     int inpol;
06013     int npnt;
06015     int npntts;
06017     int iavm;
06019     int isam;
06021     int idfc;
06023     double *vec_x_coords;
06025     double *vec_y_coords;
06027     double *vec_z_coords;
06029     double th;
06031     double thstp;
06033     double thlst;
06035     double ths;
06037     double thsstp;
06039     double thslst;
06041     double ph;
06043     double phstp;
06045     double phlst;
06047     double phs;
06049     double phsstp;
06051     double phslst;
06053     int ndirs;
06055     double exri;
06057     int nxi;
06059     int xi_block_size;
06061     int jwtm;
06063     int *vec_jxi;
06065     short *vec_ier;
06067     double *vec_vk;
06069     double *vec_xi;
06071     int configurations;
06073     double *vec_sphere_sizes;
06075     dcomplex *vec_sphere_ref_indices;
06077     double *vec_scs1;
06079     double *vec_scs2;
06081     double *vec_abs1;
06083     double *vec_abs2;
06085     double *vec_exs1;
06087     double *vec_exs2;
06089     double *vec_albeds1;
06091     double *vec_albeds2;
06093     double *vec_scsrt1;
06095     double *vec_scsrt2;
06097     double *vec_absrt1;
06099     double *vec_absrt2;
07001     double *vec_exsrt1;
07003     double *vec_exsrt2;
07005     double *vec_qschu1;
07007     double *vec_qschu2;
07009     double *vec_pschu1;
07011     double *vec_pschu2;
07013     double *vec_s0mag1;
07015     double *vec_s0mag2;
07017     double *vec_cosav1;
07019     double *vec_cosav2;
07021     double *vec_raprs1;
07023     double *vec_raprs2;
07025     double *vec_fk1;
07027     double *vec_fk2;
07029     dcomplex *vec_fsas1;
07031     dcomplex *vec_fsas2;

```



```
00733     dcomplex *vec_fsas22;
00735     dcomplex *vec_fsas12;
00737     double *vec_dir_tidg;
00739     double *vec_dir_pidg;
00741     double *vec_dir_tsdg;
00743     double *vec_dir_psdg;
00745     double *vec_dir_scand;
00747     double *vec_dir_cfmp;
00749     double *vec_dir_sfmp;
00751     double *vec_dir_cfsp;
00753     double *vec_dir_sfsp;
00755     double *vec_dir_un;
00757     double *vec_dir_uns;
00759     double *vec_dir_scs1;
00761     double *vec_dir_scs2;
00763     double *vec_dir_abs1;
00765     double *vec_dir_abs2;
00767     double *vec_dir_exs1;
00769     double *vec_dir_exs2;
00771     double *vec_dir_albeds1;
00773     double *vec_dir_albeds2;
00775     double *vec_dir_scsrt1;
00777     double *vec_dir_scsrt2;
00779     double *vec_dir_absrt1;
00781     double *vec_dir_absrt2;
00783     double *vec_dir_exsrt1;
00785     double *vec_dir_exsrt2;
00787     dcomplex *vec_dir_fsas11;
00789     dcomplex *vec_dir_fsas21;
00791     dcomplex *vec_dir_fsas12;
00793     dcomplex *vec_dir_fsas22;
00795     dcomplex *vec_dir_sas11;
00797     dcomplex *vec_dir_sas21;
00799     dcomplex *vec_dir_sas12;
00801     dcomplex *vec_dir_sas22;
00803     double *vec_dir_qschul;
00805     double *vec_dir_qschu2;
00807     double *vec_dir_pschul;
00809     double *vec_dir_pschu2;
00811     double *vec_dir_s0mag1;
00813     double *vec_dir_s0mag2;
00815     double *vec_dir_cosav1;
00817     double *vec_dir_cosav2;
00819     double *vec_dir_rapr1;
00821     double *vec_dir_rapr2;
00823     double *vec_dir_fl1;
00825     double *vec_dir_fl2;
00827     double *vec_dir_fr1;
00829     double *vec_dir_fr2;
00831     double *vec_dir_fk1;
00833     double *vec_dir_fk2;
00835     double *vec_dir_fx1;
00837     double *vec_dir_fx2;
00839     double *vec_dir_fy1;
00841     double *vec_dir_fy2;
00843     double *vec_dir_fz1;
00845     double *vec_dir_fz2;
00847     double *vec_dir_tqel1;
00849     double *vec_dir_tqel2;
00851     double *vec_dir_tqer1;
00853     double *vec_dir_tqer2;
00855     double *vec_dir_tqek1;
00857     double *vec_dir_tqek2;
00859     double *vec_dir_tqex1;
00861     double *vec_dir_tqex2;
00863     double *vec_dir_tqey1;
00865     double *vec_dir_tqey2;
00867     double *vec_dir_tqez1;
00869     double *vec_dir_tqez2;
00871     double *vec_dir_tqsl1;
00873     double *vec_dir_tqsl2;
00875     double *vec_dir_tqsr1;
00877     double *vec_dir_tqsr2;
00879     double *vec_dir_tqsk1;
00881     double *vec_dir_tqsk2;
00883     double *vec_dir_tqsx1;
00885     double *vec_dir_tqsx2;
00887     double *vec_dir_tqsy1;
00889     double *vec_dir_tqsy2;
00891     double *vec_dir_tqsx1;
00893     double *vec_dir_tqsx2;
00895     double *vec_dir_mull1;
00897     double *vec_dir_mull1r;
00898
00907     InclusionOutputInfo(
00908         ScattererConfiguration *sc, GeometryConfiguration *gc,
00909         const mixMPI *mpidata, int first_xi = 1, int xi_length = 0
```

```

00910     );
00911
00916     InclusionOutputInfo(const std::string &hdf5_name);
00917
00922     InclusionOutputInfo(const int skip_flag);
00923
00926     ~InclusionOutputInfo();
00927
00936     static long compute_size(
00937         ScattererConfiguration *sc, GeometryConfiguration *gc,
00938         int first_xi = 1, int xi_length = 0
00939     );
00940
00945     long compute_size();
00946
00952     int insert(const InclusionOutputInfo &rhs);
00953
00960     int write(const std::string &output, const std::string &format);
00961
00962 #ifdef MPI_VERSION
00974     int mpireceive(const mixMPI* mpidata, int pid);
00975
00987     int mpisend(const mixMPI *mpidata);
00988 #endif // MPI_VERSION
00989 };
00990 // >> END OF OUTPUT FOR INCLUSION <<
00991
00992 // >> OUTPUT FOR SPHERE <<
01002 class SphereOutputInfo {
01003 protected:
01005     int _skip_flag;
01007     int _num_theta;
01009     int _num_thetas;
01011     int _num_phi;
01013     int _num_phis;
01015     int _first_xi;
01016
01022     int write_hdf5(const std::string &file_name);
01023
01033     int write_legacy(const std::string &output);
01034
01035 public:
01037     const int &skip_flag = _skip_flag;
01039     const int &first_xi = _first_xi;
01041     int nsph;
01043     int lm;
01045     int inpol;
01047     int npnt;
01049     int npntts;
01051     int isam;
01053     int idfc;
01055     double th;
01057     double thstp;
01059     double thlst;
01061     double ths;
01063     double thsstp;
01065     double thslst;
01067     double ph;
01069     double phstp;
01071     double phlst;
01073     double phs;
01075     double phsstp;
01077     double phslst;
01079     int ndirs;
01081     double exri;
01083     int nxi;
01085     int xi_block_size;
01087     int jwtm;
01089     int configurations;
01091     int lcalc;
01093     dcomplex arg;
01095     int *vec_jxi;
01097     short *vec_ier;
01099     double *vec_vk;
01101     double *vec_xi;
01103     double *vec_sphere_sizes;
01105     dcomplex *vec_sphere_ref_indices;
01107     double *vec_scs;
01109     double *vec_abs;
01111     double *vec_exs;
01113     double *vec_albeds;
01115     double *vec_scsrt;
01117     double *vec_absrt;
01119     double *vec_exsrt;
01121     dcomplex *vec_fsas;
01123     double *vec_qschu;
01125     double *vec_pschu;

```

```

01127 double *vec_s0mag;
01129 double *vec_cosav;
01131 double *vec_raprs;
01133 double *vec_tqek1;
01135 double *vec_tqek2;
01137 double *vec_tqsk1;
01139 double *vec_tqsk2;
01141 dcomplex *vec_fsat;
01143 double *vec_qschut;
01145 double *vec_pschut;
01147 double *vec_s0magt;
01149 double *vec_dir_tidg;
01151 double *vec_dir_pidg;
01153 double *vec_dir_tsdg;
01155 double *vec_dir_psdg;
01157 double *vec_dir_scand;
01159 double *vec_dir_cfmp;
01161 double *vec_dir_sfmp;
01163 double *vec_dir_cfsp;
01165 double *vec_dir_sfsp;
01167 double *vec_dir_un;
01169 double *vec_dir_uns;
01171 dcomplex *vec_dir_sas11;
01173 dcomplex *vec_dir_sas21;
01175 dcomplex *vec_dir_sas12;
01177 dcomplex *vec_dir_sas22;
01179 double *vec_dir_fx;
01181 double *vec_dir_fy;
01183 double *vec_dir_fz;
01185 double *vec_dir_muls;
01187 double *vec_dir_muls1r;
01188
01197 SphereOutputInfo(
01198     ScattererConfiguration *sc, GeometryConfiguration *gc,
01199     const mixMPI *mpidata, int first_xi = 1, int xi_length = 0
01200 );
01201
01206 SphereOutputInfo(const std::string &hdf5_name);
01207
01212 SphereOutputInfo(const int skip_flag);
01213
01216 ~SphereOutputInfo();
01217
01226 static long compute_size(
01227     ScattererConfiguration *sc, GeometryConfiguration *gc,
01228     int first_xi = 1, int xi_length = 0
01229 );
01230
01235 long compute_size();
01236
01242 int insert(const SphereOutputInfo &rhs);
01243
01250 int write(const std::string &output, const std::string &format);
01251
01252 #ifdef MPI_VERSION
01264 int mpireceive(const mixMPI *mpidata, int pid);
01265
01277 int mpisend(const mixMPI *mpidata);
01278 #endif // MPI_VERSION
01279 };
01280 // >> END OF OUTPUT FOR SPHERE <<
01281
01282 // >> OUTPUT FOR TRAPPING <<
01292 class TrappingOutputInfo {
01293 protected:
01295     int _jft;
01297     int _jss;
01299     int _jtw;
01301     int _nkx;
01303     int _nxv;
01305     int _nyv;
01307     int _nzv;
01309     int _lmode;
01311     int _le;
01313     double _vk;
01315     double _exri;
01317     double _an;
01319     double _ff;
01321     double _tra;
01323     double _spd;
01325     double _frsh;
01327     double _exril;
01328
01334 int write_hdf5(const std::string &file_name);
01335
01345 int write_legacy(const std::string &output);
01346

```

```

01347 public:
01349     const int& jft = _jft;
01351     const int& jss = _jss;
01353     const int& jtw = _jtw;
01355     const int& nkx = _nkx;
01357     const int& nxv = _nxv;
01359     const int& nyv = _nyv;
01361     const int& nzv = _nzv;
01363     const int& lmode = _lmode;
01365     const int& le = _le;
01367     const double& vk = _vk;
01369     const double& exri = _exri;
01371     const double& an = _an;
01373     const double& ff = _ff;
01375     const double& tra = _tra;
01377     const double& spd = _spd;
01379     const double& frsh = _frsh;
01381     const double& exril = _exril;
01383     double *vec_x;
01385     double *vec_y;
01387     double *vec_z;
01389     double *vec_csfl;
01391     double *vec_csf2;
01393     double *vec_csf3;
01395     double *vec_cst1;
01397     double *vec_cst2;
01399     double *vec_cst3;
01400
01410     TrappingOutputInfo(
01411         int ift, int iss, int itw, int nx, int ny, int nz
01412     );
01413
01416     ~TrappingOutputInfo();
01417
01427     static long get_size(int ift, int iss, int nx, int ny, int nz);
01428
01433     long get_size();
01434
01440     void set_param(const std::string& pname, double pvalue);
01441
01448     int write(const std::string &output, const std::string &format);
01449 };
01450 // >> END OF OUTPUT FOR TRAPPING <<
01451
01452 #endif // INCLUDE_OUTPUTS_H_

```

9.31 np_tmcode/src/include/Parsers.h File Reference

A library of functions designed to parse formatted input into memory.

Macros

- `#define FILE_NOT_FOUND_ERROR 21`
Error code if a file is not found.

Functions

- `std::string * load_file (const std::string &file_name, int *count)`
Load a text file as a sequence of strings in memory.

9.31.1 Function Documentation

9.31.1.1 load_file()

```
std::string * load_file (
    const std::string & file_name,
    int * count )
```

The configuration of the field expansion code in FORTRAN uses shared memory access and file I/O operations managed by different functions. Although this approach could be theoretically replicated, it is more convenient to handle input and output to distinct files using specific functions. `load_file()` helps in the task of handling input such as configuration files or text data structures that need to be loaded entirely. The function performs a line-by line scan of the input file and returns an array of strings that can be later parsed and ingested by the concerned code blocks. An optional pointer to integer allows the function to keep track of the number of file lines that were read, if needed.

Parameters

<code>file_name</code>	string The path of the file to be read.
<code>count</code>	int* Pointer to an integer recording the number of read lines [OPTIONAL, default=NULL].

Returns

`array_lines string*` An array of strings, one for each input file line.

9.32 Parsers.h

[Go to the documentation of this file.](#)

```
00001
00007 #ifndef FILE_NOT_FOUND_ERROR
00009 #define FILE_NOT_FOUND_ERROR 21
00010 #endif
00011
00012 #ifndef INCLUDE_PARSERS_H_
00013 #define INCLUDE_PARSERS_H_
00014
00035 std::string *load_file(const std::string& file_name, int *count);
00036
00037 #endif /* INCLUDE_PARSERS_H_ */
```

9.33 np_tmcode/src/include/sph_subs.h File Reference

C++ porting of SPH functions and subroutines.

Functions

- void `aps` (double ****zpv, int li, int nsph, `ParticleDescriptor` *c1, double sqk, double *gaps)
Compute the asymmetry-corrected scattering cross-section of a sphere.
- void `cbf` (int n, `dcomplex` z, int &nm, `dcomplex` *csj)
Complex Bessel Function.
- double `cg1` (int lmpml, int mu, int l, int m)
Clebsch-Gordan coefficients.

- void [diel](#) (int npntmo, int ns, int i, int ic, double vk, [ParticleDescriptor](#) *c1)
Compute the continuous variation of the refractive index and of its radial derivative.
- void [dme](#) (int li, int i, int npnt, int npntts, double vk, double exdc, double exri, [ParticleDescriptor](#) *c1, int &jer, int &lcalc, [dcomplex](#) &arg, int last_conf=0)
Compute Mie scattering coefficients.
- double [envj](#) (int n, double x)
Bessel function calculation control parameters.
- void [mmulc](#) ([dcomplex](#) *vint, double **cmullr, double **cmul)
Compute the Mueller Transformation Matrix.
- int [msta1](#) (double x, int mp)
Starting point for Bessel function magnitude.
- int [msta2](#) (double x, int n, int mp)
Starting point for Bessel function precision.
- void [orunve](#) (double *u1, double *u2, double *u3, int iorth, double torth)
Compute the amplitude of the orthogonal unit vector.
- void [pwma](#) (double *up, double *un, [dcomplex](#) *ylm, int inpol, int lw, int isq, [ParticleDescriptor](#) *c1)
Compute incident and scattered field amplitudes.
- void [rabas](#) (int inpol, int li, int nsph, [ParticleDescriptor](#) *c1, double **tqse, [dcomplex](#) **tqspe, double **tqss, [dcomplex](#) **tqsps)
Compute radiation torques on a single sphere in Cartesian coordinates.
- void [rbf](#) (int n, double x, int &nm, double sj[])
Real Bessel Function.
- void [rk](#) (int npntmo, double step, [dcomplex](#) dcc, double &x, int lpo, [dcomplex](#) &y1, [dcomplex](#) &y2, [dcomplex](#) &dy1, [dcomplex](#) &dy2)
Soft layer radial function and derivative.
- void [rkt](#) (int npntmo, double step, double &x, int lpo, [dcomplex](#) &y1, [dcomplex](#) &y2, [dcomplex](#) &dy1, [dcomplex](#) &dy2, [ParticleDescriptor](#) *c1)
Transition layer radial function and derivative.
- void [rnf](#) (int n, double x, int &nm, double *sy)
Spherical Bessel functions.
- void [sphar](#) (double cosrth, double sinrth, double cosrph, double sinrph, int ll, [dcomplex](#) *ylm)
Spherical harmonics for given direction.
- void [sscr0](#) ([dcomplex](#) &tfsas, int nsph, int lm, double vk, double exri, [ParticleDescriptor](#) *c1)
Compute scattering, absorption and extinction cross-sections.
- void [sscr2](#) (int nsph, int lm, double vk, double exri, [ParticleDescriptor](#) *c1)
C++ Compute the scattering amplitude and the scattered field intensity.
- void [thdps](#) (int lm, double ***zpv)
Determine the geometrical asymmetry parameter coefficients.
- void [upvmp](#) (double thd, double phd, int icspnv, double &cost, double &sint, double &cosp, double &sinp, double *u, double *up, double *un)
Compute the unitary vectors onb the polarization plane and its orthogonal direction.
- void [upvsp](#) (double *u, double *upmp, double *unmp, double *us, double *upsm, double *unsm, double *up, double *un, double *ups, double *uns, double *duk, int &isq, int &ibf, double &scand, double &cfmp, double &sfmp, double &cfsp, double &sfsp)
Compute the unitary vector perpendicular to incident and scattering plane.
- void [wmamp](#) (int iis, double cost, double sint, double cosp, double sinp, int inpol, int lm, int idot, int nsph, double *arg, double *u, double *up, double *un, [ParticleDescriptor](#) *c1)
Compute meridional plane-referred geometrical asymmetry parameter coefficients.
- void [wmasp](#) (double cost, double sint, double cosp, double sinp, double costs, double sints, double cosps, double sinps, double *u, double *up, double *un, double *us, double *ups, double *uns, int isq, int ibf, int inpol, int lm, int idot, int nsph, double *argi, double *args, [ParticleDescriptor](#) *c1)
Compute the scattering plane-referred geometrical asymmetry parameter coefficients.

9.33.1 Detailed Description

This library includes a collection of functions that are used to solve the scattering problem in the case of a single sphere. Some of these functions are also generalized to the case of clusters of spheres. In most cases, the results of calculations do not fall back to fundamental data types. They are rather multi-component structures. In order to manage access to such variety of return values, most functions are declared as `void` and they operate on output arguments passed by reference.

9.33.2 Function Documentation

9.33.2.1 `aps()`

```
void aps (
    double **** zpv,
    int li,
    int nsph,
    ParticleDescriptor * c1,
    double sqk,
    double * gaps )
```

This function computes the product between the geometrical asymmetry parameter and the scattering cross-section. See Sec. 3.2.1 of Borghese, Denti & Saija (2007).

Parameters

<i>zpv</i>	double **** Geometrical asymmetry parameter coefficients matrix.
<i>li</i>	int Maximum field expansion order.
<i>nsph</i>	int Number of spheres.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.
<i>sqk</i>	double
<i>gaps</i>	double * Geometrical asymmetry parameter-corrected cross-section.

9.33.2.2 `cbf()`

```
void cbf (
    int n,
    dcomplex z,
    int & nm,
    dcomplex * csj )
```

This function computes the complex spherical Bessel functions j . It uses the auxiliary functions `msta1()` and `msta2()` to determine the starting point for backward recurrence. This is the `CSPHJ` implementation of the `specfun` library.

Parameters

<i>n</i>	int Order of the function.
<i>z</i>	complex double Argument of the function.
<i>nm</i>	int & Highest computed order.
<i>csj</i>	complex double * The desired function j .

9.33.2.3 cg1()

```
double cg1 (
    int lmpml,
    int mu,
    int l,
    int m )
```

This function computes the Clebsch-Gordan coefficients for the irreducible spherical tensors. See Sec. 1.6.3 and Table 1.1 in Borghese, Denti & Saija (2007).

Parameters

<i>lmpml</i>	int
<i>mu</i>	int
<i>l</i>	int
<i>m</i>	int

Returns

result: double Clebsch-Gordan coefficient.

9.33.2.4 diel()

```
void diel (
    int npntmo,
    int ns,
    int i,
    int ic,
    double vk,
    ParticleDescriptor * c1 )
```

This function implements the continuous variation of the refractive index and of its radial derivative through the materials that constitute the sphere and its surrounding medium. See Sec. 5.2 in Borghese, Denti & Saija (2007).

Parameters

<i>npntmo</i>	int
<i>ns</i>	int
<i>i</i>	int
<i>ic</i>	int
<i>vk</i>	double
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.

9.33.2.5 dme()

```
void dme (
    int li,
    int i,
```



```

    int npnt,
    int npntts,
    double vk,
    double exdc,
    double exri,
    ParticleDescriptor * c1,
    int & jer,
    int & lcalc,
    dcomplex & arg,
    int last_conf = 0 )

```

This function determines the L-dependent Mie scattering coefficients a_l and b_l for the cases of homogeneous spheres, radially non-homogeneous spheres and, in case of sphere with dielectric function, sustaining longitudinal waves. See Sec. 5.1 in Borghese, Denti & Saija (2007).

Parameters

<i>li</i>	int Maximum field expansion order.
<i>i</i>	int
<i>npnt</i>	int
<i>npntts</i>	int
<i>vk</i>	double Wave number in scale units.
<i>exdc</i>	double External medium dielectric constant.
<i>exri</i>	double External medium refractive index.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.
<i>jer</i>	int & Reference to integer error code variable.
<i>lcalc</i>	int & Reference to integer variable recording the maximum expansion order accounted for.
<i>arg</i>	complex double &
<i>last_conf</i>	int Last sphere configuration (used by CLUSTER)

9.33.2.6 envj()

```

double envj (
    int n,
    double x )

```

This function determines the control parameters required to calculate the Bessel functions up to the desired precision.

Parameters

<i>n</i>	int
<i>x</i>	double

Returns

result: double

9.33.2.7 mmulc()

```

void mmulc (
    dcomplex * vint,

```

```
double ** cmullr,
double ** cmul )
```

This function computes the Mueller Transformation Matrix, or Phase Matrix. See Sec. 2.8.1 of Borghese, Denti & Saija (2007).

Parameters

<i>vint</i>	complex double *
<i>cmullr</i>	double **
<i>cmul</i>	double **

9.33.2.8 msta1()

```
int msta1 (
    double x,
    int mp )
```

This function determines the starting point for backward recurrence such that the magnitude of all J and j functions is of the order of 10^{-mp} .

Parameters

<i>x</i>	double Absolute value of the argumetn to J or j .
<i>mp</i>	int Requested order of magnitude.

Returns

result: int The necessary starting point.

9.33.2.9 msta2()

```
int msta2 (
    double x,
    int n,
    int mp )
```

This function determines the starting point for backward recurrence such that all J and j functions have `mp` significant digits.

Parameters

<i>x</i>	double Absolute value of the argumetn to J or j .
<i>n</i>	int Order of the function.
<i>mp</i>	int Requested number of significant digits.

Returns

result: int The necessary starting point.

9.33.2.10 orunve()

```
void orunve (
    double * u1,
    double * u2,
    double * u3,
    int iorth,
    double torth )
```

This function computes the amplitude of the orthogonal unit vector for a geometry based either on the scattering plane or on the meridional plane. It is used by [upvsp\(\)](#) and [upvmp\(\)](#). See Sec. 2.7 in Borghese, Denti & Saija (2007).

Parameters

<i>u1</i>	double *
<i>u2</i>	double *
<i>u3</i>	double *
<i>iorth</i>	int
<i>torth</i>	double

9.33.2.11 pwma()

```
void pwma (
    double * up,
    double * un,
    dcomplex * ylm,
    int inpol,
    int lw,
    int isq,
    ParticleDescriptor * c1 )
```

This function computes the amplitudes of the incident and scattered field on the basis of the multi-polar expansion. See Sec. 4.1.1 in Borghese, Denti and Saija (2007).

Parameters

<i>up</i>	double *
<i>un</i>	double *
<i>ylm</i>	complex double * Field polar spherical harmonics.
<i>inpol</i>	int Incident field polarization type (0 - linear; 1 - circular).
<i>lw</i>	int
<i>isq</i>	int
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.

9.33.2.12 rabas()

```
void rabas (
    int inpol,
    int li,
```

```

int nsph,
ParticleDescriptor * c1,
double ** tqse,
dcomplex ** tqspe,
double ** tqss,
dcomplex ** tqspss )

```

This function computes radiation torque on a sphere unit as the result of the difference between the extinction and the scattering contributions. See Sec. 4.9 in Borghese, Denti & Saija (2007).

Parameters

<i>inpol</i>	int Incident polarization type (0 - linear; 1 - circular)
<i>li</i>	int Maximum field expansion order.
<i>nsph</i>	int Number of spheres.
<i>c1</i>	ParticleDescriptor * Pointer to ParticleDescriptor data structure.
<i>tqse</i>	double **
<i>tqspe</i>	complex double **
<i>tqss</i>	double **
<i>tqspss</i>	complex double **

9.33.2.13 rbf()

```

void rbf (
    int n,
    double x,
    int & nm,
    double sj[] )

```

This function computes the real spherical Bessel functions j . It uses the auxiliary functions `msta1()` and `msta2()` to determine the starting point for backward recurrence. This is the SPHJ implementation of the `specfun` library.

Parameters

<i>n</i>	int Order of the function.
<i>x</i>	double Argument of the function.
<i>nm</i>	int & Highest computed order.
<i>sj</i>	double[] The desired function j .

9.33.2.14 rkc()

```

void rkc (
    int npntmo,
    double step,
    dcomplex dcc,
    double & x,
    int lpo,
    dcomplex & y1,
    dcomplex & y2,

```

```

    dcomplex & dy1,
    dcomplex & dy2 )

```

This function determines the radial function and its derivative for a soft layer in a radially non homogeneous sphere. See Sec. 5.1 in Borghese, Denti & Saija (2007).

Parameters

<i>npntmo</i>	int
<i>step</i>	double
<i>dcc</i>	complex double
<i>x</i>	double &
<i>lpo</i>	int
<i>y1</i>	complex double &
<i>y2</i>	complex double &
<i>dy1</i>	complex double &
<i>dy2</i>	complex double &

9.33.2.15 rkt()

```

void rkt (
    int npntmo,
    double step,
    double & x,
    int lpo,
    dcomplex & y1,
    dcomplex & y2,
    dcomplex & dy1,
    dcomplex & dy2,
    ParticleDescriptor * c1 )

```

This function determines the radial function and its derivative for a transition layer in a radially non homogeneous sphere. See Sec. 5.1 in Borghese, Denti & Saija (2007).

Parameters

<i>npntmo</i>	int
<i>step</i>	double
<i>x</i>	double &
<i>lpo</i>	int
<i>y1</i>	complex double &
<i>y2</i>	complex double &
<i>dy1</i>	complex double &
<i>dy2</i>	complex double &
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.33.2.16 rnf()

```

void rnf (
    int n,

```

```
double x,
int & nm,
double * sy )
```

This function computes the spherical Bessel functions y . It adopts the SPHJY implementation of the `specfun` library.

Parameters

<i>n</i>	int Order of the function (from 0 up).
<i>x</i>	double Argumento of the function ($x > 0$).
<i>nm</i>	int & Highest computed order.
<i>sy</i>	double * The desired function y .

9.33.2.17 sphar()

```
void sphar (
    double cosrth,
    double sinrth,
    double cosrph,
    double sinrph,
    int ll,
    dcomplex * ylm )
```

This function computes the field spherical harmonics for a given direction. See Sec. 1.5.2 in Borghese, Denti & Saija (2007).

Parameters

<i>cosrth</i>	double Cosine of direction's elevation.
<i>sinrth</i>	double Sine of direction's elevation.
<i>cosrph</i>	double Cosine of direction's azimuth.
<i>sinrph</i>	double Sine of direction's azimuth.
<i>ll</i>	int L value expansion order.
<i>ylm</i>	complex double * The requested spherical harmonics.

9.33.2.18 sscr0()

```
void sscr0 (
    dcomplex & tfsas,
    int nsph,
    int lm,
    double vk,
    double exri,
    ParticleDescriptor * cl )
```

This function computes the scattering, absorption and extinction cross-sections in terms of Forward Scattering Amplitudes. See Sec. 4.2.1 in Borghese, Denti & Saija (2007).

Parameters

<i>tfsas</i>	complex double &
<i>nsph</i>	int Number of spheres.
<i>lm</i>	int Maximum field expansion order.
<i>vk</i>	double Wave number in scale units.
<i>exri</i>	double External medium refractive index.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.

9.33.2.19 sscr2()

```
void sscr2 (
    int nsph,
    int lm,
    double vk,
    double exri,
    ParticleDescriptor * c1 )
```

The role of this function is to compute the scattering amplitude and the intensity of the scattered field. See Sec. 4.2 in Borghese, Denti & Saija (2007).

Parameters

<i>nsph</i>	int Number of spheres.
<i>lm</i>	int Maximum field expansion order.
<i>vk</i>	double Wave number in scale units.
<i>exri</i>	double External medium refractive index.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.

9.33.2.20 thdps()

```
void thdps (
    int lm,
    double **** zpv )
```

This function computes the coefficients that enter the definition of the geometrical asymmetry parameter based on the L-value of the field expansion order. See Sec. 3.2.1 in Borghese, Denti & Saija (2007).

Parameters

<i>lm</i>	int Maximum field expansion order.
<i>zpv</i>	double **** Matrix of geometrical asymmetry parameter coefficients.

9.33.2.21 upvmp()

```
void upvmp (
    double thd,
```

```

double phd,
int icspnv,
double & cost,
double & sint,
double & cosp,
double & sinp,
double * u,
double * up,
double * un )

```

This function computes the unitary vectors lying on the polarization plane and on its orthogonal direction, to optimize the identification of the scattering geometry. See Sec. 2.3 in Borghese, Denti & Saija (2007).

Parameters

<i>thd</i>	double
<i>phd</i>	double
<i>icspnv</i>	int
<i>cost</i>	double
<i>sint</i>	double
<i>cosp</i>	double
<i>sinp</i>	double
<i>u</i>	double *
<i>up</i>	double *
<i>un</i>	double *

9.33.2.22 upvsp()

```

void upvsp (
    double * u,
    double * upmp,
    double * unmp,
    double * us,
    double * upsmmp,
    double * unsmmp,
    double * up,
    double * un,
    double * ups,
    double * uns,
    double * duk,
    int & isq,
    int & ibf,
    double & scand,
    double & cfmp,
    double & sfmp,
    double & cfsp,
    double & sfsp )

```

This function computes the unitary vector perpendicular to the incident and scattering plane in a geometry based on the scattering plane. It uses [orunve\(\)](#). See Sec. 2.7 in Borghese, Denti & Saija (2007).

Parameters

<i>u</i>	double *
----------	----------

Parameters

<i>upmp</i>	double *
<i>unmp</i>	double *
<i>us</i>	double *
<i>upsmp</i>	double *
<i>unsmp</i>	double *
<i>up</i>	double *
<i>un</i>	double *
<i>ups</i>	double *
<i>uns</i>	double *
<i>duk</i>	double *
<i>isq</i>	int &
<i>ibf</i>	int &
<i>scand</i>	double & Scattering angle in degrees.
<i>cfmp</i>	double &
<i>sfmp</i>	double &
<i>cfsp</i>	double &
<i>sfsp</i>	double &

9.33.2.23 wmamp()

```
void wmamp (
    int iis,
    double cost,
    double sint,
    double cosp,
    double sinp,
    int inpol,
    int lm,
    int idot,
    int nsph,
    double * arg,
    double * u,
    double * up,
    double * un,
    ParticleDescriptor * cl )
```

This function computes the coefficients that define the geometrical symmetry parameter as defined with respect to the meridional plane. It makes use of [sphar\(\)](#) and [pwma\(\)](#). See Sec. 3.2.1 in Borghese, Denti & Saija (2007).

Parameters

<i>iis</i>	int
<i>cost</i>	double Cosine of the elevation angle.
<i>sint</i>	double Sine of the elevation angle.
<i>cosp</i>	double Cosine of the azimuth angle.
<i>sinp</i>	double Sine of the azimuth angle.
<i>inpol</i>	int Incident field polarization type (0 - linear; 1 - circular).
<i>lm</i>	int Maximum field expansion orde.
<i>idot</i>	int

Parameters

<i>nsph</i>	int Number of spheres.
<i>arg</i>	double *
<i>u</i>	double *
<i>up</i>	double *
<i>un</i>	double *
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.

9.33.2.24 wmasp()

```

void wmasp (
    double cost,
    double sint,
    double cosp,
    double sinp,
    double costs,
    double sints,
    double cosps,
    double sinps,
    double * u,
    double * up,
    double * un,
    double * us,
    double * ups,
    double * uns,
    int isq,
    int ibf,
    int inpol,
    int lm,
    int idot,
    int nsph,
    double * argi,
    double * args,
    ParticleDescriptor * c1 )

```

This function computes the coefficients that define the geometrical asymmetry parameter based on the L-value with respect to the scattering plane. It uses [sphar\(\)](#) and [pwma\(\)](#). See Sec. 3.2.1 in Borghese, Denti and Saija (2007).

Parameters

<i>cost</i>	double Cosine of elevation angle.
<i>sint</i>	double Sine of elevation angle.
<i>cosp</i>	double Cosine of azimuth angle.
<i>sinp</i>	double Sine of azimuth angle.
<i>costs</i>	double Cosine of scattering elevation angle.
<i>sints</i>	double Sine of scattering elevation angle.
<i>cosps</i>	double Cosine of scattering azimuth angle.
<i>sinps</i>	double Sine of scattering azimuth angle.
<i>u</i>	double *
<i>up</i>	double *
<i>un</i>	double *

Parameters

<i>us</i>	double *
<i>ups</i>	double *
<i>uns</i>	double *
<i>isq</i>	int
<i>ibf</i>	int
<i>inpol</i>	int Incident field polarization (0 - linear; 1 -circular).
<i>lm</i>	int Maximum field expansion order.
<i>idot</i>	int
<i>nsph</i>	int Number of spheres.
<i>argi</i>	double *
<i>args</i>	double *
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.

9.34 sph_subs.h

[Go to the documentation of this file.](#)

```

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00014    this program in the COPYING file. If not, see: <https://www.gnu.org/licenses/>.
00015 */
00016
00030 #ifndef INCLUDE_SPH_SUBS_H_
00031 #define INCLUDE_SPH_SUBS_H_
00032
00045 void aps(double ****zpv, int li, int nsph, ParticleDescriptor *c1, double sqk, double *gaps);
00046
00058 void cbf(int n, dcomplex z, int &nm, dcomplex *csj);
00059
00071 double cgl(int lmpml, int mu, int l, int m);
00072
00086 void diel(int npntmo, int ns, int i, int ic, double vk, ParticleDescriptor *c1);
00087
00108 void dme(
00109     int li, int i, int npnt, int npntts, double vk, double exdc, double exri,
00110     ParticleDescriptor *c1, int &jer, int &lcalc, dcomplex &arg, int last_conf=0
00111 );
00112
00122 double envj(int n, double x);
00123
00133 void mmulc(dcomplex *vint, double **cmullr, double **cmul);
00134
00144 int mstal(double x, int mp);
00145
00156 int msta2(double x, int n, int mp);
00157
00170 void orunve(double *u1, double *u2, double *u3, int iorth, double torth);
00171
00185 void pwma(
00186     double *up, double *un, dcomplex *ylm, int inpol, int lw,
00187     int isq, ParticleDescriptor *c1
00188 );
00189
00205 void rabas(
00206     int inpol, int li, int nsph, ParticleDescriptor *c1, double **tqse, dcomplex **tqspe,
00207     double **tqss, dcomplex **tqsps
00208 );
00209

```

```

00221 void rbf(int n, double x, int &nm, double sj[]);
00222
00238 void rkc(
00239     int npntmo, double step, dcomplex dcc, double &x, int lpo,
00240     dcomplex &y1, dcomplex &y2, dcomplex &dy1, dcomplex &dy2
00241 );
00242
00258 void rkt(
00259     int npntmo, double step, double &x, int lpo, dcomplex &y1,
00260     dcomplex &y2, dcomplex &dy1, dcomplex &dy2, ParticleDescriptor *c1
00261 );
00262
00273 void rnf(int n, double x, int &nm, double *sy);
00274
00287 void sphar(
00288     double cosrth, double sinrth, double cosrph, double sinrph,
00289     int ll, dcomplex *ylm
00290 );
00291
00304 void sscr0(dcomplex &tfsas, int nsph, int lm, double vk, double exri, ParticleDescriptor *c1);
00305
00317 void sscr2(int nsph, int lm, double vk, double exri, ParticleDescriptor *c1);
00318
00328 void thdps(int lm, double ****zpv);
00329
00348 void upvmp(
00349     double thd, double phd, int icspnv, double &cost, double &sint,
00350     double &cosp, double &sinp, double *u, double *up, double *un
00351 );
00352
00378 void upvsp(
00379     double *u, double *upmp, double *unmp, double *us, double *upsmp, double *unsmmp,
00380     double *up, double *un, double *ups, double *uns, double *duk, int &isq,
00381     int &ibf, double &scand, double &cfmp, double &sfmp, double &cfsp, double &sfsp
00382 );
00383
00405 void wmamp(
00406     int iis, double cost, double sint, double cosp, double sinp, int inpol,
00407     int lm, int idot, int nsph, double *arg, double *u, double *up,
00408     double *un, ParticleDescriptor *c1
00409 );
00410
00441 void wmasp(
00442     double cost, double sint, double cosp, double sinp, double costs, double sints,
00443     double cosps, double sinps, double *u, double *up, double *un, double *us,
00444     double *ups, double *uns, int isq, int ibf, int inpol, int lm, int idot,
00445     int nsph, double *argi, double *args, ParticleDescriptor *c1
00446 );
00447
00448 #endif /* SRC_INCLUDE_SPH_SUBS_H_ */

```

9.35 np_tmcode/src/include/tfrfme.h File Reference

Representation of the trapping calculation objects.

Classes

- class [Swap1](#)
Class to represent the first group of trapping swap data.
- class [Swap2](#)
Class to represent the second group of trapping swap data.
- class [TFRFME](#)
Class to represent the trapping configuration.

9.36 tfrfme.h

[Go to the documentation of this file.](#)

```
00001 /* Copyright (C) 2025 INAF - Osservatorio Astronomico di Cagliari
```

```

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00012
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00014     this program in the COPYING file. If not, see: <https://www.gnu.org/licenses/>.
00015 */
00016
00022 #ifndef INCLUDE_TFRFME_H_
00023 #define INCLUDE_TFRFME_H_
00024
00027 class Swap1 {
00028 protected:
00030     int _last_index;
00032     int _nkx;
00034     int _nlmmt;
00035
00037     dcomplex *_wk;
00038
00044     static Swap1 *from_hdf5(const std::string& file_name);
00045
00051     static Swap1 *from_legacy(const std::string& file_name);
00052
00057     void write_hdf5(const std::string& file_name);
00058
00063     void write_legacy(const std::string& file_name);
00064
00065 public:
00067     const dcomplex *_wk;
00068
00074     Swap1(int lm, int nkx);
00075
00078     ~Swap1() { delete[] _wk; }
00079
00084     void append(dcomplex value) { _wk[_last_index++] = value; }
00085
00093     static Swap1* from_binary(const std::string& file_name, const std::string& mode="LEGACY");
00094
00101     static long get_size(int lm, int nkx);
00102
00105     void reset() { _last_index = 0; }
00106
00113     void write_binary(const std::string& file_name, const std::string& mode="LEGACY");
00114
00122     bool operator ==(Swap1 &other);
00123 };
00124
00127 class Swap2 {
00128 protected:
00130     int _last_vector;
00132     int _last_matrix;
00134     int _nkx;
00136     double _apfafa;
00138     double _pmf;
00140     double _spd;
00142     double _rir;
00144     double _ftcn;
00146     double _fshmx;
00148     double _vxyzmx;
00150     double _delxyz;
00152     double _vknmx;
00154     double _delk;
00156     double _delks;
00158     int _nlmmt;
00160     int _nrvc;
00161
00167     static Swap2 *from_hdf5(const std::string& file_name);
00168
00174     static Swap2 *from_legacy(const std::string& file_name);
00175
00180     void write_hdf5(const std::string& file_name);
00181
00186     void write_legacy(const std::string& file_name);
00187
00188 public:
00190     const int &last_vector = _last_vector;
00192     const int &last_matrix = _last_matrix;
00194     const int &nkv = _nkx;
00196     double *_kv;
00198     double *_vec_vkzm;

```

```

00200     const double &apfafa = _apfafa;
00202     const double &pmf = _pmf;
00204     const double &spd = _spd;
00206     const double &rir = _rir;
00208     const double &ftcn = _ftcn;
00210     const double &fshmx = _fshmx;
00212     const double &vxyzmx = _vxyzmx;
00214     const double &delxyz = _delxyz;
00216     const double &vknmx = _vknmx;
00218     const double &delk = _delk;
00220     const double &delks = _delks;
00222     const int &nlmmt = _nlmmt;
00224     const int &nrvc = _nrvc;
00225
00230     Swap2(int nk);
00231
00234     ~Swap2();
00235
00243     static Swap2* from_binary(const std::string& file_name, const std::string& mode="LEGACY");
00244
00250     static long get_size(int nk);
00251
00256     double *get_vector() { return vk; }
00257
00262     void push_matrix(double value);
00263
00268     void push_vector(double value) { vk[_last_vector++] = value; }
00269
00272     void reset_matrix() { _last_matrix = 0; }
00273
00276     void reset_vector() { _last_vector = 0; }
00277
00283     void set_param(const std::string& param_name, double value);
00284
00291     void write_binary(const std::string& file_name, const std::string& mode="LEGACY");
00292
00300     bool operator ==(Swap2 &other);
00301 };
00302
00305 class TFRFME {
00306 protected:
00308     int _nlmmt;
00310     int _nrvc;
00312     int _lmode;
00314     int _lm;
00316     int _nk;
00318     int _nx;
00320     int _ny;
00322     int _nz;
00324     double _vk;
00326     double _exri;
00328     double _an;
00330     double _ff;
00332     double _tra;
00334     double _spd;
00336     double _frsh;
00338     double _exril;
00340     double *xv;
00342     double *yv;
00344     double *zv;
00345
00352     static TFRFME *from_hdf5(const std::string& file_name);
00353
00360     static TFRFME *from_legacy(const std::string& file_name);
00361
00366     void write_hdf5(const std::string& file_name);
00367
00372     void write_legacy(const std::string& file_name);
00373
00374 public:
00376     const int& nlmmt = _nlmmt;
00378     const int& nrvc = _nrvc;
00380     const int& lmode = _lmode;
00382     const int& lm = _lm;
00384     const int& nk = _nk;
00386     const int& nx = _nx;
00388     const int& ny = _ny;
00390     const int& nz = _nz;
00392     const double& vk = _vk;
00394     const double& exri = _exri;
00396     const double& an = _an;
00398     const double& ff = _ff;
00400     const double& tra = _tra;
00402     const double& spd = _spd;
00404     const double& frsh = _frsh;
00406     const double& exril = _exril;
00408     dcomplex *vec_wsum;

```

```

00409
00419   TFRFME(int lmode, int lm, int nk, int nxv, int nyv, int nzv);
00420
00423   ~TFRFME();
00424
00433   static TFRFME* from_binary(
00434       const std::string& file_name, const std::string& mode="LEGACY"
00435   );
00436
00446   static long get_size(int lm, int nk, int nxv, int nyv, int nzv);
00447
00452   double *get_x() { return xv; }
00453
00458   double *get_y() { return yv; }
00459
00464   double *get_z() { return zv; }
00465
00471   void set_param(const std::string& param_name, double value);
00472
00479   void write_binary(const std::string& file_name, const std::string& mode="LEGACY");
00480
00488   bool operator ==(const TFRFME& other);
00489 };
00490 #endif

```

9.37 np_tmcode/src/include/tra_subs.h File Reference

C++ porting of TRAPPING functions and subroutines.

Classes

- struct [CIL](#)
CIL data structure.
- struct [CCR](#)
CCR data structure.

Functions

- void [camp](#) (dcomplex *ac, dcomplex **am0m, dcomplex *ws, [CIL](#) *cil)
- void [czamp](#) (dcomplex *ac, dcomplex **amd, int **indam, dcomplex *ws, [CIL](#) *cil)
- void [ffrf](#) (double ****zpv, dcomplex *ac, dcomplex *ws, double *fffe, double *fffs, [CIL](#) *cil, [CCR](#) *ccr)
- void [ffrt](#) (dcomplex *ac, dcomplex *ws, double *ffte, double *ffts, [CIL](#) *cil)
- dcomplex * [frfmer](#) (int nk, double vkm, double vknmx, double apfafa, double tra, double spd, double rir, double ftcn, int le, int lmode, double pmf, [Swap1](#) *tt1, [Swap2](#) *tt2)
- void [pwmalp](#) (dcomplex **w, double *up, double *un, dcomplex *ylm, int lw)
- void [samp](#) (dcomplex *ac, dcomplex *tmsm, dcomplex *tmse, dcomplex *ws, [CIL](#) *cil)
- void [sampo](#) (dcomplex *ac, dcomplex **tms, dcomplex *ws, [CIL](#) *cil)
- void [wamff](#) (dcomplex *wk, double x, double y, double z, int lm, double apfafa, double tra, double spd, double rir, double ftcn, int lmode, double pmf)

9.37.1 Detailed Description

This library includes a collection of functions that are used to solve the trapping problem. The functions that were generalized from the case of the single sphere are imported the [sph_subs.h](#) library. As it occurs with the single sphere case functions, in most cases, the results of calculations do not fall back to fundamental data types. They are rather multi-component structures. In order to manage access to such variety of return values, most functions are declared as `void` and they operate on output arguments passed by reference.

9.37.2 Function Documentation

9.37.2.1 camp()

```
void camp (
    dcomplex * ac,
    dcomplex ** am0m,
    dcomplex * ws,
    CIL * cil )
```

C++ porting of CAMP

Parameters

<i>ac</i>	complex double * ANNOTATION: vector of complex amplitudes.
<i>am0m</i>	complex double ** ANNOTATION: T-matrix of the particle.
<i>ws</i>	complex double * ANNOTATION: scattering amplitudes.
<i>cil</i>	CIL * Pointer to a CIL structure.

This function builds the AC vector using AM0M and WS.

9.37.2.2 czamp()

```
void czamp (
    dcomplex * ac,
    dcomplex ** amd,
    int ** indam,
    dcomplex * ws,
    CIL * cil )
```

C++ porting of CZAMP

Parameters

<i>ac</i>	Vector of complex. ANNOTATION: vector of complex amplitudes.
<i>amd</i>	Matrix of complex. ANNOTATION: T-matrix.
<i>indam</i>	int **. ANNOTATION: map of matrix indexes.
<i>ws</i>	Vector of complex. ANNOTATION: scattering amplitudes.
<i>cil</i>	CIL * Pointer to a CIL structure.

This function builds the AC vector using AMD, INDAM and WS.

9.37.2.3 ffrf()

```
void ffrf (
    double **** zpv,
    dcomplex * ac,
    dcomplex * ws,
    double * fffe,
```



```
double * fffs,
CIL * cil,
CCR * ccr )
```

C++ porting of FFRF

Parameters

<i>zpv</i>	double ****. ANNOTATION: asymmetry tensor.
<i>ac</i>	complex double * ANNOTATION: vector of complex amplitudes.
<i>ws</i>	complex double * ANNOTATION: scattering amplitudes.
<i>fffe</i>	double *. ANNOTATION: extinction contributions to force.
<i>fffs</i>	double *. ANNOTATION: scattering contributions to force.
<i>cil</i>	CIL * Pointer to a CIL structure.
<i>ccr</i>	CCR * Pointer to a CCR structure.

9.37.2.4 ffrt()

```
void ffrt (
    dcomplex * ac,
    dcomplex * ws,
    double * ffte,
    double * ffts,
    CIL * cil )
```

C++ porting of FFRT

Parameters

<i>ac</i>	complex double * ANNOTATION: vector of complex amplitudes.
<i>ws</i>	complex double * ANNOTATION: scattering amplitudes.
<i>ffte</i>	double *. ANNOTATION: extinction contributions to torque.
<i>ffts</i>	double *. ANNOTATION: scattering contributions to torque.
<i>cil</i>	CIL * Pointer to a CIL structure.

9.37.2.5 frfmer()

```
dcomplex * frfmer (
    int nkx,
    double vkm,
    double vknmx,
    double apfafa,
    double tra,
    double spd,
    double rir,
    double ftcn,
    int le,
    int lmode,
    double pmf,
```

```

Swap1 * tt1,
Swap2 * tt2 )

```

C++ porting of FRFMER

Parameters

<i>nk</i>	int ANNOTATION: number of wave vectors.
<i>vk</i>	double ANNOTATION: wave number.
<i>vkmax</i>	double ANNOTATION: maximum wave number.
<i>apfa</i>	double ANNOTATION: normalized aperture.
<i>tra</i>	double ANNOTATION: lens transmission.
<i>spd</i>	double ANNOTATION: cover thickness.
<i>rir</i>	double ANNOTATION: external over oil refractive index ratio.
<i>ftcn</i>	double ANNOTATION: refraction factor.
<i>le</i>	int ANNOTATION: external multipole truncation order.
<i>lmode</i>	int ANNOTATION: laser mode.
<i>pmf</i>	double ANNOTATION: aperture correction.
<i>tt1</i>	Swap1 * Pointer to first swap object.
<i>tt2</i>	Swap2 * Pointer to second swap object.

Returns

wk: complex double * ANNOTATION: incident amplitudes.

9.37.2.6 pwmalp()

```

void pwmalp (
    dcomplex ** w,
    double * up,
    double * un,
    dcomplex * ylm,
    int lw )

```

C++ porting of PWMALP

Parameters

<i>w</i>	complex double * ANNOTATION: amplitudes.
<i>up</i>	double * ANNOTATION: unit polarization vector.
<i>un</i>	double * ANNOTATION: unit normal vector.
<i>ylm</i>	complex double * Field vector spherical harmonics.
<i>lw</i>	int ANNOTATION: order of amplitudes.

9.37.2.7 samp()

```

void samp (
    dcomplex * ac,

```

```

dcomplex * tmsm,
dcomplex * tmse,
dcomplex * ws,
CIL * cil )

```

C++ porting of SAMP

Parameters

<i>ac</i>	complex double * ANNOTATION: vector of complex amplitudes.
<i>tmsm</i>	complex double * ANNOTATION: T-matrix M components.
<i>tmse</i>	complex double * ANNOTATION: T-matrix E components.
<i>ws</i>	complex double * ANNOTATION: scattering amplitudes.
<i>cil</i>	CIL * Pointer to a CIL structure.

This function builds the AC vector using TMSM, TMSE and WS.

9.37.2.8 sampoa()

```

void sampoa (
    dcomplex * ac,
    dcomplex ** tms,
    dcomplex * ws,
    CIL * cil )

```

C++ porting of SAMPOA

Parameters

<i>ac</i>	complex double * ANNOTATION: vector of complex amplitudes.
<i>tms</i>	complex double ** ANNOTATION: T-matrix.
<i>ws</i>	complex double * ANNOTATION: scattering amplitudes.
<i>cil</i>	CIL * Pointer to a CIL structure.

This function builds the AC vector using TMS and WS.

9.37.2.9 wamff()

```

void wamff (
    dcomplex * wk,
    double x,
    double y,
    double z,
    int lm,
    double apfafa,
    double tra,
    double spd,
    double rir,
    double ftcn,
    int lmode,
    double pmf )

```

C++ porting of WAMFF

Parameters

<i>wk</i>	complex double * ANNOTATION: incident amplitudes.
<i>x</i>	double ANNOTATION: X coordinate.
<i>y</i>	double ANNOTATION: Y coordinate.
<i>z</i>	double ANNOTATION: Z coordinate.
<i>lm</i>	int ANNOTATION: maximum multipole truncation order.
<i>apfafa</i>	double ANNOTATION: Normalized aperture.
<i>tra</i>	double ANNOTATION: Lens transmission.
<i>spd</i>	double ANNOTATION: Cover slip thickness.
<i>rir</i>	double ANNOTATION: External over oil refractive index ratio.
<i>ftcn</i>	double ANNOTATION: Refraction factor.
<i>lmode</i>	int ANNOTATION: Laser mode.
<i>pmf</i>	double ANNOTATION: aperture correction.

9.38 tra_subs.h

[Go to the documentation of this file.](#)

```

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00015 */
00016
00031 #ifndef INCLUDE_TRA_SUBS_H_
00032 #define INCLUDE_TRA_SUBS_H_
00033 #endif
00034
00035 // Structures for TRAPPING
00040 struct CIL {
00042     int le;
00044     int nlem;
00046     int nlemt;
00048     int mxmpo;
00050     int mxim;
00051 };
00052
00057 struct CCR {
00059     double cof;
00061     double cim;
00062 };
00063 //End of TRAPPING structures
00064
00074 void camp(dcomplex *ac, dcomplex **am0m, dcomplex *ws, CIL *cil);
00075
00086 void czamp(dcomplex *ac, dcomplex **amd, int **indam, dcomplex *ws, CIL *cil);
00087
00098 void ffrf(
00099     double ****zpv, dcomplex *ac, dcomplex *ws, double *fffe,
00100     double *fffs, CIL *cil, CCR *ccr
00101 );
00102
00111 void ffrt(dcomplex *ac, dcomplex *ws, double *ffte, double *ffts, CIL *cil);
00112
00130 dcomplex *frfmer(
00131     int nk, double vkm, double vkmx, double apfafa, double tra,
00132     double spd, double rir, double ftcn, int le, int lmode, double pmf,
00133     Swap1 *tt1, Swap2 *tt2
00134 );
00135

```

```

00144 void pwmalp(dcomplex **w, double *up, double *un, dcomplex *ylm, int lw);
00145
00156 void samp(dcomplex *ac, dcomplex *tmsm, dcomplex *tmse, dcomplex *ws, CIL *cil);
00157
00167 void sampoa(dcomplex *ac, dcomplex **tms, dcomplex *ws, CIL *cil);
00168
00184 void wamff(
00185     dcomplex *wk, double x, double y, double z, int lm, double apfafa,
00186     double tra, double spd, double rir, double ftcn, int lmode, double pmf
00187 );

```

9.39 np_tmcode/src/include/TransitionMatrix.h File Reference

Representation of the Transition Matrix.

Classes

- class [TransitionMatrix](#)

Class to represent the Transition Matrix.

9.40 TransitionMatrix.h

[Go to the documentation of this file.](#)

```

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00015 */
00016
00022 #ifndef INCLUDE_TRANSITIONMATRIX_H_
00023 #define INCLUDE_TRANSITIONMATRIX_H_
00024
00027 class TransitionMatrix {
00028 protected:
00030     int _is;
00032     int _l_max;
00034     double _vk;
00036     double _exri;
00038     double _sphere_radius;
00039
00049     static TransitionMatrix *from_hdf5(const std::string& file_name);
00050
00057     static TransitionMatrix *from_legacy(const std::string& file_name);
00058
00067     void write_hdf5(const std::string& file_name);
00068
00083     static void write_hdf5(
00084         const std::string& file_name, np_int nlemt, int lm, double vk,
00085         double exri, dcomplex **am0m
00086     );
00087
00103     static void write_hdf5(
00104         const std::string& file_name, int lm, double vk, double exri,
00105         dcomplex **rmi, dcomplex **rei, double sphere_radius
00106     );
00107
00112     void write_legacy(const std::string& file_name);
00113
00128     static void write_legacy(
00129         const std::string& file_name, np_int nlemt, int lm, double vk,

```

```

00130     double exri, dcomplex **am0m
00131 );
00132
00148 static void write_legacy(
00149     const std::string& file_name, int lm, double vk, double exri,
00150     dcomplex **rmi, dcomplex **rei, double radius
00151 );
00152
00153 public:
00155 const int& is = _is;
00157 const int& l_max = _l_max;
00159 const double& vk = _vk;
00161 const double& exri = _exri;
00163 dcomplex *elements;
00165 const double& sphere_radius = _sphere_radius;
00167 int *shape;
00168
00178 TransitionMatrix(
00179     int is, int lm, double vk, double exri, dcomplex *_elems, double radius=0.0
00180 );
00181
00194 TransitionMatrix(
00195     int lm, double vk, double exri, dcomplex **rmi, dcomplex **rei, double radius
00196 );
00197
00209 TransitionMatrix(int nlemt, int lm, double vk, double exri, dcomplex **am0m);
00210
00213 ~TransitionMatrix();
00214
00231 static TransitionMatrix* from_binary(const std::string& file_name, const std::string&
mode="LEGACY");
00232
00245 void write_binary(const std::string& file_name, const std::string& mode="LEGACY");
00246
00269 static void write_binary(
00270     const std::string& file_name, np_int nlemt, int lm, double vk,
00271     double exri, dcomplex **am0m, const std::string& mode="LEGACY"
00272 );
00273
00297 static void write_binary(
00298     const std::string& file_name, int lm, double vk, double exri,
00299     dcomplex **rmi, dcomplex **rei, double sphere_radius,
00300     const std::string& mode="LEGACY"
00301 );
00302
00312 bool operator ==(TransitionMatrix &other);
00313 };
00314
00315 #endif

```

9.41 np_tmcode/src/include/types.h File Reference

Definition of fundamental types in use.

```
#include <complex.h>
```

Macros

- **#define np_int** int32_t
Alias of the default integer type.
- **#define dconjg(z)** ((__real__ (z) - I * (__imag__ (z))))
Macro to compute the conjugate of a double precision complex number.
- **#define dcabs(z)** (std::sqrt(__real__ (z) * __real__ (z) + __imag__ (z) * __imag__ (z)))
Macro to compute the magnitude of a double precision complex number.

Typedefs

- typedef __complex__ double **dcomplex**
Short-cut to C-style double precision complex type.

Functions

- double `imag` (const `dcomplex` &z)
Get the imaginary part of double precision complex number.
- double `real` (const `dcomplex` &z)
Get the imaginary part of double precision complex number.

9.41.1 Function Documentation

9.41.1.1 `imag()`

```
double imag (
    const dcomplex & z ) [inline]
```

Parameters

<code>z</code>	const dcomplex & Reference to the complex number from which to extract the imaginary part.
----------------	--

Returns

`Im(z)`: double Imaginary part of `z`.

9.41.1.2 `real()`

```
double real (
    const dcomplex & z ) [inline]
```

Parameters

<code>z</code>	const dcomplex & Reference to the complex number from which to extract the real part.
----------------	---

Returns

`Re(z)`: double Real part of `z`.

9.42 `types.h`

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```



```

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00015 */
00016
00022 #ifndef INCLUDE_TYPES_H_
00023 #define INCLUDE_TYPES_H_
00024
00025 #include <complex.h>
00026
00028 typedef __complex__ double dcomplex;
00029
00030 #ifdef USE_MKL
00031 #ifdef USE_ILP64
00032 #ifndef MKL_INT
00033 #define MKL_INT int64_t
00034 #endif // MKL_INT
00035 #else
00036 #ifndef MKL_INT
00037 #define MKL_INT int32_t
00038 #endif // MKL_INT
00039 #endif // USE_ILP64
00040 #endif // USE_MKL
00041
00042 #ifdef USE_LAPACK
00043 #ifdef USE_MKL
00044 #include <mkl_lapacke.h>
00045 #else
00046 #include <lapacke.h>
00047 #endif // USE_MKL
00048 #endif // USE_LAPACK
00049
00050 #ifdef USE_MAGMA
00051 #include <magma_v2.h>
00052 #endif
00053
00054 #ifndef np_int
00055 #ifdef lapack_int
00057 #define np_int lapack_int
00058 #else
00059 #ifdef USE_ILP64
00061 #define np_int int64_t
00062 #else
00064 #define np_int int32_t
00065 #endif // USE_ILP64
00066 #endif // lapack_int
00067 #endif // np_int
00068
00070 #define dconjg(z) ( (__real__ (z) - I * (__imag__ (z))) )
00071
00073 #define dcabs(z) ( std::sqrt(__real__ (z) * __real__ (z) + __imag__ (z) * __imag__ (z)) )
00074
00081 double inline imag(const dcomplex &z) { return __imag__ z; }
00082
00089 double inline real(const dcomplex &z) { return __real__ z; }
00090 #endif

```

9.43 np_tmcode/src/include/utils.h File Reference

Definition of auxiliary code utilities.

Functions

- double [get_ram_overhead](#) ()
Obtain an estimate of the RAM overhead factor for optimization.
- int [write_dcomplex_matrix](#) (VirtualAsciiFile *af, dcomplex **mat, int rows, int columns, const std::string &format=" %5d %5d (%17.8IE,%17.8IE)\n", int first_index=1)
Write a double complex matrix to a text file.
- int [write_matrix_as_ppm](#) (dcomplex *A, int64_t m, int64_t n, const std::string &file_name, const std::string &mode="MAG", int row_bin=1, int col_bin=1)
Draw a PPM representation of a matrix.

9.43.1 Function Documentation

9.43.1.1 `get_ram_overhead()`

```
double get_ram_overhead ( )
```

Code speed-up optimization usually comes at the cost of increased memory requirements. While this should not generally be a serious issue, it can become such in case of models that require large amounts of memory to be handled. This function tests the code build configuration to provide an zero-order estimate of the weight of these overheads to protect the host system from saturating the RAM.

Returns

factor: double The multiplicative factor to be applied to data size.

9.43.1.2 `write_dcomplex_matrix()`

```
int write_dcomplex_matrix (
    VirtualAsciiFile * af,
    dcomplex ** mat,
    int rows,
    int columns,
    const std::string & format = " %5d %5d (%17.8lE,%17.8lE)\n",
    int first_index = 1 )
```

Parameters

<i>af</i>	VirtualAsciiFile * Pointer to an existing VirtualAsciiFile . [IN]
<i>mat</i>	dcomplex ** Pointer to the matrix. [IN]
<i>rows</i>	int Number of rows in the matrix. [IN]
<i>columns</i>	int Number of columns in the matrix. [IN]
<i>format</i>	const string& Format of the line (default is " %5d %5d (%17.8lE,%17.8lE)\n"). [IN]
<i>first_index</i>	int Index of the first element (default is 1, i.e. base 1 FORTRAN array notation). [IN]

Returns

result: int An exit code (0 if successful).

9.43.1.3 `write_matrix_as_ppm()`

```
int write_matrix_as_ppm (
    dcomplex * A,
    int64_t m,
    int64_t n,
    const std::string & file_name,
    const std::string & mode = "MAG",
    int row_bin = 1,
    int col_bin = 1 )
```

Parameters

in	<i>A</i>	const dcomplex * Vector representation of the matrix.
in	<i>m</i>	int64_t Number of rows in the matrix.
in	<i>n</i>	int64_t Number of columns in the matrix.
in	<i>file_name</i>	const string& Reference to a string with the name of the output file.
in	<i>mode</i>	const string& Reference to a string for the value to be represented ("MAG" to draw magnitudes, "RE" to draw real parts, "IM" to draw imaginary parts. Optional, default is "MAG").
in	<i>row_bin</i>	int Number of rows to bin together (optional, default is 1).
in	<i>col_bin</i>	int Number of columns to bin together (optional, default is 1).

Returns

result: int An exit code (0 if successful).

9.44 utils.h

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```

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00015 */
00016
00023 #ifndef INCLUDE_UTILS_H_
00024 #define INCLUDE_UTILS_H_
00025
00038 double get_ram_overhead();
00039
00051 int write_dcomplex_matrix(
00052     VirtualAsciiFile *af, dcomplex **mat, int rows, int columns,
00053     const std::string& format=" %5d %5d (%17.81E,%17.81E)\n", int first_index=1
00054 );
00055
00070 int write_matrix_as_ppm(
00071     dcomplex *A, int64_t m, int64_t n, const std::string& file_name, const std::string& mode="MAG",
00072     int row_bin=1, int col_bin=1
00073 );
00074 #endif

```

9.45 np_tmcode/src/inclusion/inclusion.cpp File Reference

Implementation of the calculation for a sphere with inclusions.

```

#include <chrono>
#include <cmath>
#include <cstdio>
#include <exception>
#include <fstream>
#include < hdf5.h>

```

```
#include <string>
#include "../include/types.h"
#include "../include/errors.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/Commons.h"
#include "../include/sph_subs.h"
#include "../include/clu_subs.h"
#include "../include/inclu_subs.h"
#include "../include/TransitionMatrix.h"
#include "../include/algebraic.h"
#include "../include/List.h"
#include "../include/file_io.h"
#include "../include/utils.h"
#include "../include/outputs.h"
#include "../include/IterationData.h"
```

Functions

- int [inclusion_jxi488_cycle](#) (int jxi488, [ScattererConfiguration](#) *sconf, [GeometryConfiguration](#) *gconf, [ScatteringAngles](#) *sa, [InclusionIterationData](#) *cid, [InclusionOutputInfo](#) *output, const string &output_path)

Main calculation loop.
- void [inclusion](#) (const string &config_file, const string &data_file, const string &output_path, const [mixMPI](#) *mpidata)

C++ implementation of INCLU.

9.45.1 Function Documentation

9.45.1.1 inclusion()

```
void inclusion (
    const string & config_file,
    const string & data_file,
    const string & output_path,
    const mixMPI * mpidata )
```

Parameters

<i>config_file</i>	string Name of the configuration file.
<i>data_file</i>	string Name of the input data file.
<i>output_path</i>	string Directory to write the output files in.
<i>mpidata</i>	mixMPI * Pointer to an instance of MPI data settings.

9.45.1.2 inclusion_jxi488_cycle()

```
int inclusion_jxi488_cycle (
    int jxi488,
    ScattererConfiguration * sconf,
```

```

    GeometryConfiguration * gconf,
    ScatteringAngles * sa,
    InclusionIterationData * cid,
    InclusionOutputInfo * output,
    const string & output_path )

```

The solution of the scattering problem for different wavelengths is an embarrassingly parallel task. This function, therefore, collects all the operations that can be independently executed by different processes, after the configuration stage and the first calculation loop have been executed.

Parameters

<i>jxi488</i>	int Wavelength loop index.
<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration object.
<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration object.
<i>sa</i>	ScatteringAngles * Pointer to a ScatteringAngles object.
<i>cid</i>	InclusionIterationData * Pointer to an InclusionIterationData object.
<i>output</i>	InclusionOutputInfo * Pointer to an InclusionOutputInfo object.
<i>output_path</i>	const string & Path to the output directory.

9.46 np_tmcode/src/inclusion/np_inclusion.cpp File Reference

Sphere with inclusions scattering program handler.

```

#include <stdio>
#include <string>
#include "../include/types.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/Commons.h"

```

Functions

- void [inclusion](#) (const string &config_file, const string &data_file, const string &output_path, const [mixMPI](#) *mpidata)
C++ implementation of INCLU.
- int [main](#) (int argc, char **argv)
Main program entry point.

9.46.1 Detailed Description

This program emulates the execution work-flow originally handled by the FORTRAN EDFB_INCLU and INCLU codes, which undertook the task of solving the scattering calculation for the case of a sphere with inclusions. The program is designed to work in two modes: emulation and enhanced. The emulation mode is activated by invoking the program without arguments. In this case, the code looks for hard-coded default input and writes output in the execution folder, replicating the behaviour of the original FORTRAN code. Enhanced mode, instead, is activated by passing command line arguments that locate the desired input files and a valid folder to write the output into. The emulation mode is useful for testing, while the advanced mode implements the possibility to change input and output options, without having to modify the code.

9.46.2 Function Documentation

9.46.2.1 inclusion()

```
void inclusion (
    const string & config_file,
    const string & data_file,
    const string & output_path,
    const mixMPI * mpidata ) [extern]
```

Parameters

<i>config_file</i>	string Name of the configuration file.
<i>data_file</i>	string Name of the input data file.
<i>output_path</i>	string Directory to write the output files in.
<i>mpidata</i>	mixMPI * Pointer to an instance of MPI data settings.

9.46.2.2 main()

```
int main (
    int argc,
    char ** argv )
```

This is the starting point of the execution flow. Here we may choose how to configure the code, e.g. by loading a legacy configuration file or some otherwise formatted configuration data set. The code can be linked to a launcher script or to a GUI oriented application that performs the configuration and runs the main program.

Parameters

<i>argc</i>	int The number of arguments given in command-line.
<i>argv</i>	char ** The vector of command-line arguments.

Returns

result: int An exit code passed to the OS (0 for succesful execution).

9.47 np_tmcode/src/libnptm/algebraic.cpp File Reference

Implementation of algebraic functions with different call-backs.

```
#include <string>
#include "../include/types.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/algebraic.h"
```

Functions

- void [lucin](#) ([dcomplex](#) **am, const [np_int](#) nddmst, [np_int](#) n, int &ier)
Invert the multi-centered M-matrix.
- void [invert_matrix](#) ([dcomplex](#) **mat, [np_int](#) size, int &ier, const string &output_path, int jxi488, [np_int](#) max_size, int target_device, const [RuntimeSettings](#) &rs)
Perform in-place matrix inversion.

9.47.1 Function Documentation

9.47.1.1 invert_matrix()

```
void invert_matrix (
    dcomplex ** mat,
    np\_int size,
    int & ier,
    const string & output_path,
    int jxi488,
    np\_int max_size = 0,
    int target_device = 0,
    const RuntimeSettings & rs = RuntimeSettings() )
```

Parameters

<i>mat</i>	<code>complex double **</code> The matrix to be inverted (must be a square matrix).
<i>size</i>	<code>np_int</code> The size of the matrix (i.e. the number of its rows or columns).
<i>ier</i>	<code>int &</code> Reference to an integer variable for returning a result flag.
<i>output_path</i>	<code>const string &</code> Path where the output needs to be placed.
<i>jxi488</i>	<code>int</code> Index of the current wavelength calculation.
<i>max_size</i>	<code>np_int</code> The maximum expected size (required by some call-backs, optional, defaults to 0).
<i>target_device</i>	<code>int</code> ID of target GPU, if available (defaults to 0).
<i>rs</i>	<code>const RuntimeSettings &</code> Runtime options instance (optional).

9.47.1.2 lucin()

```
void lucin (
    dcomplex ** am,
    const np\_int nddmst,
    np\_int n,
    int & ier ) [extern]
```

This function performs the inversion of the multi-centered M-matrix through LU decomposition. See Eq. (5.29) in Borghese, Denti & Saija (2007).

Parameters

<i>am</i>	<code>complex double **</code>
<i>nddmst</i>	<code>const int64_t</code>
<i>n</i>	<code>int64_t</code>
<i>ier</i>	<code>int &</code>

9.48 np_tmcode/src/libnptm/clu_subs.cpp File Reference

C++ implementation of CLUSTER subroutines.

```
#include <cstring>
#include "../include/types.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/Constants.h"
#include "../include/sph_subs.h"
#include "../include/clu_subs.h"
```

Functions

- void [apc](#) (double ****zpv, int le, [dcomplex](#) **am0m, [dcomplex](#) **w, double sqk, double **gapr, [dcomplex](#) **gapp)
Compute the asymmetry-corrected scattering cross-section of a cluster.
- void [apcra](#) (double ****zpv, const int le, [dcomplex](#) **am0m, int inpol, double sqk, double **gaprm, [dcomplex](#) **gappm)
Compute the asymmetry-corrected scattering cross-section under random average conditions.
- [dcomplex](#) [cdtp](#) ([dcomplex](#) z, [dcomplex](#) *vec_am, int i, int jf, int k, int nj, [np_int](#) istep)
Complex inner product.
- double [cgev](#) (int ipamo, int mu, int l, int m)
C++ porting of CGEV. ANNOTATION: Get weight of T-matrix element.
- void [cms](#) ([dcomplex](#) *am, [ParticleDescriptor](#) *c1)
Build the multi-centered M-matrix of the cluster.
- void [crsm1](#) (double vk, double exri, [ParticleDescriptor](#) *c1)
Compute orientation-averaged scattered field intensity.
- [dcomplex](#) [ghit](#) (int ihi, int ipamo, int nbl, int l1, int m1, int l2, int m2, [ParticleDescriptor](#) *c1, double rac3j[])
Compute the transfer vector from N2 to N1.
- void [hvj](#) (double exri, double vk, int &jer, int &lcalc, [dcomplex](#) &arg, [ParticleDescriptor](#) *c1)
Compute Hankel funtion and Bessel functions.
- void [lucin](#) ([dcomplex](#) **am, const [np_int](#) nddmst, [np_int](#) n, int &ier)
Invert the multi-centered M-matrix.
- void [mextc](#) (double vk, double exri, [dcomplex](#) **fsac, double **cextlr, double **cext)
Compute the average extinction cross-section.
- void [pcros](#) (double vk, double exri, [ParticleDescriptor](#) *c1)
Compute cross-sections and forward scattering amplitude for the cluster.
- void [pcrsm0](#) (double vk, double exri, int inpol, [ParticleDescriptor](#) *c1)
Compute orientation-averaged cross-sections and forward scattering amplitude.
- void [polar](#) (double x, double y, double z, double &r, double &cth, double &sth, double &cph, double &sph)
Transform Cartesian quantities to spherical coordinates ones.
- void [r3j000](#) (int j2, int j3, double *rac3j)
Compute the 3j symbol for Clebsch-Gordan coefficients towards J=0.
- void [r3jjr](#) (int j2, int j3, int m2, int m3, double rac3j[])
Compute the 3j symbol for Clebsch-Gordan coefficients for JJ transitions.
- void [r3jmr](#) (int j1, int j2, int j3, int m1, double *rac3j)
Compute the 3j symbol for Clebsch-Gordan coefficients for JM transitions.
- void [raba](#) (int le, [dcomplex](#) **am0m, [dcomplex](#) **w, double **tqce, [dcomplex](#) **tqcpe, double **tqcs, [dcomplex](#) **tqcps)

Compute radiation torques on a particle in Cartesian coordinates.

- void `rfr` (double *u, double *up, double *un, double *gapv, double extins, double scatts, double &rapr, double &cosav, double &fp, double &fn, double &fk, double &fx, double &fy, double &fz)

Compute the radiation force Cartesian components.

- void `scr0` (double vk, double exri, [ParticleDescriptor](#) *c1)

Compute Mie cross-sections for the sphere units in the cluster.

- void `scr2` (double vk, double vkarg, double exri, double *duk, [ParticleDescriptor](#) *c1)

Compute the scattering amplitude for a single sphere in an aggregate.

- void `str` (double **rcf, [ParticleDescriptor](#) *c1)

Transform sphere Cartesian coordinates to spherical coordinates.

- void `tqr` (double *u, double *up, double *un, double *tqev, double *tqsv, double &tep, double &ten, double &tek, double &tsp, double &tsn, double &tsk)

Compute radiation torques on particles on a k-vector oriented system.

- void `ztm` ([dcomplex](#) **am, [ParticleDescriptor](#) *c1)

Calculate the single-centered inversion of the M-matrix.

Variables

- const double `four_pi` = 8.0 * acos(0.0)

Four times PI.

9.48.1 Function Documentation

9.48.1.1 `apc()`

```
void apc (
    double **** zpv,
    int le,
    dcomplex ** am0m,
    dcomplex ** w,
    double sqk,
    double ** gapr,
    dcomplex ** gapp )
```

This function computes the product between the geometrical asymmetry parameter and the scattering cross-section, like `aps()`, but for a cluster of spheres. See Eq. (3.16) in Borghese, Denti & Saija (2007).

Parameters

<code>zpv</code>	double ****
<code>le</code>	int
<code>am0m</code>	complex double **
<code>w</code>	complex double **
<code>sqk</code>	double
<code>gapr</code>	double **
<code>gapp</code>	complex double **

9.48.1.2 apcra()

```
void apcra (
    double **** zpv,
    const int le,
    dcomplex ** am0m,
    int inpol,
    double sqk,
    double ** gaprm,
    dcomplex ** gappm )
```

This function computes the product between the geometrical asymmetry parameter and the scattering cross-section of a cluster using the random average directions.

Parameters

<i>zpv</i>	double ****
<i>le</i>	int
<i>am0m</i>	complex double **
<i>inpol</i>	int Polarization type.
<i>sqk</i>	double
<i>gaprm</i>	double **
<i>gappm</i>	complex double **

9.48.1.3 cdtp()

```
dcomplex cdtp (
    dcomplex z,
    dcomplex * vec_am,
    int i,
    int jf,
    int k,
    int nj,
    np_int istep )
```

This function performs the complex inner product. It is used by `lucin()`.

Parameters

<i>z</i>	complex double Starting element.
<i>vec_am</i>	complex double * Vectorized matrix.
<i>i</i>	int Index of first sub-matrix row.
<i>jf</i>	int Index of first submatrix column.
<i>k</i>	int Index of second sub-matrix row.
<i>nj</i>	int Range of first sub-matrix columns.
<i>istep</i>	np_int Size of rows in the matrix.

9.48.1.4 cgev()

```
double cgev (
```

```

    int ipamo,
    int mu,
    int l,
    int m )

```

Parameters

<i>ipamo</i>	int
<i>mu</i>	int
<i>l</i>	int
<i>m</i>	int

Returns

result: double

9.48.1.5 cms()

```

void cms (
    dcomplex * am,
    ParticleDescriptor * c1 )

```

This function constructs the multi-centered M-matrix of the cluster, according to Eq. (5.28) of Borghese, Denti & Saija (2007).

Parameters

in, out	<i>am</i>	complex double *
in	<i>c1</i>	ParticleDescriptor *

9.48.1.6 crsm1()

```

void crsm1 (
    double vk,
    double exri,
    ParticleDescriptor * c1 )

```

This function computes the intensity of the scattered field for the cluster, averaged on the orientations. It is invoked for IAVM=1 (ANNOTATION: geometry referred to the meridional plane).

Parameters

<i>vk</i>	double Wave number.
<i>exri</i>	double External medium refractive index.
<i>c1</i>	ParticleDescriptor *

9.48.1.7 ghit()

```
dcomplex ghit (
    int ihi,
    int ipamo,
    int nbl,
    int l1,
    int m1,
    int l2,
    int m2,
    ParticleDescriptor * c1,
    double rac3j[] )
```

This function computes the transfer vector going from N2 to N1, using either Hankel, Bessel or Bessel from origin functions.

Parameters

in	<i>ihi</i>	int Function mode.
in	<i>ipamo</i>	int
in	<i>nbl</i>	int Block identifier.
in	<i>l1</i>	int First L quantum number.
in	<i>m1</i>	int First M quantum number.
in	<i>l2</i>	int Second L quantum number.
in	<i>m2</i>	int Second M quantum number.
in	<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.
in, out	<i>rac3j</i>	double[] J connection vector.

Returns

result: dcomplex Matrix element.

9.48.1.8 hjv()

```
void hjv (
    double exri,
    double vk,
    int & jer,
    int & lcalc,
    dcomplex & arg,
    ParticleDescriptor * c1 )
```

This function constructs the Hankel function and the Bessel functions vectors. See page 331 in Borghese, Denti & Saija (2007).

Parameters

<i>exri</i>	double External medium refractive index.
<i>vk</i>	double Wave number.
<i>jer</i>	int & Reference to error code flag.
<i>lcalc</i>	int & Reference to the highest order accounted for in calculation.
<i>arg</i>	complex\<double\> &
<i>c1</i>	ParticleDescriptor *

9.48.1.9 lucin()

```
void lucin (
    dcomplex ** am,
    const np_int nddmst,
    np_int n,
    int & ier )
```

This function performs the inversion of the multi-centered M-matrix through LU decomposition. See Eq. (5.29) in Borghese, Denti & Saija (2007).

Parameters

<i>am</i>	complex double **
<i>nddmst</i>	const int64_t
<i>n</i>	int64_t
<i>ier</i>	int &

9.48.1.10 mextc()

```
void mextc (
    double vk,
    double exri,
    dcomplex ** fsac,
    double ** cextlr,
    double ** cext )
```

This function computes the average extinction cross-section starting from the definition of the scattering amplitude. See Sec. 3.2.1 of Borghese, Denti & Saija (2007).

Parameters

<i>vk</i>	double Wave number.
<i>exri</i>	double External medium refractive index.
<i>fsac</i>	Matrix of complex
<i>cextlr</i>	double **
<i>cext</i>	double **

9.48.1.11 pcros()

```
void pcros (
    double vk,
    double exri,
    ParticleDescriptor * cl )
```

This function computes the scattering, absorption and extinction cross-sections of the cluster, together with the Forward Scattering Amplitude.

This function is intended to evaluate the particle cross-section. QUESTUON: correct?

Parameters

<i>vk</i>	double Wave number.
<i>exri</i>	double External medium refractive index.
<i>c1</i>	ParticleDescriptor *

9.48.1.12 pcrsm0()

```
void pcrsm0 (
    double vk,
    double exri,
    int inpol,
    ParticleDescriptor * c1 )
```

This function computes the orientation-averaged scattering, absorption and extinction cross-sections of the cluster, together with the averaged Forward Scattering Amplitude.

Parameters

<i>vk</i>	double Wave number.
<i>exri</i>	double External medium refractive index.
<i>inpol</i>	int Incident field polarization type.
<i>c1</i>	ParticleDescriptor *

9.48.1.13 polar()

```
void polar (
    double x,
    double y,
    double z,
    double & r,
    double & cth,
    double & sth,
    double & cph,
    double & sph )
```

This function performs a conversion from the Cartesian coordinates system to the spherical one. It is used by `sphar()`.

Parameters

<i>x</i>	double X-axis Cartesian coordinate.
<i>y</i>	double Y-axis Cartesian coordinate.
<i>z</i>	double Z-axis Cartesian coordinate.
<i>r</i>	double & Reference to radial vector (output value).
<i>cth</i>	double & Reference to the cosine of the azimuth coordinate (output value).
<i>sth</i>	double & Reference to the sine of the azimuth coordinate (output value).
<i>cph</i>	double & Reference to the cosine of the elevation coordinate (output value).
<i>sph</i>	double & Reference to the sine of the elevation coordinate (output value).

9.48.1.14 r3j000()

```
void r3j000 (
    int j2,
    int j3,
    double * rac3j )
```

This function calculates the $3j(J,J2,J3;0,0,0)$ symbol for the Clebsch-Gordan coefficients. See Appendix a.3.1 in Borghese, Denti & Saija (2007).

Parameters

<i>j2</i>	int
<i>j3</i>	int
<i>rac3j</i>	double * Vector of 3j symbols.

9.48.1.15 r3jjr()

```
void r3jjr (
    int j2,
    int j3,
    int m2,
    int m3,
    double rac3j[] )
```

This function calculates the $3j(J,J2,J3;-M2-M3,M2,M3)$ symbol for the Clebsch-Gordan coefficients. See Appendix a.3.1 in Borghese, Denti & Saija (2007).

Parameters

in	<i>j2</i>	int Value of J2.
in	<i>j3</i>	int Value of J3.
in	<i>m2</i>	int Value of M2.
in	<i>m3</i>	int Value of M3.
out	<i>rac3j</i>	double[] Vector of 3j symbols.

9.48.1.16 r3jmr()

```
void r3jmr (
    int j1,
    int j2,
    int j3,
    int m1,
    double * rac3j )
```

This function calculates the $3j(J,J2,J3;M1,M,-M1-M)$ symbol for the Clebsch-Gordan coefficients. See Appendix a.3.1 in Borghese, Denti & Saija (2007).

Parameters

<i>j1</i>	int
<i>j2</i>	int
<i>j3</i>	int
<i>m1</i>	int
<i>rac3j</i>	double * Vector of 3j symbols.

9.48.1.17 raba()

```
void raba (
    int le,
    dcomplex ** am0m,
    dcomplex ** w,
    double ** tqce,
    dcomplex ** tqcpe,
    double ** tqcs,
    dcomplex ** tqcps )
```

This function computes radiation torque on on a cluster of spheres as the result of the difference between the extinction and the scattering contributions for a Cartesian coordinate system, as `rabas()`. See Sec. 4.9 in Borghese, Denti & Saija (2007).

Parameters

<i>le</i>	int
<i>am0m</i>	complex double **
<i>w</i>	complex double **
<i>tqce</i>	double **
<i>tqcpe</i>	complex double **
<i>tqcs</i>	double **
<i>tqcps</i>	complex double **

9.48.1.18 rftr()

```
void rftr (
    double * u,
    double * up,
    double * un,
    double * gapv,
    double extins,
    double scatts,
    double & rapr,
    double & cosav,
    double & fp,
    double & fn,
    double & fk,
    double & fx,
    double & fy,
    double & fz )
```


This function computes the Cartesian components of the radiation force exerted on a particle. See Sec. 3.2.1 in Borghese, Denti & Saija (2007).

Parameters

<i>u</i>	double *
<i>up</i>	double *
<i>un</i>	double *
<i>gapv</i>	double *
<i>extins</i>	double
<i>scatts</i>	double
<i>rapr</i>	double &
<i>cosav</i>	double &
<i>fp</i>	double &
<i>fn</i>	double &
<i>fk</i>	double &
<i>fx</i>	double &
<i>fy</i>	double &
<i>fz</i>	double &

9.48.1.19 scr0()

```
void scr0 (
    double vk,
    double exri,
    ParticleDescriptor * c1 )
```

This function computes the scattering, absorption and extinction cross-sections for the spheres composing the cluster, in terms of Mie coefficients, together with the Forward Scattering Amplitude. See Sec. 4.2.1 in Borghese, Denti & Saija (2007).

Parameters

<i>vk</i>	double Wave number
<i>exri</i>	double External medium refractive index.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.48.1.20 scr2()

```
void scr2 (
    double vk,
    double vkarg,
    double exri,
    double * duk,
    ParticleDescriptor * c1 )
```

This function computes the scattering amplitude for single spheres constituting an aggregate. See Sec. 4.2.1 in Borghese, Denti & Saija (2007).

Parameters

<i>vk</i>	double Wave number.
<i>vkarg</i>	double ANNOTATION: Argument of wave number.
<i>exri</i>	double External medium refractive index.
<i>duk</i>	double * ANNOTATION: variation of unit wave vector.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.48.1.21 str()

```
void str (
    double ** rcf,
    ParticleDescriptor * c1 )
```

This function transforms the Cartesian coordinates of the spheres in an aggregate to radial coordinates, then it calls `sphar()` to calculate the vector of spherical harmonics of the incident field.

Parameters

<i>rcf</i>	double ** Matrix of sphere configuration fractional radii.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.48.1.22 tqr()

```
void tqr (
    double * u,
    double * up,
    double * un,
    double * tqev,
    double * tqsv,
    double & tep,
    double & ten,
    double & tek,
    double & tsp,
    double & tsn,
    double & tsk )
```

This function computes the radiation torques resulting from the difference between absorption and scattering contributions, like `rabas()`, but for a coordinate system oriented along the wave vector and its orthogonal axis. See Sec. 4.9 in Borghese, Denti & Saija (2007).

Parameters

<i>u</i>	double *
<i>up</i>	double *
<i>un</i>	double *
<i>tqev</i>	double *
<i>tqsv</i>	double *
<i>tep</i>	double &
<i>ten</i>	double &

Parameters

<i>tek</i>	double &
<i>tsp</i>	double &
<i>tsn</i>	double &
<i>tsk</i>	double &

9.48.1.23 ztm()

```
void ztm (
    dcomplex ** am,
    ParticleDescriptor * c1 )
```

This function computes the single-centered inverted M-matrix appearing in Eq. (5.28) of Borghese, Denti & Saija (2007).

Parameters

<i>am</i>	complex double **
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.49 np_tmcode/src/libnptm/Commons.cpp File Reference

Implementation of the common data structures.

```
#include <cstring>
#include "../include/types.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/Commons.h"
```

9.50 np_tmcode/src/libnptm/Configuration.cpp File Reference

Implementation of the configuration classes.

```
#include <cmath>
#include <cstdio>
#include <exception>
#include <fstream>
#include <hdf5.h>
#include <regex>
#include <string>
#include <cstring>
#include "../include/types.h"
#include "../include/errors.h"
#include "../include/List.h"
#include "../include/Parsers.h"
```

```
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/Constants.h"
#include "../include/file_io.h"
```

Functions

- `int * check_overlaps (ScattererConfiguration *sconf, GeometryConfiguration *gconf, const double tolerance)`
Check the presence of overlapping spheres in an aggregate.
- `double get_overlap (ScattererConfiguration *sconf, GeometryConfiguration *gconf, const int index_0, const int index_1)`
Get the overlap among two spheres.

9.50.1 Function Documentation

9.50.1.1 check_overlaps()

```
int * check_overlaps (
    ScattererConfiguration * sconf,
    GeometryConfiguration * gconf,
    const double tolerance = 0.0 )
```

The theoretical implementation of the code is based on the assumption that the spherical monomers that compose the aggregate do not overlap in space. Small violations of this assumption do not critically affect the results, but they need to be reported. This function performs a scan of the aggregate and returns a summary of any potential overlap, if detected.

Parameters

<i>sconf</i>	<code>ScattererConfiguration</code> * Pointer to a <code>ScattererConfiguration</code> instance.
<i>gconf</i>	<code>GeometryConfiguration</code> * Pointer to a <code>GeometryConfiguration</code> instance.
<i>tolerance</i>	const double A tolerance value below which overlap is ignored.

Returns

overlaps: `int *` A vector containing the number of overlaps and a list of overlapping sphere indices.

9.50.1.2 get_overlap()

```
double get_overlap (
    ScattererConfiguration * sconf,
    GeometryConfiguration * gconf,
    const int index_0,
    const int index_1 )
```

This function computes the thickness of the compenetration between two spheres, in order to compare it with the current tolerance settings.

Parameters

<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration instance.
<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration instance.
<i>index_0</i>	const int Index of the first sphere.
<i>index_1</i>	const int Index of the second sphere.

Returns

overlap: double The thickness of the overlapping layer in meters.

9.51 np_tmcode/src/libnptm/cublas_calls.cpp File Reference

Implementation of the interface with CUBLAS libraries.

9.52 np_tmcode/src/libnptm/file_io.cpp File Reference

Implementation of file I/O operations.

```
#include <exception>
#include <fstream>
#include <regex>
#include <cstring>
#include <string>
#include <hdf5.h>
#include "../include/errors.h"
#include "../include/List.h"
#include "../include/types.h"
#include "../include/file_io.h"
```

9.53 np_tmcode/src/libnptm/inclu_subs.cpp File Reference

C++ implementation of INCLUSION subroutines.

```
#include "../include/types.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/Commons.h"
#include "../include/sph_subs.h"
#include "../include/clu_subs.h"
```

Functions

- void `cnf` (int `n`, `dcomplex` `z`, int `nm`, `dcomplex` *`csj`, `dcomplex` *`csy`)
C++ porting of CNF.
- void `exma` (`dcomplex` **`am`, `ParticleDescriptor` *`c1`)
C++ porting of EXMA.
- void `incms` (`dcomplex` **`am`, double `enti`, `ParticleDescriptor` *`c1`)
C++ porting of INCMS.
- void `indme` (int `i`, int `npnt`, int `npntts`, double `vk`, `dcomplex` `ent`, double `enti`, `dcomplex` `entn`, int &`jer`, int &`lcalc`, `dcomplex` &`arg`, `ParticleDescriptor` *`c1`)
C++ porting of INDME.
- void `instr` (double **`rcf`, `ParticleDescriptor` *`c1`)
C++ porting of INSTR.
- void `ospv` (`ParticleDescriptor` *`c1`, double `vk`, double `sze`, double `exri`, `dcomplex` `entn`, double `enti`, int &`jer`, int &`lcalc`, `dcomplex` &`arg`)
C++ porting of OSPV.

9.53.1 Function Documentation

9.53.1.1 `cnf()`

```
void cnf (
    int n,
    dcomplex z,
    int nm,
    dcomplex * csj,
    dcomplex * csy )
```

Parameters

<code>n</code>	int Bessel y function order.
<code>z</code>	<code>dcomplex</code> Argument of Bessel y function.
<code>nm</code>	int Maximum computed order.
<code>csj</code>	<code>dcomplex</code> * ANNOTATION: Complex spherical J vector.
<code>csy</code>	<code>dcomplex</code> * Complex spherical Bessel functions up to desired order.

9.53.1.2 `exma()`

```
void exma (
    dcomplex ** am,
    ParticleDescriptor * c1 )
```

Parameters

<code>am</code>	<code>dcomplex</code> ** Field transition coefficients matrix.
<code>c1</code>	<code>ParticleDescriptor</code> * Pointer to a <code>ParticleDescriptor</code> instance.

9.53.1.3 incms()

```
void incms (
    dcomplex ** am,
    double enti,
    ParticleDescriptor * c1 )
```

Parameters

<i>am</i>	dcomplex ** Field transition coefficients matrix.
<i>enti</i>	double ANNOTATION: imaginary part of coating dielectric function.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.53.1.4 indme()

```
void indme (
    int i,
    int npnt,
    int npntts,
    double vk,
    dcomplex ent,
    double enti,
    dcomplex entn,
    int & jer,
    int & lcalc,
    dcomplex & arg,
    ParticleDescriptor * c1 )
```

Parameters

<i>i</i>	int 1-based sphere configuration index.
<i>npnt</i>	int ANNOTATION: Number of non transition layer integration points.
<i>npntts</i>	int ANNOTATION: Number of transition layer integrtion points.
<i>vk</i>	double Vacuum wave vector magnitude.
<i>ent</i>	dcomplex ANNOTATION: coating dielectric function.
<i>enti</i>	double ANNOTATION: imaginary part of coating dielectric function.
<i>entn</i>	dcomplex ANNOTATION: coating refractive index.
<i>jer</i>	int & Error code flag.
<i>lcalc</i>	int & Maximum order achieved in calculation.
<i>arg</i>	dcomplex & ANNOTATION: argument of calculation.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.53.1.5 instr()

```
void instr (
    double ** rcf,
    ParticleDescriptor * c1 )
```

Parameters

<i>rcf</i>	double ** Pointer to matrix of fractional radii.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.53.1.6 ospv()

```
void ospv (
    ParticleDescriptor * c1,
    double vk,
    double sze,
    double exri,
    dcomplex entn,
    double enti,
    int & jer,
    int & lcalc,
    dcomplex & arg )
```

Parameters

<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.
<i>vk</i>	double ANNOTATION: vacuum wave number.
<i>sze</i>	double ANNOTATION: size.
<i>exri</i>	double External medium refractive index.
<i>entn</i>	dcomplex Outer sphere refractive index.
<i>enti</i>	double Imaginary part of the outer medium refractive index.
<i>jer</i>	int & Reference to an integer error flag.
<i>lcalc</i>	int & Maximum order achieved in calculation.
<i>arg</i>	dcomplex Complex Bessel function argument.

9.54 np_tmcode/src/libnptm/lapack_calls.cpp File Reference

Implementation of the interface with LAPACK libraries.

9.55 np_tmcode/src/libnptm/logging.cpp File Reference

Implementation of the logging system.

```
#include <cstdio>
#include <string>
#include "../include/logging.h"
```

9.56 np_tmcode/src/libnptm/magma_calls.cpp File Reference

Implementation of the interface with MAGMA libraries.

9.57 np_tmcode/src/libnptm/outputs.cpp File Reference

Implementation of the code output format system.

```
#include <cstdio>
#include <exception>
#include <string>
#include <cstring>
#include <hdf5.h>
#include "../include/errors.h"
#include "../include/List.h"
#include "../include/types.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/file_io.h"
#include "../include/outputs.h"
```

9.58 np_tmcode/src/libnptm/Parsers.cpp File Reference

Implementation of the parsing functions.

```
#include <exception>
#include <fstream>
#include <string>
#include "../include/errors.h"
#include "../include/List.h"
#include "../include/Parsers.h"
```

Functions

- `std::string * load_file (const std::string &file_name, int *count=0)`

Load a text file as a sequence of strings in memory.

9.58.1 Function Documentation

9.58.1.1 load_file()

```
std::string * load_file (
    const std::string & file_name,
    int * count )
```

The configuration of the field expansion code in FORTRAN uses shared memory access and file I/O operations managed by different functions. Although this approach could be theoretically replicated, it is more convenient to handle input and output to distinct files using specific functions. [load_file\(\)](#) helps in the task of handling input such as configuration files or text data structures that need to be loaded entirely. The function performs a line-by line scan of the input file and returns an array of strings that can be later parsed and ingested by the concerned code blocks. An optional pointer to integer allows the function to keep track of the number of file lines that were read, if needed.

Parameters

<i>file_name</i>	string The path of the file to be read.
<i>count</i>	int* Pointer to an integer recording the number of read lines [OPTIONAL, default=NULL].

Returns

array_lines string* An array of strings, one for each input file line.

9.59 np_tmcode/src/libnptm/sph_subs.cpp File Reference

C++ implementation of SPHERE subroutines.

```
#include "../include/types.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/Commons.h"
#include "../include/sph_subs.h"
```

Functions

- void [aps](#) (double ****zpv, int li, int nsph, [ParticleDescriptor](#) *c1, double sqk, double *gaps)
Compute the asymmetry-corrected scattering cross-section of a sphere.
- void [cbf](#) (int n, [dcomplex](#) z, int &nm, [dcomplex](#) *csj)
Complex Bessel Function.
- double [cg1](#) (int lmpml, int mu, int l, int m)
Clebsch-Gordan coefficients.
- void [diel](#) (int npntmo, int ns, int i, int ic, double vk, [ParticleDescriptor](#) *c1)
Compute the continuous variation of the refractive index and of its radial derivative.
- void [dme](#) (int li, int i, int npnt, int npntts, double vk, double exdc, double exri, [ParticleDescriptor](#) *c1, int &jer, int &lcalc, [dcomplex](#) &arg, int last_conf)
Compute Mie scattering coefficients.
- double [envj](#) (int n, double x)
Bessel function calculation control parameters.
- void [mmulc](#) ([dcomplex](#) *vint, double **cmullr, double **cmul)
Compute the Mueller Transformation Matrix.
- int [msta1](#) (double x, int mp)
Starting point for Bessel function magnitude.
- int [msta2](#) (double x, int n, int mp)
Starting point for Bessel function precision.
- void [orunve](#) (double *u1, double *u2, double *u3, int iorth, double torth)
Compute the amplitude of the orthogonal unit vector.
- void [pwma](#) (double *up, double *un, [dcomplex](#) *ylm, int inpol, int lw, int isq, [ParticleDescriptor](#) *c1)
Compute incident and scattered field amplitudes.
- void [rabas](#) (int inpol, int li, int nsph, [ParticleDescriptor](#) *c1, double **tqse, [dcomplex](#) **tqspe, double **tqss, [dcomplex](#) **tqsps)
Compute radiation torques on a single sphere in Cartesian coordinates.
- void [rbf](#) (int n, double x, int &nm, double sj[])

Real Bessel Function.

- void `rkc` (int npntmo, double step, `dcomplex` dcc, double &x, int lpo, `dcomplex` &y1, `dcomplex` &y2, `dcomplex` &dy1, `dcomplex` &dy2)

Soft layer radial function and derivative.

- void `rkt` (int npntmo, double step, double &x, int lpo, `dcomplex` &y1, `dcomplex` &y2, `dcomplex` &dy1, `dcomplex` &dy2, `ParticleDescriptor` *c1)

Transition layer radial function and derivative.

- void `rnf` (int n, double x, int &nm, double *sy)

Spherical Bessel functions.

- void `sphar` (double cosrth, double sinrth, double cosrph, double sinrph, int ll, `dcomplex` *ylm)

Spherical harmonics for given direction.

- void `sscr0` (`dcomplex` &tfsas, int nsph, int lm, double vk, double exri, `ParticleDescriptor` *c1)

Compute scattering, absorption and extinction cross-sections.

- void `sscr2` (int nsph, int lm, double vk, double exri, `ParticleDescriptor` *c1)

C++ Compute the scattering amplitude and the scattered field intensity.

- void `thdps` (int lm, double ****zpv)

Determine the geometrical asymmetry parameter coefficients.

- void `upvmp` (double thd, double phd, int icspnv, double &cost, double &sint, double &cosp, double &sinp, double *u, double *up, double *un)

Compute the unitary vectors onb the polarization plane and its orthogonal direction.

- void `upvsp` (double *u, double *upmp, double *unmp, double *us, double *upsmp, double *unsmp, double *up, double *un, double *ups, double *uns, double *duk, int &isq, int &ibf, double &scand, double &cfmp, double &sfmp, double &cfsp, double &sfsp)

Compute the unitary vector perpendicular to incident and scattering plane.

- void `wmamp` (int iis, double cost, double sint, double cosp, double sinp, int inpol, int lm, int idot, int nsph, double *arg, double *u, double *up, double *un, `ParticleDescriptor` *c1)

Compute meridional plane-referred geometrical asymmetry parameter coefficients.

- void `wmasp` (double cost, double sint, double cosp, double sinp, double costs, double sints, double cosps, double sinps, double *u, double *up, double *un, double *us, double *ups, double *uns, int isq, int ibf, int inpol, int lm, int idot, int nsph, double *argi, double *args, `ParticleDescriptor` *c1)

Compute the scattering plane-referred geometrical asymmetry parameter coefficients.

9.59.1 Function Documentation

9.59.1.1 `aps()`

```
void aps (
    double **** zpv,
    int li,
    int nsph,
    ParticleDescriptor * c1,
    double sqk,
    double * gaps )
```

This function computes the product between the geometrical asymmetry parameter and the scattering cross-section. See Sec. 3.2.1 of Borghese, Denti & Saija (2007).

Parameters

<code>zpv</code>	double **** Geometrical asymmetry parameter coefficients matrix.
<code>li</code>	int Maximum field expansion order.
<code>nsph</code>	int Number of spheres.
<code>c1</code>	<code>ParticleDescriptor</code> * Pointer to a <code>ParticleDescriptor</code> data structure.
<code>sqk</code>	double Scattering wave vector.
<code>gaps</code>	double * Geometrical asymmetry parameter-corrected cross-section.

9.59.1.2 cbf()

```
void cbf (
    int n,
    dcomplex z,
    int & nm,
    dcomplex * csj )
```

This function computes the complex spherical Bessel functions j . It uses the auxiliary functions `msta1()` and `msta2()` to determine the starting point for backward recurrence. This is the CSPHJ implementation of the `specfun` library.

Parameters

<i>n</i>	int Order of the function.
<i>z</i>	complex double Argument of the function.
<i>nm</i>	int & Highest computed order.
<i>csj</i>	complex double * The desired function j .

9.59.1.3 cg1()

```
double cg1 (
    int lmpml,
    int mu,
    int l,
    int m )
```

This function computes the Clebsch-Gordan coefficients for the irreducible spherical tensors. See Sec. 1.6.3 and Table 1.1 in Borghese, Denti & Saija (2007).

Parameters

<i>lmpml</i>	int
<i>mu</i>	int
<i>l</i>	int
<i>m</i>	int

Returns

result: double Clebsh-Gordan coefficient.

9.59.1.4 diel()

```
void diel (
    int npntmo,
    int ns,
    int i,
    int ic,
    double vk,
    ParticleDescriptor * cl )
```

This function implements the continuous variation of the refractive index and of its radial derivative through the materials that constitute the sphere and its surrounding medium. See Sec. 5.2 in Borghese, Denti & Saija (2007).

Parameters

<i>npntmo</i>	int
<i>ns</i>	int
<i>i</i>	int
<i>ic</i>	int
<i>vk</i>	double
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.

9.59.1.5 dme()

```
void dme (
    int li,
    int i,
    int npnt,
    int npntts,
    double vk,
    double exdc,
    double exri,
    ParticleDescriptor * c1,
    int & jer,
    int & lcalc,
    dcomplex & arg,
    int last_conf = 0 )
```

This function determines the L-dependent Mie scattering coefficients a_l and b_l for the cases of homogeneous spheres, radially non-homogeneous spheres and, in case of sphere with dielectric function, sustaining longitudinal waves. See Sec. 5.1 in Borghese, Denti & Saija (2007).

Parameters

<i>li</i>	int Maximum field expansion order.
<i>i</i>	int
<i>npnt</i>	int
<i>npntts</i>	int
<i>vk</i>	double Wave number in scale units.
<i>exdc</i>	double External medium dielectric constant.
<i>exri</i>	double External medium refractive index.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.
<i>jer</i>	int & Reference to integer error code variable.
<i>lcalc</i>	int & Reference to integer variable recording the maximum expansion order accounted for.
<i>arg</i>	complex double &
<i>last_conf</i>	int Last sphere configuration (used by CLUSTER)

9.59.1.6 envj()

```
double envj (
    int n,
    double x )
```

This function determines the control parameters required to calculate the Bessel functions up to the desired precision.

Parameters

<i>n</i>	int
<i>x</i>	double

Returns

result: double

9.59.1.7 mmulc()

```
void mmulc (
    dcomplex * vint,
    double ** cmullr,
    double ** cmul )
```

This function computes the Mueller Transformation Matrix, or Phase Matrix. See Sec. 2.8.1 of Borghese, Denti & Saija (2007).

Parameters

<i>vint</i>	complex double *
<i>cmullr</i>	double **
<i>cmul</i>	double **

9.59.1.8 msta1()

```
int msta1 (
    double x,
    int mp )
```

This function determines the starting point for backward recurrence such that the magnitude of all J and j functions is of the order of 10^{-mp} .

Parameters

<i>x</i>	double Absolute value of the argumetn to J or j .
<i>mp</i>	int Requested order of magnitude.

Returns

result: int The necessary starting point.

9.59.1.9 msta2()

```
int msta2 (
```

```
double x,
int n,
int mp )
```

This function determines the starting point for backward recurrence such that all J and j functions have `mp` significant digits.

Parameters

<i>x</i>	double Absolute value of the argumethn to J or j .
<i>n</i>	int Order of the function.
<i>mp</i>	int Requested number of significant digits.

Returns

result: int The necessary starting point.

9.59.1.10 orunve()

```
void orunve (
    double * u1,
    double * u2,
    double * u3,
    int iorth,
    double torth )
```

This function computes the amplitude of the orthogonal unit vector for a geometry based either on the scattering plane or on the meridional plane. It is used by `upvsp ()` and `upvmp ()`. See Sec. 2.7 in Borghese, Denti & Saija (2007).

Parameters

<i>u1</i>	double *
<i>u2</i>	double *
<i>u3</i>	double *
<i>iorth</i>	int
<i>torth</i>	double

9.59.1.11 pwma()

```
void pwma (
    double * up,
    double * un,
    dcomplex * ylm,
    int inpol,
    int lw,
    int isq,
    ParticleDescriptor * cl )
```

This function computes the amplitudes of the incident and scattered field on the basis of the multi-polar expansion. See Sec. 4.1.1 in Borghese, Denti and Saija (2007).

Parameters

<i>up</i>	double *
<i>un</i>	double *
<i>ylm</i>	complex double * Field polar spherical harmonics.
<i>inpol</i>	int Incident field polarization type (0 - linear; 1 - circular).
<i>lw</i>	int
<i>isq</i>	int
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.

9.59.1.12 rabas()

```
void rabas (
    int inpol,
    int li,
    int nsph,
    ParticleDescriptor * c1,
    double ** tqse,
    dcomplex ** tqspe,
    double ** tqss,
    dcomplex ** tqspss )
```

This function computes radiation torque on a sphere unit as the result of the difference between the extinction and the scattering contributions. See Sec. 4.9 in Borghese, Denti & Saija (2007).

Parameters

<i>inpol</i>	int Incident polarization type (0 - linear; 1 - circular)
<i>li</i>	int Maximum field expansion order.
<i>nsph</i>	int Number of spheres.
<i>c1</i>	ParticleDescriptor * Pointer to ParticleDescriptor data structure.
<i>tqse</i>	double **
<i>tqspe</i>	complex double **
<i>tqss</i>	double **
<i>tqspss</i>	complex double **

9.59.1.13 rbf()

```
void rbf (
    int n,
    double x,
    int & nm,
    double sj[] )
```

This function computes the real spherical Bessel functions j . It uses the auxiliary functions [msta1\(\)](#) and [msta2\(\)](#) to determine the starting point for backward recurrence. This is the SPHJ implementation of the `specfun` library.

Parameters

<i>n</i>	int Order of the function.
----------	----------------------------

Parameters

<i>x</i>	double Argument of the function.
<i>nm</i>	int & Highest computed order.
<i>sj</i>	double[] The desired function <i>j</i> .

9.59.1.14 rkc()

```
void rkc (
    int npntmo,
    double step,
    dcomplex dcc,
    double & x,
    int lpo,
    dcomplex & y1,
    dcomplex & y2,
    dcomplex & dy1,
    dcomplex & dy2 )
```

This function determines the radial function and its derivative for a soft layer in a radially non homogeneous sphere. See Sec. 5.1 in Borghese, Denti & Saija (2007).

Parameters

<i>npntmo</i>	int
<i>step</i>	double
<i>dcc</i>	complex double
<i>x</i>	double &
<i>lpo</i>	int
<i>y1</i>	complex double &
<i>y2</i>	complex double &
<i>dy1</i>	complex double &
<i>dy2</i>	complex double &

9.59.1.15 rkt()

```
void rkt (
    int npntmo,
    double step,
    double & x,
    int lpo,
    dcomplex & y1,
    dcomplex & y2,
    dcomplex & dy1,
    dcomplex & dy2,
    ParticleDescriptor * c1 )
```

This function determines the radial function and its derivative for a transition layer in a radially non homogeneous sphere. See Sec. 5.1 in Borghese, Denti & Saija (2007).

Parameters

<i>npntmo</i>	int
<i>step</i>	double
<i>x</i>	double &
<i>lpo</i>	int
<i>y1</i>	complex double &
<i>y2</i>	complex double &
<i>dy1</i>	complex double &
<i>dy2</i>	complex double &
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor instance.

9.59.1.16 rnf()

```
void rnf (
    int n,
    double x,
    int & nm,
    double * sy )
```

This function computes the spherical Bessel functions y . It adopts the SPHJY implementation of the `specfun` library.

Parameters

<i>n</i>	int Order of the function (from 0 up).
<i>x</i>	double Argument of the function ($x > 0$).
<i>nm</i>	int & Highest computed order.
<i>sy</i>	double * The desired function y .

9.59.1.17 sphar()

```
void sphar (
    double cosrth,
    double sinrth,
    double cosrph,
    double sinrph,
    int ll,
    dcomplex * ylm )
```

This function computes the field spherical harmonics for a given direction. See Sec. 1.5.2 in Borghese, Denti & Saija (2007).

Parameters

<i>cosrth</i>	double Cosine of direction's elevation.
<i>sinrth</i>	double Sine of direction's elevation.
<i>cosrph</i>	double Cosine of direction's azimuth.
<i>sinrph</i>	double Sine of direction's azimuth.
<i>ll</i>	int L value expansion order.
<i>ylm</i>	complex double * The requested spherical harmonics.

9.59.1.18 sscr0()

```
void sscr0 (
    dcomplex & tfsas,
    int nsph,
    int lm,
    double vk,
    double exri,
    ParticleDescriptor * c1 )
```

This function computes the scattering, absorption and extinction cross-sections in terms of Forward Scattering Amplitudes. See Sec. 4.2.1 in Borghese, Denti & Saija (2007).

Parameters

<i>tfsas</i>	complex double &
<i>nsph</i>	int Number of spheres.
<i>lm</i>	int Maximum field expansion order.
<i>vk</i>	double Wave number in scale units.
<i>exri</i>	double External medium refractive index.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.

9.59.1.19 sscr2()

```
void sscr2 (
    int nsph,
    int lm,
    double vk,
    double exri,
    ParticleDescriptor * c1 )
```

The role of this function is to compute the scattering amplitude and the intensity of the scattered field. See Sec. 4.2 in Borghese, Denti & Saija (2007).

Parameters

<i>nsph</i>	int Number of spheres.
<i>lm</i>	int Maximum field expansion order.
<i>vk</i>	double Wave number in scale units.
<i>exri</i>	double External medium refractive index.
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.

9.59.1.20 thdps()

```
void thdps (
    int lm,
    double **** zpv )
```

This function computes the coefficients that enter the definition of the geometrical asymmetry parameter based on the L-value of the field expansion order. See Sec. 3.2.1 in Borghese, Denti & Saija (2007).

Parameters

<i>lm</i>	int Maximum field expansion order.
<i>zpv</i>	double **** Matrix of geometrical asymmetry parameter coefficients.

9.59.1.21 upvmp()

```
void upvmp (
    double thd,
    double phd,
    int icspnv,
    double & cost,
    double & sint,
    double & cosp,
    double & sinp,
    double * u,
    double * up,
    double * un )
```

This function computes the unitary vectors lying on the polarization plane and on its orthogonal direction, to optimize the identification of the scattering geometry. See Sec. 2.3 in Borghese, Denti & Saija (2007).

Parameters

<i>thd</i>	double
<i>phd</i>	double
<i>icspnv</i>	int
<i>cost</i>	double
<i>sint</i>	double
<i>cosp</i>	double
<i>sinp</i>	double
<i>u</i>	double *
<i>up</i>	double *
<i>un</i>	double *

9.59.1.22 upvsp()

```
void upvsp (
    double * u,
    double * upmp,
    double * unmp,
    double * us,
    double * upsm,
    double * unsm,
    double * up,
    double * un,
    double * ups,
    double * uns,
    double * duk,
    int & isq,
```

```

int & ibf,
double & scand,
double & cfmp,
double & sfmp,
double & cfsp,
double & sfsp )

```

This function computes the unitary vector perpendicular to the incident and scattering plane in a geometry based on the scattering plane. It uses [orunve\(\)](#). See Sec. 2.7 in Borghese, Denti & Saija (2007).

Parameters

<i>u</i>	double *
<i>upmp</i>	double *
<i>unmp</i>	double *
<i>us</i>	double *
<i>upsmp</i>	double *
<i>unsmp</i>	double *
<i>up</i>	double *
<i>un</i>	double *
<i>ups</i>	double *
<i>uns</i>	double *
<i>duk</i>	double *
<i>isq</i>	int &
<i>ibf</i>	int &
<i>scand</i>	double & Scattering angle in degrees.
<i>cfmp</i>	double &
<i>sfmp</i>	double &
<i>cfsp</i>	double &
<i>sfsp</i>	double &

9.59.1.23 wmamp()

```

void wmamp (
    int iis,
    double cost,
    double sint,
    double cosp,
    double sinp,
    int inpol,
    int lm,
    int idot,
    int nsph,
    double * arg,
    double * u,
    double * up,
    double * un,
    ParticleDescriptor * cl )

```

This function computes the coefficients that define the geometrical symmetry parameter as defined with respect to the meridional plane. It makes use of [sphar\(\)](#) and [pwma\(\)](#). See Sec. 3.2.1 in Borghese, Denti & Saija (2007).

Parameters

<i>iis</i>	int
<i>cost</i>	double Cosine of the elevation angle.
<i>sint</i>	double Sine of the elevation angle.
<i>cosp</i>	double Cosine of the azimuth angle.
<i>sinp</i>	double Sine of the azimuth angle.
<i>inpol</i>	int Incident field polarization type (0 - linear; 1 - circular).
<i>lm</i>	int Maximum field expansion orde.
<i>idot</i>	int
<i>nsph</i>	int Number of spheres.
<i>arg</i>	double *
<i>u</i>	double *
<i>up</i>	double *
<i>un</i>	double *
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.

9.59.1.24 wmasp()

```

void wmasp (
    double cost,
    double sint,
    double cosp,
    double sinp,
    double costs,
    double sints,
    double cosps,
    double sinps,
    double * u,
    double * up,
    double * un,
    double * us,
    double * ups,
    double * uns,
    int isq,
    int ibf,
    int inpol,
    int lm,
    int idot,
    int nsph,
    double * argi,
    double * args,
    ParticleDescriptor * c1 )

```

This function computes the coefficients that define the geometrical asymmetry parameter based on the L-value with respect to the scattering plane. It uses [sphar\(\)](#) and [pwma\(\)](#). See Sec. 3.2.1 in Borghese, Denti and Saija (2007).

Parameters

<i>cost</i>	double Cosine of elevation angle.
<i>sint</i>	double Sine of elevation angle.
<i>cosp</i>	double Cosine of azimuth angle.

Parameters

<i>sinp</i>	double Sine of azimuth angle.
<i>costs</i>	double Cosine of scattering elevation angle.
<i>sints</i>	double Sine of scattering elevation angle.
<i>cosps</i>	double Cosine of scattering azimuth angle.
<i>sinps</i>	double Sine of scattering azimuth angle.
<i>u</i>	double *
<i>up</i>	double *
<i>un</i>	double *
<i>us</i>	double *
<i>ups</i>	double *
<i>uns</i>	double *
<i>isq</i>	int
<i>ibf</i>	int
<i>inpol</i>	int Incident field polarization (0 - linear; 1 -circular).
<i>lm</i>	int Maximum field expansion order.
<i>idot</i>	int
<i>nsph</i>	int Number of spheres.
<i>argi</i>	double *
<i>args</i>	double *
<i>c1</i>	ParticleDescriptor * Pointer to a ParticleDescriptor data structure.

9.60 np_tmcode/src/libnptm/tfrfme.cpp File Reference

Implementation of the trapping calculation objects.

```
#include <exception>
#include <fstream>
#include <hdf5.h>
#include <string>
#include "../include/types.h"
#include "../include/errors.h"
#include "../include/List.h"
#include "../include/tfrfme.h"
#include "../include/file_io.h"
```

9.61 np_tmcode/src/libnptm/tra_subs.cpp File Reference

C++ implementation of TRAPPING subroutines.

```
#include <cmath>
#include <fstream>
#include "../include/types.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/Commons.h"
#include "../include/sph_subs.h"
#include "../include/tfrfme.h"
#include "../include/tra_subs.h"
```


Functions

- void `camp` (`dcomplex` *ac, `dcomplex` **am0m, `dcomplex` *ws, `CIL` *cil)
- void `czamp` (`dcomplex` *ac, `dcomplex` **amd, int **indam, `dcomplex` *ws, `CIL` *cil)
- void `ffr` (double ****zpv, `dcomplex` *ac, `dcomplex` *ws, double *fffe, double *fffs, `CIL` *cil, `CCR` *ccr)
- void `frr` (`dcomplex` *ac, `dcomplex` *ws, double *fffe, double *fffs, `CIL` *cil)
- `dcomplex` * `frfmer` (int nk, double vkm, double vknmx, double apfafa, double tra, double spd, double rir, double ftn, int le, int lmode, double pmf, `Swap1` *tt1, `Swap2` *tt2)
- void `pwmalp` (`dcomplex` **w, double *up, double *un, `dcomplex` *ylm, int lw)
- void `samp` (`dcomplex` *ac, `dcomplex` *tmsm, `dcomplex` *tmse, `dcomplex` *ws, `CIL` *cil)
- void `sampo` (`dcomplex` *ac, `dcomplex` **tms, `dcomplex` *ws, `CIL` *cil)
- void `wamff` (`dcomplex` *wk, double x, double y, double z, int lm, double apfafa, double tra, double spd, double rir, double ftn, int lmode, double pmf)

9.61.1 Function Documentation

9.61.1.1 `camp()`

```
void camp (
    dcomplex * ac,
    dcomplex ** am0m,
    dcomplex * ws,
    CIL * cil )
```

C++ porting of CAMP

Parameters

<i>ac</i>	complex double * ANNOTATION: vector of complex amplitudes.
<i>am0m</i>	complex double ** ANNOTATION: T-matrix of the particle.
<i>ws</i>	complex double * ANNOTATION: scattering amplitudes.
<i>cil</i>	<code>CIL</code> * Pointer to a <code>CIL</code> structure.

This function builds the AC vector using AM0M and WS.

9.61.1.2 `czamp()`

```
void czamp (
    dcomplex * ac,
    dcomplex ** amd,
    int ** indam,
    dcomplex * ws,
    CIL * cil )
```

C++ porting of CZAMP

Parameters

<i>ac</i>	Vector of complex. ANNOTATION: vector of complex amplitudes.
<i>amd</i>	Matrix of complex. ANNOTATION: T-matrix.
<i>indam</i>	int **. ANNOTATION: map of matrix indexes.
<i>ws</i>	Vector of complex. ANNOTATION: scattering amplitudes.
Generated by Doxygen	<code>CIL</code> * Pointer to a <code>CIL</code> structure.

This function builds the AC vector using AMD, INDAM and WS.

9.61.1.3 ffrf()

```
void ffrf (
    double **** zpv,
    dcomplex * ac,
    dcomplex * ws,
    double * fffe,
    double * fffs,
    CIL * cil,
    CCR * ccr )
```

C++ porting of FFRF

Parameters

<i>zpv</i>	double ****. ANNOTATION: asymmetry tensor.
<i>ac</i>	complex double * ANNOTATION: vector of complex amplitudes.
<i>ws</i>	complex double * ANNOTATION: scattering amplitudes.
<i>ffe</i>	double *. ANNOTATION: extinction contributions to force.
<i>ffs</i>	double *. ANNOTATION: scattering contributions to force.
<i>cil</i>	CIL * Pointer to a CIL structure.
<i>ccr</i>	CCR * Pointer to a CCR structure.

9.61.1.4 ffrt()

```
void ffrt (
    dcomplex * ac,
    dcomplex * ws,
    double * ffte,
    double * ffts,
    CIL * cil )
```

C++ porting of FFRF

Parameters

<i>ac</i>	complex double * ANNOTATION: vector of complex amplitudes.
<i>ws</i>	complex double * ANNOTATION: scattering amplitudes.
<i>fte</i>	double *. ANNOTATION: extinction contributions to torque.
<i>fts</i>	double *. ANNOTATION: scattering contributions to torque.
<i>cil</i>	CIL * Pointer to a CIL structure.

9.61.1.5 frfmer()

```
dcomplex * frfmer (
    int nk,
    int nkv,
```

```

double vkm,
double vknmx,
double apfafa,
double tra,
double spd,
double rir,
double ftcn,
int le,
int lmode,
double pmf,
Swap1 * tt1,
Swap2 * tt2 )

```

C++ porting of FRFMER

Parameters

<i>nk</i>	int ANNOTATION: number of wave vectors.
<i>vkm</i>	double ANNOTATION: wave number.
<i>vknmx</i>	double ANNOTATION: maximum wave number.
<i>apfafa</i>	double ANNOTATION: normalized aperture.
<i>tra</i>	double ANNOTATION: lens transmission.
<i>spd</i>	double ANNOTATION: cover thickness.
<i>rir</i>	double ANNOTATION: external over oil refractive index ratio.
<i>ftcn</i>	double ANNOTATION: refraction factor.
<i>le</i>	int ANNOTATION: external multipole truncation order.
<i>lmode</i>	int ANNOTATION: laser mode.
<i>pmf</i>	double ANNOTATION: aperture correction.
<i>tt1</i>	Swap1 * Pointer to first swap object.
<i>tt2</i>	Swap2 * Pointer to second swap object.

Returns

wk: complex double * ANNOTATION: incident amplitudes.

9.61.1.6 pwmalp()

```

void pwmalp (
    dcomplex ** w,
    double * up,
    double * un,
    dcomplex * ylm,
    int lw )

```

C++ porting of PWMALP

Parameters

<i>w</i>	complex double * ANNOTATION: amplitudes.
<i>up</i>	double * ANNOTATION: unit polarization vector.
<i>un</i>	double * ANNOTATION: unit normal vector.
<i>ylm</i>	complex double * Field vector spherical harmonics.
<i>lw</i>	int ANNOTATION: order of amplitudes.

9.61.1.7 samp()

```
void samp (
    dcomplex * ac,
    dcomplex * tmsm,
    dcomplex * tmse,
    dcomplex * ws,
    CIL * cil )
```

C++ porting of SAMP

Parameters

<i>ac</i>	complex double * ANNOTATION: vector of complex amplitudes.
<i>tmsm</i>	complex double * ANNOTATION: T-matrix M components.
<i>tmse</i>	complex double * ANNOTATION: T-matrix E components.
<i>ws</i>	complex double * ANNOTATION: scattering amplitudes.
<i>cil</i>	CIL * Pointer to a CIL structure.

This function builds the AC vector using TMSM, TMSE and WS.

9.61.1.8 sampoa()

```
void sampoa (
    dcomplex * ac,
    dcomplex ** tms,
    dcomplex * ws,
    CIL * cil )
```

C++ porting of SAMPOA

Parameters

<i>ac</i>	complex double * ANNOTATION: vector of complex amplitudes.
<i>tms</i>	complex double ** ANNOTATION: T-matrix.
<i>ws</i>	complex double * ANNOTATION: scattering amplitudes.
<i>cil</i>	CIL * Pointer to a CIL structure.

This function builds the AC vector using TMS and WS.

9.61.1.9 wamff()

```
void wamff (
    dcomplex * wk,
    double x,
    double y,
    double z,
    int lm,
    double apfafa,
    double tra,
```

```
double spd,
double rir,
double ftcn,
int lmode,
double pmf )
```

C++ porting of WAMFF

Parameters

<i>wk</i>	complex double * ANNOTATION: incident amplitudes.
<i>x</i>	double ANNOTATION: X coordinate.
<i>y</i>	double ANNOTATION: Y coordinate.
<i>z</i>	double ANNOTATION: Z coordinate.
<i>lm</i>	int ANNOTATION: maximum multipole truncation order.
<i>apfafa</i>	double ANNOTATION: Normalized aperture.
<i>tra</i>	double ANNOTATION: Lens transmission.
<i>spd</i>	double ANNOTATION: Cover slip thickness.
<i>rir</i>	double ANNOTATION: External over oil refractive index ratio.
<i>ftcn</i>	double ANNOTATION: Refraction factor.
<i>lmode</i>	int ANNOTATION: Laser mode.
<i>pmf</i>	double ANNOTATION: aperture correction.

9.62 np_tmcode/src/libnptm/TransitionMatrix.cpp File Reference

Implementation of the Transition Matrix structure.

```
#include <exception>
#include <fstream>
#include <hdf5.h>
#include <string>
#include "../include/types.h"
#include "../include/errors.h"
#include "../include/List.h"
#include "../include/TransitionMatrix.h"
#include "../include/file_io.h"
```

9.63 np_tmcode/src/libnptm/utils.cpp File Reference

Implementation of auxiliary code utilities.

```
#include <fstream>
#include <cstring>
#include <hdf5.h>
#include "../include/types.h"
#include "../include/errors.h"
#include "../include/List.h"
#include "../include/file_io.h"
#include "../include/utils.h"
```

Functions

- double `get_ram_overhead()`
Obtain an estimate of the RAM overhead factor for optimization.
- int `write_dcomplex_matrix` (`VirtualAsciiFile` *af, `dcomplex` **mat, int rows, int columns, const std::string &format, int first_index)
Write a double complex matrix to a text file.
- int `write_matrix_as_ppm` (`dcomplex` *A, int64_t m, int64_t n, const std::string &file_name, const std::string &mode, int row_bin, int col_bin)
Draw a PPM representation of a matrix.

9.63.1 Function Documentation

9.63.1.1 `get_ram_overhead()`

```
double get_ram_overhead ( )
```

Code speed-up optimization usually comes at the cost of increased memory requirements. While this should not generally be a serious issue, it can become such in case of models that require large amounts of memory to be handled. This function tests the code build configuration to provide an zero-order estimate of the weight of these overheads to protect the host system from saturating the RAM.

Returns

factor: double The multiplicative factor to be applied to data size.

9.63.1.2 `write_dcomplex_matrix()`

```
int write_dcomplex_matrix (
    VirtualAsciiFile * af,
    dcomplex ** mat,
    int rows,
    int columns,
    const std::string & format = " %5d %5d (%17.81E,%17.81E)\n",
    int first_index = 1 )
```

Parameters

<i>af</i>	<code>VirtualAsciiFile</code> * Pointer to an existing <code>VirtualAsciiFile</code> . [IN]
<i>mat</i>	<code>dcomplex</code> ** Pointer to the matrix. [IN]
<i>rows</i>	int Number of rows in the matrix. [IN]
<i>columns</i>	int Number of columns in the matrix. [IN]
<i>format</i>	const string& Format of the line (default is " %5d %5d (%17.81E,%17.81E)\n"). [IN]
<i>first_index</i>	int Index of the first element (default is 1, i.e. base 1 FORTRAN array notation). [IN]

Returns

result: int An exit code (0 if successful).

9.63.1.3 write_matrix_as_ppm()

```
int write_matrix_as_ppm (
    dcomplex * A,
    int64_t m,
    int64_t n,
    const std::string & file_name,
    const std::string & mode = "MAG",
    int row_bin = 1,
    int col_bin = 1 )
```

Parameters

in	<i>A</i>	const dcomplex * Vector representation of the matrix.
in	<i>m</i>	int64_t Number of rows in the matrix.
in	<i>n</i>	int64_t Number of columns in the matrix.
in	<i>file_name</i>	const string& Reference to a string with the name of the output file.
in	<i>mode</i>	const string& Reference to a string for the value to be represented ("MAG" to draw magnitudes, "RE" to draw real parts, "IM" to draw imaginary parts. Optional, default is "MAG").
in	<i>row_bin</i>	int Number of rows to bin together (optional, default is 1).
in	<i>col_bin</i>	int Number of columns to bin together (optional, default is 1).

Returns

result: int An exit code (0 if successful).

9.64 np_tmcode/src/sphere/np_sphere.cpp File Reference

Single sphere scattering problem handler.

```
#include <stdio>
#include <string>
#include "../include/types.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/Commons.h"
```

Functions

- void [sphere](#) (const string &config_file, const string &data_file, const string &output_path, const [mixMPI](#) *mpidata)
C++ implementation of SPH.
- int [main](#) (int argc, char **argv)
Main program entry point.

9.64.1 Detailed Description

This program emulates the execution work-flow originally handled by the FORTRAN EDFB and SPH codes, which undertook the task of solving the scattering calculation for the case of a single, possibly multi-layered sphere. The program is designed to work in two modes: emulation and enhanced. The emulation mode is activated by invoking the program without arguments. In this case, the code looks for hard-coded default input and writes output in the execution folder, replicating the behaviour of the original FORTRAN code. Advanced mode, instead, is activated by passing command line arguments that locate the desired input files and a valid folder to write the output into. The emulation mode is useful for testing, while the advanced mode implements the possibility to change input and output options, without having to modify the code.

9.64.2 Function Documentation

9.64.2.1 main()

```
int main (
    int argc,
    char ** argv )
```

This is the starting point of the execution flow. Here we may choose how to configure the code, e.g. by loading a legacy configuration file or some otherwise formatted configuration data set. The code can be linked to a launcher script or to a GUI oriented application that performs the configuration and runs the main program.

Parameters

<i>argc</i>	int The number of arguments given in command-line.
<i>argv</i>	char ** The vector of command-line arguments.

Returns

result: int An exit code passed to the OS (0 for succesful execution).

9.64.2.2 sphere()

```
void sphere (
    const string & config_file,
    const string & data_file,
    const string & output_path,
    const mixMPI * mpidata ) [extern]
```

Parameters

<i>config_file</i>	string Name of the configuration file.
<i>data_file</i>	string Name of the input data file.
<i>output_path</i>	string Directory to write the output files in.
<i>mpidata</i>	const mixMPI * Pointer to a mixMPI data structure.

9.65 np_tmcode/src/sphere/sphere.cpp File Reference

Implementation of the single sphere calculation.

```
#include <cstdio>
#include <exception>
#include <fstream>
#include <hdf5.h>
#include <string>
#include "../include/types.h"
#include "../include/errors.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/Constants.h"
#include "../include/sph_subs.h"
#include "../include/TransitionMatrix.h"
#include "../include/List.h"
#include "../include/file_io.h"
#include "../include/outputs.h"
#include "../include/IterationData.h"
```

Functions

- int [sphere_jxi488_cycle](#) (int jxi488, [ScattererConfiguration](#) *sconf, [GeometryConfiguration](#) *gconf, [ScatteringAngles](#) *sa, [SphereIterationData](#) *sid, [SphereOutputInfo](#) *oi, const string &output_path)
Main calculation loop.
- void [sphere](#) (const string &config_file, const string &data_file, const string &output_path, const [mixMPI](#) *mpidata)
C++ implementation of SPH.

9.65.1 Function Documentation

9.65.1.1 sphere()

```
void sphere (
    const string & config_file,
    const string & data_file,
    const string & output_path,
    const mixMPI * mpidata )
```

Parameters

<i>config_file</i>	string Name of the configuration file.
<i>data_file</i>	string Name of the input data file.
<i>output_path</i>	string Directory to write the output files in.
<i>mpidata</i>	const mixMPI * Pointer to a mixMPI data structure.

9.65.1.2 sphere_jxi488_cycle()

```
int sphere_jxi488_cycle (
    int jxi488,
    ScattererConfiguration * sconf,
    GeometryConfiguration * gconf,
    ScatteringAngles * sa,
    SphereIterationData * sid,
    SphereOutputInfo * oi,
    const string & output_path )
```

Parameters

<i>jxi488</i>	int Wavelength loop index.
<i>sconf</i>	ScattererConfiguration * Pointer to a ScattererConfiguration object.
<i>gconf</i>	GeometryConfiguration * Pointer to a GeometryConfiguration object.
<i>sa</i>	ScatteringAngles * Pointer to a ScatteringAngles object.
<i>sid</i>	SphereIterationData * Pointer to a SphereIterationData object.
<i>oi</i>	SphereOutputInfo * Pointer to a SphereOutputInfo object.
<i>output_path</i>	const string & Path to the output directory.

9.66 np_tmcode/src/testing/test_file_io.cpp File Reference

```
#include <cstdio>
#include <exception>
#include <hdf5.h>
#include <string>
#include "../include/types.h"
#include "../include/errors.h"
#include "../include/List.h"
#include "../include/file_io.h"
```

Functions

- int **main** ()
Main program execution body.

9.67 np_tmcode/src/testing/test_TEDF.cpp File Reference

```
#include <cstdio>
#include <exception>
#include <hdf5.h>
#include <string>
#include "../include/types.h"
#include "../include/errors.h"
#include "../include/List.h"
#include "../include/file_io.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
```

Functions

- `int main (int argc, char **argv)`
Main program execution body.

9.67.1 Function Documentation

9.67.1.1 main()

```
int main (
    int argc,
    char ** argv )
```

This program executes a test to compare whether three configuration instances, loaded respectively from an EDFB configuration file, a legacy binary, or a HDF5 binary are actually equivalent. The test writes a result message to `stdout` then it returns 0 (OS flag for a successful process) or some kind of error code, depending on whether the test files were found all equal or not. The test accepts three command line arguments: the name of the EDFB configuration file, the name of the legacy binary file and the name of the HDF5 binary configuration file.

Parameters

<code>argc</code>	<code>int</code> Number of command line arguments
<code>argv</code>	<code>char **</code> Array of command argument character strings.

Returns

result: `int` Can be: 0 (all files equal); 1 (EDFB and legacy binary are different); 10 (EDFB and HDF5 are different); 100 (legacy and HDF5 are different). In case more differences are found, the error codes sum up together (e.g. 111 means all files are different).

9.68 np_tmcode/src/testing/test_TTMS.cpp File Reference

```
#include <cstdio>
#include <exception>
#include <hdf5.h>
#include <string>
#include "../include/types.h"
#include "../include/errors.h"
#include "../include/List.h"
#include "../include/file_io.h"
#include "../include/TransitionMatrix.h"
```

Functions

- `int main (int argc, char **argv)`
Main program execution body.

9.68.1 Function Documentation

9.68.1.1 main()

```
int main (
    int argc,
    char ** argv )
```

This program executes a test to compare whether two transition matrix instances, loaded respectively from a legacy and a HDF5 binary file are actually equivalent. The test writes a result message to `stdout` then it returns 0 (OS flag for a successful process) or 1 (OS flag for failing process) depending on whether the two instances were considered equivalent or not.

Parameters

<code>argc</code>	int Number of command line arguments
<code>argv</code>	char ** Array of command argument character strings.

Returns

result: int Can be 0 (files are equal) or 1 (files are different).

9.69 np_tmcode/src/trapping/cfrfme.cpp File Reference

```
#include <chrono>
#include <cstdio>
#include <exception>
#include <fstream>
#include <regex>
#include <string>
#include "../include/types.h"
#include "../include/Parsers.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/Commons.h"
#include "../include/sph_subs.h"
#include "../include/tfrfme.h"
#include "../include/tra_subs.h"
#include "../include/errors.h"
```

Functions

- void [frfme](#) (string data_file, string output_path)
C++ implementation of FRFME.

9.69.1 Detailed Description

C++ implementation of FRFME functions.

9.69.2 Function Documentation

9.69.2.1 frfme()

```
void frfme (
    string data_file,
    string output_path )
```

Parameters

<i>data_file</i>	string Name of the input data file.
<i>output_path</i>	string Directory to write the output files in.

9.70 np_tmcode/src/trapping/clffft.cpp File Reference

C++ implementation of LFFFT functions.

```
#include <chrono>
#include <cmath>
#include <cstdio>
#include <exception>
#include <fstream>
#include <hdf5.h>
#include <regex>
#include <string>
#include "../include/types.h"
#include "../include/Parsers.h"
#include "../include/logging.h"
#include "../include/Configuration.h"
#include "../include/Commons.h"
#include "../include/sph_subs.h"
#include "../include/tfrfme.h"
#include "../include/tra_subs.h"
#include "../include/errors.h"
#include "../include/List.h"
#include "../include/TransitionMatrix.h"
#include "../include/file_io.h"
#include "../include/outputs.h"
```

Functions

- void [lffft](#) (string data_file, string output_path)
C++ implementation of LFFFT.

9.70.1 Function Documentation

9.70.1.1 lffft()

```
void lffft (
    string data_file,
    string output_path )
```

Parameters

<i>data_file</i>	string Name of the input data file.
<i>output_path</i>	string Directory to write the output files in.

9.71 np_tmcode/src/trapping/np_trapping.cpp File Reference

Trapping problem handler.

```
#include <chrono>
#include <cstdio>
#include <string>
#include "../include/logging.h"
```

Functions

- void [frfme](#) (string data_file, string output_path)
C++ implementation of FRFME.
- void [lffft](#) (string data_file, string output_path)
C++ implementation of LFFFT.
- int [main](#) (int argc, char **argv)
Main program execution body.

9.71.1 Function Documentation

9.71.1.1 frfme()

```
void frfme (
    string data_file,
    string output_path ) [extern]
```

Parameters

<i>data_file</i>	string Name of the input data file.
<i>output_path</i>	string Directory to write the output files in.

9.71.1.2 lffft()

```
void lffft (
    string data_file,
    string output_path ) [extern]
```

Parameters

<i>data_file</i>	string Name of the input data file.
<i>output_path</i>	string Directory to write the output files in.

9.71.1.3 main()

```
int main (
    int argc,
    char ** argv )
```

Parameters

<i>argc</i>	int
<i>argv</i>	char **

Returns

result: int

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